MILITARY STANDARD

SWITCHES AND ASSOCIATED HARDWARE, SELECTION AND USE OF

TO ALL HOLDERS OF MIL-STD-1132A:

1. THE FOLLOWING SECTIONS (REVISED AND NEW) ARE TO BE ADDED TO MIL-STD-1132A

SECTION	DATE	SUPERSEDED SECTION	DATE
104B (Rev)		104A	20 June 1977
114A (Rev)		114	6 June 1979
115 (New)			

2. THE FOLLOWING PAGES OF MIL-STD-1132A ARE NEW OR HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
iii	23 March 1976	iii	REPRINTED WITHOUT CHANGE
iv		iv	6 June 1979
1	23 March 1976	1	REPRINTED WITHOUT CHANGE
2		2	6 June 1979
5		5	23 March 1976
6	23 March 1976	6	REPRINTED WITHOUT CHANGE
7	23 March 1976	7	REPRINTED WITHOUT CHANGE
8		8	23 March 1976
9		9	6 June 1979
10		10	23 March 1976
13		13	23 March 1976
100.1		100.1	6 June 1979
102.1		102.1	20 June 1977
102.2	20 June 1977	102.2	REPRINTED WITHOUT CHANGE
102.5	20 June 1977	102.5	REPRINTED WITHOUT CHANGE
102.5A			- - -
102.5B			- - -
102.5C			- - -
102.5D			- - -
102.6	20 June 1977	102.6	REPRINTED WITHOUT CHANGE
102.24A			- - -
102.31	20 June 1977	102.31	REPRINTED WITHOUT CHANGE
102.31A			- - -
102.31B			- - -
102.32	20 June 1977	102.32	REPRINTED WITHOUT CHANGE
102.35 thru 102.73			- - -
107.1		107.1	20 June 1977
107.1A			- - -
107.2	23 March 1976	107.2	REPRINTED WITHOUT CHANGE
107.3		107.3	23 March 1976
107.4	23 March 1976	107.4	REPRINTED WITHOUT CHANGE
107.5	23 March 1976	107.5	REPRINTED WITHOUT CHANGE
107.6		107.6	23 March 1976
107.7		107.7	23 March 1976
107.8	23 March 1976	107.8	REPRINTED WITHOUT CHANGE
107.11		107.11	23 March 1976
107.12	23 March 1976	107.12	REPRINTED WITHOUT CHANGE
107.13	23 March 1976	107.13	REPRINTED WITHOUT CHANGE
107.14		107.14	23 March 1976

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
107.15		107.15	23 March 1976
107.16	23 March 1976	107.16	REPRINTED WITHOUT CHANGE
107.29		107.29	23 March 1976
107.30	20 June 1977	107.30	REPRINTED WITHOUT CHANGE
107.33	20 June 1977	107.33	REPRINTED WITHOUT CHANGE
107.34 thru 107.59			- - -
109.1		109.1	23 March 1976
109.2	23 March 1976	109.2	REPRINTED WITHOUT CHANGE
109.39 thru 109.44			- - -
110.1		110.1	23 March 1976
110.2		110.2	20 June 1977
110.9		110.9	23 March 1976
110.10		110.10	23 March 1976
110.99	23 March 1976	110.99	REPRINTED WITHOUT CHANGE
110.100		110.100	23 March 1976
110.101		110.101	23 March 1976
110.102	20 June 1977	110.102	REPRINTED WITHOUT CHANGE
110.105	20 June 1977	110.105	REPRINTED WITHOUT CHANGE
110.106 thru 110.110			- - -
111.1		111.1	20 June 1977
111.2	20 June 1977	111.2	REPRINTED WITHOUT CHANGE
111.29	20 June 1977	111.29	REPRINTED WITHOUT CHANGE
111.30			- - -
111.31			- - -
112.1	23 March 1976	112.1	REPRINTED WITHOUT CHANGE
112.2		112.2	6 June 1979
112.27	6 June 1979	112.27	REPRINTED WITHOUT CHANGE
112.27A thru 112.27H			- - -
112.28	23 March 1976	112.28	REPRINTED WITHOUT CHANGE
112.33	20 June 1977	112.33	REPRINTED WITHOUT CHANGE
112.34		112.34	6 June 1979
112.35		112.35	6 June 1979
112.36			- - -
112.37			- - -
113.1		113.1	23 March 1976
113.2	23 March 1976	113.2	REPRINTED WITHOUT CHANGE
113.9			- - -
113.10			- - -
113.11			- - -

3. RETAIN THIS NOTICE AND INSERT BEFORE THE TABLE OF CONTENTS.

4. Holders of MIL-STD-1132A will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or canceled.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Air Force - 85, 17
DLA - ES

User activity:
Navy - CG

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5930-1191)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

FOREWORD

This standard provides selected standard switches and associated hardware which are mandatory for use in the design of military equipment.

The application information contained in this standard is offered for guidance only and is not to be considered as mandatory. Additional application information will be added when coordinated with the three military departments.

Additional sections of this standard will be developed as the standard switches and associated hardware of a given specification family are selected and coordinated with the three military departments.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

CONTENTS

			Page
Paragraph	1.	SCOPE	1
	1.1	Scope	1
	1.2	Purpose of standard	1
	2.	REFERENCED DOCUMENTS	2
	3.	DEFINITIONS	3
	4.	GENERAL REQUIREMENTS	6
	4.1	Choice of switch types and associated hardware	6
	4.2	Conflict of requirements	6
	4.3	Criteria for inclusion in this standard	6
	5.	DETAILED REQUIREMENTS	7
	6.	APPLICATION INFORMATION	8
	6.1	General	8
	6.2	Principal applications	8
	6.3	Switch selection	9
	6.4	Switch parameters	10
	6.5	Environmental considerations	11
	7.	SUPPLEMENTARY INFORMATION	13
	7.1	Drawing notes	13

SECTIONS

Section	100	SWITCHES	100.1
	200	ASSOCIATED SWITCH HARDWARE	200.1

CROSS REFERENCE
(Section number to specification number)

Switches

Section		Applicable specification(s)
101	(Reed)	MIL-S-55433
102	(Push button, Illuminated)	MIL-S-22885
103	(Interlock)	MIL-S-8805
104	(Multistation)	MIL-S-24317
105	(Pressure)	MIL-S-9395
106	(Push Button)	MIL-S-8805
107	(Rotary)	MIL-S-3786, MIL-S-6807, MIL-S-15291, and MIL-S-21604
108	(Rotary, Electro-Mechanical Actuated, Solenoid Type)	MIL-S-3786
109	(Rotary, Printed Circuit, Thumbwheel and Pushbutton)	MIL-S-22710
110	(Sensitive)	MIL-S-8805
111	(Thermostatic)	MIL-S-24236 and MIL-S-28827
112	(Toggle)	MIL-S-3950, MIL-S-8834, and MIL-S-8805
113	(Modular Switch Assemblies)	MIL-S-22885
114	(Air and Liquid Flow)	MIL-S-28788
115	(Dual In-line Package)	MIL-S-83504

Supersedes page iv
of 6 June 1979

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This standard establishes requirements for selection of switches and their associated hardware, shown in detail in sections 100 and 200, for use in military equipment. Included are the following:

(a) Selected standard switch types and associated hardware chosen jointly by the Departments of the Army, the Navy, and the Air Force for use in the design and manufacture of military equipment under the jurisdiction of the Department of Defense.

(b) Guides for the choice and application of switches and associated hardware for use in military equipment.

Complete detailed requirements for switches and associated hardware listed in this standard are covered in the applicable specification sheets or MS sheets (see 4.2). (The complete part number or type designation, and the number of the applicable specification sheet or MS sheet should be referenced in procurement documents for the switches or associated hardware.) When it has been determined that circuit requirements cannot be met by using the switch styles or characteristics listed in this standard, the design engineer with the approval of the cognizant military activity should select from the applicable switch specification, styles or characteristics not listed herein.

1.2 Purpose of standard. The purpose of this standard is to:

(a) Provide the equipment designer with a list of switches and their associated hardware that are considered to be the most flexible for military applications.

(b) Control and minimize the variety of switches used in military equipment in order to facilitate logistic support of the equipment in the field.

(c) Establish criteria pertinent to the choice, application, and use of switches and their associated hardware in military equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-3786	-	Switches, Rotary (Circuit Selector, Low-Current Capacity), General Specification For.
MIL-S-3950	-	Switches, Toggle, Environmentally Sealed, General Specification For.
MIL-B-5423	-	Boots, Dust and Water Seal (For Toggle and Pushbutton Switches, Circuit Breakers, and Rotary-Actuated Parts), General Specification For.
MIL-S-6807	-	Switches, Rotary, Selector Power, General Specification For.
MIL-G-7703	-	Guard, Switch, General Specification For.
MIL-S-8805	-	Switches and Switch Assemblies, Sensitive and Push (Snap Action), General Specification For.
MIL-S-8834	-	Switches, Toggle, Positive Break, General Specification For.
MIL-S-9395	-	Switches, Pressure, (Absolute, Gage, and Differential), General Specification For.
MIL-S-15291	-	Switches, Rotary, Snap Action and Detent/Spring Return, General Specification For.
MIL-S-21604	-	Switches, Rotary, Multipole and Selector, General Specification For.
MIL-S-22710	-	Switches, Rotary (Printed Circuit), (Thumbwheel, In-line, and Pushbutton), General Specification For.
MIL-S-22885	-	Switch, PushButton, Illuminated, General Specification For.
MIL-S-24236	-	Switches, Thermostatic, (Metallic and Bimetallic), General Specification For.
MIL-S-24317	-	Switches, Multistation, Pushbutton, (Illuminated and Non-Illuminated), General Specification For.
MIL-S-28788	-	Switches, Air and Liquid Flow, Sensing, General Specification For.
MIL-S-28827	-	Switches, Thermostatic, (Volatile Liquid), Hermetically Sealed, General Specification For.
MIL-S-55433	-	Switches, Reed, General Specification For.
MIL-S-83504	-	Switches, Dual In-line Package (DIP), General Specification For.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Supersedes page 2
of 6 June 1979

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

(ak) Non-shorting type switch. Also called break-before-make. When switching from one position to the next, the first contact fully breaks before the second contact is made.

(al) Spring force. The force which the actuating mechanism exerts at the maximum pre-travel position. (The minimum spring force permissible will be equal to the maximum actuating force permitted for the push or sensitive switch used.)

(am) Stator. The stationary insulating portion of the switch section.

(an) Switch life. The number of cycles of operation during which the electrical and mechanical performance will meet predetermined and stated life-limiting criteria.

(ao) Switch rating. The load-carrying-and-breaking ability of a switch. (The current rating in the title is usually the resistive load rating at the lowest dc voltage.)

(ap) Terminal lug. The lug extending outward from the switch used for attaching the external connections.

(aq) Watertight. An enclosure which prevents leakage when water strikes or covers enclosure. No ventilation or other openings are permitted in the enclosure.

(ar) Dual in-line package switch (DIP). A DIP switch is a miniature high density switch for easy printed circuit board mounting, consisting of individual switch stations.

(as) Switch stations. A switch station is the section containing the electrical circuitry, including the actuator. The switch station is operated with a rocker, toggle, slide, or pushbutton actuator.

(at) Closed base switch. A switch in which the contact mechanism is neither visible nor accessible through openings in the envelope.

(av) Open base switch. A switch in which the contact mechanism is visible or accessible, or both, through openings in the envelope.

Supersedes page 5
of 23 March 1976

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

4. GENERAL REQUIREMENTS

4.1 Choice of switch types and associated hardware. The equipment designer should consider the following characteristics and parameters in choosing the most suitable switch design and associated hardware design:

(a) Application (type of switch).

(b) Flexibility of circuitry.

(c) Type of action.

(d) Electrical data (contact ratings, etc.).

(e) Type of contacts.

(f) Environmental capabilities such as shock and vibration.

(g) Mechanical data and safety features.

(h) Panel layout.

(i) Quality and reliability.

(j) Economics.

4.1.1 Procurement information. After a preliminary selection of the desired switch or associated hardware has been made, reference should be made to the applicable specification and qualified products list. Procurement documents for switches or associated hardware shall reference the applicable specification sheet or MS sheet, and the complete part number, style, or type designation.

4.2 Conflict of requirements. After selection of a particular switch type or associated hardware from this document, the design engineer should refer to the applicable specification sheet or MS sheet for any technical conflicts. In the event of a conflict between the technical requirements described in this standard and those of the applicable specification sheet or MS sheet, the latter shall govern.

4.3 Criteria for inclusion in this standard. The criteria for the inclusion of switch types and associated hardware in this standard are as follows:

(a) The switch and associated hardware shall be the best type available for general use in military equipment.

(b) Switches and associated hardware shall be covered by military specifications (see 2.1).

(c) Switches and associated hardware shall be listed on a qualified products list.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

5. DETAILED REQUIREMENTS

5.1 The detailed requirements for standard switch types and associated hardware are contained in the applicable specification, and in the applicable section of this standard.

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:41Z
Check the source to verify that this is the current version before use.

6. APPLICATION INFORMATION

6.1 General. The application information in this standard is intended to help the engineer select the switches and associated hardware he will specify. (The application information pertaining to specific switch types is contained in the applicable section of this standard.) The most important thing a user must decide is which of the numerous types of switches and associated hardware will be the most suitable for use in the particular military equipment he is designing. Proper selection, in its broadest sense, is the first step in building reliable equipment. To effectively and properly select the switches and associated hardware to be used, the user must know as much as possible about the types from which he can choose. He should know the advantages and disadvantages, the behavior under various environmental conditions, construction, the effect upon the circuit, and the effect of the circuit upon the switch. He should know what causes switches to fail. He should have also an intimate working knowledge of the applicable military specification.

6.1.1 Switch types. All switches of the types widely used in electronic equipment can be grouped into one of three general basic types. They are rotary, non-rotary, and sensing types. These basic types differ from each other in size, cost, actuation, construction, and general characteristics. Some are better than others for a particular purpose; no one type has all of the best characteristics. The choice among them, therefore, depends on the requirements, both initial and long-term, the environment in which they must exist, and numerous other factors which the designer must understand. The designer must realize that the summaries of the following general characteristics and cost are relative, not absolute, and that all the requirements of a particular application must be taken into consideration and compared with the advantages and drawbacks of each of the several types, before a final choice is made.

6.2 Principal applications:

(a) MIL-S-3786 - Switches, Rotary. Used primarily for low-power, alternating current (ac) or direct current (dc) switching applications (capable of making and breaking a resistive load of 2 amperes or less). Includes both manually and solenoid actuated switches.
(b) MIL-S-3950 - Switches, Toggle, Environmentally Sealed. Used where simple make-and-break actions are required, and are suitable for use on ac and dc circuits.
(c) MIL-B-5423 - Boots, Dust and Water Seal. Used on toggle, push-button, and rotary switches to protect the switch-actuating mechanism from sand, dust, water, and other contaminants, and to seal the panel on which the switches are mounted.
(d) MIL-S-6807 - Switches, Rotary, Selector Power. Used in power circuits capable of making, carrying, and breaking electrical loads of 10 amperes or less.
(e) MIL-S-8805 - Switches and Switch Assemblies, Sensitive and Push (Snap Action). Used in ac and dc applications, where predetermined small and accurately controlled characteristics are required. Various means of actuation by toggle levers, pushbuttons, cams, and other light pressure devices. These switches have snap-action which eliminates teasing.
(f) MIL-S-8834 - Switches, Toggle, Positive Break. Used in ac and dc circuits where a positive make-and-break action is required. Positive break actuation causes minimum contact "tease".
(g) MIL-S-9395 - Switches, Pressure. Used primarily to detect changes in pressure, liquid and gas applications.
(h) MIL-S-15291 - Switches, Rotary, Snap Action. Used in high current circuitry capable of making and breaking 10 - 200 amperes, 120 - 250 volts dc, 125 - 500 volts ac, utilizing snap action positioning mechanism.
(i) MIL-S-21604 - Switches, Rotary, Multipole and Selector. Used in power, communication, and control equipment. Capable of making and breaking loads up to 10 amperes.

Supersedes page 8 of
23 March 1976.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

(j) MIL-S-22710 - Switches, Rotary, (Printed Circuit), (Thumbwheel, Inline, and Pushbutton). Used primarily for low power ac or dc switching applications. Thumbwheel switches provide a numerical or other legend readout tied to a particular switch position. Also provided for is logic circuitry for computer operation.
(k) MIL-S-22885 - Switch, Push Button, Illuminated. Used as panel displays and switching devices in ac and dc applications. Panel displays include various combinations of colors and legends.
(l) MIL-S-24236 - Switches, Thermostatic (Metallic and Bimetallic). Used primarily in ac and dc applications where temperature protection or accurate temperature control of an enclosure is required.
(m) MIL-S-24317 - Switches, Multistation, Pushbutton (Illuminated and Non-Illuminated). Used as panel displays and switching devices in ac and dc applications.
(n) MIL-S-28788 - Switches, Air and Liquid Flow, Sensing. Used primarily in equipment to sense velocity and flow of air or liquid.
(o) MIL-S-28827 - Switches, Thermostatic (Volatile Liquid), Hermetically Sealed. Used primarily in ac and dc applications that require rapid temperature response.
(p) MIL-S-55433 - Switches, Reed. Used in circuits as a separate element where magnetically-actuated, hermetically sealed switch contacts are required. These reed switches are capable of millions of operations and operations with low-level (dry circuit) loads.
(q) MIL-S-83504 - Switches, Dual In-line Package. A miniature high density switch for easy printed circuit board mounting, consisting of individual switch stations.

6.3 Switch selection. The designer, in selecting a switch type for a particular function to be performed, must weigh numerous factors before coming to a final decision. Selection normally starts with the most important characteristic for the application, then selecting and compromising other characteristics.

6.3.1 Selection factors. Some of the most important of these factors are noted below:

(a) Design and construction:
  (1) Open or enclosed construction.
  (2) Sealed (hermetically, environmentally (resilient), dust-tight, water-tight, explosion).
  (3) Mounting (bushing, multihole, mounting bracket, mounting plate).
  (4) Illuminated or non-illuminated.

(b) Operating characteristics:
  (1) Actuation - pushbutton, toggle, rotary, sensing.
  (2) Switching action - momentary action, maintained action, alternate action, snap action.

(c) Contacts:
  (1) Type of contacts and arrangement.
  (2) Contact ratings - resistive, inductive, lamp, motor, low-level (dry circuit), and frequency.
  (3) Contact bounce time.
  (4) Contact resistance.
  (5) Make-before-break, break-before-make; shorting or non-shorting contacts.

(d) Environmental considerations:
  (1) Temperature range.
  (2) Moisture.
  (3) Altitude.
  (4) Shock and vibration.
  (5) Acceleration.
  (6) Sand and dust.
  (7) Explosion.

Supersedes page 9
of 6 June 1979

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.

(e) Insulation requirements:
(1) Insulation resistance.
(2) Breakdown voltage.

(f) Switching speed:
(1) Actuation speed.

(g) Life:
(1) Mechanical.
(2) Electrical at elevated temperature.

(h) Terminals:
(1) Solder.
(2) Screw.
(3) Wire.
(4) Plug-in termination.
(5) Integrated wire termination.
(6) Printed-circuit board terminations.

6.4 Switch parameters.

6.4.1 Design and construction.

(a) Design. Design of the switch, related to available space, mounting dimensions, handling of the actuating element, type of connections to the switch, and aspects relating to ruggedness and safety, is an important selection factor.
(b) Construction. Many types of enclosures are available to protect the switch from external conditions, particularly high humidity and dirt. Switches may be classified according to the degree of protection offered by the enclosure. Such classifications include the following: Open, sealed, enclosed, environmentally (resilient), and hermetically sealed. With the open construction switch, no effort is made to protect the switch or its parts from atmospheric conditions. The enclosed switch is one in which the contacts are enclosed in a closed case made of plastic or metal and plastic. The environmentally (resilient) sealed switch is one in a completely sealed case where any portion of the seal is a resilient material such as a gasket or a seal in the bushing of a panel-mounted switch. The hermetically sealed switch is made airtight by a sealing process which involves fusing or soldering and does not use a gasket. The hermetically sealed enclosure offers the greatest protection because it insulates against such elements as moisture, harmful gases, and dirt. It also eliminates the increased arcing by low atmospheric pressures at high altitudes.

6.4.2 Contacts. The contacts which carry the electrical load can be classified by function, current-carrying capacity, and application. The contact arrangements vary in complexity from a simple make or break, through make-before-break, break-before-make, make-make, break-break, etc, and from single throw to multiple throw, and single pole to multipole and various combinations of these features.

6.4.2.1 Contact ratings. Contacts are usually given multiple ratings dependent on the type of load being switched. These ratings consist of resistive, low-level (dry circuit), lamp, motor, or inductive loads. Most switches are given the resistive load rating and in most instances at least one additional of the above ratings.

6.4.2.2 Contact operate and bounce times. In many applications, critical operate and bounce times of the contacts are important. Operate time in a double-throw switch is defined as the time it takes the moving contact to separate from the normally closed contact, travel to the normally open contact and make the circuit (this does not include bounce time). Bounce time is the interval between first make of the contact until the uncontrolled making and breaking of the contact ceases. In many electronic circuits, a millisecond is a long time and operate and bounce times become critical parameters.

Supersedes page 10
of 23 March 1976

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

7. SUPPLEMENTAL INFORMATION

7.1 Drawing notes. The following notes are applicable to each figure contained in Section 100 and 200 of this standard:

NOTES:

1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Air Force - 85, 17
DLA - ES

User activities:
Navy - CG

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5930-0997)

Supersedes page 13
of 23 March 1976

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

SECTION 100

SWITCHES

Section

101 Switches, Reed
102 Switches, Push Button, Illuminated
103 Switches, Interlock
104 Switches, Multistation
105 Switches, Pressure
106 Switches, Push Button
107 Switches, Rotary
108 Switches, Rotary, Electro-Mechanical Actuated (Solenoid Type)
109 Switches, Rotary, Printed Circuit (Thumbwheel and Pushbutton)
110 Switches, Sensitive
111 Switches, Thermostatic
112 Switches, Toggle
113 Switch Assemblies, Modular
114 Switches, Air and Liquid Flow
115 Switches, Dual In-line Package (DIP)

Supersedes page 100.1
of 6 June 1979

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

SECTION 102A

SWITCHES, PUSH BUTTON, ILLUMINATED
(Applicable specification: MIL-S-22885)

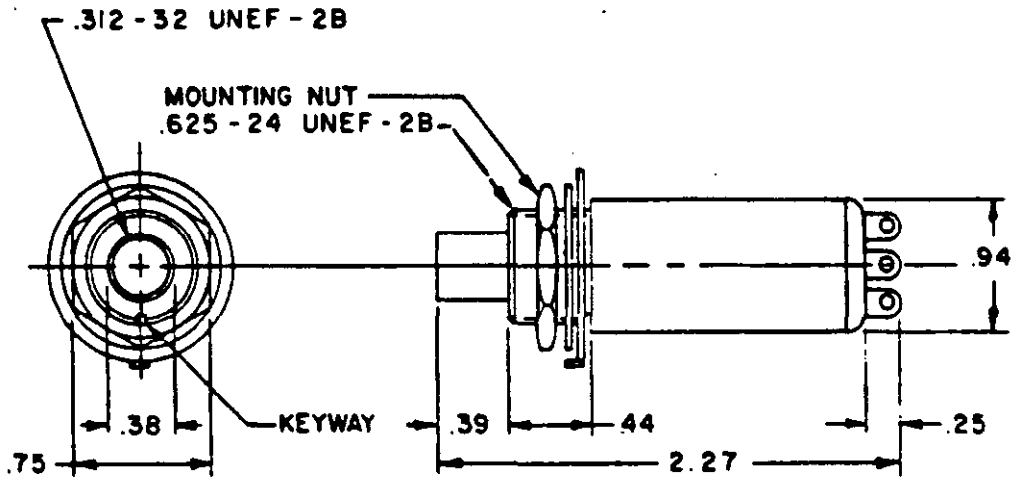
Figure number	Applicable specification sheet	Switch characteristics: Operation	Circuit	Electrical rating (28 Vdc resistive)	Remarks	Page
102-1	MIL-S-22885/8	Momentary & push-pull	2-circuit	5 amperes	Single lamp, drip-proof seal	102.2
102-1A*	MIL-S-22885/1					
102-2	MIL-S-22885/9 & MIL-S-22885/10	Momentary & alternate	2PDT & 4PDT	5 amperes	4-lamp, rectangular, modular construction	102.4
102-2A*	MIL-S-22885/85					
102-3	MIL-S-22885/15 & MIL-S-22885/16	Momentary & alternate	2PDT & 4PDT	5 amperes	4-lamp, rectangular, clip mounted, modular construction	102.6
102-4	MIL-S-22885/18	Momentary & alternate	2PDT & 2-circuit	2 amperes	Single lamp, drip-proof seal	102.8
102-4A*	MIL-S-22885/22					
102-4B*	MIL-S-22885/19					
102-5	MIL-S-22885/27	Momentary & alternate	4- & 8-circuit	10 amperes	Single lamp	102.10
102-5A*	MIL-S-22885/28					
102-6	MIL-S-22885/56	Momentary & alternate	2PDT & 4PDT	2 amperes	4-lamp, square, RFI shielding option	102.12
102-7	MIL-S-22885/67	Momentary & alternate	2PDT & 3PDT	7 amperes	4-lamp, square, RFI shielding option	102.14
102-8	MIL-S-22885/74	Momentary & alternate	2PDT & 4PDT	5 amperes	4-lamp, square, integrated wire termination system	102.16
102-9	MIL-S-22885/75	Momentary & alternate	2PDT, 3PDT, & 4PDT	7 amperes	4-lamp, square, RFI shielding option	102.18
102-10	MIL-S-22885/77	Momentary, alternate, & magnetic hold	2PDT, 3PDT, & 4PDT	7.5 amperes	4-lamp, square, RFI shielding option, integrated wire termination system	102.20
102-11	MIL-S-22885/79	Momentary	SPDT	7 (silver) or 1 (gold) amperes	Single lamp, drip-proof seal option	102.22
102-12	MIL-S-22885/58	Momentary & alternate	2PDT & 4PDT	3 amperes	2-lamp, rectangular	102.24
102-12A*	MIL-S-22885/59					
102-13	MIL-S-22885/81	Momentary & alternate	2PDT & 4PDT	5 amperes	Single lamp, drip-proof seal	102.25
102-14	MIL-S-22885/71	Momentary & alternate	2PDT	7 amperes	4-lamp, square, RFI shielded	102.26
102-15	MIL-S-22885/72 & MIL-S-22885/73	Momentary & alternate	4- & 8-circuit	10 amperes	4-lamp, square, modular construction	102.28
102-16	MIL-S-22885/80	Momentary	2PDT & 4PDT	5 amperes	4-lamp, square, RFI shielded, moisture seal, high shock	102.30
102-16A*	MIL-S-22885/82					
102-17	MIL-S-22885/86	Momentary & alternate	2PDT & 4PDT	2 amperes	4-lamp, square, RFI shielded, modular construction	102.32
102-18	MIL-S-22885/87	Momentary & alternate	2PDT & 4PDT	5 amperes	4-lamp, solid mount, modular constructed, high shock	102.35
102-19	MIL-S-22885/89	Momentary	SPDT & 2PDT	7 amperes	Single lamp	102.38
102-20	MIL-S-22885/90	Momentary & alternate	SPDT, 2PDT, & 3PDT	7 amperes	4-lamp, square sunlight readable display	102.41
102-21	MIL-S-22885/91	Momentary & alternate	1- & 2-circuit	Hall effect	Rocker, paddle, solid state hall effect, unsealed	102.45
102-23	MIL-S-22885/92	Momentary & alternate	2 to 4 circuits	10 amperes	Rocker, paddle, snap-in	102.57
102-24	MIL-S-22885/93	Momentary & alternate	1 to 4 poles	Logic to 3 amperes	Rocker, paddle, snap-in	102.64

* Applicable lens

Supersedes page 102.1
of 20 June 1977

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.

LOAD	ELECTRICAL RATINGS (amperes)	
	28 Vdc	115 Vac
Resistive - - - - -	5	5
Inductive - - - - -	3	3



INCHES	MM
.25	6.35
.312	7.92
.38	9.65
.39	9.91
.44	11.18
.625	15.88
.75	19.05
.94	23.88
2.27	57.66

Switch Part No. M22885/8-		Lens (screw-in type) (see figure 102-1A)				
Momentary	Push-pull	Configuration	Color	Brightness (3/4 face) ft L	Part No. M22885/1-	Dimension "A" inch (mm)
01	02	Switch housings without lens				
03	28	A	Yellow	225	01	.81 (20.57)
04	29		White	250	02	
05	30		Red	20	03	
06	31		Green	11	04	
07	32		Blue	7	05	
08	33	B	Yellow	150	06	.75 (19.05)
09	34		White	200	07	
10	35		Red	15	08	
11	36		Green	18	09	
12	37		Blue	4	10	
13	38	C	Yellow	300	11	.67 (17.02)
14	39		White	300	12	
15	40		Red	25	13	
16	41		Green	40	14	
17	42		Blue	10	15	
18	43	D	Yellow	150	16	.62 (15.75)
19	44		White	200	17	
20	45		Red	15	18	
21	46		Green	18	19	
22	47		Blue	4	20	
23	48	E	Yellow	120	21	.88 (22.35)
24	49		White	135	22	
25	50		Red	14	23	
26	51		Green	14	24	
27	52		Blue	2	25	

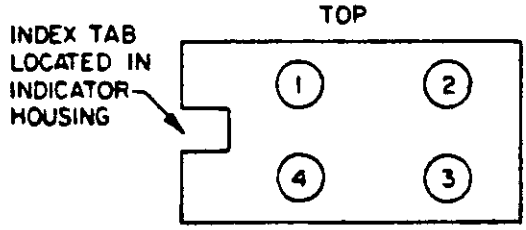
Operating force:
Momentary - 4 ± 1 pounds initial, 2.5 pounds minimum after life.
Push-pull - 1.5 pounds.
Circuit characteristics: Two circuit 1 NO.

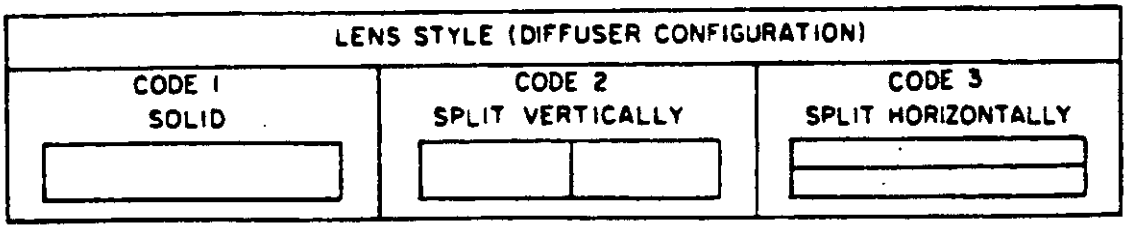
FIGURE 102-1. Switch housings, push button, illuminated, momentary and push-pull (MIL-S-22885/8).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

1/ Part No. M22885/9-						
Momentary Switch module M22885/11		Alternate Switch module M22885/11		Lens style code	Lamp wiring code	Indicator light module M22885/10-
-01 2PDT	-02 4PDT	-03 2PDT	-04 4PDT			
NARROW FRONT LENS PLATE, M22885/85-01						
01	02	03	04	A	1	01
09	10	11	12	B	1	03
13	14	15	16	B	2	04
17	18	19	20	C	1	05
21	22	23	24	C	3	06
25	26	27	28	A	3	07
167	168	169	170	A	2	08
EXTENDED FRONT LENS PLATE, M22885/85-02						
29	30	31	32	A	1	09
33	34	35	36	B	1	10
37	38	39	40	B	2	11
41	42	43	44	C	1	12
45	46	47	48	C	3	13
49	50	51	52	A	3	14
171	172	173	174	A	2	15
WITHOUT LENS PLATE						
53	54	55	56	A	1	16
57	58	59	60	B	1	17
61	62	63	64	B	2	18
65	66	67	68	C	1	19
69	70	71	72	C	3	20
73	74	75	76	A	3	21
175	176	177	178	A	2	22

1/ Part numbers are for units without color filters. When color filters are required (see section 203 - MIL-S-22885/12), the color symbols of MIL-S-22885 will be added following the dash number will the location sequence as shown in the following:





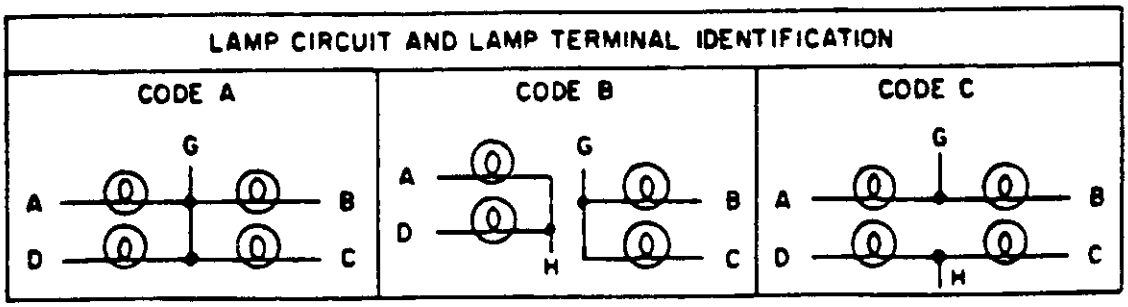


FIGURE 102-2. Switches, pushbutton, illuminated, 4-lamp, solid mount (MIL-S-22885/9 and /10 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

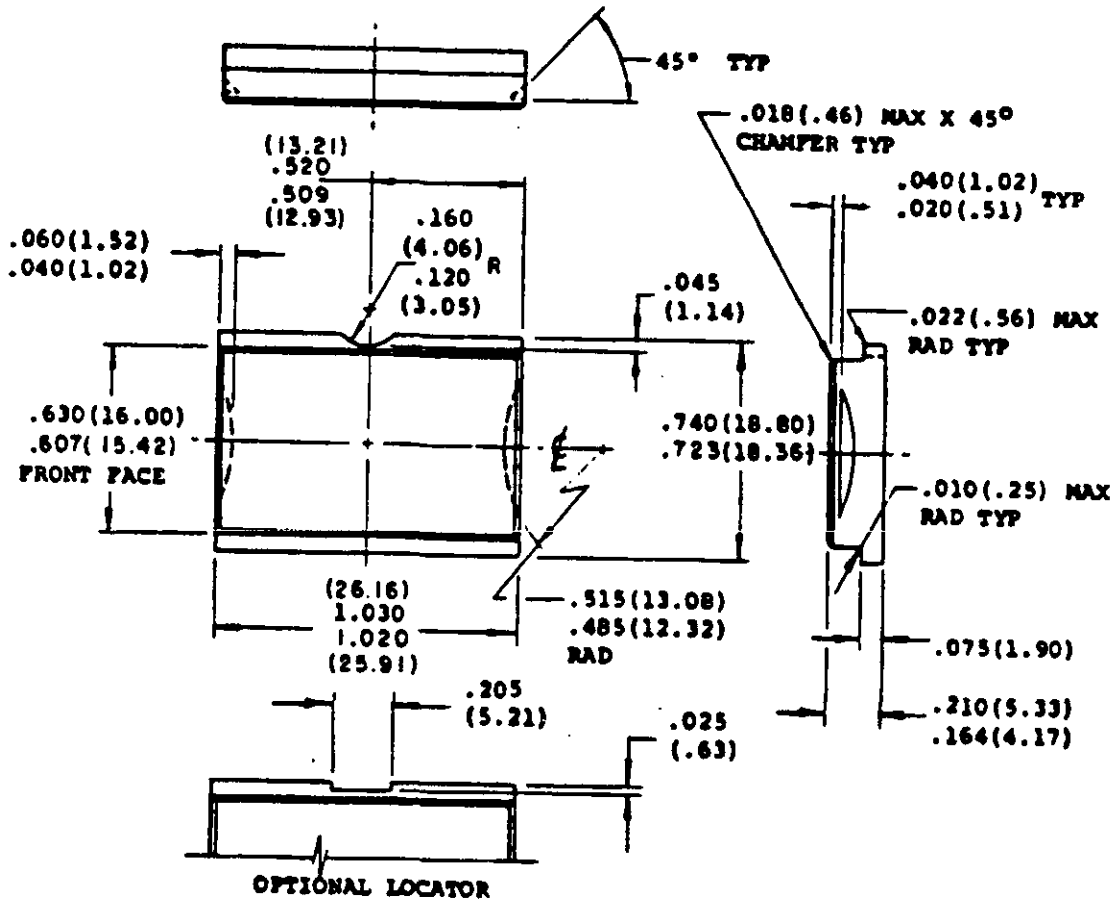


FIGURE 102-2A. Switch, pushbutton, illuminated, lens plate for, 1.030 long X .730 wide (MIL-S-22885/85).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

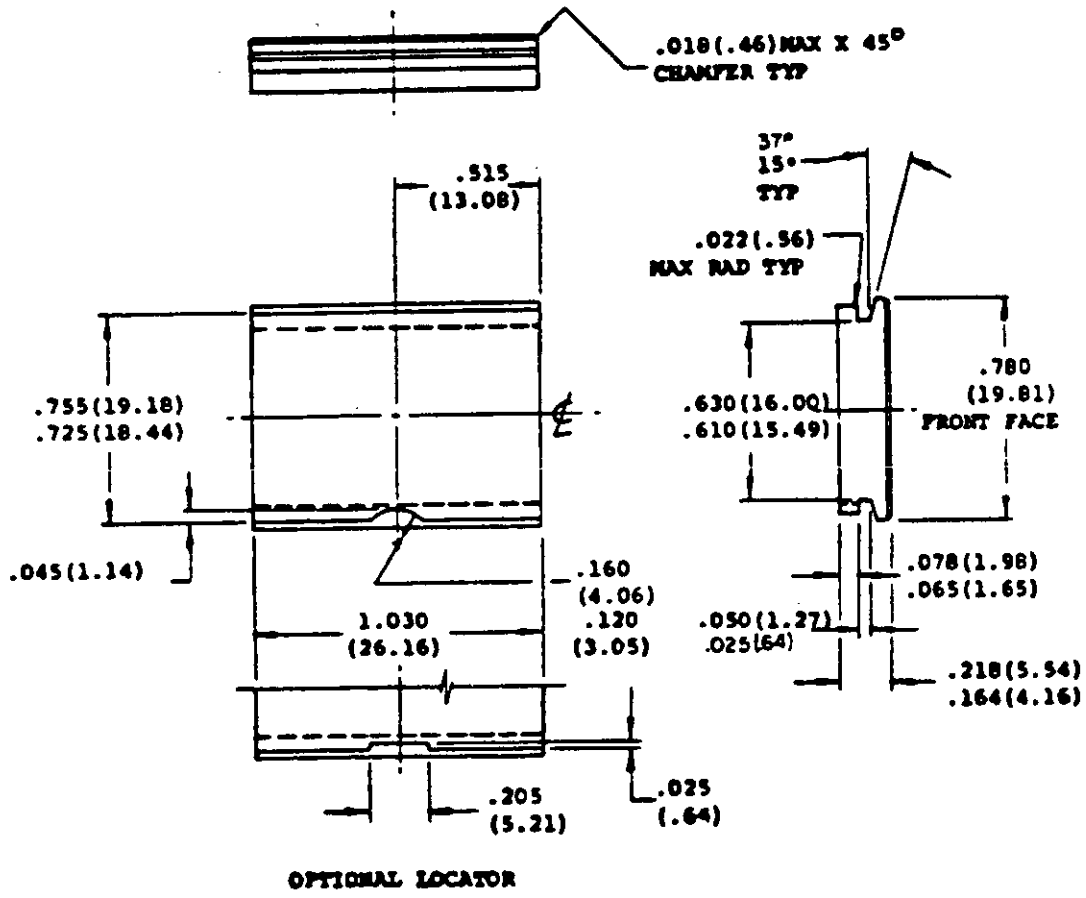
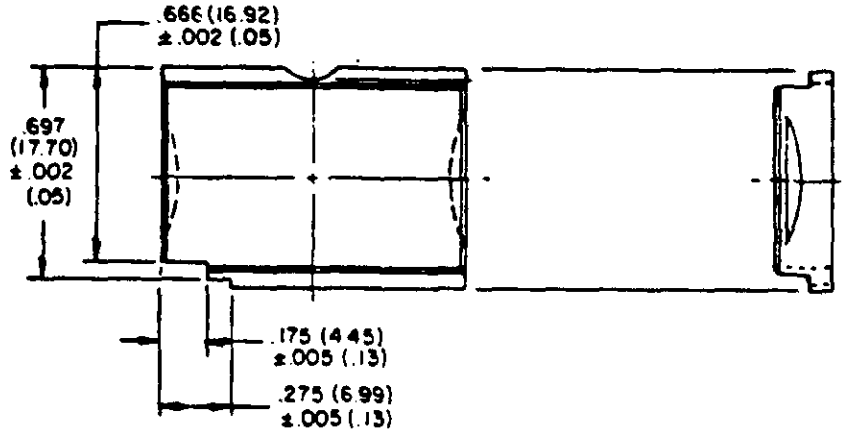


FIGURE 102-2A. Switch, pushbutton, illuminated, lens plate for, 1.030 long X .730 wide (MIL-S-22885/85) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



Part number	Description	Switch/indicator 1/ reference
M22885/85-01	Narrow face	MIL-S-22885/9, /10
M22885/85-02	Extended face	MIL-S-22885/9, /10
M22885/85-03	Narrow face with cut-out	MIL-S-22885/87

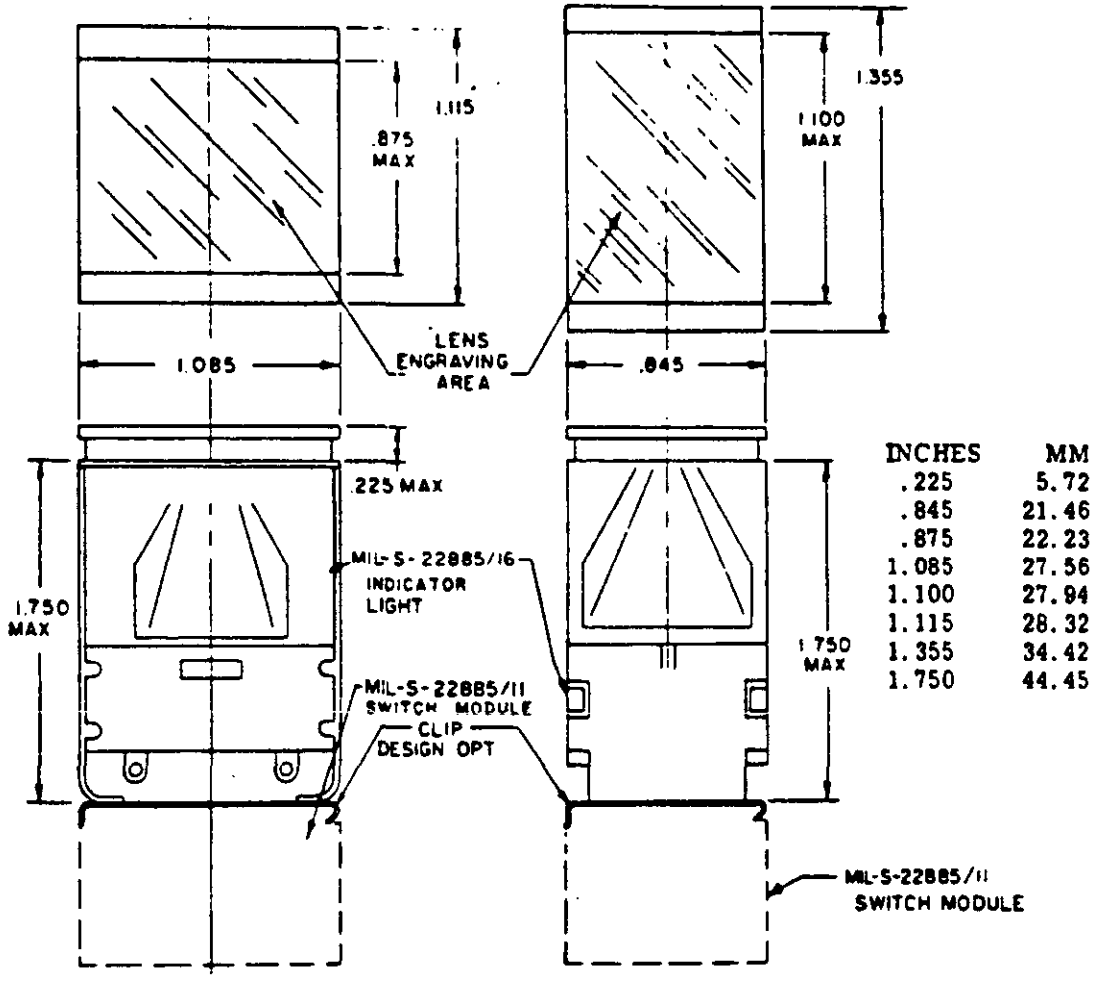
FIGURE 102-2A. Switch, pushbutton, illuminated, lens plate for, 1.030 long X .730 wide (MIL-S-22885/85) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

INTENTIONALLY BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

LOAD	ELECTRICAL RATINGS (amperes)		
	Sea level		50,000 feet
	26 Vdc	115 Vac	28 Vdc
Resistive - - -	5.0	5.0	5.0
Inductive - - -	3.0	5.0	2.5
Lamp - - - - -	2.4	1.5	2.4



Configuration A Configuration B

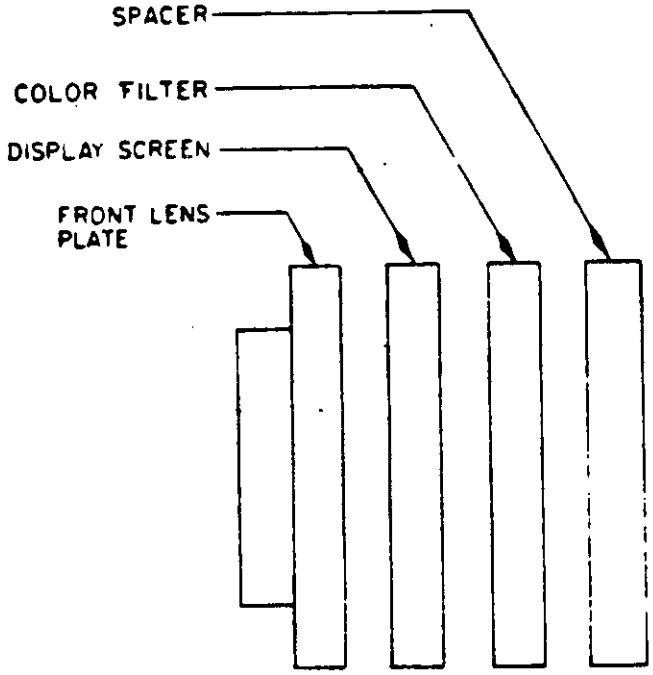
Complete switch shall consist of the following:
1 - Indicator light module, MIL-S-22885/16
1 - Switch module, MIL-S-22885/11 (see section 113)

Additional parts that may be ordered, if required:
4 - Color filters, MIL-S-22885/12 (see section 203)
4 - Lamps (T-1-3/4 midget-flange-base incandescent type)

Operating force:
Momentary - 4 pounds, maximum
Alternate - 6 pounds, maximum (2PDT) or 7 pounds, maximum (4PDT)

FIGURE 102-3. Switches, push button, illuminated, 4-lamp, clip mounted (MIL-S-22885/15 and /16)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



STYLE	LENS CONFIGURATION FOR DISPLAY SCREEN, COLOR FILTER, AND SPACER
A	SOLID
B	SPLIT HORIZONTALLY
C	SPLIT VERTICALLY

Part number	Style	Color	Description	Part number 1/	Style	Color	Description
M22885/59-01	A	Blue	Color filter	M22885/59-14	C	Green	Color filter
M22885/59-02	A	Green	Color filter	M22885/59-15	C	Red	Color filter
M22885/59-03	A	Red	Color filter	M22885/59-16	C	White	Color filter
M22885/59-04	A	White	Color filter	M22885/59-17	C	Yellow	Color filter
M22885/59-05	A	Yellow	Color filter	M22885/59-22	A,B,C	Clear	Front lens plate
M22885/59-07	B	Blue	Color filter	M22885/59-23	A	White	Display screen
M22885/59-08	B	Green	Color filter	M22885/59-24	B	White	Display screen
M22885/59-09	B	Red	Color filter	M22885/59-25	C	White	Display screen
M22885/59-10	B	White	Color filter	M22885/59-26	A	Clear	Spacer
M22885/59-11	B	Yellow	Color filter	M22885/59-27	B	Clear	Spacer
M22885/59-13	C	Blue	Color filter	M22885/59-28	C	Clear	Spacer

FIGURE 102-12A. Switch, pushbutton, illuminated, 2-lamp, lens arrangement for (MIL-S-22885/59).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

1/ Part No. M22885/80-			Circuit characteristics, operation, and part	Type
Lamp code 1 Lens code A	Lamp code 2 Lens code B	Lamp code 3 Lens code C		
COMPLETE SWITCH OR INDICATOR ASSEMBLY				
01	02	03	2PDT momentary switch	I
04	05	06	4PDT momentary switch	I
07	08	09	Indicator	I
10	11	12	2PDT momentary switch	II
13	14	15	4PDT momentary switch	II
16	17	18	Indicator	II
PLUG-IN SWITCH OR INDICATOR MODULES WITHOUT HOUSINGS				
19	20	21	2PDT momentary switch	I
22	23	24	4PDT momentary switch	I
25	26	27	Indicator	I
28	29	30	2PDT momentary switch	II
31	32	33	4PDT momentary switch	II
34	35	36	Indicator	II

1/ Part numbers are for units without color filters. When color filters are required, the color symbols of MIL-S-22885 will be added following the dash number; the location sequence is shown in Lens Code diagram.

LENS CODE

CODE	A	B	C
LENS STYLE	FULL	HORIZONTAL SPLIT (1 / 2)	VERTICAL SPLIT (1 | 2)

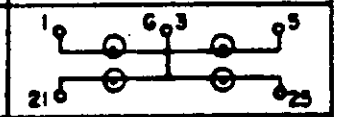
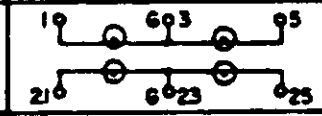
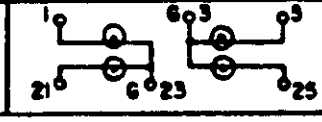
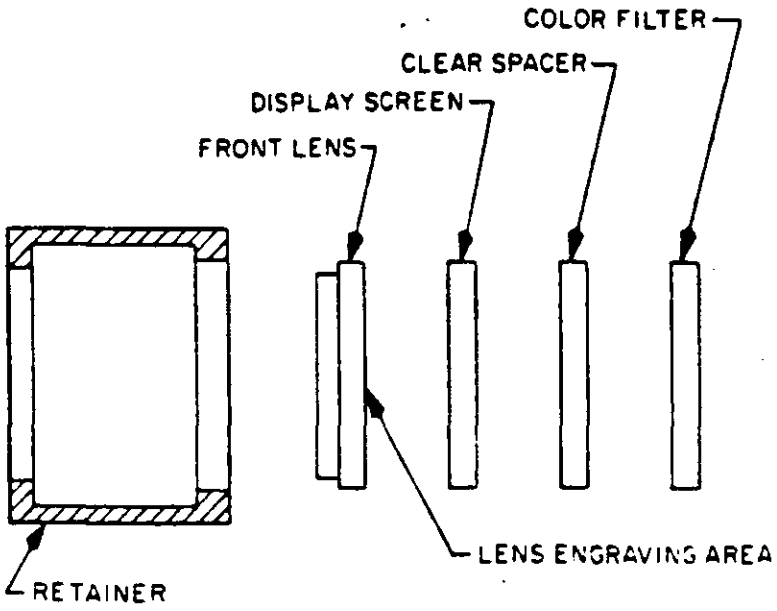
CODE	LAMP CIRCUITRY
1	
2	
3	

FIGURE 102-16. Switches, push button, illuminated, 4-lamp, RFI shielded, Moisture sealed, high impact shock, modular construction (MIL-S-22885/80) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



CODE	DISPLAY AND STYLE NUMBER ARRANGEMENT
A	FULL
B	HORIZONAL SPLIT
C	VERTICAL SPLIT
D	3-WAY SPLIT
E	3-WAY SPLIT
F	4-WAY SPLIT

Lens assemblies:

Code A consists of:
- 1 - Retainer
- 1 - Front lens plate
- 1 - Display screen
- 1 - Spacer
- 1 - Color filter

Codes B and C consist of:
- 1 - Retainer
- 2 - Display screens
- 2 - Spacers
- 2 - Color filters

Codes D and E consist of:
- 1 - Retainer
- 1 - Front lens plate
- 3 - Display screens
- 3 - Spacers
- 3 - Color filters

Code F consist of:
- 1 - Retainer
- 1 - Front lens plate
- 4 - Display screens
- 4 - Spacers
- 4 - Color filters

FIGURE 102-16A. Switch, pushbutton, illuminated, lens assemblies for, .665 sq and .916 lg x .665 wide (MIL-S-22885/82) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

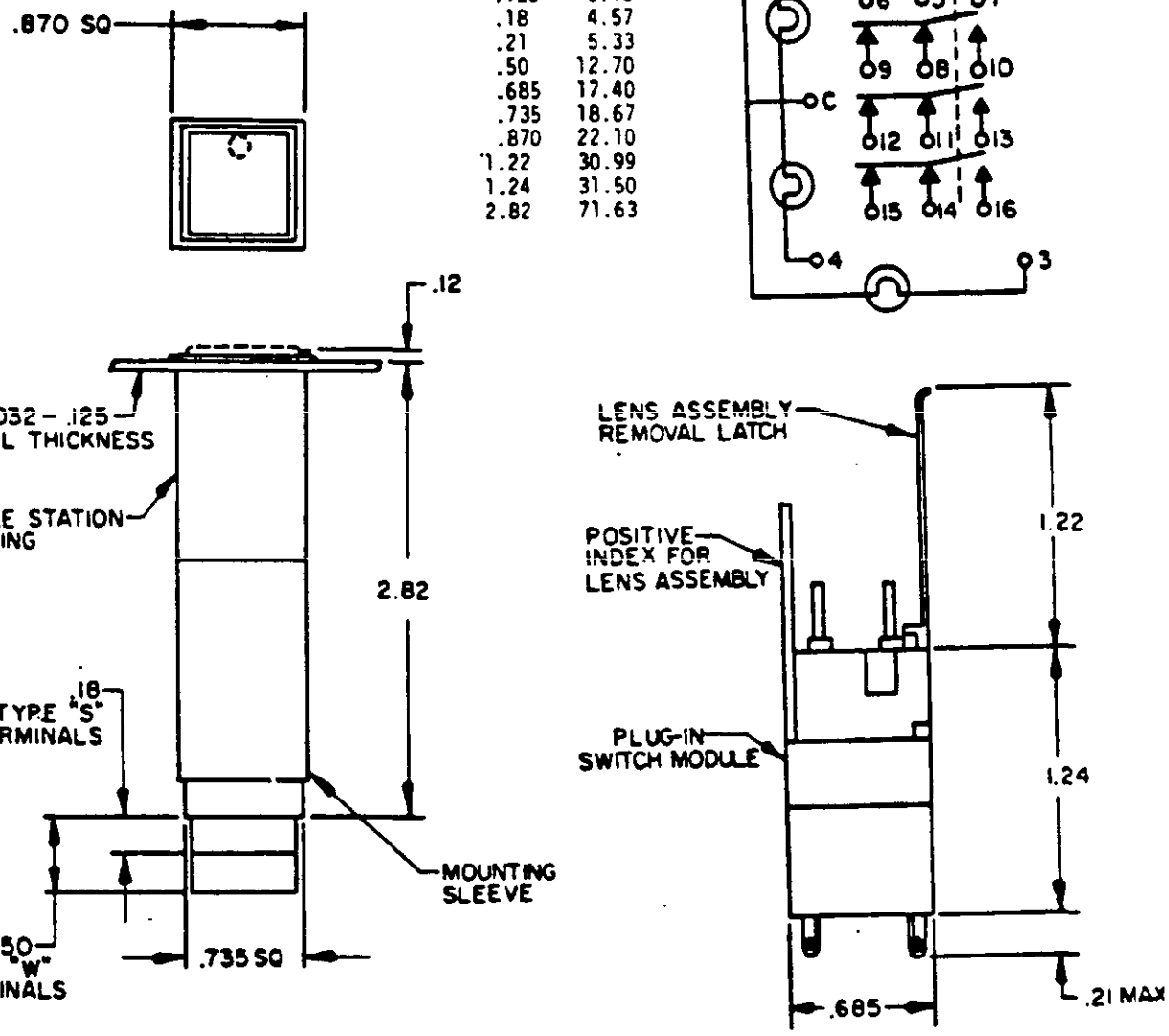
Part number M22885/82	Lens code	Style number	Color	Description
Square display				
-01	All	- - -	Clear	Front lens plate
-02	A	1	White	Display screen
-03	B, C, D, or E	2 or 3	White	Display screen
-04	A	1	Clear	Spacer
-05	B, C, D, or E	2 or 3	Clear	Spacer
-06	A	1	Blue	Color filter
-07	A	1	Green	Color filter
-08	A	1	Red	Color filter
-09	A	1	White	Color filter
-10	A	1	Yellow	Color filter
-12	B, C, D, or E	2 or 3	Blue	Color filter
-13	B, C, D, or E	2 or 3	Green	Color filter
-14	B, C, D, or E	2 or 3	Red	Color filter
-15	B, C, D, or E	2 or 3	White	Color filter
-16	B, C, D, or E	2 or 3	Yellow	Color filter
-43	D, E, or F	4	White	Display screen
-44	D, E, or F	4	Clear	Spacer
-45	D, E, or F	4	Blue	Color filter
-46	D, E, or F	4	Green	Color filter
-47	D, E, or F	4	Red	Color filter
-48	D, E, or F	4	White	Color filter
-49	D, E, or F	4	Yellow	Color filter
-50	All	- - -	- - -	Retainer
Rectangular display				
-18	All	- - -	Clear	Front lens plate
-19	A	1	White	Display screen
-20	B, D, or E	2	White	Display screen
-21	C	3	White	Display screen
-22	A	1	Clear	Spacer
-23	B, D, or E	2	Clear	Spacer
-24	C	3	Clear	Spacer
-25	A	1	Blue	Color filter
-26	A	1	Green	Color filter
-27	A	1	Red	Color filter
-28	A	1	White	Color filter
-29	A	1	Yellow	Color filter
-31	B, D, or E	2	Blue	Color filter
-32	B, D, or E	2	Green	Color filter
-33	B, D, or E	2	Red	Color filter
-34	B, D, or E	2	White	Color filter
-35	B, D, or E	2	Yellow	Color filter
-37	C	3	Blue	Color filter
-38	C	3	Green	Color filter
-39	C	3	Red	Color filter
-40	C	3	White	Color filter
-41	C	3	Yellow	Color filter
-51	D, E, or F	4	White	Display screen
-52	D, E, or F	4	Clear	Spacer
-53	D, E, or F	4	Blue	Color filter
-54	D, E, or F	4	Green	Color filter
-55	D, E, or F	4	Red	Color filter
-56	D, E, or F	4	White	Color filter
-57	D, E, or F	4	Yellow	Color filter
-58	All	- - -	- - -	Retainer

FIGURE 102-16A. Switch, pushbutton, illuminated, lens assemblies for, .665 SQ and .916 LG X .665 wide. (MIL-S-22885/82) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

LOAD	ELECTRICAL RATINGS (amperes) 28 Vdc	115 Vac
Resistive	2.0	2.0
Inductive	1.0	1.0
Lamp	0.75	0.75

INCHES	MM
.032	.81
.12	3.05
.125	3.18
.18	4.57
.21	5.33
.50	12.70
.685	17.40
.735	18.67
.870	22.10
1.22	30.99
1.24	31.50
2.82	71.63

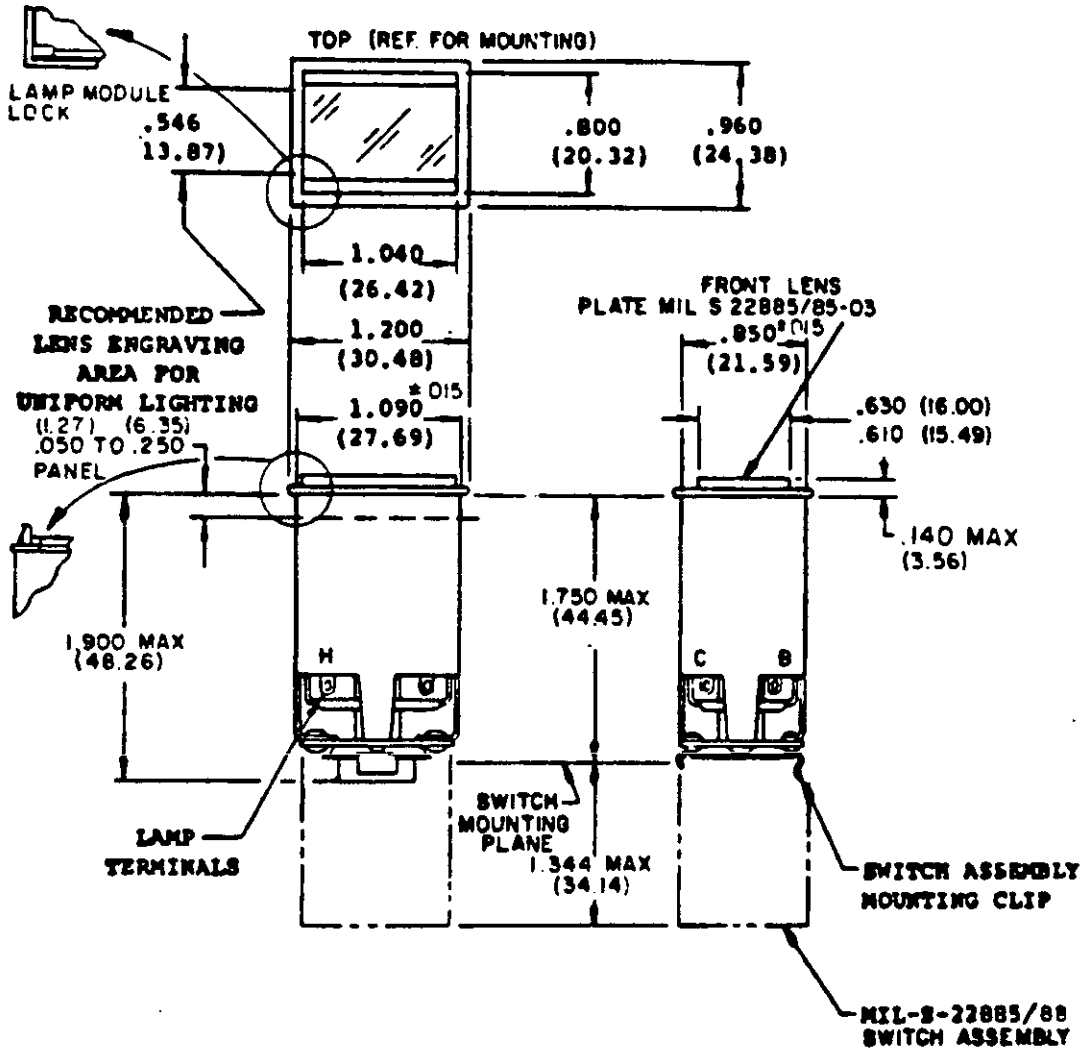


Complete switch consists of:
- 1 - Lens assembly (MIL-S-22885/52)
- 1 - Plug-in switch module
- 1 - Housing
- 4 - Lamps (T-1-3/4 midget-flange base incandescent type). Not included, order separately.

Operating force: 3.5 ± 1.0 pounds.

FIGURE 102-17. Switches, push button, illuminated, 4-lamp, square, RFI shielded, modular construction (MIL-S-22885/86).

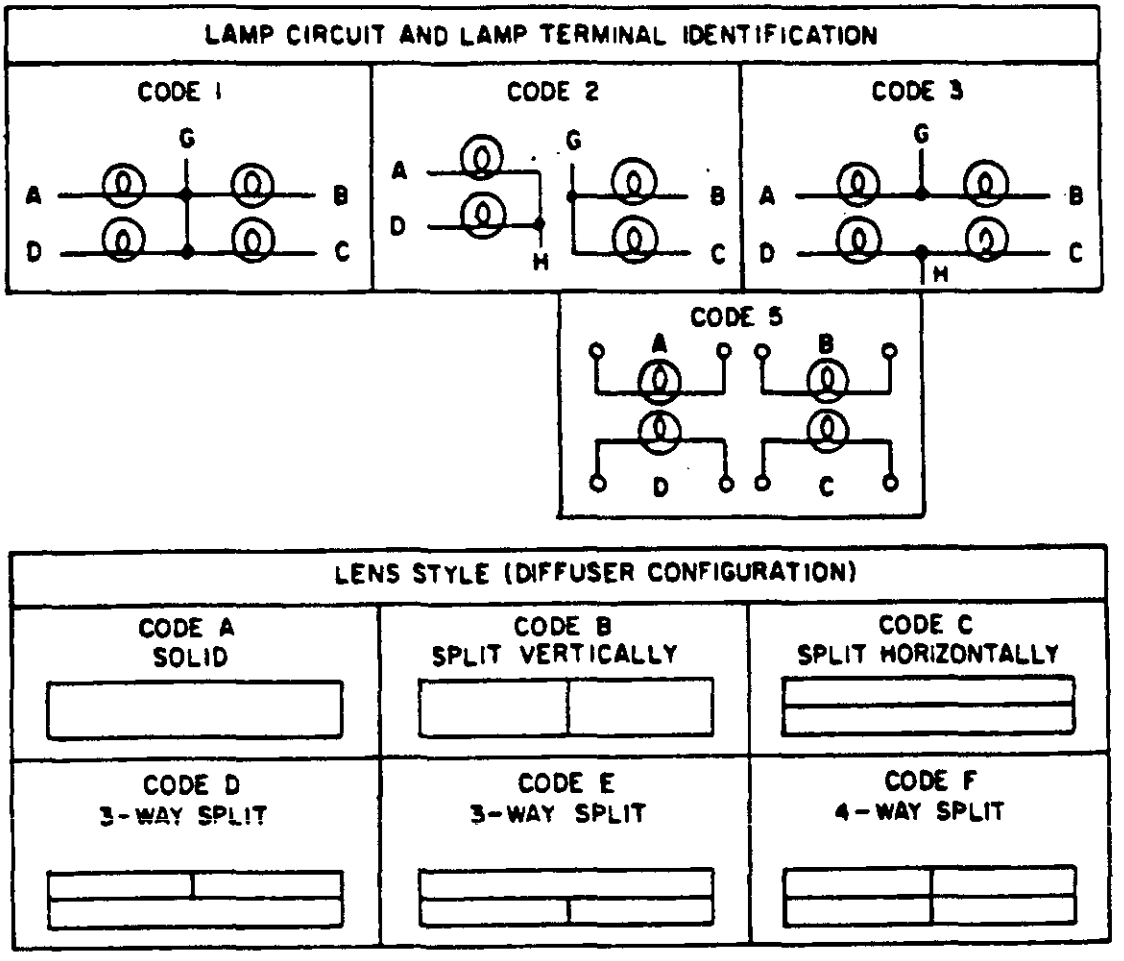
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



Load	Sea level		50,000 feet
	28 Vdc	115 Vac	28 Vdc
	(amperes)	(amperes)	(amperes)
Resistive	5.0	5.0	5.0
Inductive	3.0	5.0	2.5
Lamp	2.4	1.5	2.4

FIGURE 102-18. Switch, pushbutton, illuminated, 4-lamp, solid mount, modular constructed, high impact shock, (MIL-S-22885/87) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



Part numbers: M22885/87- (and two-digit alpha-numeric dash number from tables I and II. Add color filter symbols when required). For direct government stocking, do not use color filter symbols; color filters should be procured separately.

EXAMPLE: M22885/87- D E G Y Y G

- Table I (lens style, lamp wiring code)
- TABLE II (switch action, EMI screen).
- Color of lamp filters in positions 1,2, 3, and 4, respectively.

This part number describes a complete switch (including front lens plate) with vertical split lens style and lamp circuit, 4PDT alternate switch action, green color filters in positions 1 and 4, and yellow color filters in positions 2 and 3.

FIGURE 102-18. Switch, pushbutton, illuminated, 4-lamp, solid mount, modular constructed, high impact shock, (MIL-S-22885/87) - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE I. Code letter for combination of lens style and lamp wiring code.

Code letter	Lens style	Lamp wiring code	Front lens plate
A	A	1	Included
C	A	3	"
D	B	2	"
E	C	3	"
L	B	1	"
M	C	1	"
N	D	1	"
P	D	3	"
Q	D	5	"
R	E	1	"
S	E	3	"
T	E	5	"
U	F	1	"
V	F	5	"
F	A	1	Not included
H	A	3	"
J	B	2	"
K	C	3	"
W	B	1	"
X	C	1	"
Y	D	1	"
Z	D	3	"
1	D	5	"
2	E	1	"
3	E	3	"
4	E	5	"
5	F	1	"
6	F	5	"

TABLE II. Code letter for switch action and EMI screen.

Code letter		Switch action	EMI screen
A		None 1/	No
B	2/	2PDT momentary	"
C	2/	4PDT momentary	"
D	2/	2PDT alternate	"
E	2/	4PDT alternate	"
F		None 1/	Yes
H	2/	2PDT momentary	"
J	2/	4PDT momentary	"
K	2/	2PDT alternate	"
L	2/	4PDT alternate	"

1/ Indicator only.
2/ For other than direct government stocking, procurement of assembled units may be made with the code letters indicated.

FIGURE 102-18. Switch, pushbutton, illuminated, 4-lamp, solid mount, modular construction, high impact shock (MIL-S-22885/87) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Electrical ratings: See tables I and II.

TABLE I. Electrical ratings - MS24547-1 basic switch (silver contacts).

Load	Sea level		50,000
	28 Vdc (amperes)	115 Vac, 60 Hz (amperes)	28 Vdc (amperes)
Resistive	7	7	4
Inductive	4	7	2.5
Lamp	2.5	2	2.5

TABLE II. Electrical ratings - MS24547-2 basic switch (gold contacts).

Load	Sea level	50,000
	28 Vdc (amperes)	28 Vdc (amperes)
Resistive	1	1
Inductive	.5	.5

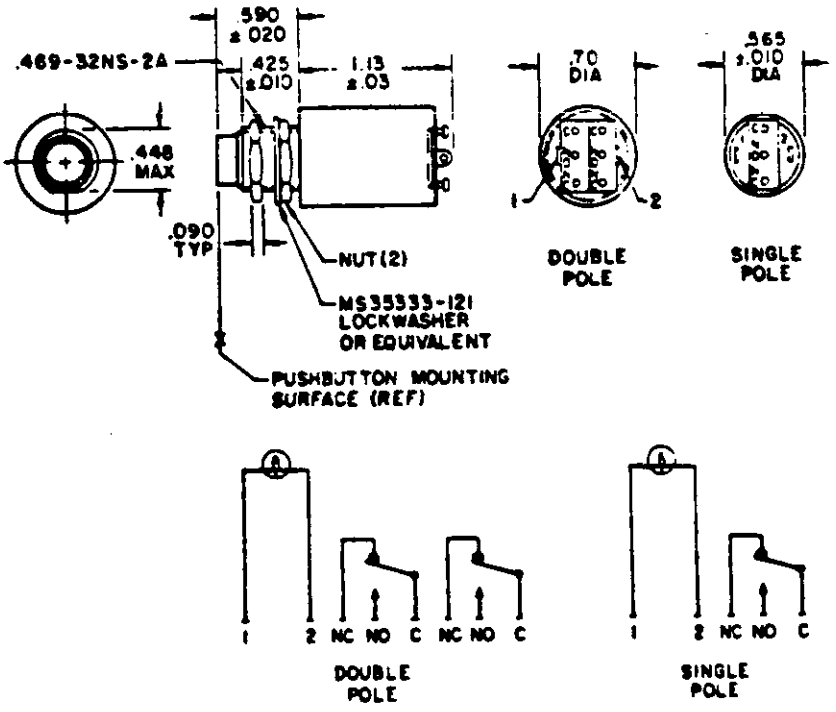


FIGURE 102-19. Switch, pushbutton, illuminated, single lamp, 7 amperes, momentary action (MIL-S-22885/89) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

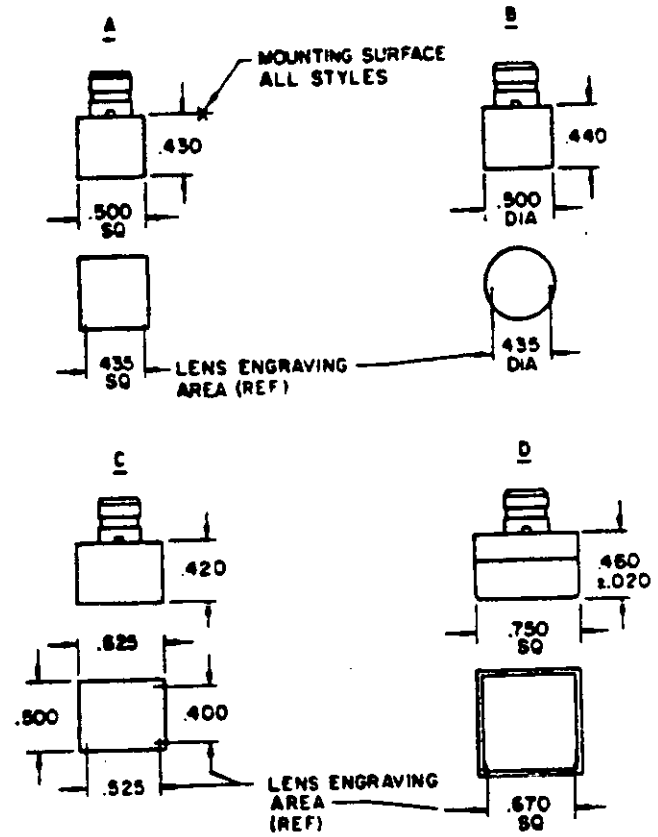
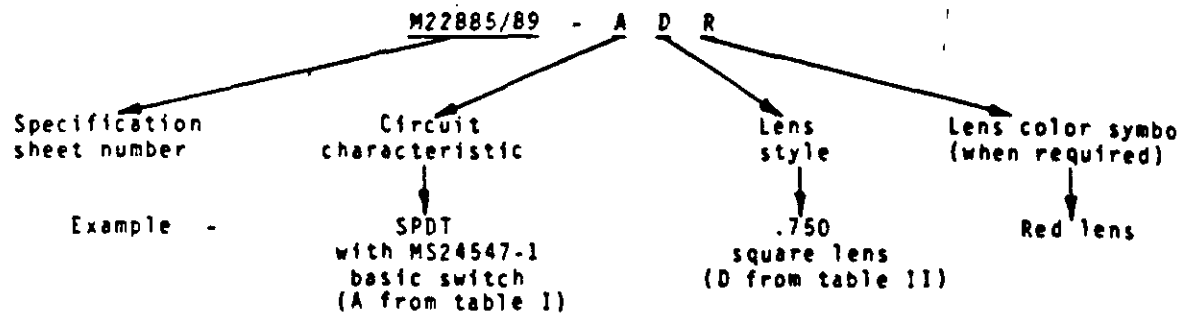


FIGURE 102-19. Switch, pushbutton, illuminated, single lamp, 7 amperes, momentary action (MIL-S-22885/89) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Part number: Part numbers shall be assigned as illustrated below:



Part number M22885/89-ADR identifies a SPDT switch with silver contacts and a .750 square red lens.

TABLE I. Circuit characteristics.

Code	Basic switch	Circuit
A	MS24547-1	SPDT
B	MS24547-1	DPDT
C	MS24547-2	SPDT
D	MS24547-2	DPDT
E 1/	None	---

1/ Use code E to specify lens portion of switch only.

TABLE II. Lens style.

Code	Lens style
A	A
B	B
C	C
D	D
Z 1/	None

1/ Use code Z to specify switch body only (less lens).

FIGURE 102-19. Switch, pushbutton, illuminated, single lamp, 7 amperes, momentary action (MIL-S-22885/89) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

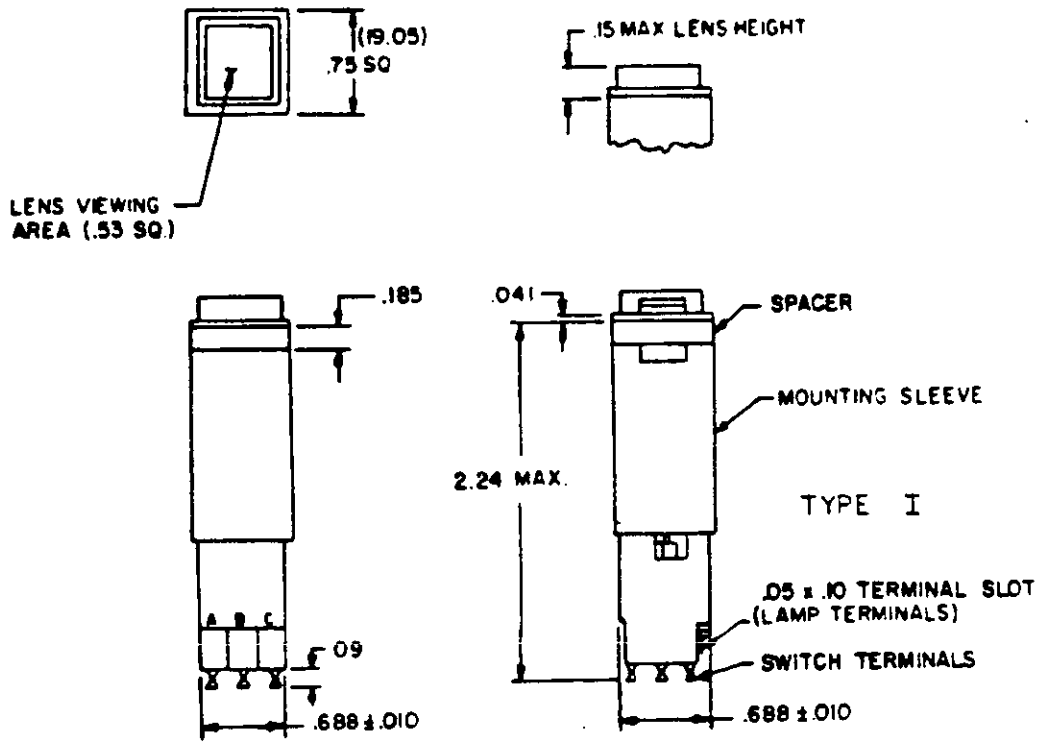


FIGURE 102-20. Switch, pushbutton, illuminated, 4-lamp, 0.750 square, 7 amperes, sunlight readable display (MIL-S-22885/90)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

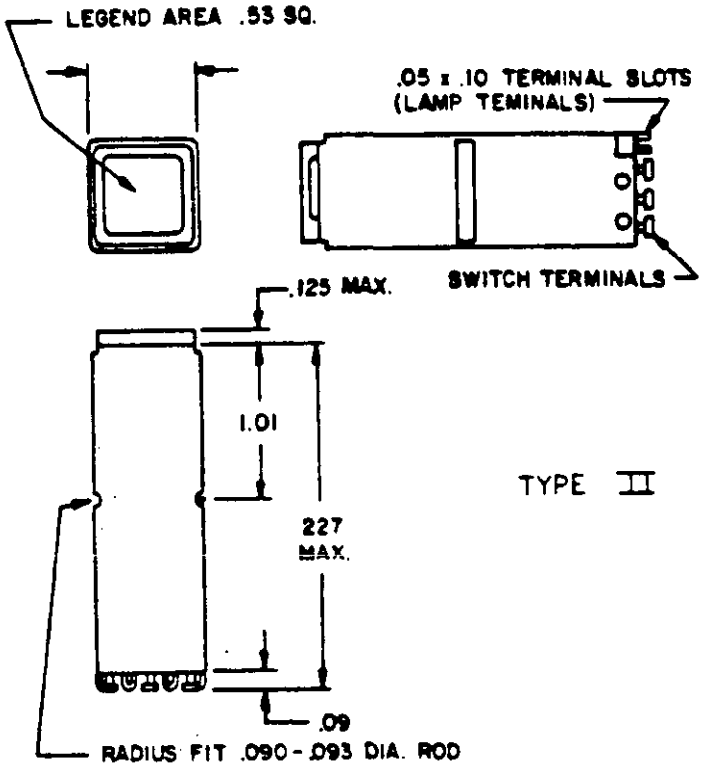


FIGURE 102-20. Switch, pushbutton, illuminated, 4-lamp, 0.750 square, 7 amperes, sunlight readable display (MIL-S-22885/90) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Contact material	Load	Sea level		50,000
		28 Vdc (amperes)	115 Vac, 60 Hz (amperes)	28 Vdc (amperes)
Silver (MS24547-1)	Resistive	7	7	4
	Inductive	4	7	2.5
	Lamp	2.5	2	2.5
Gold (MS24547-6)	Resistive	1	-	1
	Inductive	0.5	-	0.5
	Low level life applicable: 30 millivolts maximum or peak ac at 10 milliamperes maximum			

FIGURE 102-20. Switch, pushbutton, illuminated, 4-lamp, 0.750 square, 7 amperes, sunlight readable display (MIL-S-22885/90) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Part number: Part numbers shall be assigned as illustrated below:

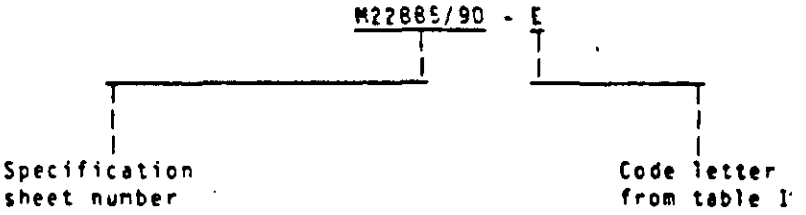


TABLE I. Part number codes. 1/

Color letters		Circuit	Operation	Contact material
Type I	Type II			
A	G	1PDT	Momentary	Silver
B	H	2PDT	"	"
C	J	3PDT	"	"
				"
				"
D	K	1PDT	Alternate	"
E	L	2PDT	"	"
F	M	3PDT	"	"
AA	GA	1PDT	Momentary	Gold
BA	HA	2PDT	"	"
CA	JA	3PDT	"	"
				"
				"
DA	KA	1PDT	Alternate	"
EA	LA	2PDT	"	"
FA	MA	3PDT	"	"

1/ Part number codes are shown for switch body portion only. Lens modules, with required legends, dripproof seals, or EMI shielding shall be procured separately from a source listed on the qualified products list. Available lens module configuration are shown for information only:

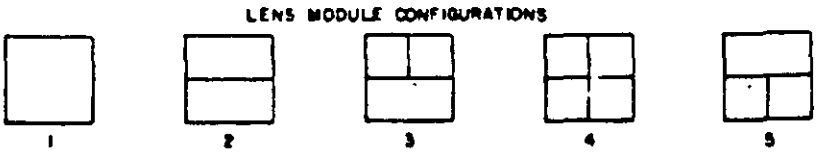
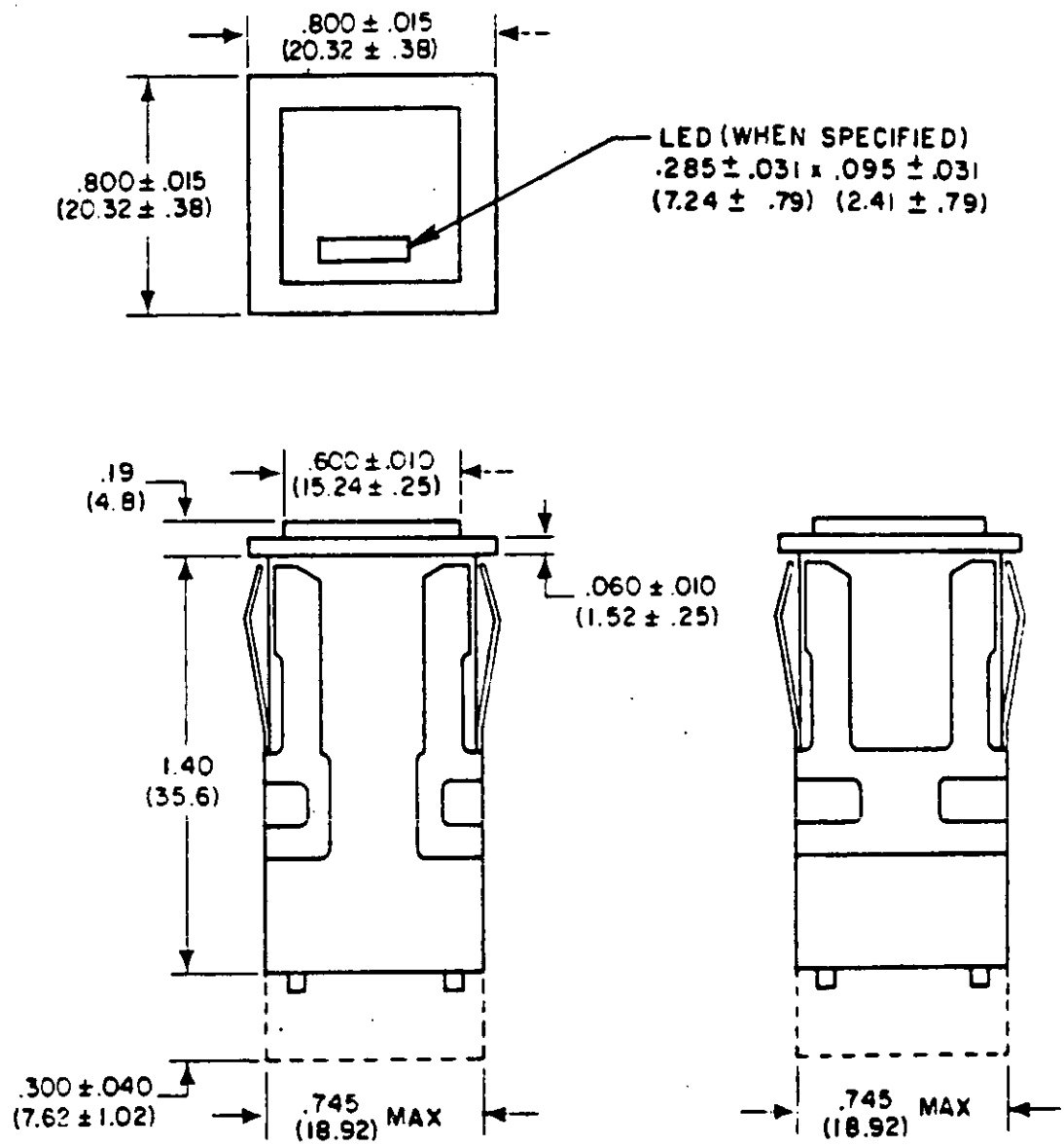


FIGURE 102-20. Switch, pushbutton, illuminated, 4-lamp, 0.750 square, 7 amperes, sunlight readable display (MIL-S-22885/90) - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.



PUSHBUTTON - SQUARE

FIGURE 102-21. Switch, pushbutton, illuminated, rocker, paddle, solid-state hall effect, unsealed (MIL-S-22885/91).

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.

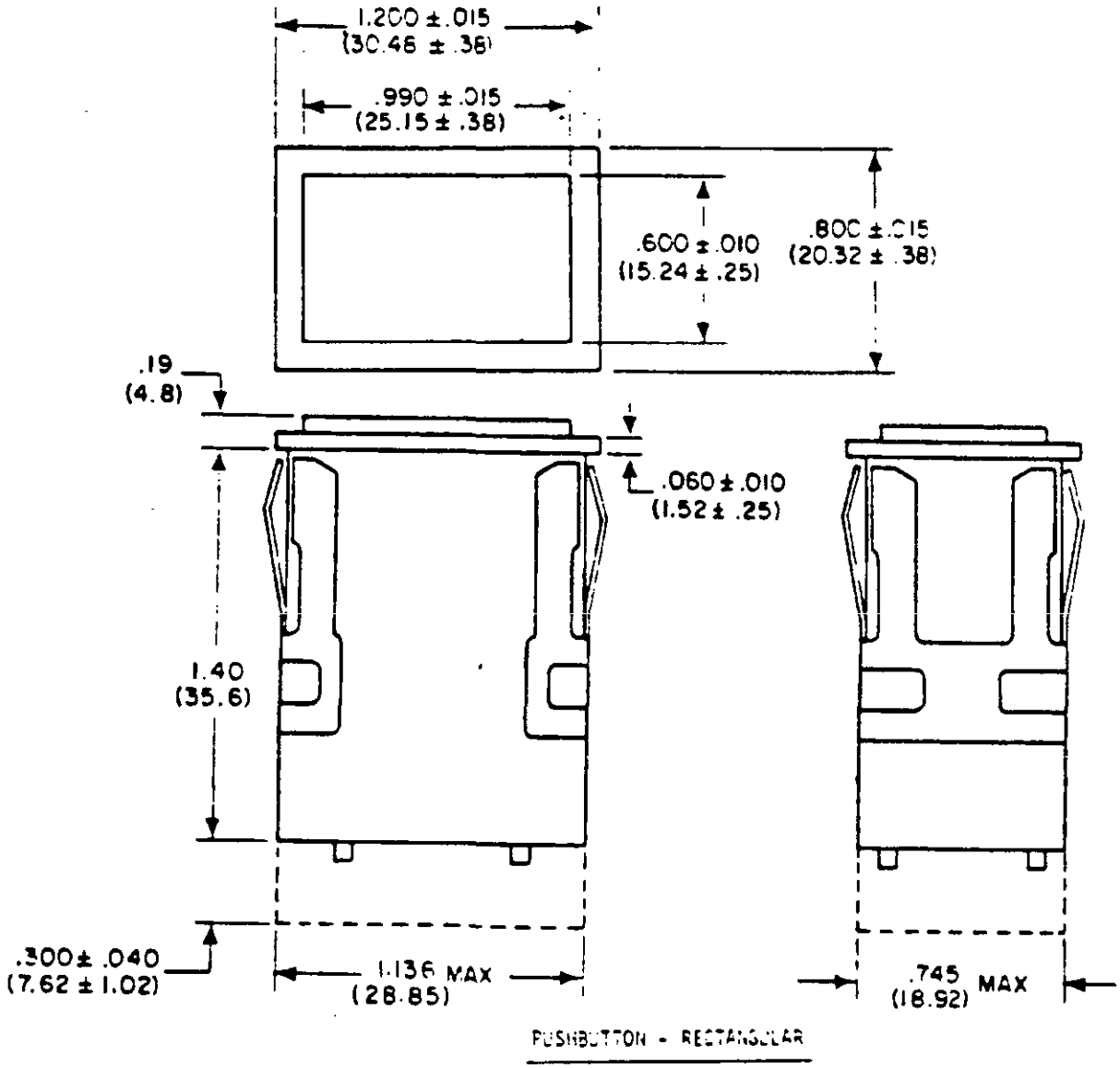


FIGURE 102-21. Switch, pushbutton, illuminated, rocker, paddle, solid-state hall effect, unsealed (MIL-S-22885/91) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

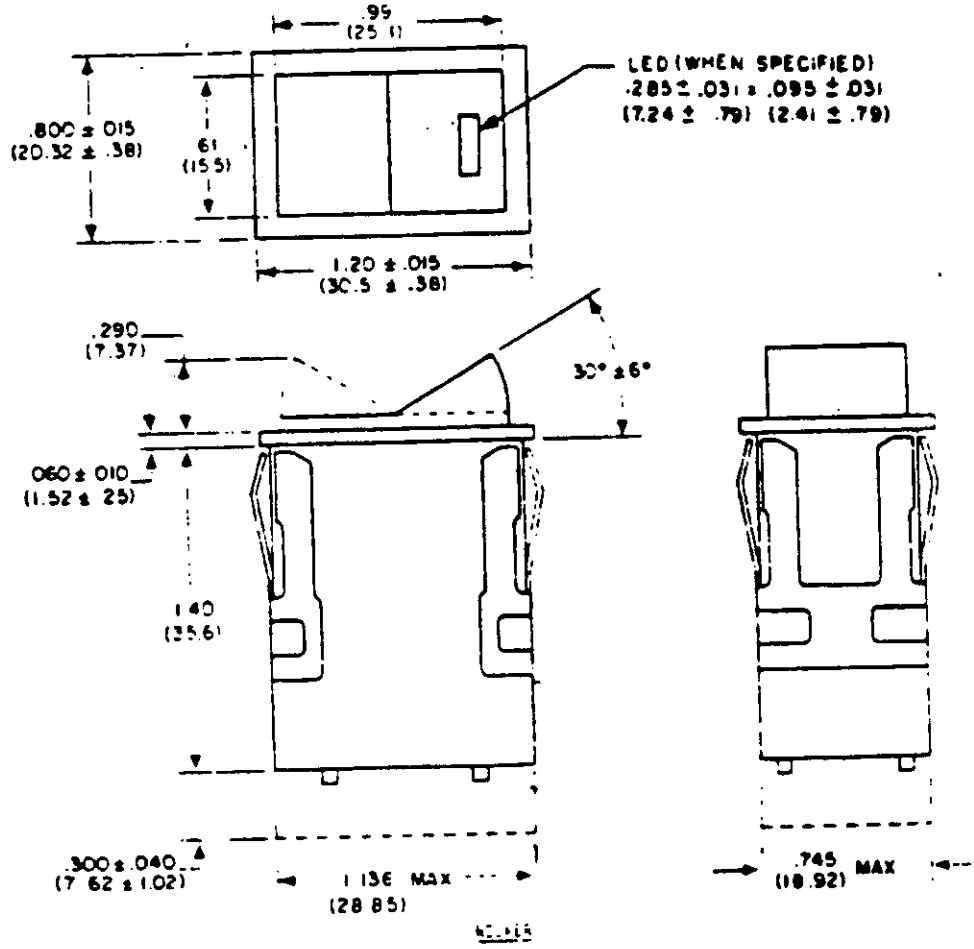


FIGURE 102-21. Switch, pushbutton, illuminated, rocker, paddle, solid-state hall effect, unsealed (MIL-S-22885/91) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

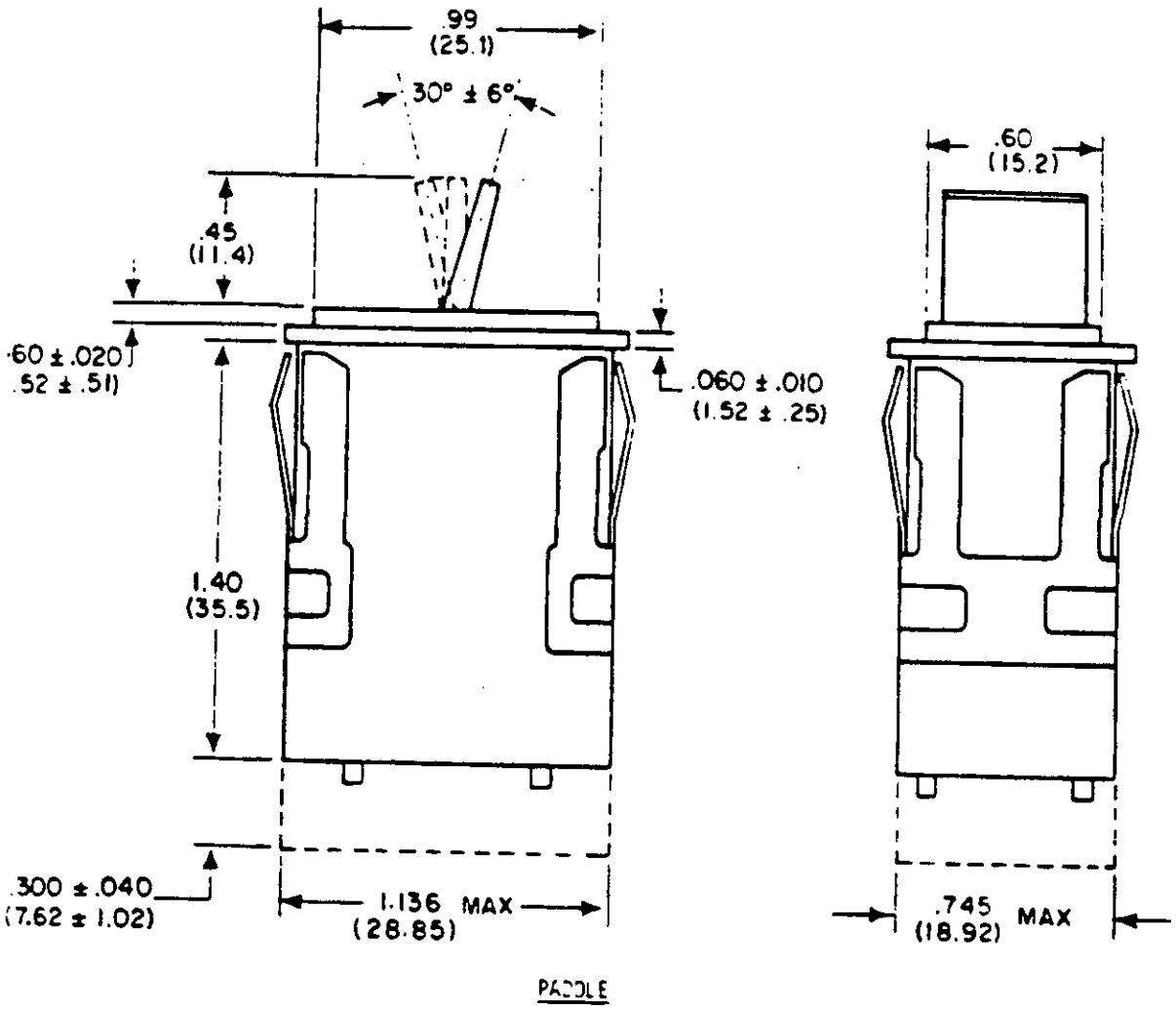


FIGURE 102-21. Switch, pushbutton, illuminated, rocker, paddle, solid-state hall effect, unsealed (MIL-S-22885/91) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE I. Circuitries.

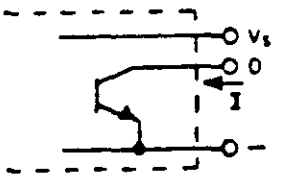
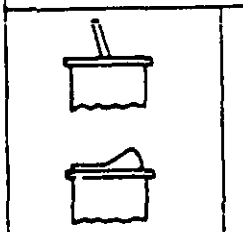
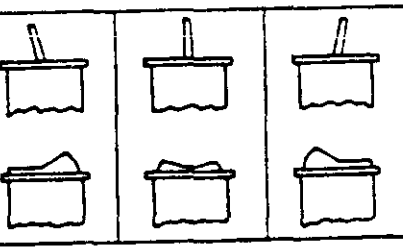
<table>
<tr><td colspan="6">HALL EFFECT INTEGRATED CIRCUIT</td></tr>
<tr><td colspan="6"> 
</td></tr>
<tr><td colspan="6">CIRCUIT OUTPUT STATES</td></tr>
<tr><td colspan="6">PUSHBUTTON
ONE INTEGRATED CIRCUIT ONLY</td></tr>
<tr><td colspan="3">BUTTON POSITION</td><td colspan="3">CIRCUIT CONDITION</td></tr>
<tr><td colspan="3">FREE POSITION</td><td colspan="3">CIRCUIT RELEASED</td></tr>
<tr><td colspan="3">FULL OVERTRAVEL OR 2ND LEVEL ALTERNATE ACTION</td><td colspan="3">CIRCUIT OPERATED</td></tr>
<tr><td colspan="6">PADDLE AND ROCKER</td></tr>
<tr><td rowspan="2">CIRCUITRY</td><td colspan="3">2 POSITION
1 OR 2 INTEGRATED CIRCUITS</td><td colspan="2">3 POSITION
2 INTEGRATED CIRCUITS ONLY</td><td></td></tr>
<tr><td colspan="3">  </td><td colspan="2">  </td><td></td></tr>
<tr><td rowspan="2">ONE CIRCUIT</td><td>CIRCUIT A</td><td>RELEASED</td><td>OPERATED</td><td colspan="3" rowspan="2"></td></tr>
<tr><td>CIRCUIT B</td><td>NONE</td><td>NONE</td></tr>
<tr><td rowspan="2">TWO CIRCUITS</td><td>CIRCUIT A</td><td>RELEASED</td><td>OPERATED</td><td>OPERATED</td><td>RELEASED</td><td>RELEASED</td></tr>
<tr><td>CIRCUIT B</td><td>RELEASED</td><td>OPERATED</td><td>RELEASED</td><td>RELEASED</td><td>OPERATED</td></tr>
</table>

FIGURE 102-21. Switch, pushbutton, illuminated, rocker, paddle, solid-state hall effect, unsealed (MIL-S-22885/91) - Continued.

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:41Z
Check the source to verify that this is the current version before use.

Integrated circuit type	Parameter	Min.	Nom.	Max.	Conditions
Electrical characteristics code A					
5 Vdc sinking	Supply voltage	4.5 Vdc	5.0 Vdc	5.5 Vdc	0° to 55°C
	Supply current released	---	---	.0035 A	Circuits released excludes load current supply voltage 4.5 to 5.5 Vdc
	Supply current operated	---	---	.0065 A	Circuit operated excludes load current supply voltage 4.5 to 5.5 Vdc
	Output voltage operated	---	---	+.4 Vdc	Circuit operated load current = .008 A supply voltage 4.5 to 5.5 Vdc
	Output leakage	---	---	5.0 micro amperes	Output terminated to +10.0 Vdc supply voltage 4.5 to 5.5 Vdc circuit released
	Fall time	---	---	0.5 micro second	Supply voltage 4.5 to 5.5 Vdc 90% to 10% of the output voltage swing when sinking .008 A
	Rise time	---	---	1.0 micro second	Supply voltage 4.5 to 5.5 Vdc 10% to 90% of the output voltage swing when sinking .008 A
Absolute maximum ratings (beyond which damage may occur)					
5 Vdc sinking	Ambient temperature	-40°C	---	+65°C	Storage
		0°C	---	+55°C	Operating
	Supply voltage	-5 Vdc	---	+7.0 Vdc	0° to +55°C
	Externally applied output voltage	-.5 Vdc	---	+15.0 Vdc	Circuit released
	Sinking output current	---	---	.020 A	Circuit operated

FIGURE 102-21. Switch, pushbutton, illuminated, rocker, paddle, solid-state hall effect, unsealed (MIL-S-22885/91) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Integrated circuit type	Parameter	Min.	Nom.	Max.	Conditions
Electrical characteristic code B					
6-16 Vdc sinking	Supply voltage	6.0 Vdc	12.0 Vdc	16.0 Vdc	0°C to 55°C
	Supply current released	---	---	.007 A at 6 Vdc .0115 A at 16 Vdc	Circuit released excludes load current
	Supply current operated	---	---	.0105 A at 6 Vdc .015 A at 16 Vdc	Circuit operated excludes load current
	Output voltage operated	---	---	+.4 Vdc	Circuit operated load current .020 A supply voltage 6 to 16 Vdc
	Output leakage	---	---	20.0 micro amperes	Output terminated to +16 Vdc supply voltage 6 to 16 Vdc circuit released
	Fall time	---	---	0.5 micro second	Supply voltage 6 to 16 Vdc 90% to 10% of the output voltage swing when sinking .020 A
	Rise time	---	---	1.5 micro second	Supply voltage 6 to 16 Vdc 10% to 90% of the output voltage swing when sinking .020 A
Absolute maximum ratings (beyond which damage may occur)					
6-16 Vdc sinking	Ambient temperature	-40°C	---	+65°C	Storage
		C°C	---	+55°C	Operating
	Supply voltage	-1.2 Vdc	---	+20.0 Vdc	0°C to +55°C
	Externally applied output voltage	-.5 Vdc	---	+20.0 Vdc	Circuit released
	Sinking output current	---	---	.040 A	Circuit operated

FIGURE 102-21. Switch, pushbutton, illuminated, rocker, paddle, solid-state hall effect, unsealed (MIL-S-22885/91) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Operating characteristics: See table I.

TABLE I. Operating characteristics.

Configuration type	Characteristics	Value
Pushbuttons	Total actuator travel	.180 in. min.
Rocker		30° ±6°
Paddle		30° ±6°

Part number for pushbutton switches:

EXAMPLE: M22885/91-C E B 2 A W

- Configuration (See Table II)
- Illumination (See Table II)
- Terminals (See Table II)
- Electrical characteristic and action (See Table III)
- Display type and button style (See Table III)
- Button color (leave blank when specifying housing only)

TABLE II. Configuration, illumination and terminal codes for pushbutton switches.

Configuration	Illumination	Terminals
A - Square, nonilluminated	A - Nonilluminated or incandescent (no lamp installed)	A - Solder or quick-connect
B - Square, 1 lamp circuit	B - *V Red LED	B - Printed circuit
C - Square, 1 LED	C - 5.0 V Red LED	
D - Rectangular, non-illuminated	D - 10.0 V Red LED	
E - Rectangular, 2 lamp circuits	E - 15.0 V Red LED	
	F - *V Yellow LED	
	G - 5.0 V Yellow LED	
	H - 10.0 V Yellow LED	
	J - 15.0 V Yellow LED	
	K - *V Green LED	
	L - 5.0 V Green LED	
	M - 10.0 V Green LED	
	N - 15.0 V Green LED	
	* Refer to application information for non-resistor	

FIGURE 102-21. Switch, pushbutton, illuminated, rocker, paddle, solid-state hall effect, unsealed (MIL-S-22885/91) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE III. Electrical characteristics, action, display and button style for pushbutton switches.

Electrical characteristic/action	Display type/ button style 1/ 2/	Button color
1 - 5 Vdc/momentary 2 - 6 to 16 Vdc/momentary 3 - 5 Vdc/alternate 4 - 6 to 16 Vdc/alternate	A - Transmitted color - Square B - Projected color - Square C - Hidden color - Square D - LED window - Square E - Transmitted color - Rectangular F - Projected color - Rectangular G - Hidden Color - Rectangular J - Switch body only. No button, caps, or color inserts supplied. Procure separately from source listed on QPL	R - Red Y - Yellow G - Green B - Blue W - White K - Black** L - Gray** **For non-illuminated and LED configuration types only

1/ Use code A, E or J, as required, for nonilluminated switches.
2/ Button styles are shown as viewed from the front of the panel.

FIGURE 102-21. Switch, pushbutton, illuminated, rocker, paddle, solid-state hall effect, unsealed (MIL-S-22885/91) - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.

Part number for rocker and paddle switches:

EXAMPLE: M22885/91- H C A 6 3 J K K

	Configuration	
See Table IV	Illumination	
	Terminals	
See Table V	Electrical characteristic and circuitry	
	Action	
	Display type and actuator style	
See Table VI	Color (side A) leave blank when specifying switch body only	
	Color (side B) leave blank when display types A ? B or when specifying switch body only	

TABLE IV. Configuration, illumination and terminal codes for rocker and paddle switches.

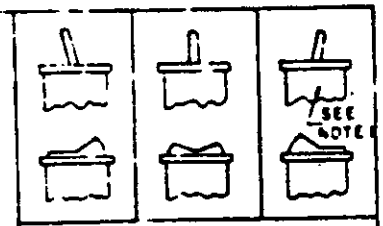
Configuration	Illumination	Terminals
F - Rocker, non-illuminated G - Rocker, 2 lamp circuits H - Rocker, 1 LED J - Paddle, non-illuminated K - Paddle, 2 lamp circuits L - Paddle, 1 LED	A - Non-illuminated or incandescent (no lamp installed) B - *V Red LED C - 5.0 V Red LED D - 10.0 V Red LED E - 15.0 V Red LED F - *V Yellow LED G - 5.0 V Yellow LED H - 10.0 V Yellow LED J - 15.0 V Yellow LED K - *V Green LED L - 5.0 V Green LED M - 10.0 V Green LED N - 15.0 V Green LED * Refer to application information for non-resistor	A - Solder or quick-connect B - Printed circuit

FIGURE 102-21. Switch, pushbutton, illuminated, rocker, paddle, solid-state hall effect, unsealed (MIL-S-22885/91) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE V. Electrical characteristic, circuitry and action codes for rocker and paddle switch.

Electrical characteristic/ circuitry	
5	5 Vdc/1 circuit
6	5 Vdc/2 circuits
7	6 to 16 Vdc/1 circuit
8	6 to 16 Vdc/2 circuits

ACTION CODE	
	2 - POSITION
1	MAIN. MOM. MAIN.
2	(NONE) (NONE) (NONE)
3	MAIN. MAIN. MOM
	3 - POSITION 2/
4	MAIN. MOM. MAIN MOM
5	MAIN. MAIN. MAIN.
6	MAIN. MOM MOM
7	MAIN. MOM MAIN

NOTES:

1/ Circuit action is shown with the switch positioned such that the manufacturer's logo is located on the front side of the switch.

2/ Available with 2 circuit units only.

FIGURE 102-21. Switch, pushbutton, illuminated, rocker, paddle, solid-state hall effect, unsealed (MIL-S-22885/91) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

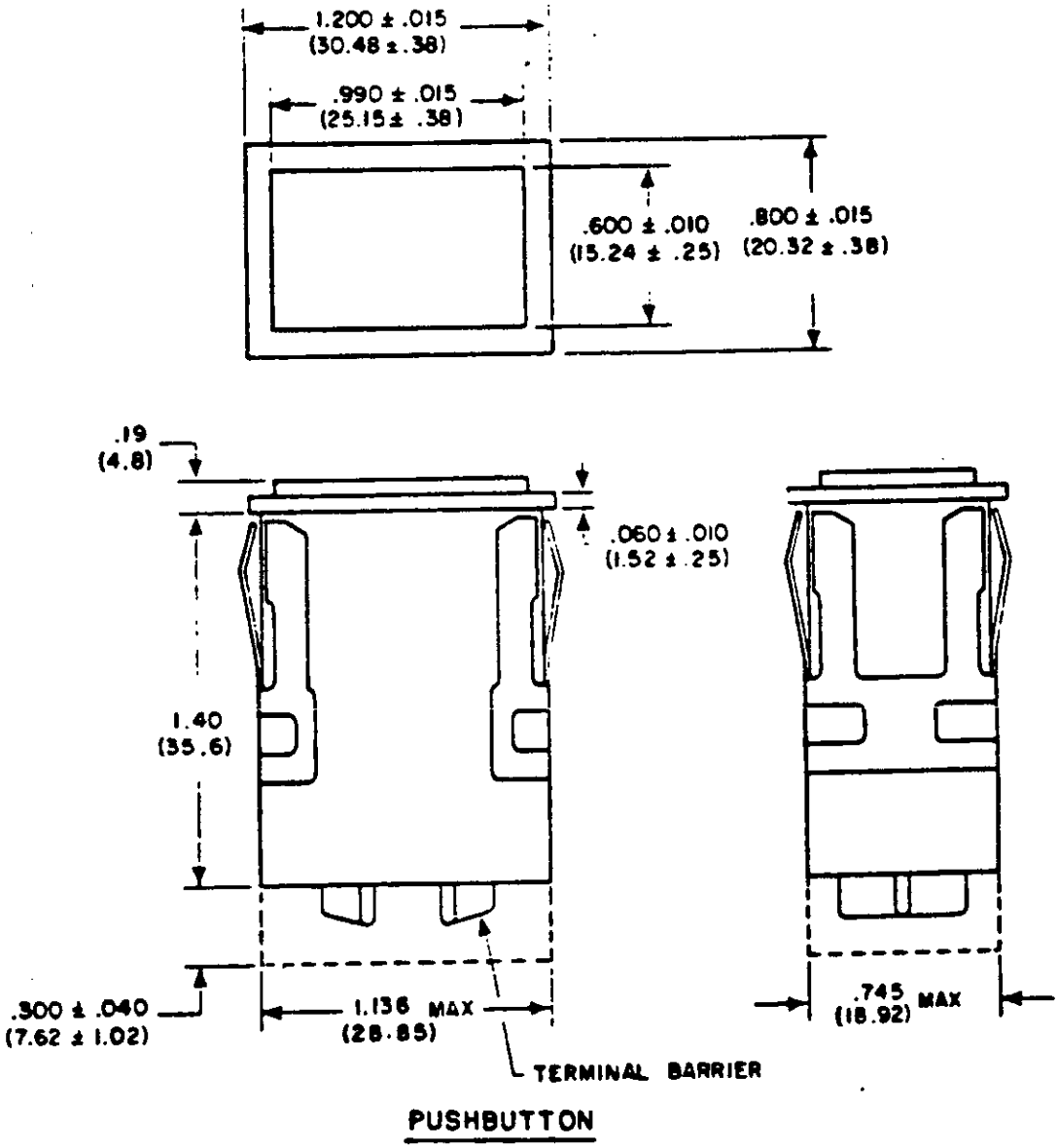
TABLE VI. Display type/actuator style and color for rocker and paddle switches.

Display type/actuator style 1/	Color (rockers, covers, or inserts)
A - Non-illuminated/1 pc. rocker (one color selection only) B - Transmitted color/1 pc. rocker (one color selection only) C - Projected color/1 pc. rocker D - Hidden color/1 pc. rocker E - Non-illuminated/2 pc. rocker F - Transmitted color/2 pc. rocker G - Projected color/2 pc. rocker H - Hidden color/2 pc. rocker J - 1 LED window/2 pc. rocker L - Non-illuminated/2 pc. paddle cover M - Transmitted color/2 pc. paddle cover N - Projected color/2 pc. paddle cover P - Hidden color/2 pc. paddle cover R - 1 LED window/2 pc. paddle cover S - Switch body only. No rockers, covers, or color inserts supplied. Procure separately from a source listed on the QPL.	Side A / Side B R - Red / R - Red Y - Yellow / Y - Yellow G - Green / G - Green B - Blue / B - Blue W - White / W - White K - Black** / K - Black** L - Gray** / L - Gray** **For non-illuminated and LED configurations only.

1/ Button styles are shown as viewed from front of the panel.

FIGURE 102-21. Switch, pushbutton, illuminated, rocker, paddle, solid-state hall effect, unsealed [MIL-S-22885/91] - Continued.

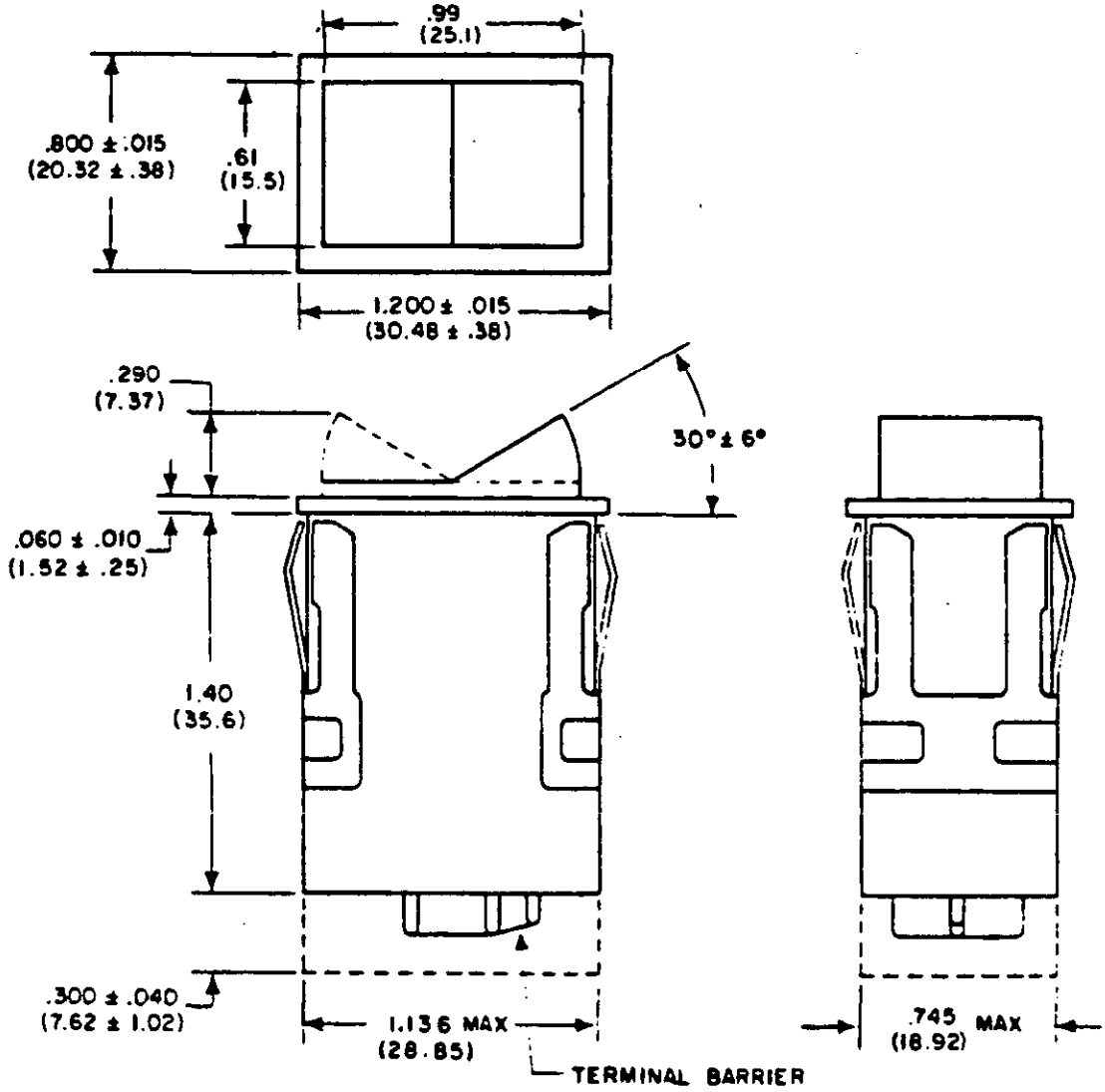
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



PUSHBUTTON

FIGURE 102-22. Switch, pushbutton, illuminated, rocker, paddle, power duty, unsealed (MIL-S-22885/92).

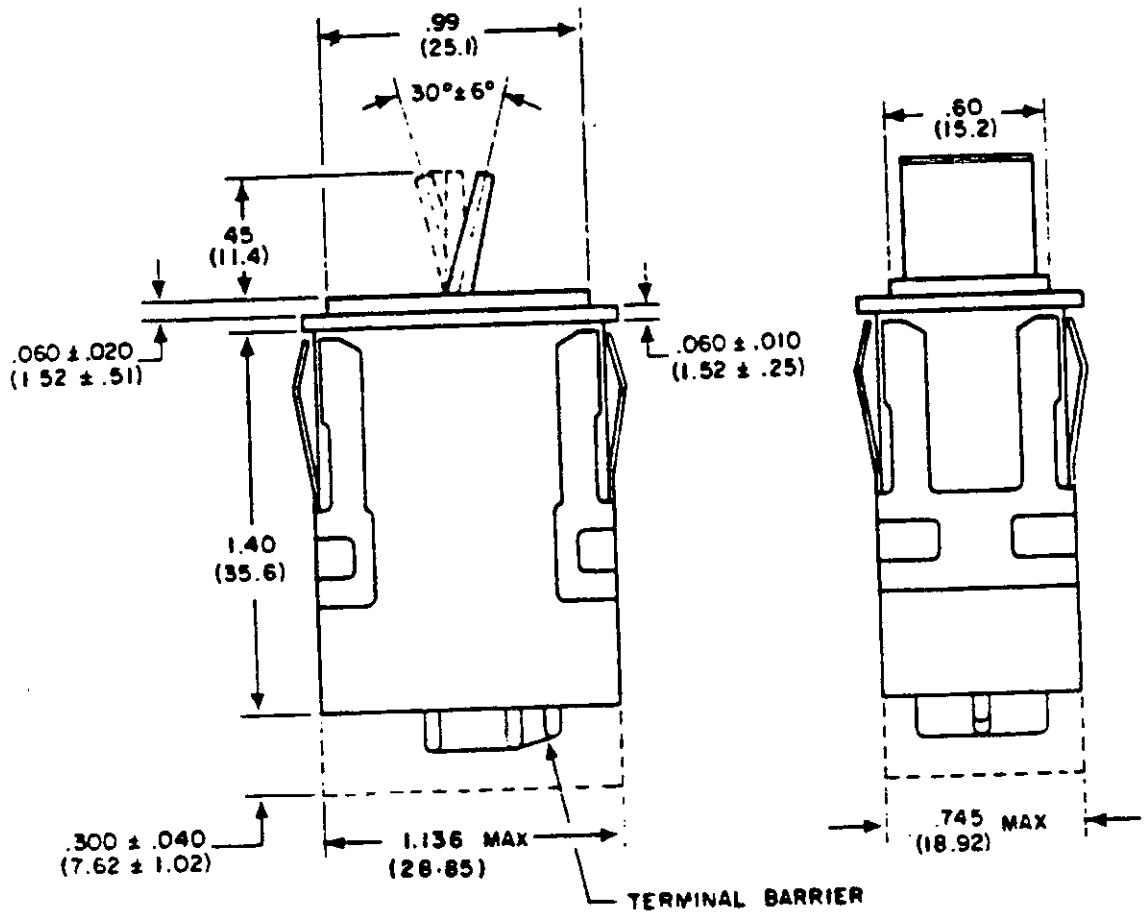
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



ROCKER

FIGURE 102-22. Switch, pushbutton, illuminated, rocker, paddle, power duty, unsealed (MIL-S-22885/92) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

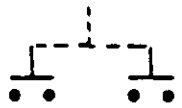


PADDLE

FIGURE 102-22. Switch, pushbutton, illuminated, rocker, paddle, power duty, unsealed (MIL-S-22885/92) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE I. Circuitries

PUSHBUTTON	
POLES	CIRCUITRIES
2 (FORM X)	

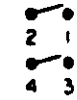
ROCKER AND PADDLE		
POLES		
1 (FORM A)		
2 (FORM A)		

FIGURE 102-22. Switch, pushbutton, illuminated, rocker, paddle, power duty, unsealed (MIL-S-22885/92) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Part number for pushbutton switches:

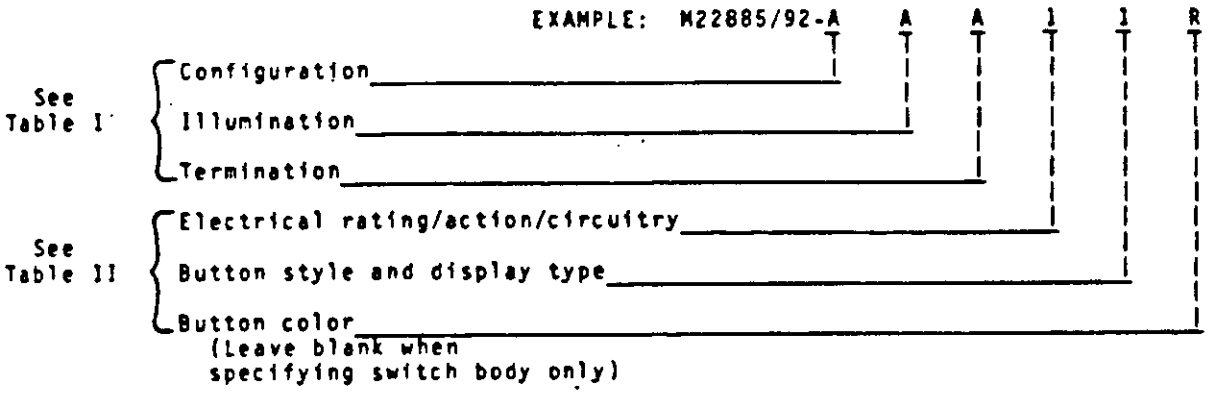


TABLE I. Configuration illumination and terminal codes for pushbutton switches.

Configuration	Illumination	Termination
A - Rectangular pushbutton nonilluminated B - Rectangular pushbutton 1 lamp circuit	A - No lamps installed	A - Solder/quick connect

TABLE II. Electrical rating, action, circuitry, display and button style for pushbutton switches.

Electrical rating/action/circuitry	Button style and display type	Button color
1 - 10 Amperes, 125 Vac (60% pF) 8 Amperes, 28 Vdc resistive Alternate Action/2 Pole 2 - 10 Amperes, 125 Vac (60% pF) 8 Amperes, 28 Vdc resistive Momentary Action/2 Pole	1 - Rectangular pushbutton, transmitted color 1/ 2 - Rectangular pushbutton, projected color 3 - Rectangular pushbutton, hidden color 4 - Switch body only. No buttons, caps, or color inserts supplied. (Procure separately from source listed on QPL)	R - Red Y - Yellow G - Green B - Blue W - White K - Black** L - Gray** **For non-illuminated configuration type only

1/ Use transmitted color for nonilluminated switches.

FIGURE 102-22. Switch, pushbutton, illuminated, rocker, paddle, power duty, unsealed (MIL-S-22885/92) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Part number for rocker and paddle switches:

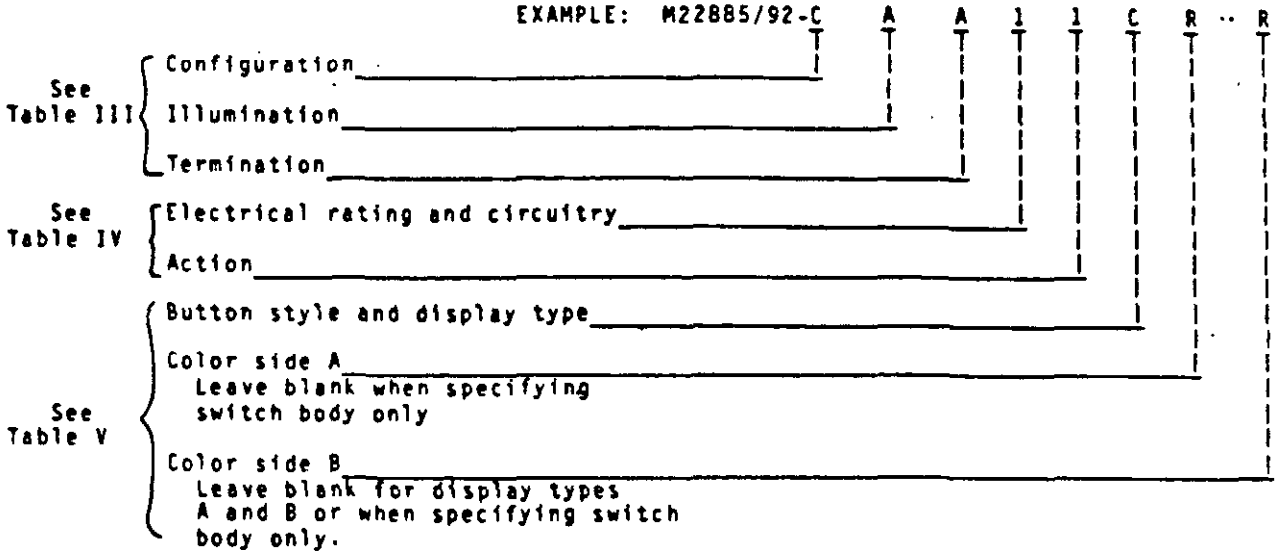


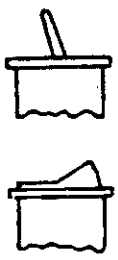
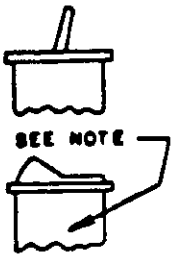
TABLE III. Configuration, illumination and terminal codes for rocker and paddle switches.

Configuration	Illumination	Termination
C - Rocker, nonilluminated	A - No lamp installed	A - Solder/quick connect
D - Rocker, 1 lamp circuit		
E - Paddle, nonilluminated		
F - Paddle, 1 lamp circuit		

FIGURE 102-22. Switch, pushbutton, illuminated, rocker, paddle, power duty, unsealed (MIL-S-22885/92) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE IV. Electrical rating, circuitry and action codes for rocker and paddle switches.

Electrical rating/circuitry	ACTION CODE		
1 - 15 Amperes, 125 Vac (60% pF) 10 Amperes, 28 Vdc resistive 1 Pole		MAINTAINED	MAINTAINED
2 - 15 Amperes, 125 Vac (60% pF) 10 Amperes, 28 Vdc resistive 2 Pole			

NOTE: Actions are shown with the switch positioned such that the manufacturer's logo is located on this side of the switch.

TABLE V. Button style, display type and color for rocker and paddle switches.

Button style/display type	Color	
	Side A	Side B
A - Nonilluminated, 1 piece rocker (one color only)	R - Red	R - Red
B - Transmitted color, 1 piece rocker (one color only)	Y - Yellow	Y - Yellow
C - Nonilluminated, 2 piece rocker	G - Green	G - Green
D - Transmitted color, 2 piece rocker	B - Blue	B - Blue
E - Nonilluminated, 2 piece paddle	W - White	W - White
F - Transmitted color, 2 piece paddle	K - Black*	K - Black*
G - Projected color, 2 piece paddle	L - Gray*	L - Gray*
H - Hidden color, 2 piece paddle		
J - Switch body only - no covers or inserts supplied. Procure from source listed on OPL.	*For nonilluminated configuration type.	



FIGURE 102-22. Switch, pushbutton, illuminated, rocker, paddle, power duty, unsealed (MIL-S-22885/92) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

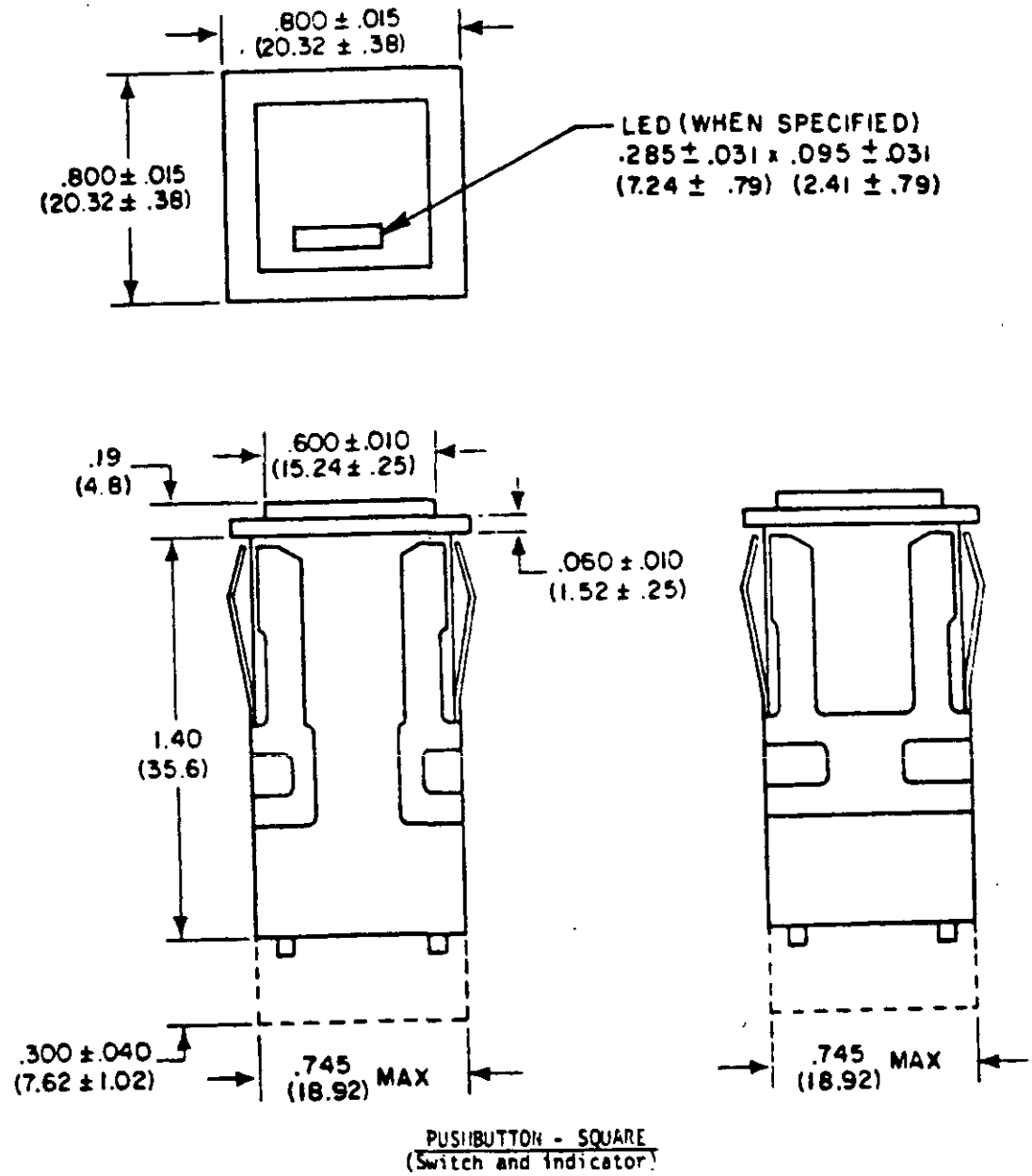
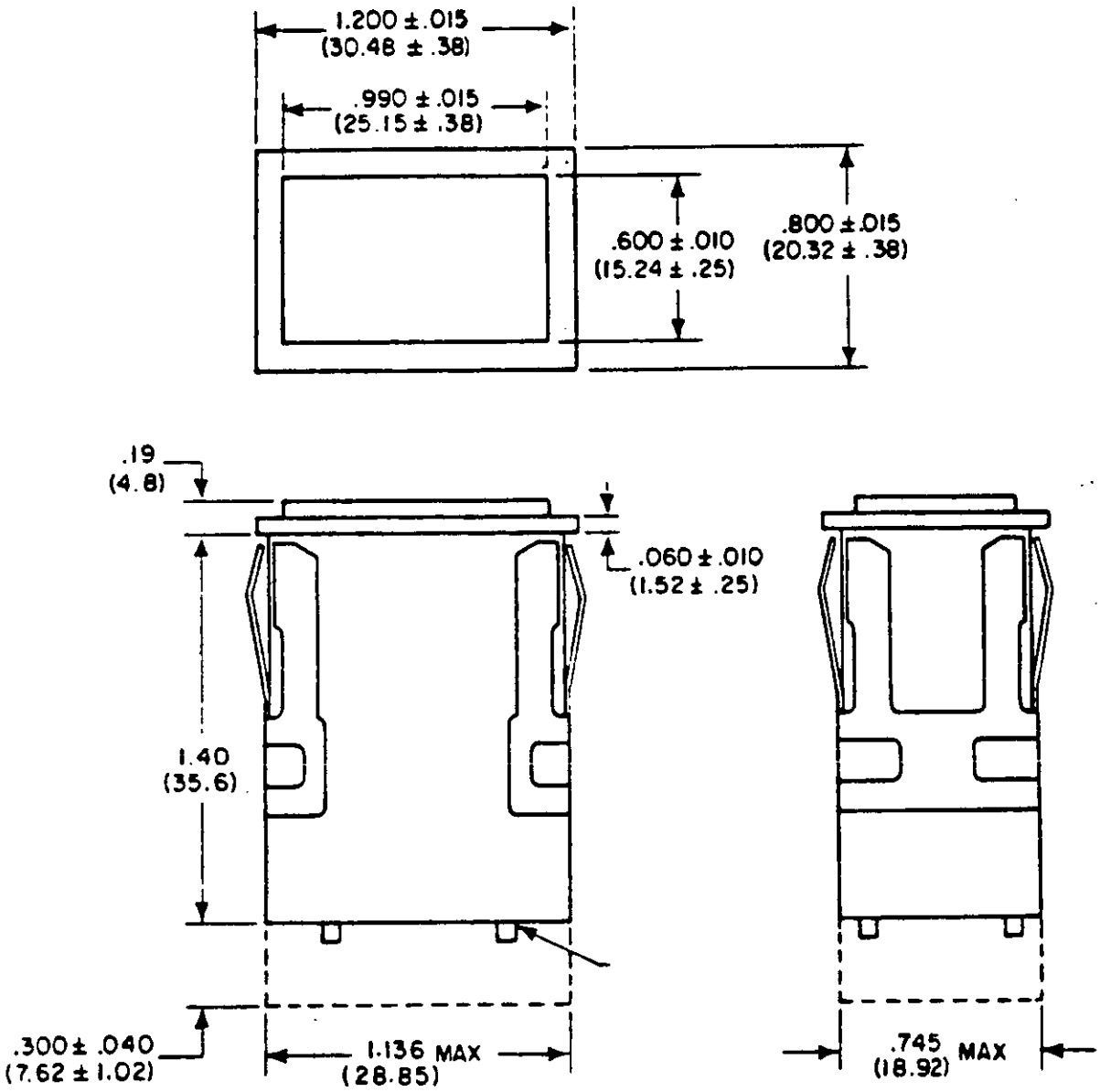


FIGURE 102-23. Switch, pushbutton, illuminated, rocker, paddle, logic load to 3 amperes, unsealed (MIL-S-22885/93).

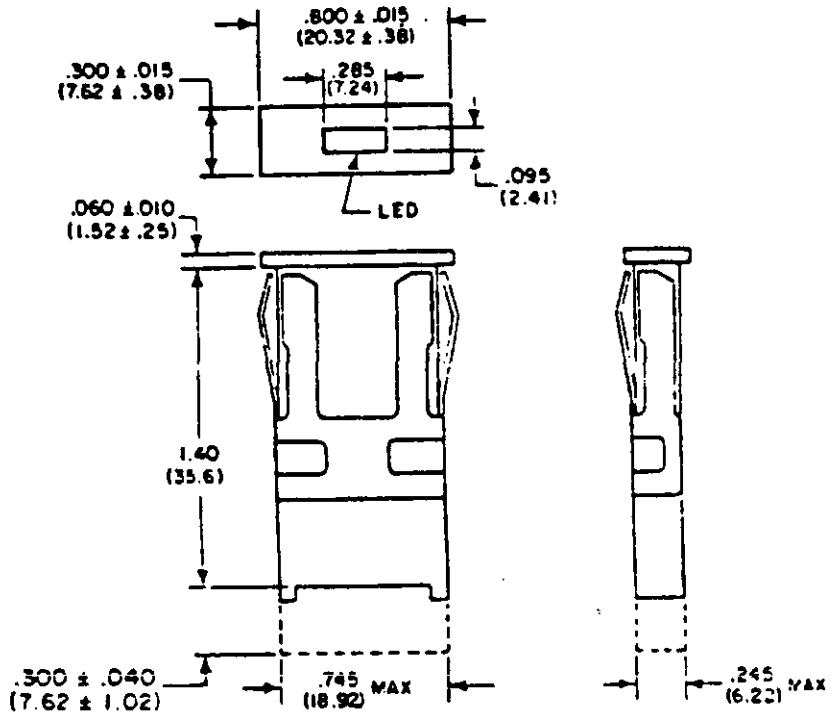
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



PUSHBUTTON-RECTANGULAR
(Switch and indicator)

FIGURE 102-23. Switch, pushbutton, illuminated, rocker, paddle, logic load to 3 amperes, unsealed (MIL-S-22885/93) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



MINIATURE INDICATOR

FIGURE 102-23. Switch, pushbutton, illuminated, rocker, paddle, logic load to 3 amperes, unsealed (MIL-S-22885/93) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

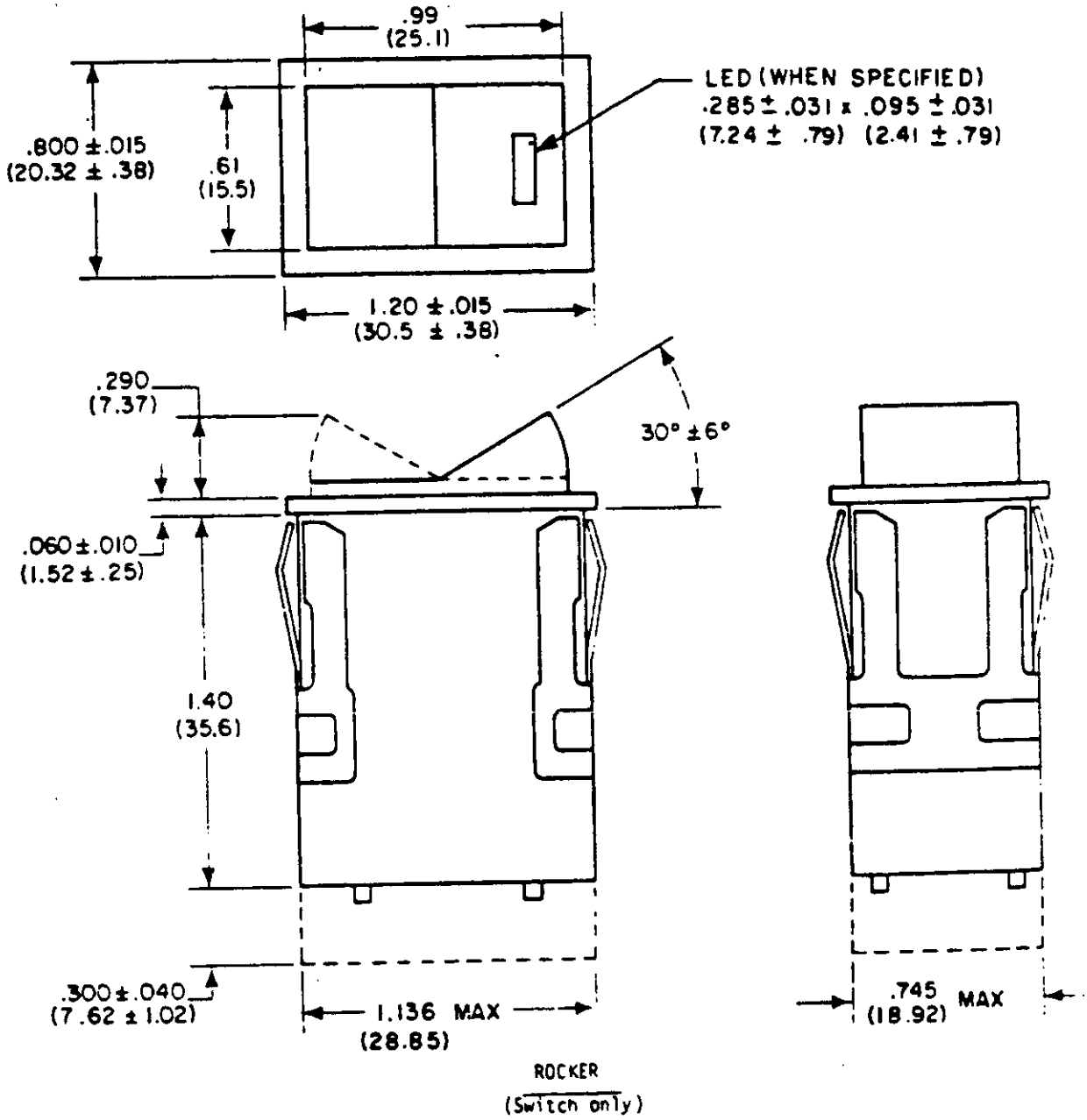


FIGURE 102-23. Switch, pushbutton, illuminated, rocker, paddle, logic load to 3 amperes, unsealed (MIL-S-22885/93) - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.

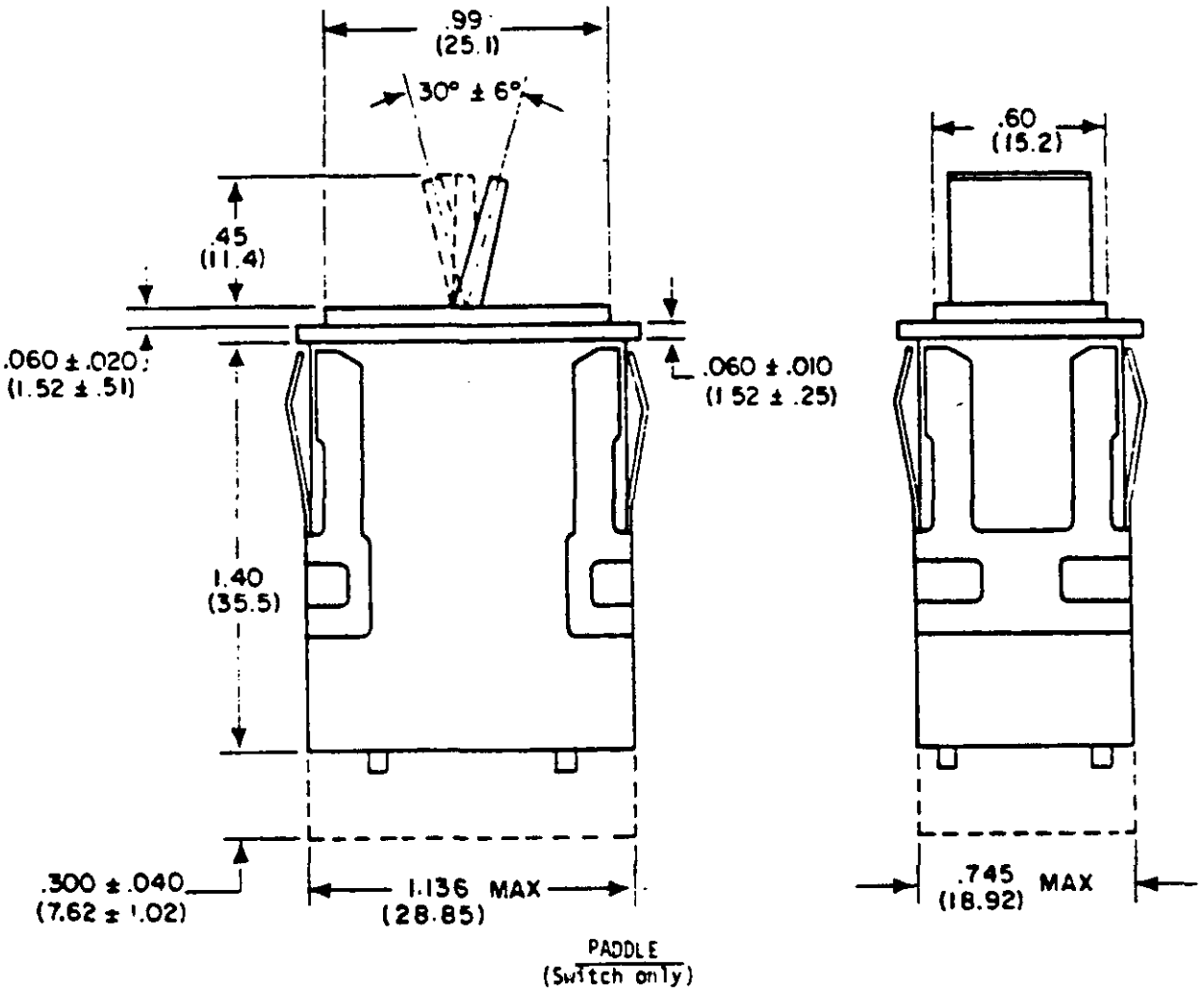


FIGURE 102-23. Switch, pushbutton, illuminated, rocker, paddle, logic load to 3 amperes, unsealed (MIL-S-22885/93) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Part number for indicators and pushbutton switches: EXAMPLE: M22885/93 A A A 01 A R

SEE TABLE I
- Configuration
- Illumination
- Terminals

SEE TABLE II
- Action/rating and circuitry
- Display type and button style
- Button color (leave blank when specifying housing only)

TABLE I. Configuration, illumination and terminal codes for indicators and pushbutton switches.

Configuration	Illumination	Terminals
A - Square, Switch, Nonilluminated	A - Nonilluminated or Incandescent (No Lamp Installed)	A - Solder or Quick Connect
B - Square, Switch, 1 Lamp Circuit	B - * V Red LED	B - Printed circuit
C - Square, Switch, 1 LED	C - 5.0 V Red LED	
D - Square, Indicator, 1 Lamp Circuit	D - 10.0 V Red LED	
E - Square, Indicator, 2 Lamp Circuits	E - 15.0 V Red LED	
F - Square, Indicator, 1 LED	F - * V Yellow LED	
G - Rectangular, Switch, Nonilluminated	G - 5.0 V Yellow LED	
H - Rectangular, Switch, 2 Lamp Circuits	H - 10.0 V Yellow LED	
J - Rectangular, Indicator, 1 Lamp Circuit	J - 15.0 V Yellow LED	
K - Rectangular, Indicator, 2 Lamp Circuits	K - * V Green LED	
L - Miniature Indicator, 1 LED	L - 5.0 V Green LED	
	M - 10.0 V Green LED	
	N - 15.0 V Green LED	

FIGURE 102-23. Switch, pushbutton, illuminated, rocker, paddle, logic load to 3 amperes, unsealed, (MIL-S-22885/93) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE II. Action/rating and circuitry, display and button style for indicators and pushbutton switches

Action/rating and circuitry	Display type button style 1/ 2/	Button color
01 - Momentary, 1 Pole, 3 A, 120 Vac Ind - 2 A, 28 Vdc, Res.	A - Transmitted Color - Square	R - Red
02 - Momentary, 2 Pole 3 A, 120 Vac Ind - 2 A, 28 Vdc, Res.	B - Projected Color - Square	Y - Yellow
03 - Momentary, 4 Pole 3 A, 120 Vac Ind - 2 A, 28 Vdc, Res.	C - Hidden Color - Square	G - Green
04 - Alternate, 1 Pole, 3 A, 120 Vac Ind - 2 A, 28 Vdc, Res.	D LED Window - Square	B - Blue
05 - Alternate, 2 Pole, 3 A, 120 Vac Ind - 2 A, 28 Vdc, Res.	E - Transmitted Color Rectangular	W - White
06 - Alternate, 4 Pole, 3 A, 120 Vac Ind - 2 A, 28 Vdc, Res.	F - Projected Color - Rectangular	A - Miniature Indicator No Button
07 - Momentary, 1 Pole 10 mA, 5 Vdc Res.	G - Hidden Color-Rectangular	K - Black 3/
08 - Momentary, 2 Pole 10 mA, 5 Vdc Res.	H - Miniature Indicator (No button required)	L - Gray 3/
09 - Momentary, 4 Pole, 10 mA, 5 Vdc, Res.	J - Housing only. No button, caps, or color inserts supplied. Acquire separately from source listed on QPL	
11 - Alternate, 1 Pole, 10 mA, 5 Vdc Res.		
12 - Alternate, 2 Pole 10 mA, 5 Vdc Res.		
13 - Alternate, 4 Pole 10 mA, 5 Vdc Res.		
14 - No Action, Indicator Only No Switching Mechanism		

1/ Use Code A, E or J, as required for nonilluminated switches.

2/ Button styles are shown as viewed from the front of the panel:

3/ For nonilluminated and LED configuration types only.

FIGURE 102-23. Switch, pushbuton, illuminated, rocker, paddle, logic load to 3 amperes, unsealed (MIL-S-22885/93) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Part Number for rocker and paddle switches:

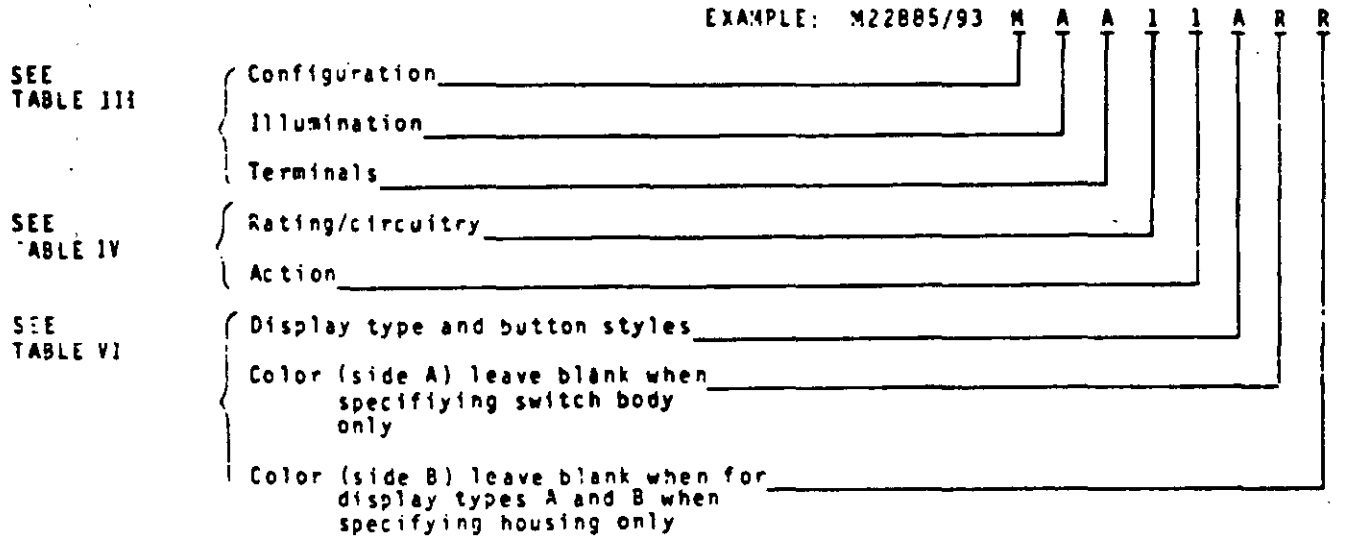


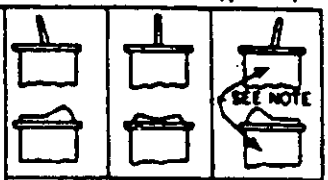
TABLE III. Configuration, illumination and terminal codes for rocker and paddle switches.

Configuration	Illumination	Terminals
M - Rocker, Nonilluminated	A - Nonilluminated or Incandescent (No lamp installed)	A - Solder or Quick-connect
N - Rocker, 2 Lamp Circuits	B - * V Red LED	B - Printed Circuit
P - Rocker, 1 LED (Side A)	C - 5.0 V Red LED	
R - Paddle, Nonilluminated	D - 10.0 V Red LED	
S - Paddle, 2 Lamp Circuits	E - 15.0 V Red LED	
T - Paddle, 1 LED (Side A)	F - * V Yellow LED	
	G - 5.0 V Yellow LED	
	H - 10.0 V Yellow LED	
	J - 15.0 V Yellow LED	
	K - * V Green LED	
	L - 5.0 V Green LED	
	M - 10.0 V Green LED	
	N - 15.0 V Green LED	

FIGURE 102-23. Switch, pushbutton, illuminated, rocker, paddle, logic load to 3 amperes, unsealed, (MIL-S-22885/93) - Continued.

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:41Z
Check the source to verify that this is the currentversion before use.

TABLE IV. Rating/circuitry and action for rocker and paddle switches.

	Rating/circuitry	ACTION CODE			
					
					SEE NOTE
			2 - POSITION		
1	1/ - 3 A, 120 Vac Ind. - 2 A, 28 Vdc res. 1 Pole (Code D Circuitry - See table XII)	1	MAIN.	(NONE)	MAIN.
2	2/ - 3 A, 120 Vac Ind. - 2 A, 28 Vdc res. 2 Pole (Code E Circuitry - See table XII)	2	MOM.	(NONE)	MAIN.
3	1/ 3/ 3 A, 120 Vac Ind. - 2 A, 28 Vdc res. 2 Pole (Code F Circuitry - See table XII)	3	MAIN.	(NONE)	MOM.
4	2/ - 3 A, 120 Vac Ind. - 2 A, 28 Vdc res. 4 Pole (Code G Circuitry - See table XII)		3 - POSITION		
5	1/ - 10 mA, 5 Vdc Res. 1 Pole (Code D Circuitry - See table XII)	4	MAIN.	MAIN.	MAIN.
6	2/ - 10 mA, 5 Vdc Res. 2 Pole (Code E Circuitry - See table XII)	5	MOM.	MAIN.	MOM.
7	1/ 3/ - 10 mA, 5 Vdc Res. 2 Pole (Code F Circuitry - See table XII)	6	MAIN.	MAIN.	MOM.
8	2/ - 10 mA, 5 Vdc Res. 4 Pole (Code G Circuitry - See table XII)	7	MOM.	MAIN.	MAIN.

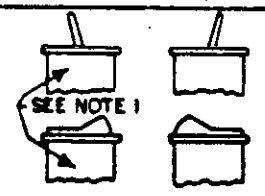
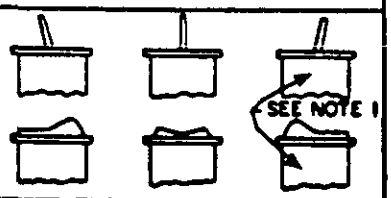
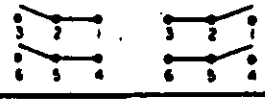
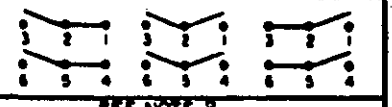
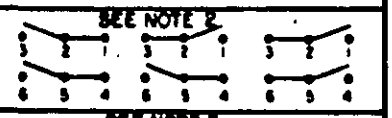
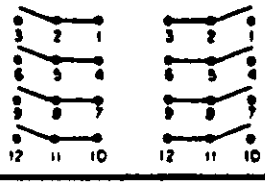
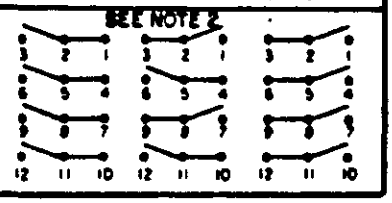
1/ LED versions available only in these circuitries.

2/ Nonilluminated switches only.

3/ For use with three position switches only.

NOTE: Actions are shown with the switch positioned such that the manufacturer's logo is located on the front side of the switch.

TABLE V. Paddle and rocker circuitry.

Paddle and Rocker Circuitries		
Code	2 Position	3 Position
D see note 4		
E See note 3		
F see note 4	For use with 3 position unit only	
G See note 3		

NOTES:
1. Circuitries are shown with the switch positioned such that the manufacturer's logo is located on the front side of the switch.
2. Refer to application information for how to wire for ON-ON-ON circuitry.
3. For nonilluminated switches only.
4. LED version available only in these circuitries.

FIGURE 102-23. Switch, pushbutton, illuminated, rocker, paddle, logic load to 3 amperes, unsealed, (MIL-S-22885/93) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE VI. Button style/display type and color for rocker and paddle switches.

Button style/display type 1/	Color	
	Side A	Side B
A - Nonilluminated/1 pc. rocker (one color selection only)	R - Red	R - Red
B - Transmitted color/1 pc. rocker (one color selection only)	Y - Yellow	Y - Yellow
C - Projected color/1 pc. rocker	G - Green	G - Green
D - Hidden color/1 pc. rocker	B - Blue	B - Blue
E - nonilluminated/2 pc. rocker	W - White	W - White
F - Transmitted color/2 pc. rocker	K - Black2/	K - Black2/
G - Projected color/2 pc. rocker	L - Gray2/7	L - Gray2/7
H - Hidden color/2 pc. rocker		
J - 1 LED window/2 pc. rocker		
K - Nonilluminated/2 pc. paddle cover		
L - Transmitted color/2 pc. paddle cover		
M - Projected color/2 pc. paddle cover		
N - Hidden color/2 pc. paddle cover		
P - 1 LED window/2 pc. paddle cover		
R - Housing only. No rocker, covers, or color inserts supplied. Acquire separately from source listed on QPL.		

1/ Button styles are shown as viewed from front of the panel:

2/ For nonilluminated and LED configurations only.

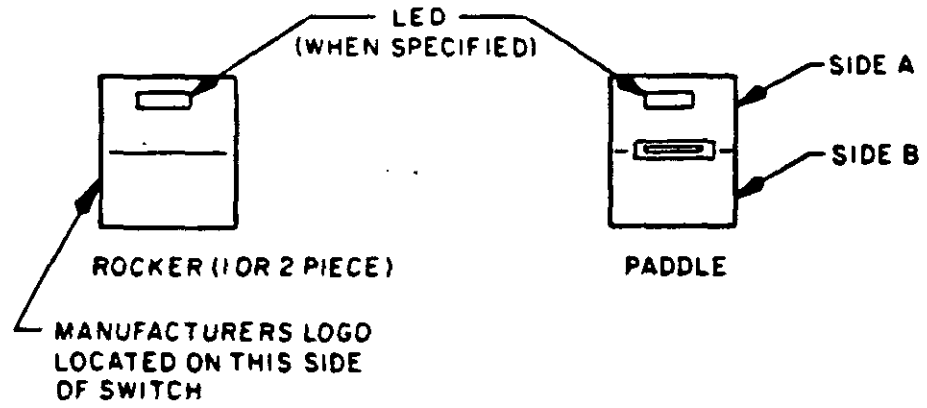


FIGURE 102-23. Switch, pushbutton, illuminated, rocker, paddle, logic load to 3 amperes, unsealed (MIL-S-22885/93) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

SECTION 104B

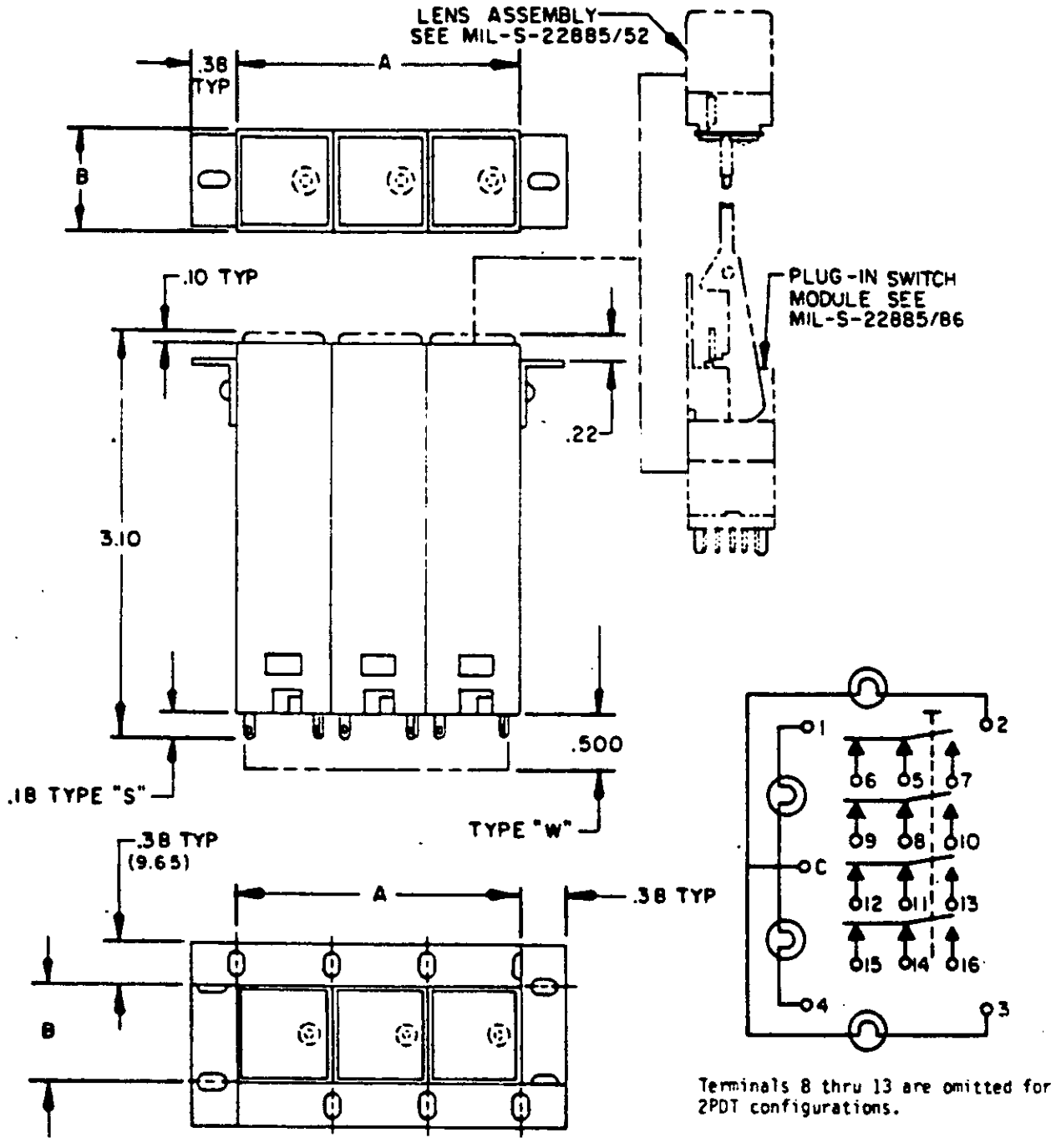
SWITCHES, MULTISTATION
(Applicable specification: MIL-S-24317)

Figure number	Applicable specification sheet	Switch characteristics				Page
		Switch module	Type	Operation	Circuit	
104-1	MIL-S-24317/9	MIL-S-22885/86	Illuminated	Momentary and alternate	2PDT 4PDT	104.2
104-2	MIL-S-24317/8	MIL-S-22885/80	Illuminated	Momentary and alternate	2PDT 4PDT	104.4

This Section supersedes
Section 104A dated 20 June 1977

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

LOAD	ELECTRICAL RATINGS (amperes)	
	28 Vdc	115 Vac
Resistive	2.0	2.0
Inductive	1.0	1.0
Lamp	0.75	0.75

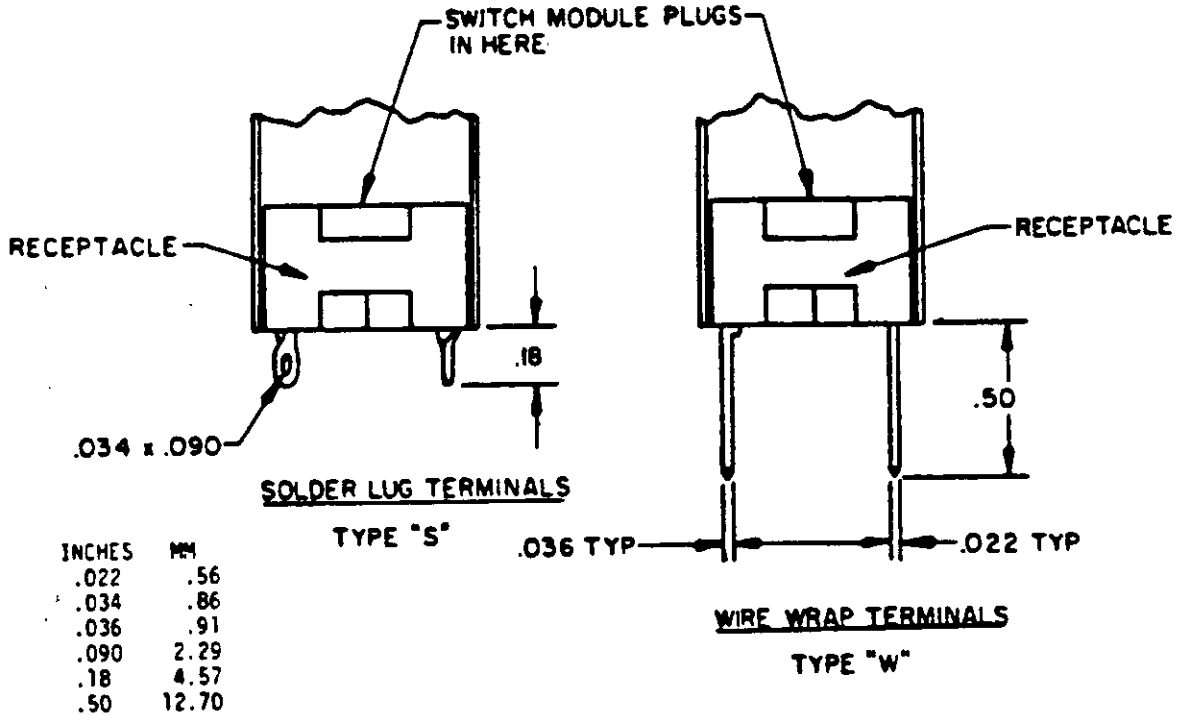


Terminals 8 thru 13 are omitted for 2PDT configurations.

FLANGE ARRANGEMENT FOR RFI SHIELDING

FIGURE 104-1. Switches, multistation, illuminated, matrix arrangement for (MIL-S-24317/9).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



INCHES	MM
.022	.56
.034	.86
.036	.91
.090	2.29
.18	4.57
.50	12.70

OPERATING FORCE: 3.5 ±1.0 pounds.

PART NUMBER: Part numbers in the tabulation are for matrix frame with flanges and receptacles and do not include the switch/indicator or lens assemblies (see MIL-S-22885/86 and /52).

Without RFI shielding			With RFI shielding			No. of stations	Dimension	
Part No. M24317/9-		No. of flanges	Part No. M24317/9-		No. of flanges		A ±.015 (.38)	B ±.015 (.38)
Type S	Type W		Type S	Type W				
001	011	2V	021	031	2V2H	1	.755 (19.18)	.755 (19.18)
002	012		022	032		2	1.510 (38.35)	
003	013		023	033		3	2.265 (57.53)	
004	014		024	034		4	3.020 (76.71)	
005	015		025	035		5	3.775 (95.89)	
006	016		026	036		6	4.530 (115.06)	
007	017	2H	027	037		7	5.285 (134.24)	
008	018		028	038		8	6.040 (153.42)	
009	019		029	039		9	6.795 (172.59)	
010	020		030	040		10	7.550 (191.77)	
041	051	2V	061	071		2	.755 (19.18)	1.510 (38.35)
042	052		062	072		4	1.510 (38.35)	
043	053		063	073		6	2.265 (57.53)	
044	054		064	074		8	3.020 (76.71)	
045	055		065	075		10	3.775 (95.89)	
046	056		066	076		12	4.530 (115.06)	
047	057	2H	067	07		14	5.285 (134.24)	
048	058		068	078		16	6.040 (153.42)	
049	059		069	079		18	6.795 (172.59)	
050	060		070	080		20	7.550 (191.77)	

FIGURE 104-1. Switches, multistation, illuminated, matrix arrangement for (MIL-S-24317/9) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

LOAD	ELECTRICAL RATINGS SEA LEVEL	
	28 Vdc	115 Vac
	(amperes)	(amperes)
Resistive	5.0	5.0
Inductive	2.5	2.5

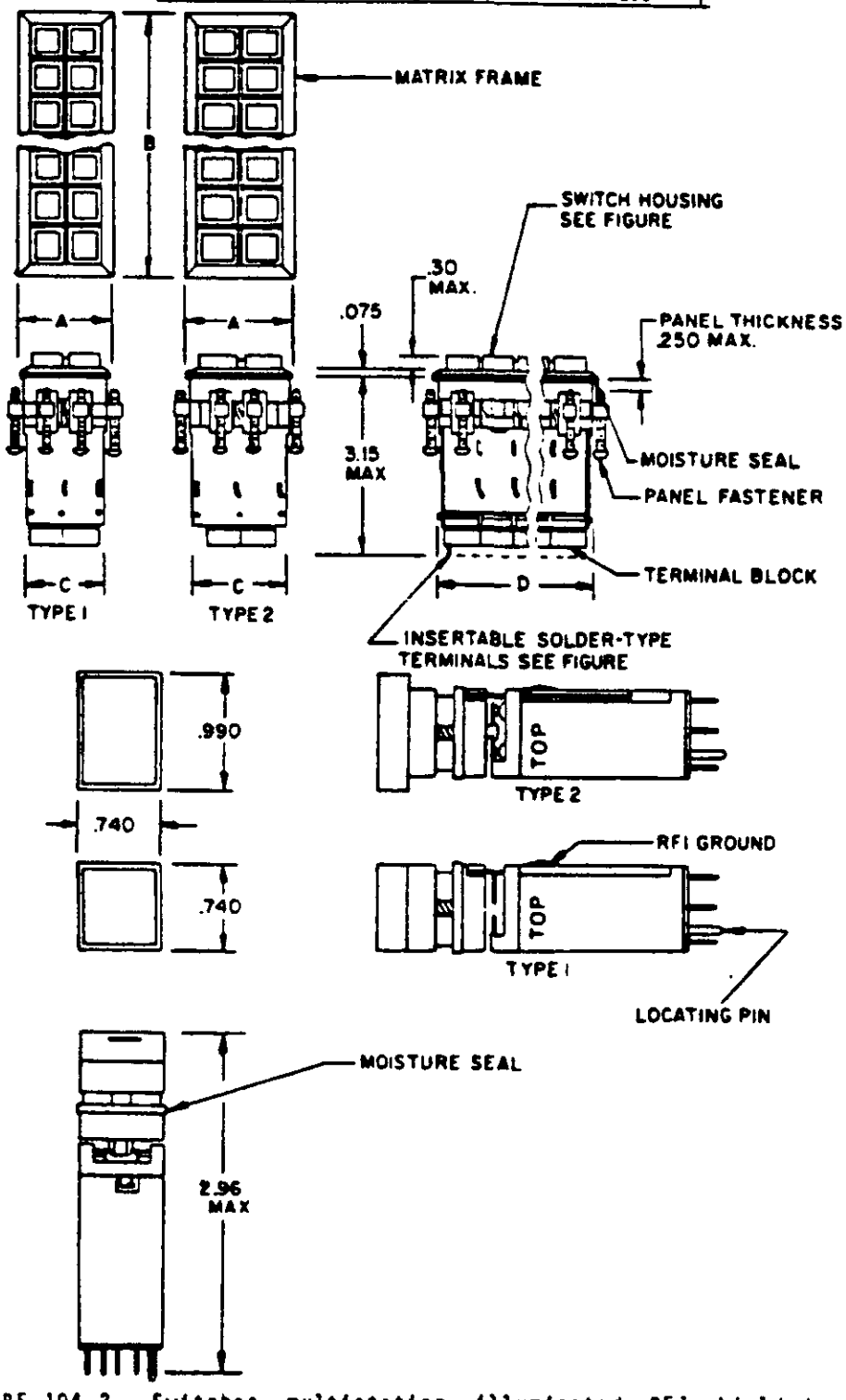
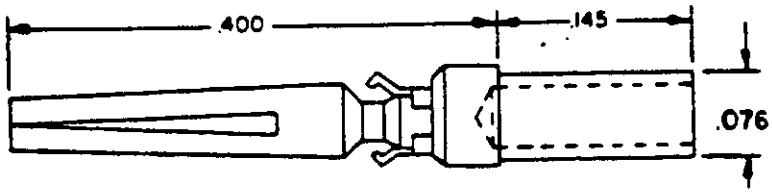
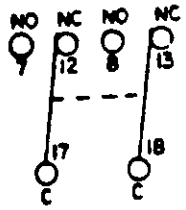


FIGURE 104-2. Switches, multistation, illuminated, RFI shielded, matrix arrangement for (MIL-S-24317/8).

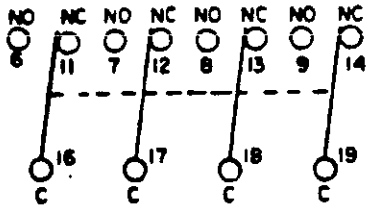
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



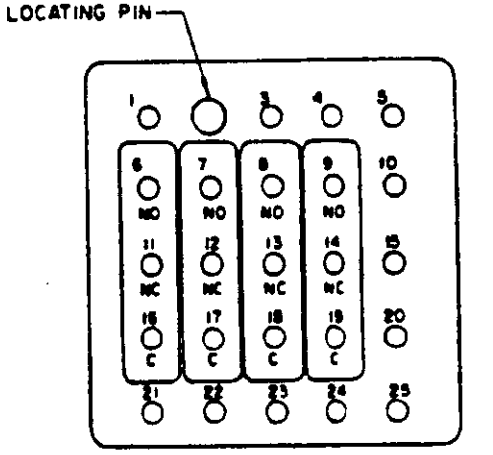
Terminal will accept one No. 18 AWG or smaller stranded wire size and are supplied 25 each in a bag under part number M22885/80-50.



Switch Circuitry
2 PDT Momentary Action



Switch Circuitry
4 PDT Momentary Action



(Rear View - Terminal Block)
Switch Terminal Identification

INCHES	MM
.075	1.91
.250	6.35
.30	7.62
.740	18.80
.990	25.15
2.96	75.18
3.15	80.01

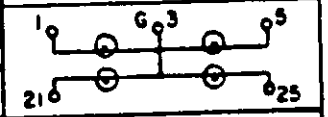
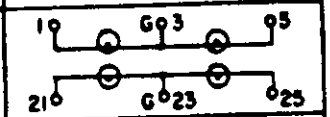
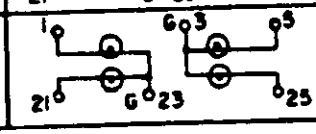
CODE	LAMP CIRCUITRY
1	
2	
3	

FIGURE 104-2. Switches, multistation, illuminated, RFI shielded, matrix arrangement for (MIL-S-24317/8) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Part number, matrix housings, number of stations and dimensions for fabricated matrix configurations, to accept type I switches.

Part number M24317/8 -	Matrix arrangement H V	Number of stations	Dimensions				Number of fasteners per side	
			A Horizontal	B Vertical	Recommended panel cutout C ±.010	Recommended panel cutout D ±.010	Horizontal side	Vertical side
01	1 x 1	1	1.150 (29.21)	1.150 (29.21)	.990 (25.15)	.990 (25.15)	1	1
02	1 x 2	2	1.150 (29.21)	1.950 (49.53)	.990 (25.15)	1.790 (45.46)	1	2
03	1 x 3	3	1.150 (29.21)	2.750 (69.85)	.990 (25.15)	2.590 (65.79)	1	3
04	1 x 4	4	1.150 (29.21)	3.550 (90.17)	.990 (25.15)	3.390 (86.11)	1	4
05	1 x 5	5	1.150 (29.21)	4.350 (110.49)	.990 (25.15)	4.190 (106.43)	1	5
06	1 x 6	6	1.150 (29.21)	5.150 (130.81)	.990 (25.15)	4.990 (126.75)	1	6
07	1 x 7	7	1.150 (29.21)	5.950 (150.13)	.990 (25.15)	5.790 (147.07)	1	7
08	1 x 8	8	1.150 (29.21)	6.750 (170.45)	.990 (25.15)	6.590 (167.39)	1	8
09	1 x 9	9	1.150 (29.21)	7.550 (190.77)	.990 (25.15)	7.390 (187.71)	1	9
10	1 x 10	10	1.150 (29.21)	8.350 (211.09)	.990 (25.15)	8.190 (208.03)	1	10
11	2 x 1	2	1.950 (49.53)	1.150 (29.21)	1.790 (45.46)	.990 (25.15)	2	1

FIGURE 104-2. Switches, multistation, illuminated RFI shielded, matrix arrangement for (MIL-S-24317/8) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Part number, matrix housings, number of stations, and dimensions for fabricated matrix configurations, to accept type I switches. - Continued

Part number M24317/8	Matrix arrangement H V	Number of stations	Dimensions				Number of fasteners per side	
					Recommended panel cutout			
			A Horizontal	B Vertical	C ±.010	D ±.010	Horizontal side	Vertical side
12	2 x 2	4	1.950 (49.53)	1.950 (49.53)	1.790 (45.46)	1.790 (45.46)	2	2
13	2 x 3	6	1.950 (49.53)	2.750 (68.85)	1.790 (45.46)	2.590 (65.79)	2	3
14	2 x 4	8	1.950 (49.53)	3.550 (90.17)	1.790 (45.46)	3.390 (86.11)	2	4
15	2 x 5	10	1.950 (49.53)	4.350 (110.49)	1.790 (45.46)	4.190 (106.43)	2	5
16	2 x 6	12	1.950 (49.53)	5.150 (130.81)	1.790 (45.46)	4.990 (126.75)	2	6
17	2 x 7	14	1.950 (49.53)	5.950 (150.13)	1.790 (45.46)	5.790 (147.07)	2	7
18	2 x 8	16	1.950 (49.53)	6.750 (170.45)	1.790 (45.46)	6.590 (167.39)	2	8
19	2 x 9	18	1.950 (49.53)	7.550 (190.77)	1.790 (45.46)	7.390 (187.71)	2	9
20	2 x 10	20	1.950 (49.53)	8.350 (211.09)	1.790 (45.46)	8.190 (208.03)	2	10
21	3 x 1	3	2.750 (69.85)	1.150 (29.21)	2.590 (65.79)	.990 (25.15)	3	1
22	3 x 2	6	2.750 (69.85)	1.950 (49.53)	2.590 (65.79)	1.790 (45.46)	3	2

FIGURE 104-2. Switches, multistation, illuminated RFI shielded, matrix arrangement for (MIL-S-24317/8) - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.

Part number, matrix housings, number of stations, and dimensions for fabricated matrix configurations, to accept type I switches. - Continued

Part number M24317/8 -	Matrix arrangement H V	Number of stations	Dimensions 2/, 3/				Number of fasteners per side	
			A Horizontal	B Vertical	Recommended panel cutout C ±.010	Recommended panel cutout D ±.010	Horizontal side	Vertical side
23	4 x 1	4	3.550 (90.17)	1.150 (29.21)	3.390 (86.11)	.990 (25.15)	4	1
24	4 x 2	8	3.550 (90.17)	1.950 (49.53)	3.390 (86.11)	1.790 (45.46)	4	2
25	5 x 1	5	4.350 (110.49)	1.150 (29.21)	4.190 (106.43)	.990 (25.15)	5	1
26	5 x 2	10	4.350 (110.49)	1.950 (49.53)	4.190 (106.43)	1.790 (45.46)	5	2
27	6 x 1	6	5.150 (130.81)	1.150 (29.21)	4.990 (126.75)	.990 (25.15)	6	1
28	6 x 2	12	5.150 (130.81)	1.950 (49.53)	4.990 (126.75)	1.790 (45.46)	6	2
29	7 x 1	7	5.950 (150.13)	1.150 (29.21)	5.790 (147.07)	.990 (25.15)	7	1
30	7 x 2	14	5.950 (150.13)	1.950 (49.53)	5.790 (147.07)	1.790 (45.46)	7	2
31	8 x 1	8	6.750 (170.45)	1.150 (29.21)	6.590 (167.39)	.990 (25.15)	8	1
32	8 x 2	16	6.750 (170.45)	1.950 (49.53)	6.590 (167.39)	1.790 (45.46)	8	2
33	9 x 1	9	7.550 (190.77)	1.150 (29.21)	7.390 (187.71)	.990 (25.15)	9	1

FIGURE 104-7. Switches, multistation, illuminated RFI shielded, matrix arrangement for (MIL-S-24317/8) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T07:41Z
Check the source to verify that this is the current version before use.

Part number, matrix housings, number of stations, and dimensions for fabricated matrix configurations, to accept type I switches. - Continued

Part number M24317/8 - 1/	Matrix arrangement H V	Number of stations	Dimensions				Number of fasteners per side	
			A Horizontal	B Vertical	Recommended panel cutout C ±.010	Recommended panel cutout D ±.010	Horizontal side	Vertical side
34	9 x 2	18	7.550 (190.77)	1.950 (49.53)	7.390 (187.71)	1.790 (45.46)	9	2
35	10 x 1	10	8.350 (211.09)	1.150 (29.21)	8.190 (208.03)	.990 (25.15)	10	1
36	10 x 2	20	8.350 (211.09)	1.950 (49.53)	8.190 (208.03)	1.790 (45.46)	10	2
TYPE II								
37	1 x 1	1	1.410 (35.81)	1.150 (29.21)	1.250 (31.75)	.990 (25.15)	1	1
38	1 x 2	2	1.410 (35.81)	1.950 (49.53)	1.250 (31.75)	1.790 (45.46)	1	2
39	1 x 3	3	1.410 (35.81)	2.750 (69.85)	1.250 (31.75)	2.590 (65.79)	1	3
40	1 x 4	4	1.410 (35.81)	3.550 (90.17)	1.250 (31.75)	3.390 (86.11)	1	4
41	1 x 5	5	1.410 (35.81)	4.350 (110.49)	1.250 (31.75)	4.190 (106.43)	1	5
42	1 x 6	6	1.410 (35.81)	5.150 (130.81)	1.250 (31.75)	4.990 (126.75)	1	6
43	1 x 7	7	1.410 (35.81)	5.950 (150.13)	1.250 (31.75)	5.790 (147.07)	1	7

FIGURE 104-2. Switches, multistation, illuminated RFI shielded, matrix arrangement for (MIL-S-24317/8) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Part number, matrix housings, number of stations and dimensions for fabricated matrix configurations, to accept type II switches. - Continued

Part number M24317/8 -	Matrix arrangement H V	Number of stations	Dimensions				Number of fasteners per side	
			A Horizontal	B Vertical	Recommended panel cutout C ±.010	Recommended panel cutout D ±.010	Horizontal side	Vertical side
44	1 x 8	8	1.410 (35.81)	6.750 (170.45)	1.250 (31.75)	6.590 (167.39)	1	8
45	1 x 9	9	1.410 (35.81)	7.550 (190.77)	1.250 (31.75)	7.390 (187.71)	1	9
46	1 x 10	10	1.410 (35.81)	8.350 (211.09)	1.250 (31.75)	8.190 (208.03)	1	10
47	2 x 1	2	2.460 (62.48)	1.150 (29.21)	2.300 (58.42)	.990 (25.15)	2	1
48	2 x 2	4	2.460 (62.48)	1.950 (49.53)	2.300 (58.42)	1.790 (45.46)	2	2
49	2 x 3	6	2.460 (62.48)	2.750 (68.85)	2.300 (58.42)	2.590 (65.79)	2	3
50	2 x 4	8	2.460 (62.48)	3.550 (90.17)	2.300 (58.42)	3.390 (86.11)	2	4
51	2 x 5	10	2.460 (62.48)	4.350 (110.49)	2.300 (58.42)	4.190 (106.43)	2	5
52	2 x 6	12	2.460 (62.48)	5.150 (130.81)	2.300 (58.42)	4.990 (126.75)	2	6
53	2 x 7	14	2.460 (62.48)	5.950 (150.13)	2.300 (58.42)	5.790 (147.07)	2	7
54	2 x 8	16	2.460 (62.48)	6.750 (170.45)	2.300 (58.42)	6.590 (167.39)	2	8

FIGURE 104-7. Switches, multistation, illuminated RFI shielded, matrix arrangement for (MIL-S-24317/8) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026[illegible]-26T07:41Z
Check the source to verify that this is the current version before use.

Part number, matrix housings, number of stations, and dimensions for fabricated matrix configurations, to accept type II switches. - Continued

Part number M24317/8 -	Matrix arrangement H V	Number of stations	Dimensions				Number of fasteners per side	
			A Horizontal	B Vertical	Recommended panel cutout C ±.010	Recommended panel cutout D ±.010	Horizonial side	Vertical side
55	2 x 9	18	2.460 (62.48)	7.550 (190.77)	2.300 (58.42)	7.390 (187.71)	2	9
56	2 x 10	20	2.460 (62.48)	8.350 (211.09)	2.300 (58.42)	8.190 (208.03)	2	10
57	3 x 1	3	3.510 (89.15)	1.150 (29.21)	3.350 (85.09)	.990 (25.15)	3	1
58	3 x 2	6	3.510 (89.15)	1.950 (49.53)	3.350 (85.09)	1.790 (45.46)	3	2
59	4 x 1	4	4.560 (115.82)	1.150 (29.21)	4.400 (111.76)	.990 (25.15)	4	1
60	4 x 2	8	4.560 (115.82)	1.950 (49.53)	4.400 (111.76)	1.790 (45.46)	4	2
61	5 x 1	5	5.610 (142.49)	1.150 (29.21)	5.450 (138.43)	.990 (25.15)	5	1
62	5 x 2	10	5.610 (142.49)	1.950 (49.53)	5.450 (138.43)	1.790 (45.46)	5	2
63	6 x 1	6	6.660 (169.16)	1.150 (29.21)	6.500 (165.10)	.990 (25.15)	6	1
64	6 x 2	12	6.660 (169.16)	1.950 (49.53)	6.500 (165.10)	1.790 (45.46)	6	2
65	7 x 1	7	7.710 (195.83)	1.150 (29.21)	7.550 (191.77)	.990 (25.15)	7	1

FIGURE 104-2. Switches, multistation, illuminated RFI shielded, matrix arrangement for (MIL-S-24317/8) - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the currentversion before use.

Part number, matrix housings, number of stations, and dimensions for fabricated matrix configurations, to accept type II switches. - Continued

Part number M24317/8 -	Matrix arrangement H V	Number of stations	Dimensions				Number of fasteners per side	
			A Horizontal	B Vertical	Recommended panel cutout C ±.010	Recommended panel cutout D ±.010	Horizontal side	Vertical side
66	7 x 2	14	7.710 (195.83)	1.950 (49.53)	7.550 (191.77)	1.790 (45.46).	7	2
67	8 x 1	8	8.760 (222.50)	1.150 (29.21)	8.600 (218.44)	.990 (25.15)	8	1
68	8 x 2	16	8.760 (222.50)	1.950 (49.53)	8.600 (218.44)	1.790 (45.46)	8	2
69	9 x 1	9	9.810 (249.17)	1.150 (29.21)	9.650 (245.11)	.990 (25.15)	9	1
70	9 x 2	18	9.810 (249.17)	1.950 (49.53)	9.650 (245.11)	1.790 (45.46)	9	2
71	10 x 1.	10	10.860 (275.84)	1.150 (29.21)	10.700 (271.78)	.990 (25.15)	10	1
72	10 x 2	20	10.860 (275.84)	1.950 (49.53)	10.700 (271.78)	1.790 (45.46)	10	2

FIGURE 104-7. Switches, multistation, illuminated RFI shielded, matrix arrangement for (MIL-S-24317/8) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

SECTION 107

SWITCHES, ROTARY
(Applicable specifications: MIL-S-3786, MIL-S-6807, MIL-S-15291, and MIL-S-21604)

Figure number	Switch characteristics						Page
	Style/class	Specification sheet number or specification number	Section size, excluding terminals (inches)	Type of construction	Current rating (amperes)	Shaft	
107-1	SR01 SR02 SR03 SR09 SR10	MIL-S-3786/1 MIL-S-3786/2 MIL-S-3786/3 MIL-S-3786/9 MIL-S-3786/10	2.094 max dia 1.594 max dia 1.375 max dia 2.344 max dia 2.328 max dia	Open	1/2	Single	107.2
107-2	SR15	MIL-S-3786/15	2.531 square max	Open	1/2	Single	107.3
107-3	SR05	MIL-S-3786/5	1.047 max dia	Open	1/5	Single	107.4
107-5	SR28	MIL-S-3786/28	1.328 max dia	Open	1	Single	107.8
107-6	SR19	MIL-S-3786/19	.500 dia	Closed	1/10	Single	107.9
107-7	SR20	MIL-S-3786/20	.500 dia	Closed	1/5	Single	107.10
107-8	SR35	MIL-S-3786/35	.562 dia	Closed	1/5	Single	107.12
107-11	SR36	MIL-S-3786/36	1.350 dia (15°) 1.190 dia (18° & 22-1/2°)	Closed	1/4	Single	107.16
107-12	SR13	MIL-S-3786/13	.800 max dia (including terminals)	Closed	1/2	Single	107.20
107-13	SR04-1 SR04-2 SR04-3	MIL-S-3786/4	1.306 max dia (including terminals)	Closed	2	Single Concentric ADD-A-POT	107.23
107-14	---	MIL-S-6807	1.950 max dia (including terminals)	Closed	5	Single	107.29
107-15	SR41	MIL-S-3786/41	.765 max dia	Closed	1/2	Single	107.30
107-16	SR40	MIL-S-3786/40	1.781 square max	Open	Low level & 1/2	Single	107.32
107-17	SR32-1 SR32-2	MIL-S-3786/32	1.562 max dia	Open	1/5	Single	107.34

Supersedes page 107.1
of 20 June 1977

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Figure number	Switch characteristics - Continued. Style/class	Specification sheet number or specification number	Section size, excluding terminals (inches)	Type of construction	Current rating (amperes)	Shaft	Page
107-18	1SR	MIL-S-15291/2	1 19/32 dia	Closed	10	Single	107.38
107-19	3SR	MIL-S-15291/6	2 3/4 dia	Closed	30	Single	107.42
107-20	6SR	MIL-S-15291/9	4 19/32 dia	Closed	60	Single	107.45
107-21	20SR	MIL-S-15291/12	6 3/4 sq	Closed	200	Single	107.48
107-22	JF JFM	MIL-S-21604/1	1.12 sq	Closed	1	Single	107.51
107-23	JK	MIL-S-21604/2	2.22 dia	Closed	5	Single	107.53
107-24	JL	MIL-S-21604/3	2.00 dia	Closed	5	Single	107.55
107-25	JM	MIL-S-21604/4	2.22 dia	Closed	7.5	Single	107.57
107-26	JR	MIL-S-21604/5	2.22 dia	Closed	10	Single	107.59

NOTES:

1. For the styles listed under MIL-S-3786, the following variations are considered standard:

   (a) Operating-shaft length, diameter, shape, and special construction.

   (b) Deck-to-deck spacing dimensions.

   (c) Optional assemblies, such as spring return and push-to-turn actions.

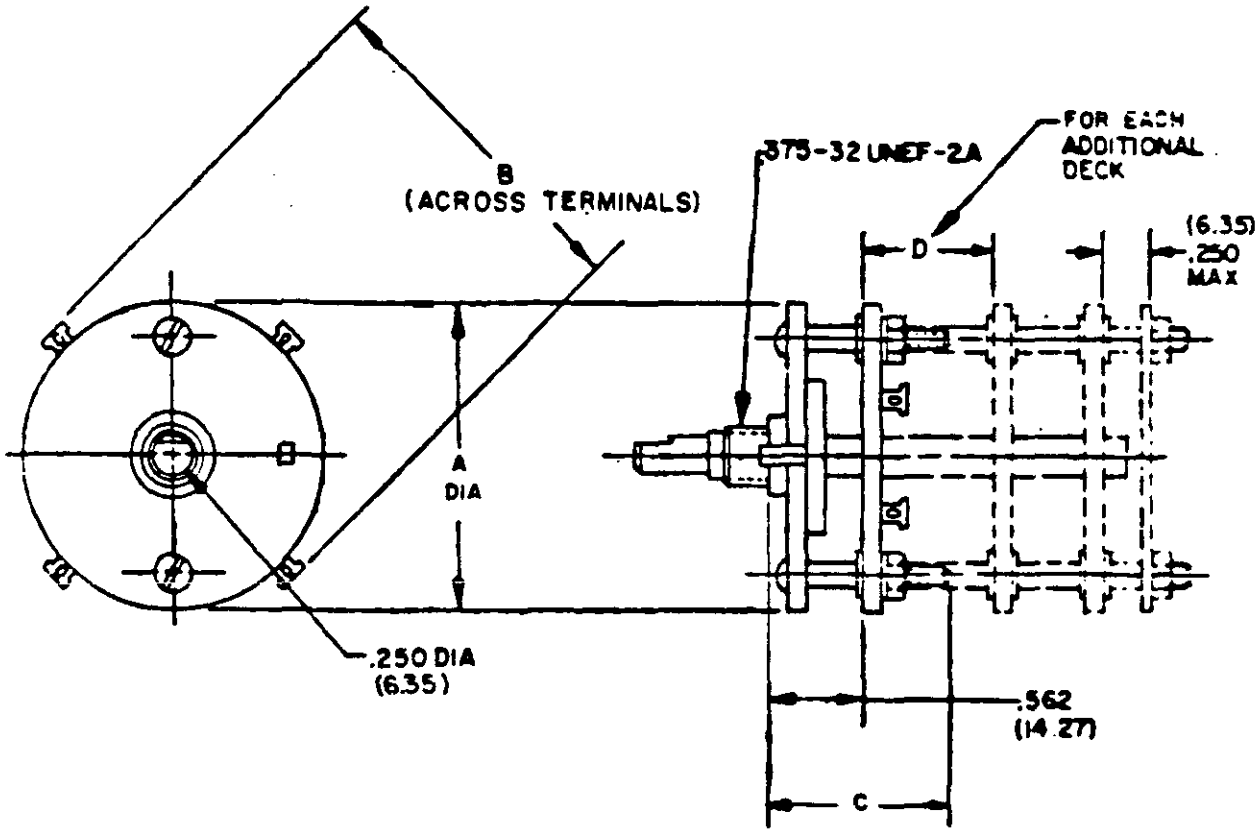
   (d) Terminal angles.

   (e) Bushing thread size and diameter, and thread length.

   (f) Addition of a shield.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

LOAD	ELECTRICAL RATINGS (milliamperes)			
	28 Vdc	115 Vrms		300 Vrms
		SR01 & 03	SR02, 09, & 10	
Resistive:				
At atmospheric pressure - - - - - -	500	225	250	50
At reduced barometric pressure- - -	350	100	100	---
Inductive (2.8 Henries):				
At atmospheric pressure - - - - - -	50	---	---	---



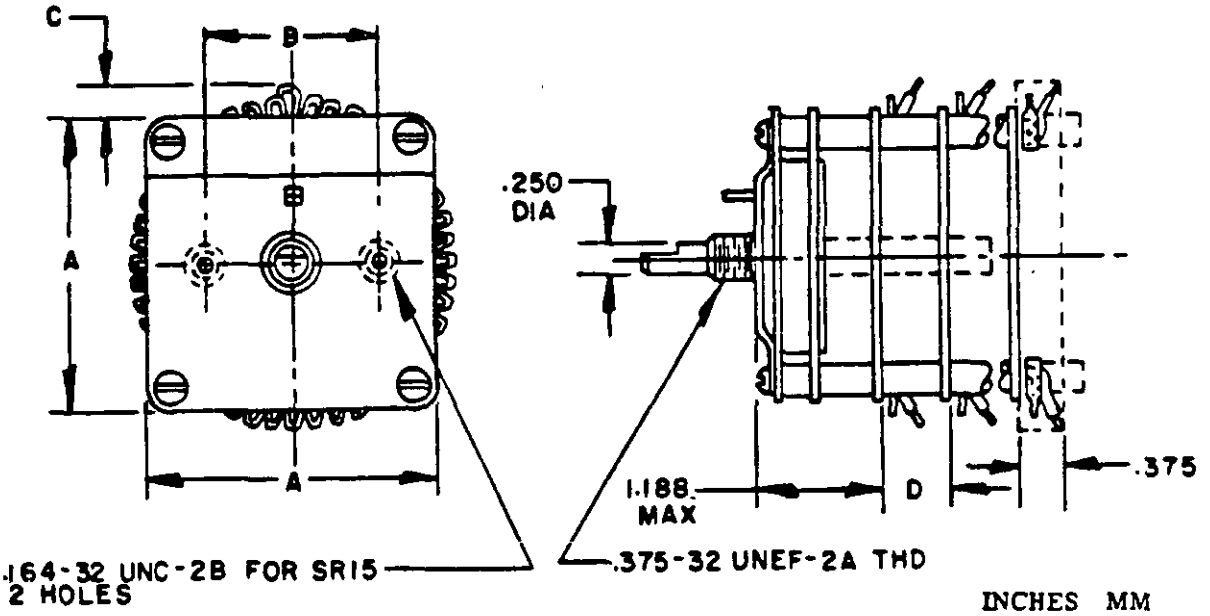
STYLE	PROCUREMENT DOCUMENT	DIMENSIONS			
		A max	B max	C max	D nom
SR01	MIL-S-3786/1	2.094 (53.19)	2.328 (59.13)	1.188 (30.20)	.750 (19.05)
SR02	MIL-S-3786/2	1.594 (40.49)	1.750 (44.45)	1.188 (30.20)	.500 (12.70)
SR03	MIL-S-3786/3	1.375 (34.93)	1.625 (41.28)	1.062 (26.97)	.500 (12.70)
SR09	MIL-S-3786/9	2.344 (59.54)	2.062 (52.37)	1.312 (33.32)	.500 (12.70)
SR10	MIL-S-3786/10	2.328 (59.13)	2.625 (66.68)	1.312 (33.32)	.500 (12.70)

NOTES:
1. For shipboard applications, switches must withstand the high impact shock test of MIL-S-3786.
2. Mounting brackets are recommended for switches that protrude over 3.50 inches (88.90 mm) behind panel or have more than five sections.
3. These switches are identified by a type designation (see MIL-S-3786).

FIGURE 107-1. SR01, SR02, SR03, SR09, and SR10 switches, rotary, open construction, 1/2 ampere, single shaft (MIL-S-3786/1, /2, /3, /9, and /10).

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:41Z
Check the source to verify that this is the current version before use.

LOAD (At atmospheric pressure)	ELECTRICAL RATINGS (milliamperes)		
	28 Vdc	115 Vrms	300 Vrms
Resistive - - - - - - - - - -	500	125	50
Inductive (2.8 Henries) - - -	50	---	---



INCHES	MM
.138	3.51
.164	4.17
.250	6.35
.375	9.53
1.188	30.23

FRONT PLATE

Style	Procurement document	Dimensions			
		A max	B	C max	D max
SR15	MIL-S-3786/15	2.531 (64.29)	1.500 (38.10)	.312 (7.92)	1.062 (26.97)

NOTES:

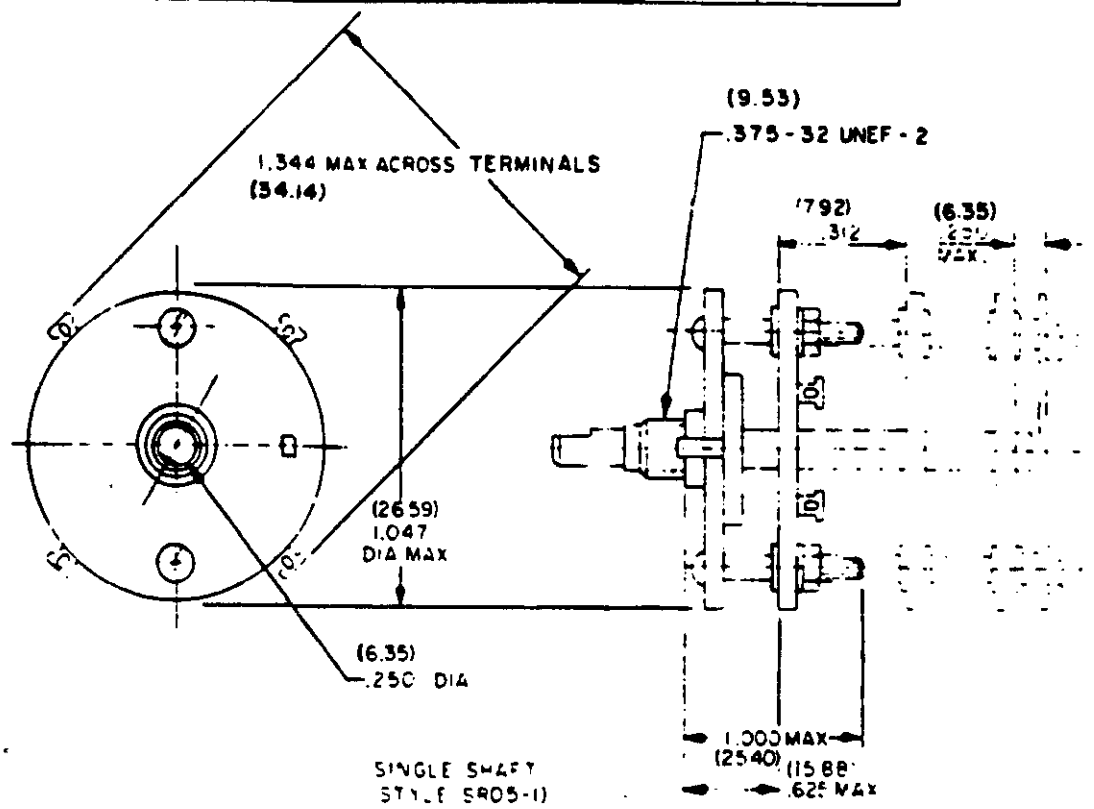
1. For shipboard applications, switches must withstand the high impact shock test of MIL-S-3786.
2. Mounting brackets are recommended for switches that protrude over 3.50 inches (88.90 mm) benind panel or have more than five sections.
3. These switches are identified by a type designation (see MIL-S-3786).

FIGURE 107-2. SR15 switches, rotary, open construction, 1/2 ampere, single shaft (MIL-S-3786/15).

Supersedes page 107.3
of 23 March 1976

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.

LOAD	ELECTRICAL RATINGS (milliamperes)	
	28 Vdc	115 Vrms
Resistive:		
At atmospheric pressure- - - - - -	200	100
At reduced barometric pressure - -	100	50
Inductive:		
At atmospheric pressure- - - - - -	20	---



NOTES:
1. For shipboard applications, switches must withstand the high impact shock test of MIL-S-3786.
2. Mounting brackets are recommended for switches that protrude over 3.50 inches (88.90 mm) behind panel or have more than five sections.

PART NUMBER: Consists of "M3786/5" followed by a dash and a four-letter code. The four-letter code used in the following example identifies a switch of construction type N, temperature-life characteristic B, shock type H and M, ceramic or glass bonded mica insulation material, 36 degree angle of throw, 1 pole per deck, 5 positions, 2 decks, and nonshorting switching characteristic.

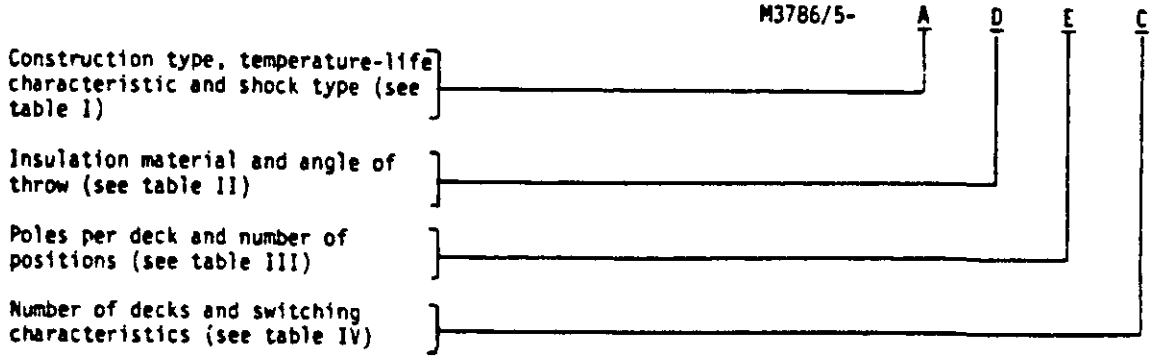


FIGURE 107-3. SR05 switches, rotary, open construction, 1/5 ampere, single shaft (MIL-S-3786/5).

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE I. Code letters for combinations of construction type, temperature-life characteristic, and shock type.

Code letter	Construction type	Temperature-life characteristic	Shock type	Code letter	Construction type	Temperature-life characteristic	Shock type
A - - -	N	B	H and M	G - - -	N	B	M
B - - -	S	B	↓	H - - -	S	B	↓
C - - -	N	C		J - - -	N	C	
D - - -	S	C		K - - -	S	C	
E - - -	N	D		L - - -	N	D	
F - - -	S	D	H and M	M - - -	S	D	M

TABLE II. Code letters for combinations of insulation material and angle of throw.

Code letter	Insulation material	Angle of throw	Code letter	Insulation material	Angle of throw	Code letter	Insulation material	Angle of throw
A - - -	C	30	F - - -	P	36	L - - -	G	60
B - - -	G	30	G - - -	C	45	M - - -	P	60
C - - -	P	30	H - - -	G	45	N - - -	C	90
D - - -	C	36	J - - -	P	45	P - - -	G	90
E - - -	G	36	K - - -	C	60	Q - - -	P	90

TABLE III. Code letters for combinations of poles per deck and number of positions.

Code letter	Poles per deck	Number of positions	Code letter	Poles per deck	Number of positions	Code letter	Poles per deck	Number of positions
A - - -	1	C 1/	J - - -	1	9	S - - -	2	6
B - - -	1	2	K - - -	1	10	T - - -	3	2
C - - -	1	3	L - - -	1	11	U - - -	3	3
D - - -	1	4	M - - -	1	12	V - - -	3	4
E - - -	1	5	N - - -	2	2	W - - -	4	2
F - - -	1	6	P - - -	2	3	X - - -	4	3
G - - -	1	7	Q - - -	2	4			
H - - -	1	8	R - - -	2	5			

1/ Switch is continuous rotation type (no stops), and number of positions is dependent on angle of throw as follows:

Angle of throw	Positions
30°	12
36°	10
45°	8
60°	6
90°	4

TABLE IV. Code letters for combinations of number of decks and switching characteristics.

Code letter	Number of decks	Switching characteristics					Code letter	Number of decks	Switching characteristics				
		First deck	Second deck	Third deck	Fourth deck	Fifth deck			First deck	Second deck	Third deck	Fourth deck	Fifth deck
A - - -	1	NS	---	---	---	---	K - - -	4	NS	NS	NS	NS	---
B - - -	1	S	---	---	---	---	L - - -	4	S	S	S	S	---
C - - -	2	NS	NS	---	---	---	M - - -	4	NS	NS	NS	S	---
D - - -	2	S	S	---	---	---	N - - -	4	NS	S	S	S	---
E - - -	2	NS	S	---	---	---	P - - -	5	NS	NS	NS	NS	NS
F - - -	3	NS	NS	NS	---	---	Q - - -	5	S	S	S	S	S
G - - -	3	S	S	S	---	---	R - - -	5	NS	NS	NS	NS	S
H - - -	3	NS	NS	S	---	---	S - - -	5	NS	S	S	S	S
J - - -	3	NS	S	S	---	---							

FIGURE 107-3. SR05 switches, rotary, open construction, 1/5 ampere, single shaft (MIL-S-3786/5) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-04-26T07:41Z
Check the source to verify that this is the current version before use.

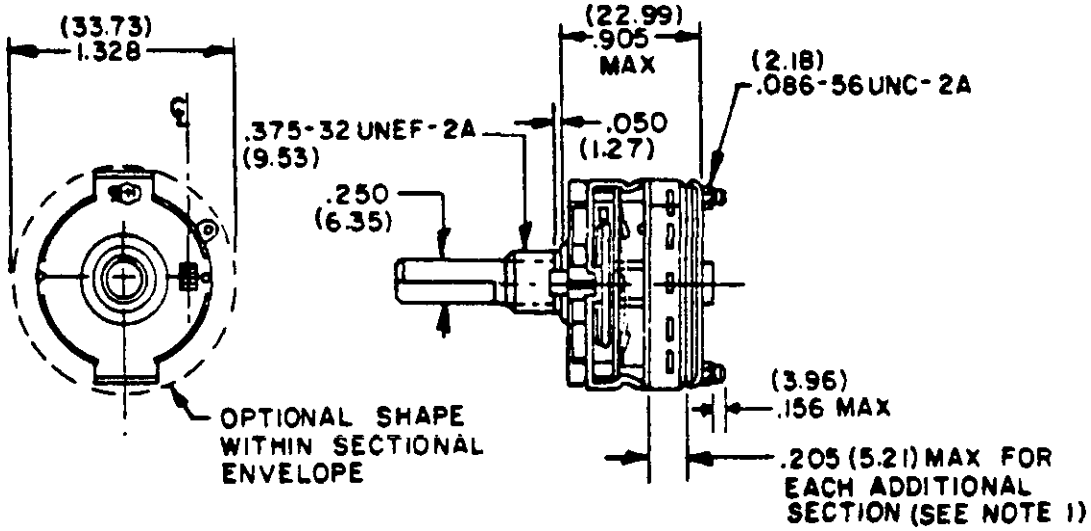
THIS PAGE INTENTIONALLY BLANK

Supersedes page 107.7
of 23 March 1976

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:41Z
Check the source to verify that this is the currentversion before use.

LOAD	ELECTRICAL RATINGS (milliamperes)		
	28 Vdc	115 Vac	115 Vac, 60 Hz
Resistive:			
At atmospheric pressure (125°C)- - - - - -	750	250	---
At atmospheric pressure (25°C) - - - - - -	1000	---	350
At reduced barometric pressure (25°C)- - -	250	40	---
Inductive (2.8 Henries):			
At atmospheric pressure (125°C)- - - - - -	100	---	---
At atmospheric pressure (25°C) - - - - - -	100 1/	---	---
At reduced barometric pressure (25°C)- - -	40	---	---

1/ *Reference only.*



NOTES:
1. For 7 through 12 positions, two sections may be joined together for each pole. An external pole strap (not shown) is used for wiring together commons of both sections of each related pole.
2. Switches having more than 2.750 inch length from the mounting surface are provided with through-bolts extended .250 inch minimum beyond the nuts securing the last switch section at the rear of the switch to provide means to support the rear of the switch.

No. of poles	Switching characteristic	Part number M3786/28-										
		Position					Position					
		2	3	4	5	6	7	8	9	10	11	12
1	Shorting	021	022	023	024	025	101	102	103	104	105	106
1	Nonshorting	026	027	028	029	030	107	108	109	110	111	112
2	Shorting	031	032	033	034	035	113	114	115	116	117	118
2	Nonshorting	036	037	038	039	040	119	120	121	122	123	124
3	Shorting	041	042	043	044	045	125	126	127	128	129	130
3	Nonshorting	046	047	048	049	050	131	132	133	134	135	136
4	Shorting	051	052	053	054	055	137	138	139	140	141	142
4	Nonshorting	056	057	058	059	060	143	144	145	146	147	148
5	Shorting	061	062	063	064	065	149	150	151	152	153	154
5	Nonshorting	066	067	068	069	070	155	156	157	158	159	160
6	Shorting	071	072	073	074	075	161	162	163	164	165	166
6	Nonshorting	076	077	078	079	080	167	168	169	170	171	172
7	Shorting	081	082	083	084	085	173	174	175	176	177	178
7	Nonshorting	086	087	088	089	090	179	180	181	182	183	184
8	Shorting	091	092	093	094	095	185	186	187	188	189	190
8	Nonshorting	096	097	098	099	100	191	192	193	194	195	196

FIGURE 107-5. SR2B switches, rotary, open construction, 1 ampere, single shaft (MIL-S-3786/28).

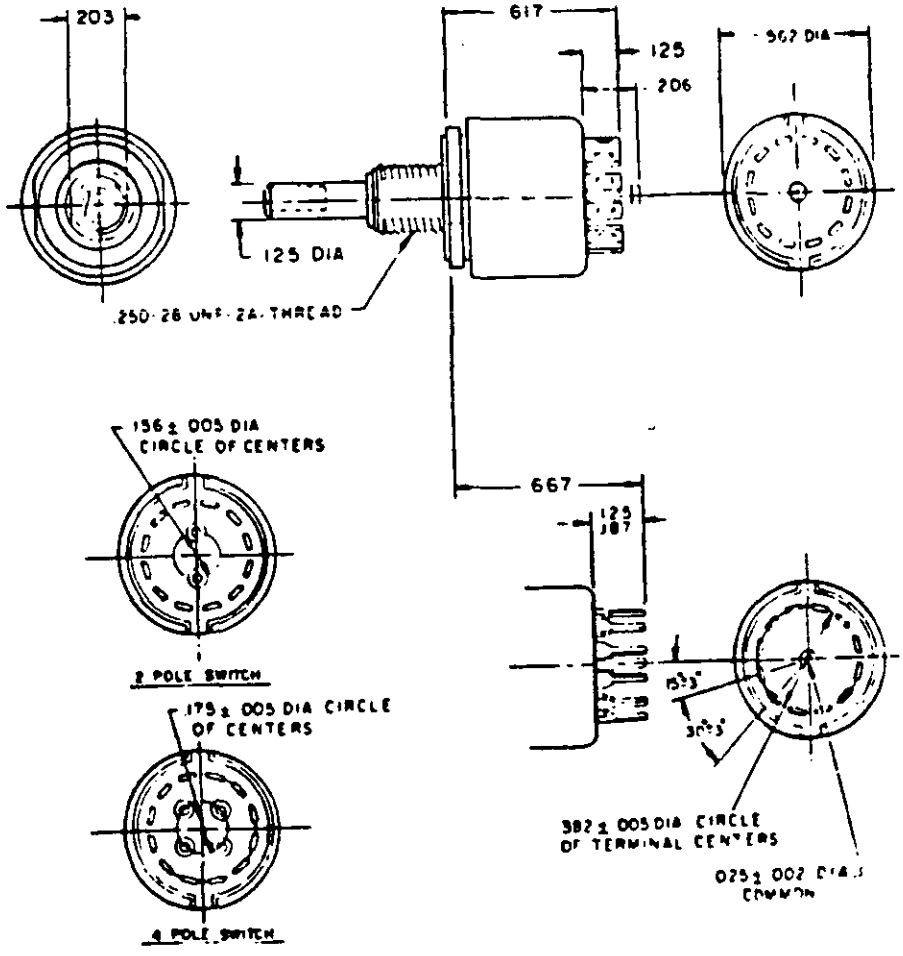
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY BLANK

Supersedes page 107.11
of 23 March 1976

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

LOAD	ELECTRICAL RATINGS (milliamperes)	
	28 Vdc	115 Vrms, 60 Hz
Lamp (tungsten):		
At atmospheric pressure- - - - - -	100	---
Resistive:		
At atmospheric pressure- - - - - -	200	150
At reduced barometric pressure - -	200	150
Inductive (2.8 Henries):		
At atmospheric pressure- - - - - -	30	---



INCHES	MM
.002	.05
.005	.13
.025	.64
.125	3.18
.156	3.96
.175	4.45
.187	4.75
.203	5.16
.206	5.23
.250	6.35
.382	9.70
.562	14.27
.617	15.67
.667	16.94

FIGURE 107-9. SR35 switches, rotary, closed construction, explosion proof, 1/5 ampere, single shaft (MIL-S-3786/35).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

No. of poles	No. of positions per pole	Type of stops	Part no. M3786/35-			
			Terminations			
			Solder lug		Printed circuit	
			Nonshorting	Shorting	Nonshorting	Shorting
1	2 thru 12	Adjustment	001	---	004	---
2	2 thru 6	Adjustment	002	---	005	---
4	2 thru 3	Adjustment	003	---	006	---
1	2	Fixed	007	043	025	061
1	3	Fixed	008	044	026	062
1	4	Fixed	009	045	027	063
1	5	Fixed	010	046	028	064
1	6	Fixed	011	047	029	065
1	7	Fixed	012	048	030	066
1	8	Fixed	013	049	031	067
1	9	Fixed	014	050	032	068
1	10	Fixed	015	051	033	069
1	11	Fixed	016	052	034	070
1	12	Fixed	017	053	035	071
2	2	Fixed	018	054	036	072
2	3	Fixed	019	055	037	073
2	4	Fixed	020	056	038	074
2	5	Fixed	021	057	039	075
2	6	Fixed	022	058	040	076
4	2	Fixed	023	059	041	077
4	3	Fixed	024	060	042	078
1	12	None	079 1/	081 1/	080 1/	082 1/

1/ Continuous rotation.

FIGURE 107-9. SR35 switches, rotary, closed construction, explosion proof, 1/5 ampere, single shaft (MIL-S-3786/35) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

LOAD	ELECTRICAL RATINGS (milliamperes)	
	28 Vdc	115 Vrms, 60 Hz
Resistive:		
At 70,000 ft. altitude - - - - - - - -	50	20
At atmospheric pressure and 125°C -	75	25
At atmospheric pressure and room temperature (approx 25°C)- - - - -	250	150
Inductive (250 millihenries):		
At 70,000 ft. altitude - - - - - - - -	10	---
At atmospheric pressure and 125°C -	25	---
At atmospheric pressure and room temperature (approx 25°C) - -	75	---

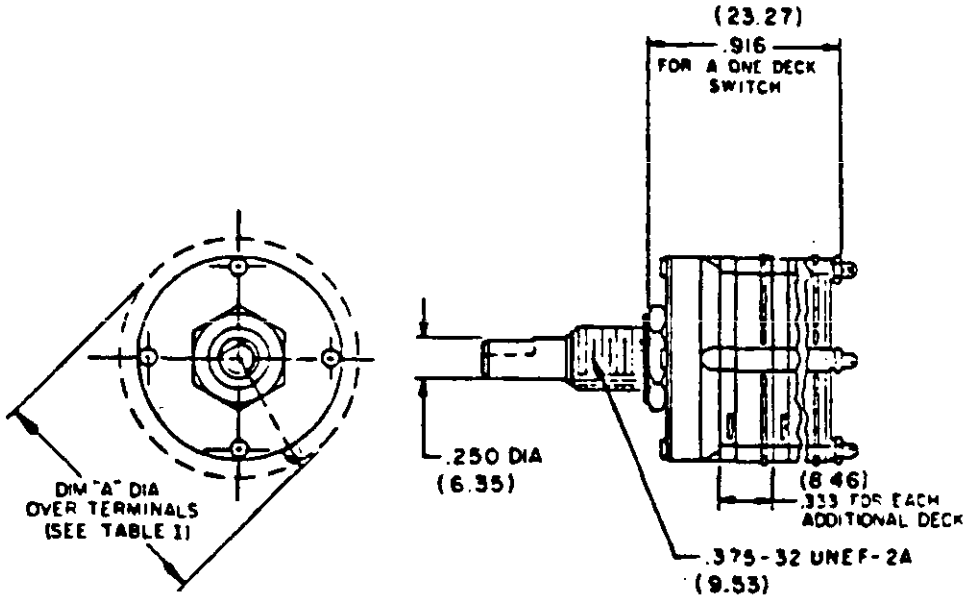


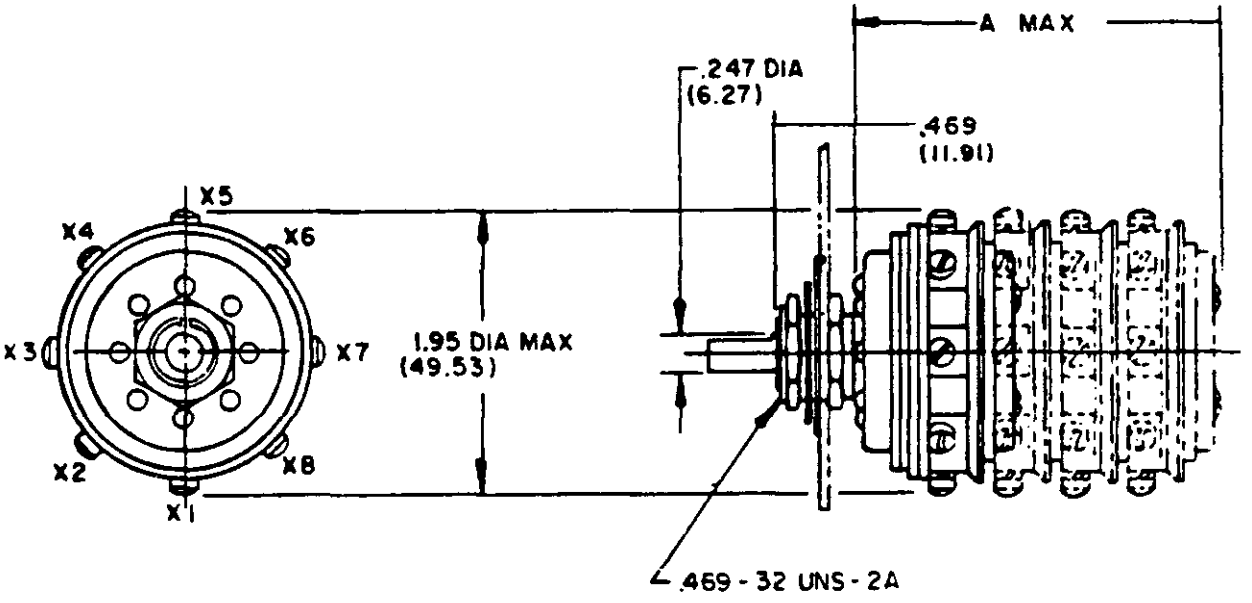
TABLE I

ANGLE OF THROW	DIM. "A"	
15°	1.350	(34.29)
18°, 22 1/2°	1.190	(30.23)

FIGURE 107-11. SR36 switches, rotary, closed construction, 1/4 ampere, single shaft (MIL-S-3786/36).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

LOAD	ELECTRICAL RATINGS (amperes)	
	28 Vdc	115 Vac, 400 Hz
Resistive - - - - - - - -	5	5
Inductive - - - - - - - -	2.5	---
Lamp- - - - - - - - - -	2	---
Power factor, 50% - - -	---	2



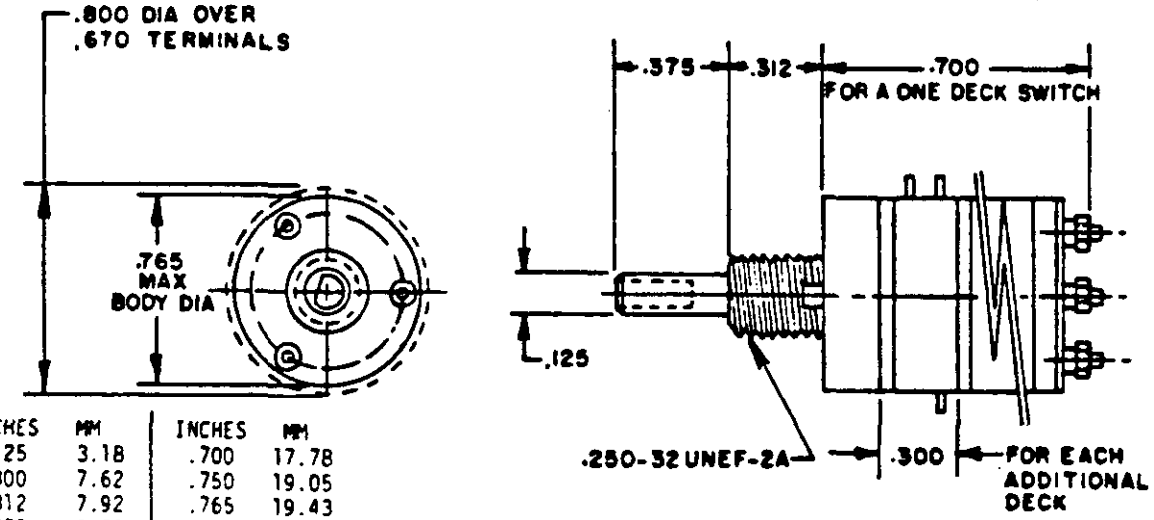
Part No. MS 25002 Shock Method I and II	Number of Section	Torque (lb-in) Min	Torque (lb-in) Max	Wt Lb Max	A DIM
-1	1	2.0	5.0	.31	1.18
-2	2	3.0	6.0	.37	1.57
-3	3	3.5	7.0	.43	1.94
-4	4	4.0	8.0	.50	2.31
-5	5	4.0	9.0	.56	2.69
-6	6	4.0	10.5	.62	3.06

FIGURE 107-14. Switches, rotary, closed construction, 5 amperes, single shaft (MIL-S-6807).

Supersedes page 107.29 of 23 March 1976

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

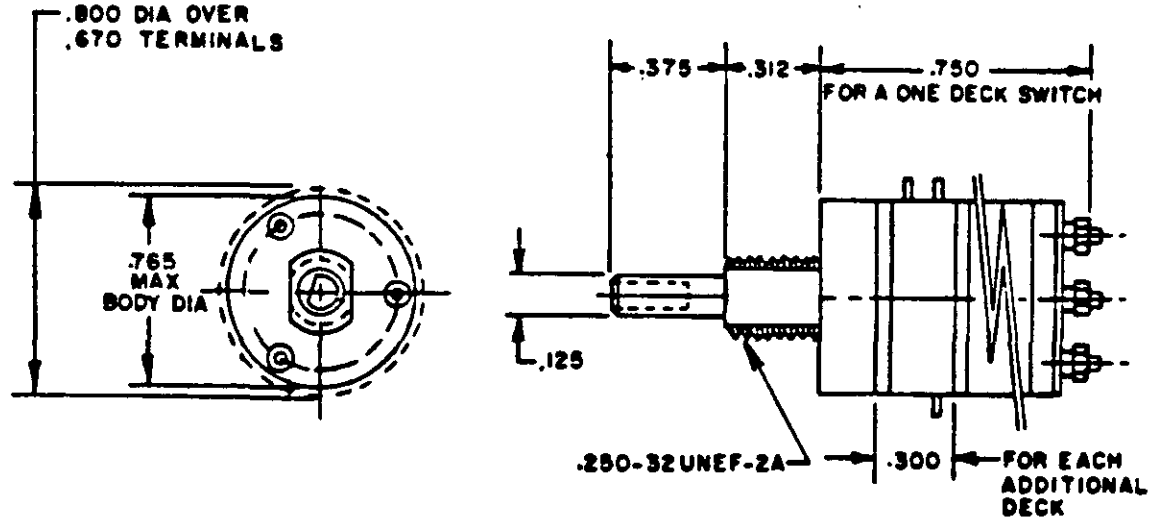
LOAD	ELECTRICAL RATINGS (milliamperes)	
	28 Vdc	115 Vrms, 60 Hz
Resistive:		
At atmospheric pressure	125 to 500	75 to 500
At reduced barometric pressure	50	20
Inductive (2.8 Henries):		
At atmospheric pressure	50 to 250	---
At reduced barometric pressure	10	---



INCHES	MM	INCHES	MM
.125	3.18	.700	17.78
.300	7.62	.750	19.05
.312	7.92	.765	19.43
.375	9.52	.800	20.32
.670	17.02		

"E" CONSTRUCTION STYLE

(EXPLOSION-PROOF, CLOSED CONSTRUCTION WITHOUT SEALED SHAFT AND BUSHING)



"F" CONSTRUCTION STYLE

(EXPLOSION-PROOF, CLOSED CONSTRUCTION WITH SEALED SHAFT AND BUSHING)

FIGURE 107-15. SR41 switches, rotary, closed construction, 1/2 ampere, single shaft (MIL-S-3786/41).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

PART NUMBER: Consists of "M3786/40" followed by a dash and a three-character code. The three-character code used in the following example identifies a switch of construction type S, 30° angle of throw, with nonshorting switching characteristic; 3 decks, 1 pole per deck, solder-lug terminals, and has 10 positions.

EXAMPLE:

M3786/40- M C K

CONSTRUCTION TYPE, ANGLE OF THROW, AND SWITCHING CHARACTERISTIC (see table I)

NUMBER OF DECKS, NUMBER OF POLES PER DECK, AND TYPE OF TERMINALS (see table II)

NUMBER OF POSITIONS (see table III)

TABLE I. Code letters for construction type, angle of throw, and switching characteristic.

	Construction type		Angle of throw	Switching characteristic
	N	S		
Code - - -	A	G	11.25°	S
Code - - -	B	H	15°	S
Code - - -	C 1/	J 1/	22.5°	S
Code - - -	D 1/	K 1/	30°	S
Code - - -	E	L	22.5°	NS
Code - - -	F	M	30°	NS

1/ Adjacent terminals must be connected together in pairs to obtain shorting characteristic.

TABLE II. Code characters for number of decks, number of poles per deck, and type of terminals.

	Number of decks										Poles per deck	Type of terminals
	1	2	3	4	5	6	7	8	9	10		
Code - - -	A	B	C	D	E	F	G	H	J	K	1	Solder lug
Code - - -	L	M	N	P	Q	R	-	-	-	-	2	Solder lug
Code - - -	S	T	U	V	-	-	-	-	-	-	4	Solder lug
Code - - -	W	-	-	-	-	-	-	-	-	-	1	Printed circuit (PC)
Code - - -	-	X	Y	Z	2	-	-	-	-	-	1	Rear deck PC 1/
Code - - -	3	-	-	-	-	-	-	-	-	-	2	Printed circuit
Code - - -	-	4	5	6	-	-	-	-	-	-	2	Rear deck PC 1/
Code - - -	7	-	-	-	-	-	-	-	-	-	4	Printed circuit
Code - - -	-	8	-	-	-	-	-	-	-	-	4	Rear deck PC 1/

1/ Multi-deck switches with PC terminals on the rear deck have solder lug terminals on all other decks.

TABLE III. Code characters for number of positions.

Code	Number of positions	Code	Number of positions	Code	Number of positions
A - - -	1/ Continuous	M - - -	12 (with stops)	Y - - -	23 (with stops)
B - - -	2 (with stops)	N - - -	13	Z - - -	24
C - - -	3	P - - -	14	2 - - -	25
D - - -	4	Q - - -	15	3 - - -	26
E - - -	5	R - - -	16	4 - - -	27
F - - -	6	S - - -	17	5 - - -	28
G - - -	7	T - - -	18	6 - - -	29
H - - -	8	U - - -	19	7 - - -	30
J - - -	9	V - - -	20	8 - - -	31 (with stops)
K - - -	10	W - - -	21		
L - - -	11(with stops)	X - - -	22 (with stops)		

1/ Switch is continuous rotation type (no stops), and number of positions is dependent on angle of throw: 12 positions for 30° angle; 16 positions for 22.5° angle; 24 positions for 15° angle; and 32 positions for 11.25° angle.

FIGURE 107-16. SR40 switches, rotary, open construction, low level and 1/2 ampere (MIL-S-3786/40) - Continued.

Source: https://assist.dla.mil -- Downloaded: 20261010-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical loads.

Environmental condition	Inductive load (2.8 Henries)		Resistive loads	
	Milliamperes	Volts, dc	Milliamperes	Volts
At atmospheric pressure	20	28	200 50	28 dc 115 Vrms
At reduced barometric pressure	---	---	100 25	28 dc 115 Vrms

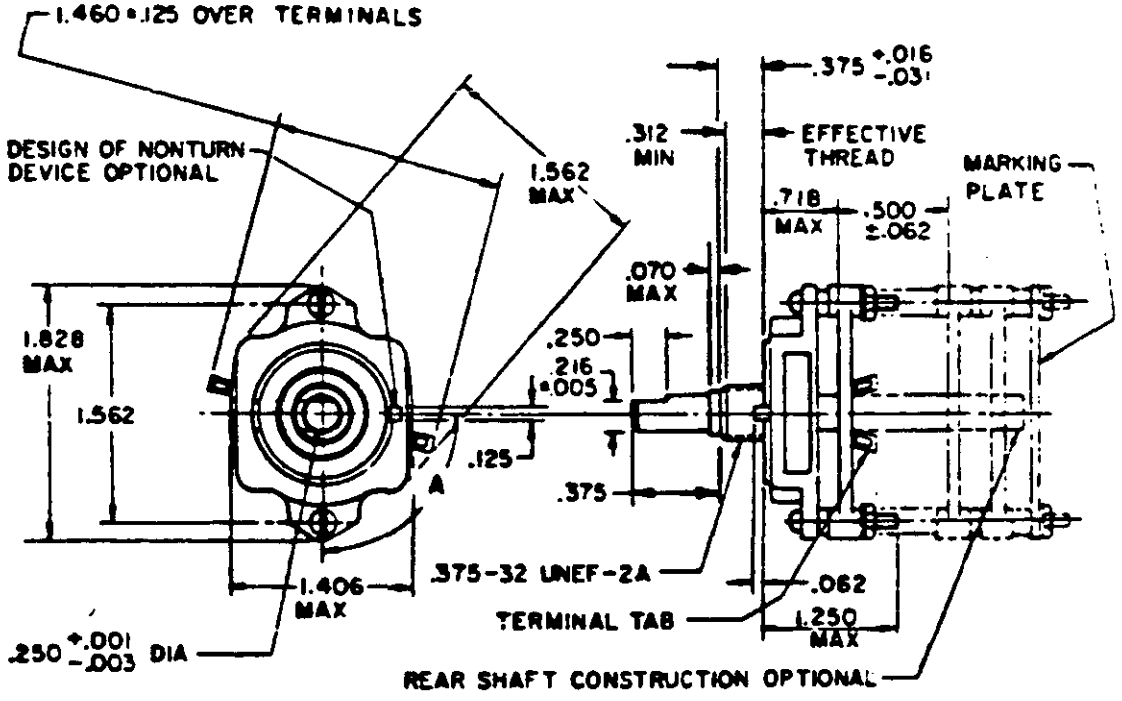
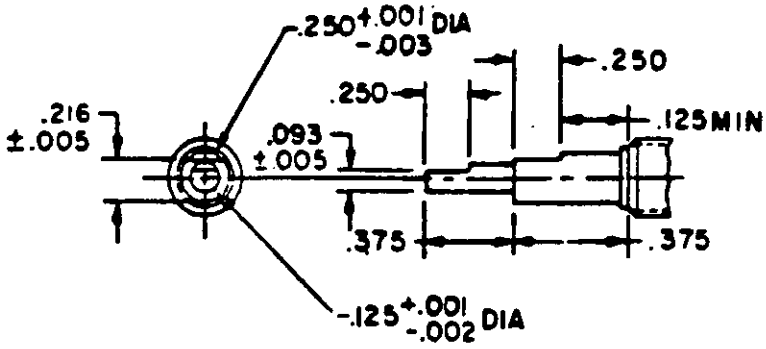
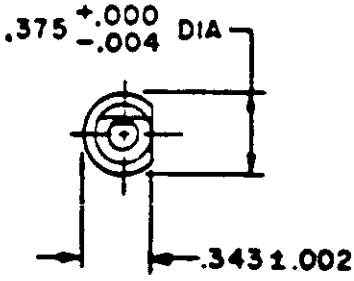


FIGURE 107-17. SR32 switches, rotary, open construction, 1/5 ampere (MIL-S-3786/32).

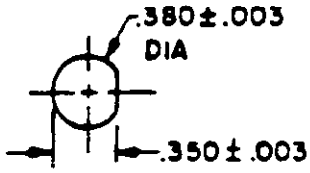
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



CONCENTRIC SHAFT
(ALTERNATE CONSTRUCTION ALLOWED ON SR 32-2 SWITCHES)



BUSHING FLAT FOR S CONSTRUCTION



RECOMMENDED PANEL PIERCING FOR S CONSTRUCTION

INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.093	2.36	.500	12.70
.002	.05	.125	3.18	.718	18.24
.003	.08	.216	5.49	1.250	31.75
.004	.10	.250	6.35	1.406	35.71
.005	.13	.312	7.92	1.460	37.08
.016	.41	.343	8.71	1.562	39.67
.031	.79	.350	8.89	1.609	40.87
.062	1.57	.375	9.53	1.828	46.43
.070	1.78	.380	9.65		

FIGURE 107-17. SR32 switches, rotary, open construction, 1/5 ampere (MIL-S-3786/32) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Part number 1/: The military part number shall consist of the prefix "M3786/32-" followed by one character each from tables II, III, and IV, as applicable. An example is shown below:

Example: M3786/32- C K G

- Military designator and specification sheet no.
- Construction type N, two decks, both nonshorting (see table II)
- Two poles per deck, 22.5 angle of throw (see table III)
- Seven positions (see table IV)

1/ Part numbers shall be used to identify only switches shown in figure 2. Switches not identified by part numbers shall be procured in accordance with the ordering data of MIL-S-3786.

TABLE II. Code character for combinations of construction type, number of decks, and switching characteristics.

Code character	Construction type	Number of decks	Switching characteristics				
			First deck	Second deck	Third deck	Fourth deck	Fifth deck
A	N	1	NS				
B	"	1	S				
C	"	2	NS	NS			
D	"	2	S	S			
E	"	2	NS	S			
F	"	3	NS	NS	NS		
G	"	3	S	S	S		
H	"	3	NS	NS	S		
J	"	3	NS	S	S		
K	"	4	NS	NS	NS	NS	
L	"	4	S	S	S	S	
M	"	4	NS	NS	NS	S	
N	"	4	NS	S	S	S	
P	"	5	NS	NS	NS	NS	NS
Q	"	5	S	S	S	S	S
R	"	5	NS	NS	NS	NS	S
S	"	5	NS	S	S	S	S
T	S	1	NS				
U	"	1	S				
V	"	2	NS	NS			
W	"	2	S	S			
X	"	2	NS	S			
Y	"	3	NS	NS	NS		
Z	"	3	S	S	S		
1	"	3	NS	NS	S		
2	"	3	NS	S	S		
3	"	4	NS	NS	NS	NS	
4	"	4	S	S	S	S	
5	"	4	NS	NS	NS	S	
6	"	4	NS	S	S	S	
7	"	5	NS	NS	NS	NS	NS
8	"	5	S	S	S	S	S
9	"	5	NS	NS	NS	NS	S
0	"	5	NS	S	S	S	S

FIGURE 107-17. SR32 switches, rotary, open construction, 1/5 ampere (MIL-S-3786/32) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE III. Code letters for combinations of poles per deck and angle of throw.

Code letter	Poles per deck	Angle of throw	Code letter	Poles per deck	Angle of throw
A	1	15°	M	2	25.7°
B	2	15°	N	1	30°
C	3	15°	P	2	30°
D	4	15°	Q	3	30°
E	1	18°	R	4	30°
F	2	18°	S	1	36°
G	1	20°	T	2	36°
H	2	20°	U	1	45°
J	1	22.5°	V	2	45°
K	2	22.5°			
L	1	25.7°			

TABLE IV. Code letters for number of positions.

Code letter	Number of positions	Code letter	Number of positions
A	1/	N	13 with stops
B	2 with stops	P	14 with stops
C	3 with stops	Q	15 with stops
D	4 with stops	R	16 with stops
E	5 with stops	S	17 with stops
F	6 with stops	T	18 with stops
G	7 with stops	U	19 with stops
H	8 with stops	V	20 with stops
J	9 with stops	W	21 with stops
K	10 with stops	X	22 with stops
L	11 with stops	Y	23 with stops
M	12 with stops	Z	24 with stops

1/ Switch is continuous rotation type (no stops), and number of positions is dependent on angle of throw as follows:

Angle of throw	Positions
15°	24
18°	20
20°	18
22.5°	16
25.7°	14
30°	12
36°	10
45°	8

FIGURE 107-17. SR32 switches, rotary, open construction, 1/5 ampere (MIL-S-3786/32) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

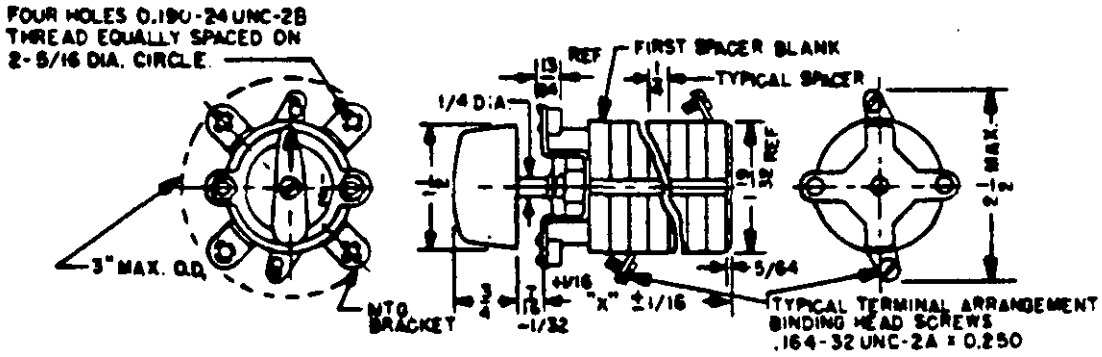


TABLE I. Electrical and endurance ratings.

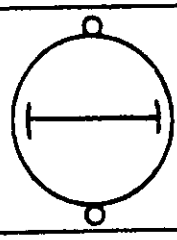
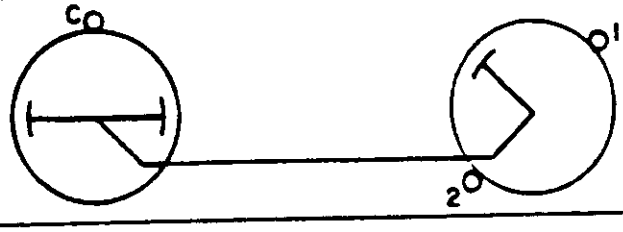
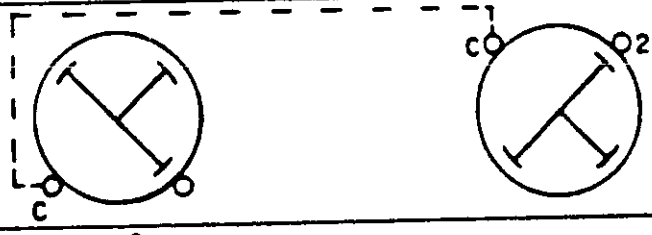
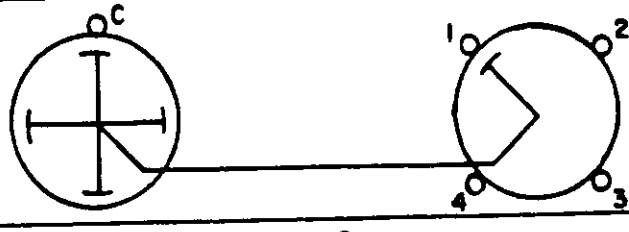
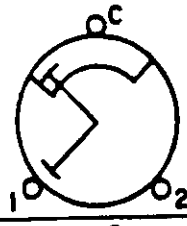
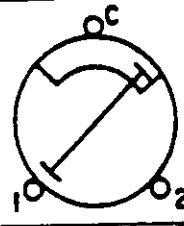
Test	Current (amperes)	Voltage (volts)	Electrical operations (Number of operations)	Test rate (operations per minute)	De-energized operations (Number of operations)
Alternating current (rms)	10	125	30,000	30	20,000
Direct current	10	120	20,000	15	

Part number: M15291/2- (dash number from table III).

FIGURE 107-18. Class 1SR switches, rotary, snap-action, 10 amperes (MIL-S-15291/2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE II. Switching characteristics.

Characteristic symbol	Operation (C indicates common)	Typical circuit 1/
A	off-on-off-on (single throw)	
B	off-on 1 off-on 2 (double throw)	
E	off-on 1-on 2-on 3 (triple throw)	
F	off-on 1-on 1& 2-on 2 (triple throw)	
G	on 1-on 2-on 3-on 4 (triple throw)	
H	on 1-off-on 2 (double throw)	
J	on 1-on 2-on 1-on 2 (double throw)	

1/ Terminal numbers shown do not necessarily correspond to actual configurations (see detail specification sheet).

FIGURE 107-18. Class 1SR switches, rotary, snap-action, 10 amperes (MIL-S-15291/2) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE III. Dash numbers and configurations.

M15291/2- Dash number	Dimension "X"	Switching characteristic (see table II)	Number of circuits	Number of spacers	Total number of decks
002	1 5/16	A	1	1	2
004	1 9/16	"	2	"	3
006	1 13/16	"	3	"	4
008	2 1/16	"	4	"	5
010	2 5/16	"	5	"	6
012	2 9/16	"	6	"	7
014	2 13/16	"	7	"	8
016	3 1/16	"	8	"	9
018	3 7/16	"	9	"	10
020	3 11/16	"	10	"	11
022	1 9/16	B	1	"	3
024	2 1/16	"	2	"	5
026	2 9/16	"	3	"	7
028	3 1/16	"	4	"	9
030	3 7/16	"	5	"	11
032	4 1/16	"	6	"	13
040	1 9/16	E	1	"	3
042	2 1/16	"	2	"	5
044	2 9/16	"	3	"	7
046	3 1/16	"	4	"	9
047	3 9/16	"	5	"	11
048	4 1/16	"	6	"	13
050	1 9/16	F 1/	1	"	3
052	2 1/16	F 1/	2	"	5
054	2 9/16	F 1/	3	"	7
056	1 9/16	G	1	"	3
058	2 1/16	"	2	"	5
060	2 7/16	"	3	"	7
062	3 1/16	"	4	"	9
063	3 9/16	"	5	"	11
064	4 1/16	"	6	"	13
065	1 5/16	H	1	"	2
066	1 9/16	"	2	"	3
067	1 13/16	"	3	"	4
068	2 1/16	"	4	"	5
069	2 5/16	"	5	"	6
070	2 9/16	"	6	"	7
071	2 15/16	"	7	"	8
072	3 3/16	"	8	"	9
073	1 5/16	J	1	"	2
074	1 9/16	"	2	"	3
075	1 13/16	"	3	"	4
076	2 1/16	"	4	"	5
077	2 5/16	"	5	"	6
078	2 9/16	"	6	"	7
079	2 15/16	"	7	"	8
080	3 3/16	"	8	"	9

1/ Make before break.

FIGURE 107-18. Class ISR switches, rotary, snap-action, 10 amperes (MIL-S-15291/2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

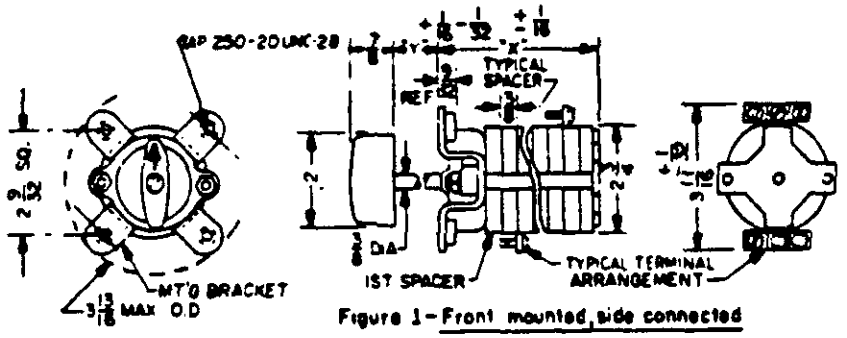


Figure 1 - Front mounted, side connected

TABLE I. Electrical and endurance ratings.

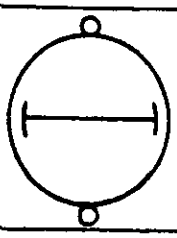
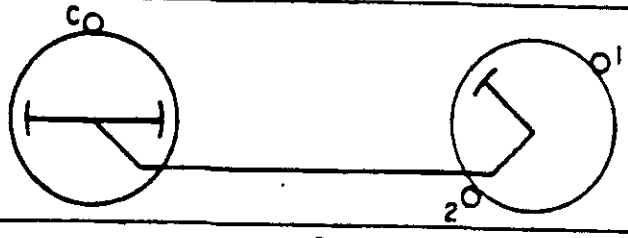
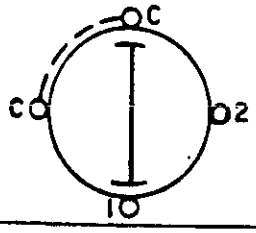
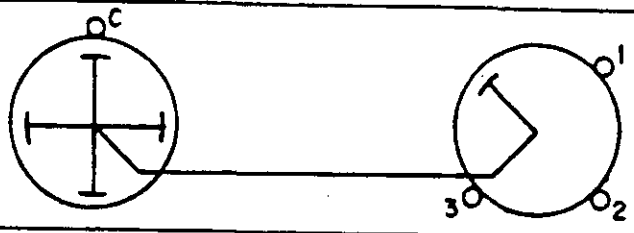
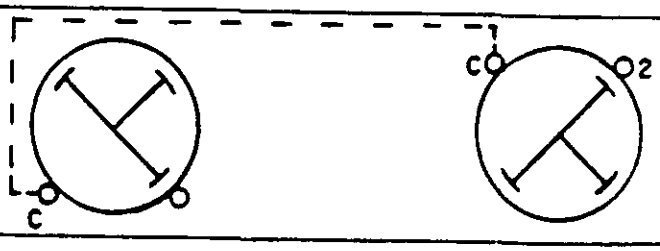
Test	Current (amperes)	Voltage (volts)	Electrical operations (Number of operations)	Test rates (operations per minute)	De-energized operations (Number of operations)
Alternating current (rms)	30	500	20,000	30	10,000
Direct current	30	250	15,000	15	-

Part number: M15291/6- (dash number from table III).

FIGURE 107-19. Class 3SR switches, rotary, snap-action, 30 amperes (MIL-S-15291/6).

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE II. Switching characteristics.

Characteristic symbol	Operation (C indicates common)	Typical circuit 1/
A	off-on-off-on (single throw)	
B	off-on 1 off-on 2 (double throw)	
C	on 1-on 2-on 1-on 2 (double throw)	
E	off-on 1-on 2-on 3 (triple throw)	
F	off-on 1-on 18 2-on 2 (triple throw)	

1/ Terminal numbers shown do not necessarily correspond to actual configurations (see detail specification sheet).

FIGURE 107-19. Class 3SR switches, rotary, snap-action, 30 amperes (MIL-S-15291/6) - Continued.

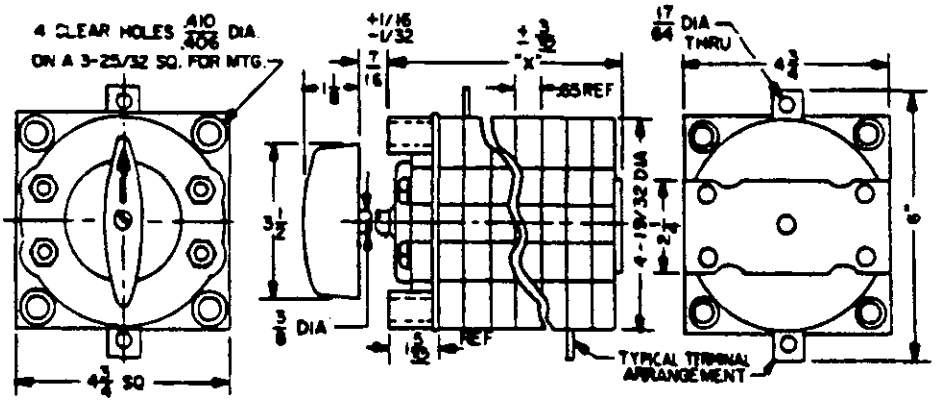
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE III. Dash numbers and configurations.

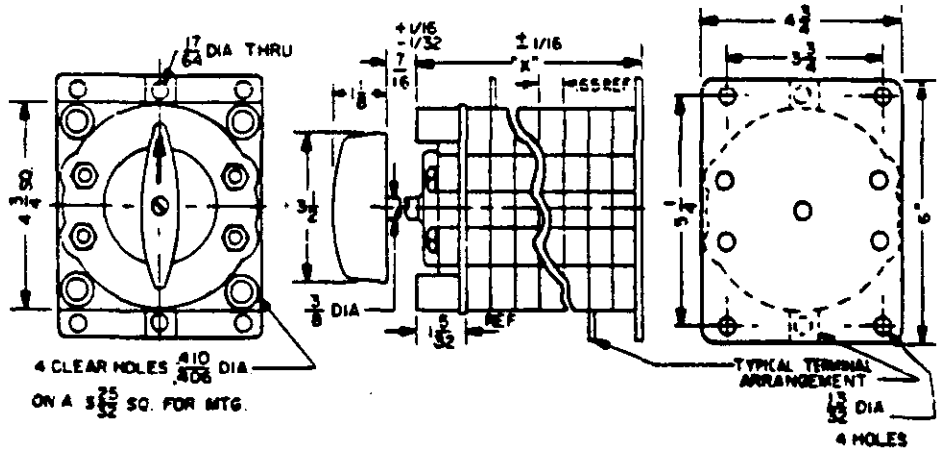
M15291/6- Dash number	Dimension "X"	Dimension "Y"	Switching characteristic (see table II)	Number of circuits	Number of spacers	Total number of decks
001	1 7/8	3/8	A	1	1	2
002	2 1/4	"	"	2	"	3
003	2 5/8	"	"	3	"	4
004	3	"	"	4	"	5
005	3 3/8	"	"	5	"	6
006	3 3/4	7/16	"	6	"	7
007	4 1/8	"	"	7	"	8
008	4 1/2	"	"	8	"	9
009	5 1/8	"	"	9	"	10
010	5 1/2	"	"	10	"	11
011	2 1/4	"	B	1	"	3
012	3	"	"	2	"	5
013	3 3/4	"	"	3	"	7
014	4 1/2	"	"	4	"	9
015	5 1/4	"	"	5	"	11
016	6	"	"	6	"	13
017	1 7/8	"	C	1	"	2
018	2 5/8	"	"	2	2	4
019	3 3/8	"	"	3	3	6
020	4 1/8	"	"	4	4	8
021	4 7/8	"	"	5	5	10
022	5 5/8	"	"	6	6	12
023	2 1/4	"	E	1	1	3
024	3	"	"	2	"	5
025	3 3/4	"	"	3	"	7
[illegible]6	4 1/2	"	"	4	"	9
[illegible]7	2 1/4	"	F	1	"	3
28	3	"	"	2	"	5
029	3 3/4	"	"	3	"	7
030	4 3/4	"	"	4	"	9
031	5 1/2	"	"	5	"	11
032	6 1/4	"	"	6	"	13

FIGURE 107-19. Class 35R switches, rotary, snap-action, 30 amperes (MIL-S-15291/6) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



Configuration for less than 10 decks



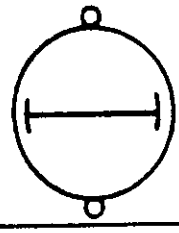
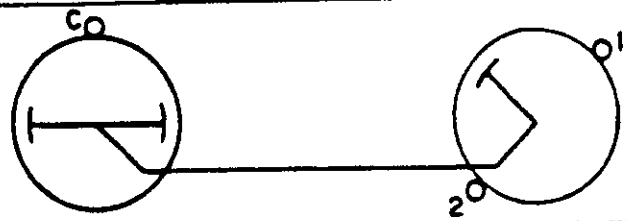
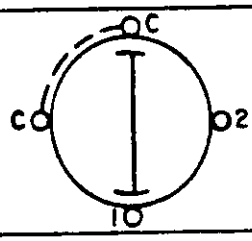
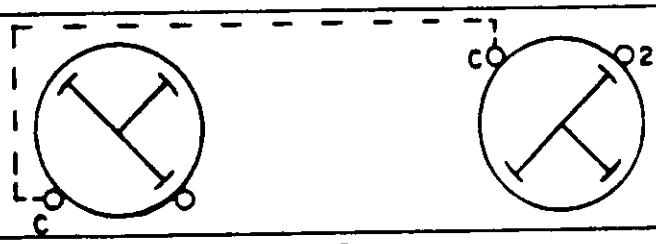
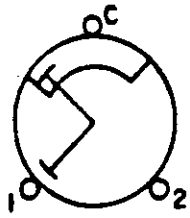
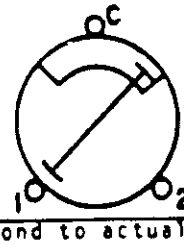
Configuration for 10 or more decks

Part number: M15291/9- (dash number from table III).

FIGURE 107-20. Class 6SR switches, rotary, snap-action, 60 amperes (MIL-S-15291/9).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE I. Switching characteristics.

Characteristic symbol	Operation (C indicates common)	Typical circuit 1/
A	off-on-off-on (single throw)	
B	off-on 1 off-on 2 (double throw)	
C	on 1-on 2-on 1-on 2 (double throw)	
E	off-on 1-on 2-on 3 (triple throw)	
F	off-on 1-on 1& 2-on 2 (triple throw)	
H	on 1-off-on 2 (double throw)	
J	on 1-on 2-on 1-on 2 (double throw)	

1/ Terminal numbers shown do not necessarily correspond to actual configurations (see detail specification sheet).

FIGURE 107-20. Class 6SR, switches, rotary, snap-action, 60 amperes (MIL-S-15291/9) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE II. Electrical and endurance ratings.

Test	Current (amperes)	Voltage (volts)	Electrical operations (Number of operations)	Test rate (operations per minute)	De-energized operations (Number of operations)
Alternating current (rms)	60	500	6,000	10	4,000
Direct current	60	250	4,000	5	-

TABLE III. Dash numbers and configurations.

M15291/9- Dash number	Dimension "X"	Switching characteristic (see table I)	Number of circuits	Number of spacers	Total number of decks
001	3 3/32	A	2	1	3
002	3 15/16	"	3	"	4
003	4 9/16	"	4	"	5
004	5 7/32	"	5	"	6
005	5 7/8	"	6	"	7
006	7 3/16	"	8	"	9
007	8 1/2	"	10	"	11
008	4 9/16	B	2	"	5
009	5 7/8	B	3	"	7
010	3 15/16	C	2	2	4
011	3 9/32	E	1	1	3
012	4 9/16	"	2	"	5
013	5 7/8	"	3	"	7
014	7 7/32	"	4	"	9
015	8 17/32	"	5	"	11
016	5 7/32	F	2	2	6
017	6 9/16	"	3	"	8
018	7 27/32	"	4	"	10
019	3 15/16	H	2	"	4
020	4 9/16	H	3	"	5
021	3 15/16	J	2	"	4
022	4 9/16	J	3	"	5

FIGURE 107-20. Class 6SR switches, rotary, snap-action, 60 amperes (MIL-S-15291/9) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

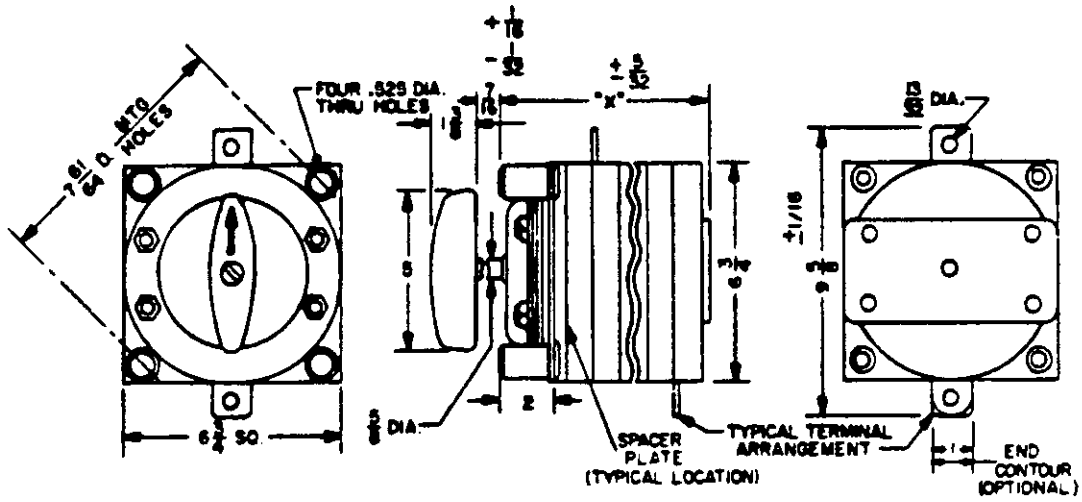


TABLE I. Electrical and endurance ratings.

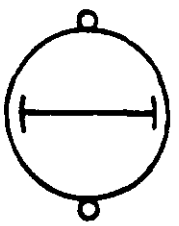
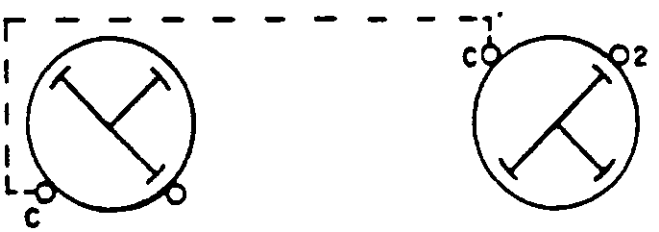
Test	Current (amperes)	Voltage (volts)	Electrical operations (Number of operations)	Test rate (operations per minute)	De-energized operations (Number of operations)
Alternating current (rms)	200	500	6,000	8	4,000
Direct current	200	250	4,000	3	-

Part number: M15291/12- (dash number from table III).

FIGURE 107-21. Class 20SR switches, rotary, snap-action, 200 amperes (MIL-S-15291/12).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE II. Switching characteristics.

Characteristic symbol	Operation (C indicates common)	Typical circuit 1/
A	off-on-off-on (single throw)	
B	off-on 1 off-on 2 (double throw)	
E	off-on 1-on 2-on 3 (triple throw)	
F	off-on 1-on 1& 2-on 2 (triple throw)	

1/ Terminal numbers shown do not necessarily correspond to actual configurations (see detail specification sheet).

FIGURE 107-21. Class 20ST switches, rotary, snap-action, 200 amperes (MIL-S-15291/12).

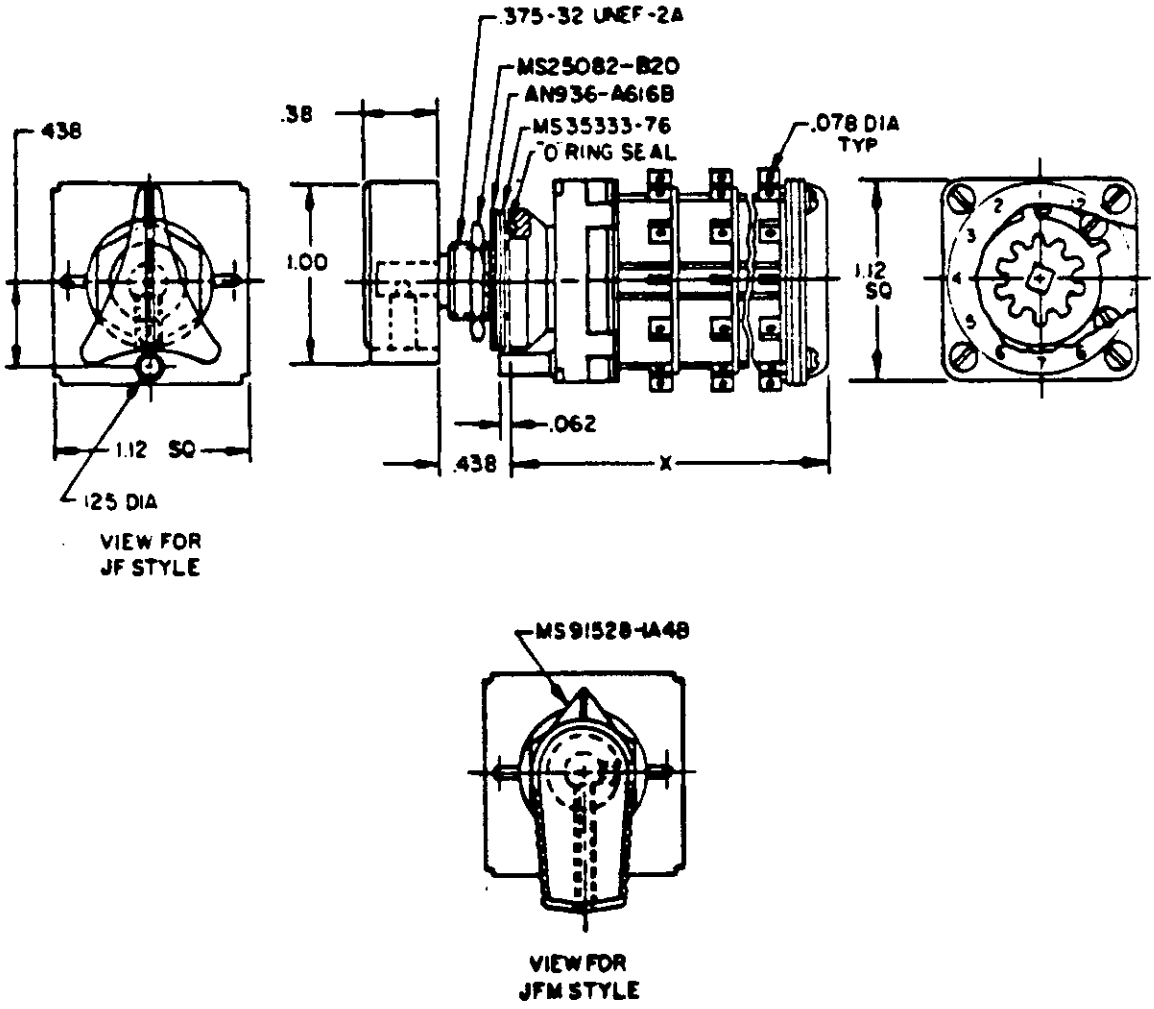
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE III. Dash numbers and configurations.

M15291/9- Dash number	Dimension "X"	Switching characteristic (see table II)	Number of circuits	Number of spacers	Total number of decks
001	4 13/16	A	2	1	3
002	5 11/16	A	3	"	4
003	6 9/16	B	2	"	5
004	8 5/16	B	3	"	7
005	6 9/16	E	2	"	5
006	8 5/16	E	3	"	7
007	7 7/16	F	2	2	6
008	9 3/16	F	3	2	8

FIGURE 107-21. Class 20SR switches, rotary, snap-action, 200 amperes (MIL-S-15291/12).

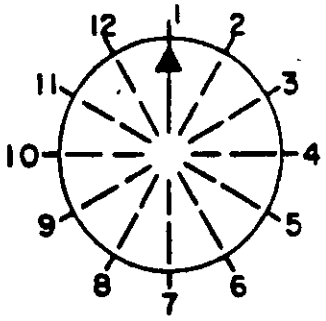
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



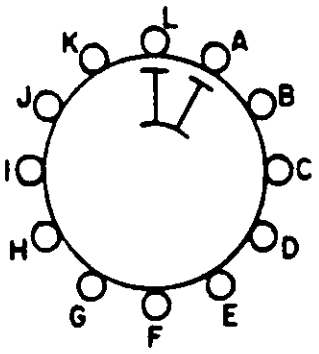
Angle of throw: 30 degrees.
Switching action: Detent (style JF), momentary (style JFM). These switches have single break contact action.
Electrical loads: A.C. current: 1 ampere.
Voltage: 125 volts (at 60 Hz, 0.5 power factor inductive).

FIGURE 107-22. Switches, rotary, multipole and selector, 1 ampere (MIL-S-21604/1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

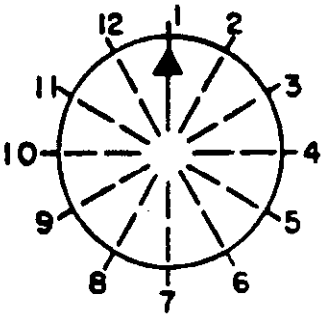


HANDLE POSITIONS

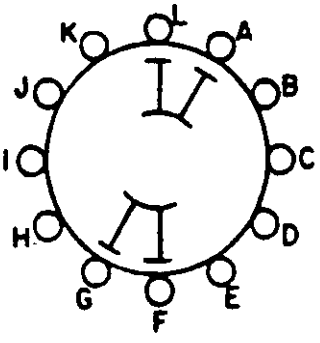


TYPICAL SECTION
SINGLE POLE

ROTOR DESIGNATION 1



HANDLE POSITIONS



TYPICAL SECTION
DOUBLE POLE

ROTOR DESIGNATION 2

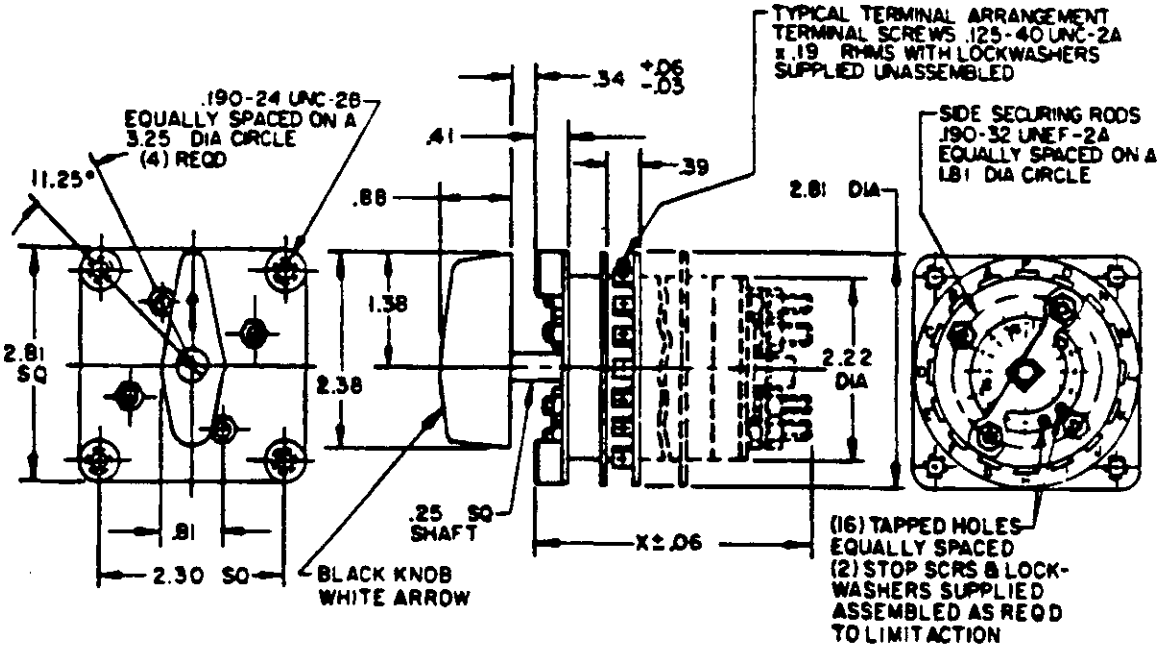
TABLE I - Type designation and switching characteristics.

Type designation				Dimension "X" ±.062	Approx. weight oz.	Maximum rotational torque pound-inches
Component	Rotor designation	Style	No. of sections			
S	1	JF	4	1.960	4.2	8
S	1	JF	5	2.250	4.4	8
S	2	JF	1	1.090	3.5	5
S	2	JF	2	1.380	3.7	5
S	2	JF	3	1.670	3.9	5
S	2	JF	4	1.960	4.2	8
S	2	JF	5	2.250	4.4	8
S	2	JFM 1/	3	1.950	4.4	5
S	2	JFM 1/	4	2.240	4.6	5.5
S	2	JFM 1/	5	2.530	4.9	5.5

1/ Type JFM switches shall be limited to three positions, (0 degrees and ±30 degrees) by a stop assembly located in the spring case. Either of the 30 degree positions can be eliminated by adjustment of the female gears in the rear stop shown on figure 1. Both of the 30 degree positions are momentary.

FIGURE 107-22. Switches, rotary, multipole and selector, 1 ampere (MIL-S-21604/1) - Continued.

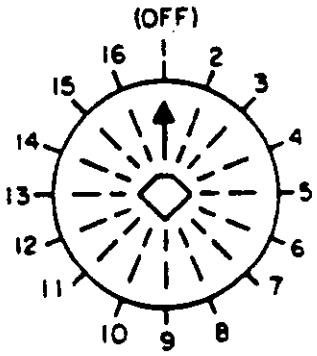
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



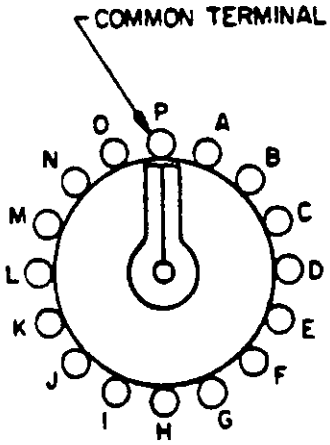
Angle of throw: 22.5 (degrees).
Switching action: Detent.
Electrical loads: A.C. current: 5 ampere.
Voltage: 125 volts (at 60 and 400 Hz, 0.80 inductive power factor).

FIGURE 107-23. Switches, rotary, multipole and selector, 5 amperes (MIL-S-21604/2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



HANDLE POSITIONS



TYPICAL POSITIONS
SINGLE POLE
SELECTOR

ROTOR DESIGNATION 3

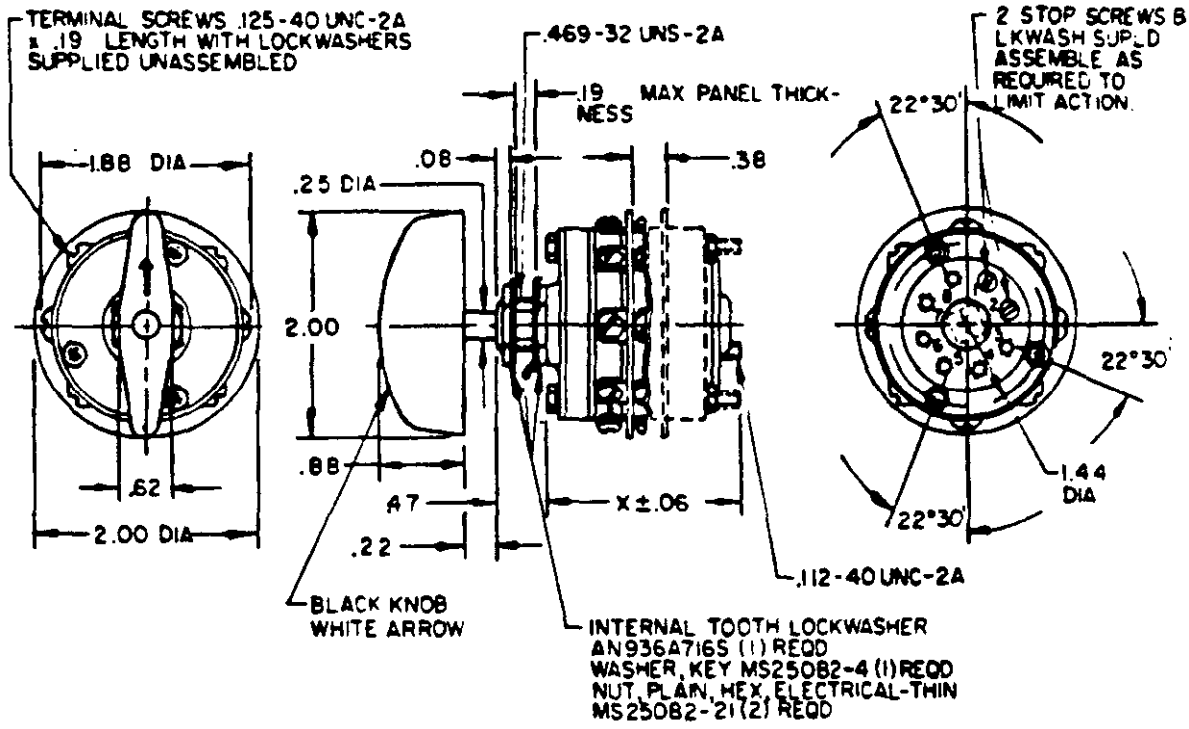
BREAK-BEFORE-MAKE CONTACTS

TABLE I - Type designation and switching characteristics.

Type designation				Dimension "X" 2/ ±.062	Approx. weight oz.	Maximum rotational torque pounds-inches
Component	Rotor designation	Style	No. of sections			
S	3	JK	3	3.03	22	15
S	3	JK	5	3.81	28	20
S	3	JK	10	6.34	47	35
S	3	JK	15	8.50	56	40

FIGURE 107-23. Switches, rotary, multipole and selector, 5 amperes (MIL-S-21604/2) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



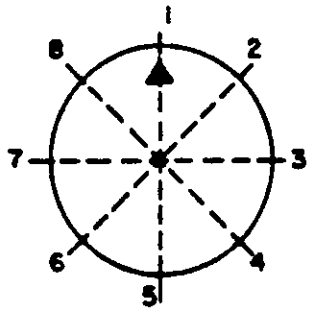
Angle of throw: 45 degrees.
Switching action: Detent.
Electrical loads: A.C. current: 5 ampere.
Voltage: 125 volts (at 60 and 400 Hz, 0.80 inductive power factor).

TABLE I - Type designation and switching characteristics.

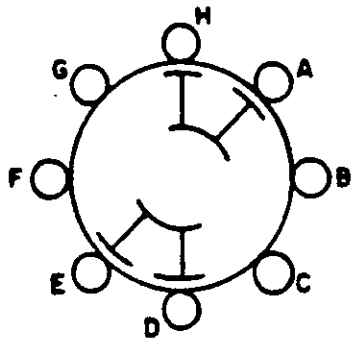
Type designation				Dimension "X" ±.062	Approx. weight oz.	Maximum rotational torque pound-inches
Component	Rotor designation	Style	No. of sections			
S	2	JL	3	1.91	8	9
S	2	JL	5	2.66	10	11
S	2	JL	10	5.28	16	16
S	3	JL	3	1.91	8	9
S	3	JL	5	2.66	10	11
S	3	JL	10	5.28	16	16
S	4	JL	3	1.91	8	9
S	4	JL	5	2.66	10	11
S	4	JL	10	5.28	16	16

FIGURE 107-24. Switches, rotary, multipole and selector, 5 amperes (MIL-S-21604/3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

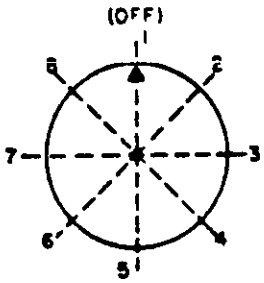


HANDLE POSITIONS

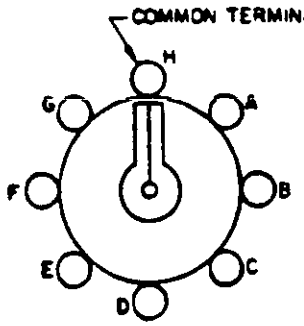


TYPICAL SECTION
DOUBLE POLE
DOUBLE BREAK

ROTOR DESIGNATION 2

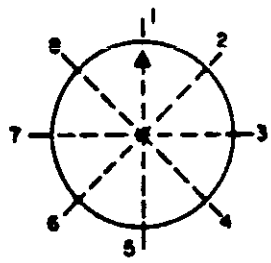


HANDLE POSITIONS

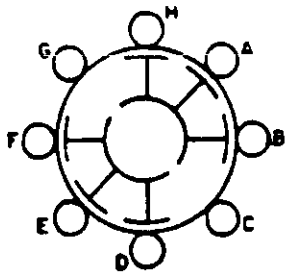


TYPICAL SECTION
SINGLE POLE
SELECTOR

ROTOR DESIGNATION 3



HANDLE POSITIONS



TYPICAL SECTION
DOUBLE POLE
TRIPLE BREAK

ROTOR DESIGNATION 4 ✓

✓ WITH ROTOR CONFIGURATION 4, CONTACT H-1 BREAKS BEFORE C-3 MAKES, AND CONTACTS A-1 AND B-2 MAKE BEFORE A-2 AND B-3 BREAK

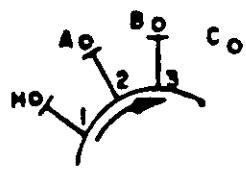
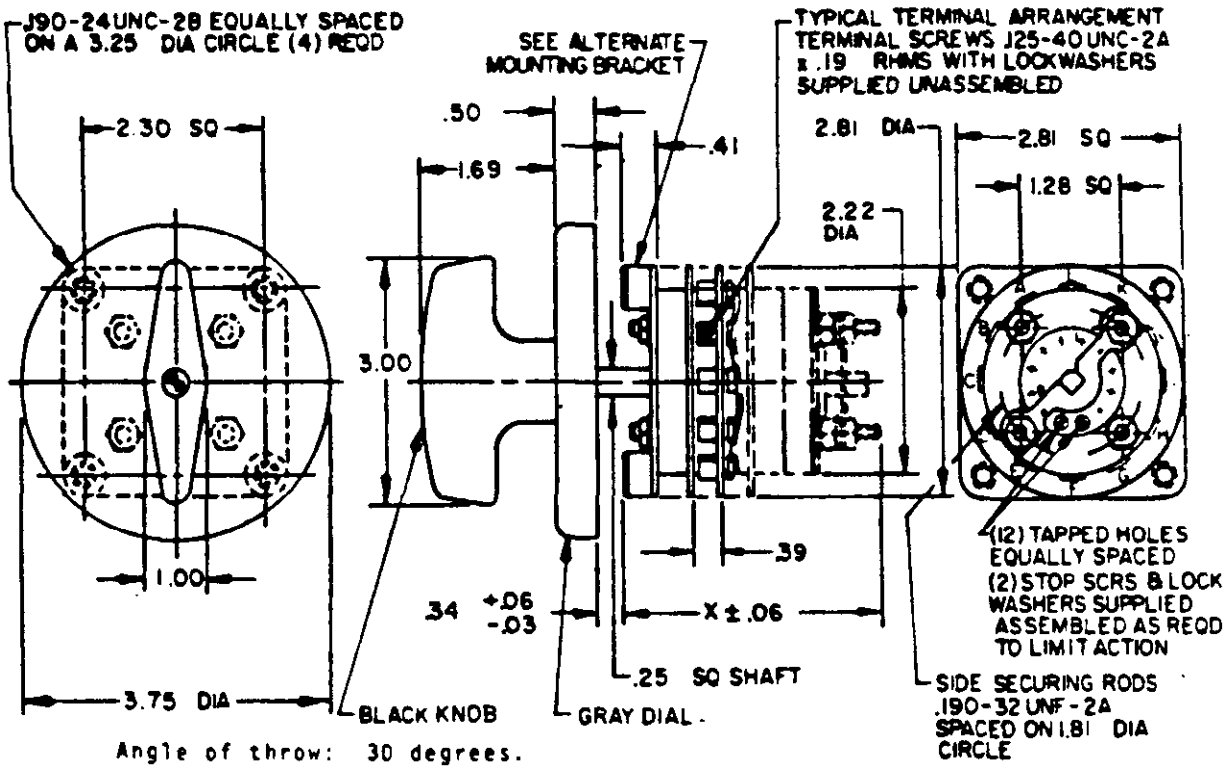


FIGURE 107-24. Switches, rotary, multipole and selector, 5 amperes (MIL-S-21604/3) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



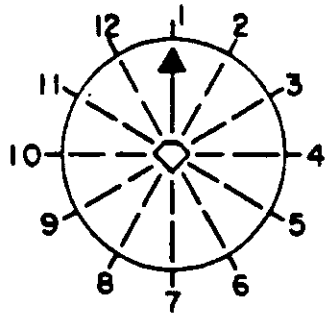
Angle of throw: 30 degrees.
Switching action: Detent.
Electrical loads: A.C. current: 7-1/2 amperes.
Voltage: 125 volts (at 60 and 400 Hz, 0.80 inductor power factor).

TABLE I - *Type designation and switching characteristics.*

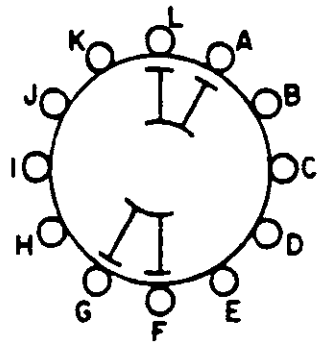
Type designation				Dimension "X" ±.062	Approx. weight oz.	Maximum rotational torque pound-inches
Component	Rotor designation	Style	No. of sections			
S	2	JM	5	3.81	25	20
S	2	JM	10	5.75	40	25
S	2	JM	15	8.09	59	45
S	2	JM	20	10.41	75	50
S	2	JM	25	12.34	90	55
S	3	JM	5	3.81	25	20
S	3	JM	10	5.75	40	25
S	3	JM	15	8.09	59	45
S	3	JM	20	10.41	75	50
S	3	JM	25	12.34	90	55
S	5	JM	5	3.81	25	20
S	5	JM	10	5.75	40	25
S	5	JM	15	8.09	59	45
S	5	JM	20	10.41	75	50
S	5	JM	25	12.34	90	55

FIGURE 107-25. Switch, rotary, multipole and selector, 7.5 amperes (MIL-S-21604/4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



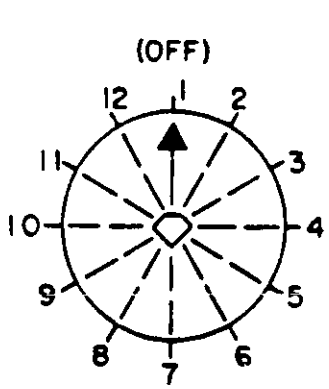
HANDLE POSITIONS



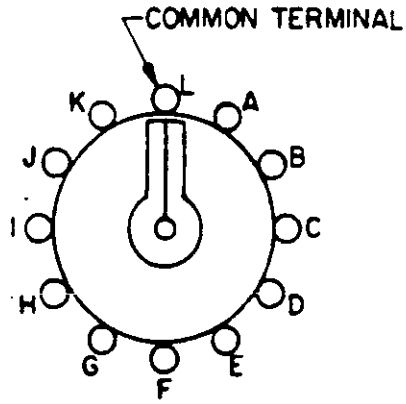
TYPICAL SECTION
DOUBLE POLE
DOUBLE BREAK

BREAK BEFORE MAKE CONTACTING

ROTOR DESIGNATION 2



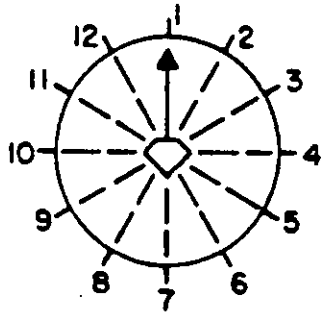
HANDLE POSITIONS



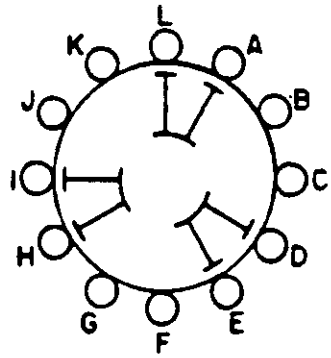
TYPICAL SECTIONS
SINGLE POLE
SELECTOR

BREAK BEFORE MAKE CONTACTING

ROTOR DESIGNATION 3



HANDLE POSITIONS



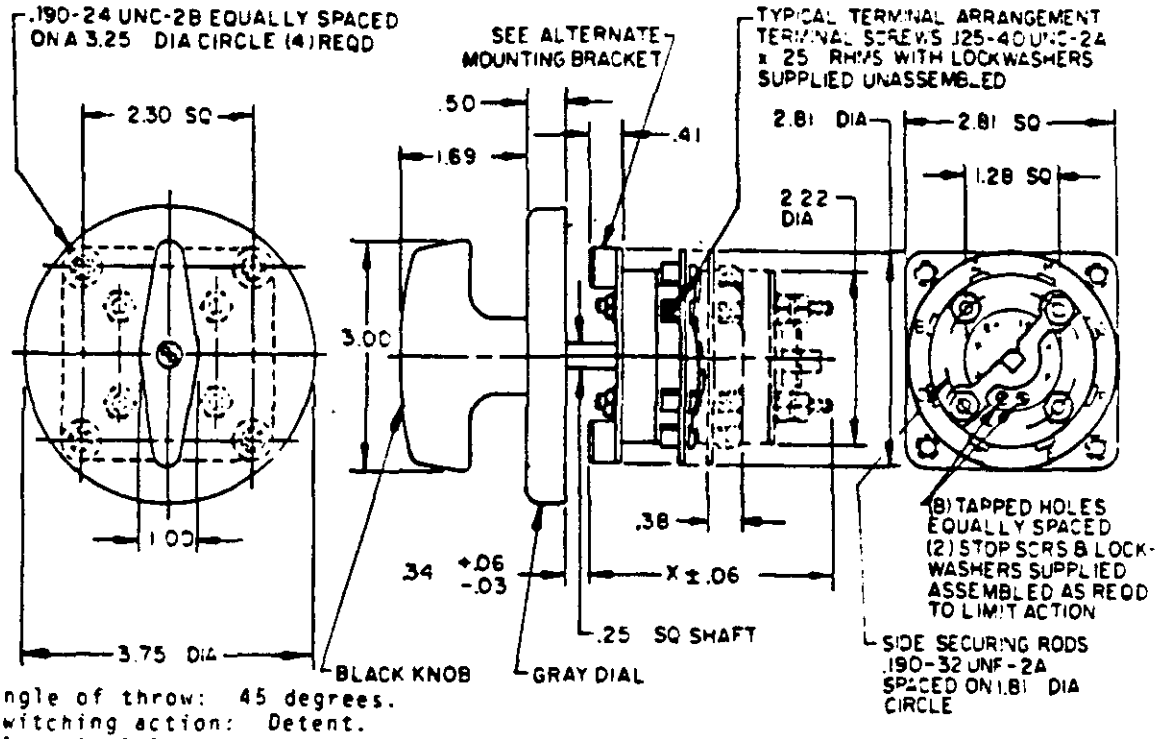
TYPICAL SECTION
TRIPLE POLE
DOUBLE BREAK

BREAK BEFORE MAKE CONTACTING

ROTOR DESIGNATION 5

FIGURE 107-25. Switch, rotary, multipole and selector, 7.5 amperes (MIL-S-21604/4) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



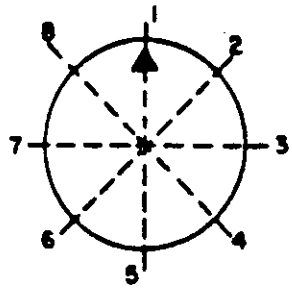
Angle of throw: 45 degrees.
Switching action: Detent.
Electrical loads: A.C. current: 10 amperes.
Voltage: 125 volts (at 60 Hz, 0.80 inductive power factor).

TABLE I - Type designation and switching characteristics.

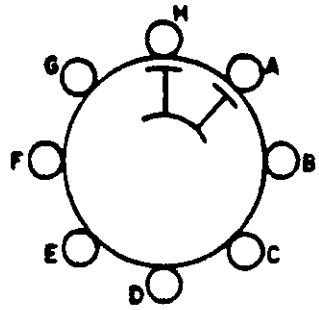
Type designation				Dimension "X" ±.062	Approx. weight oz.	Maximum rotational torque pound-inches
Component	Rotor designation	Style	No. of sections			
S	1	JR	5	3.75	29	25
S	1	JR	10	5.62	39	30
S	1	JR	15	8	48	45
S	1	JR	20	9.94	60	50
S	1	JR	25	11.81	70	55
S	2	JR	5	3.75	29	25
S	2	JR	10	5.62	39	30
S	2	JR	15	8	48	45
S	2	JR	20	9.94	60	50
S	2	JR	25	11.81	70	55
S	3	JR	5	3.75	29	25
S	3	JR	10	5.62	39	30
S	3	JR	15	8	48	45
S	3	JR	20	9.94	60	50
S	3	JR	25	11.81	70	55
S	4	JR	5	3.75	29	25
S	4	JR	10	5.62	39	30
S	4	JR	15	8	48	45
S	4	JR	20	9.94	60	50
S	4	JR	25	11.81	70	55

FIGURE 107-26. Switch, rotary, multipole and selector, 10 amperes (MIL-S-21604/5).

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:41Z
Check the source to verify that this is the current version before use.

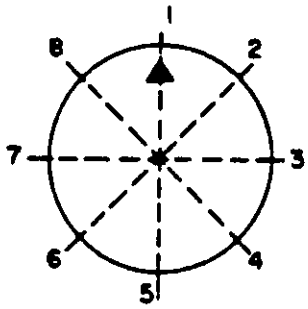


HANDLE POSITIONS

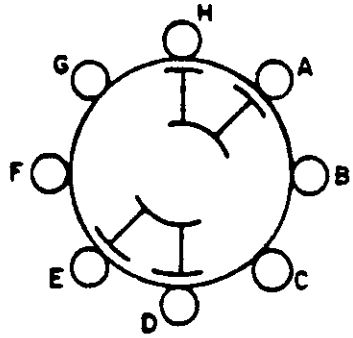


TYPICAL SECTION
SINGLE POLE
DOUBLE BREAK

ROTOR DESIGNATION 1

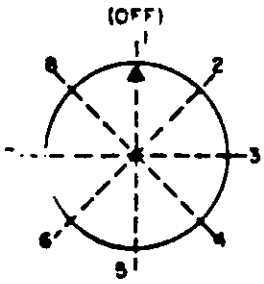


HANDLE POSITIONS

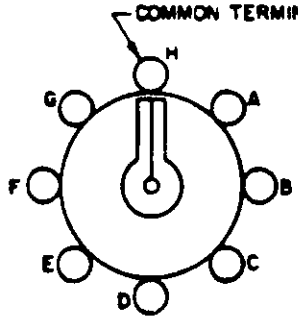


TYPICAL SECTION
DOUBLE POLE
DOUBLE BREAK

ROTOR DESIGNATION 2

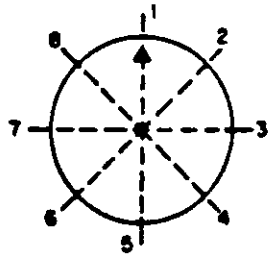


HANDLE POSITIONS

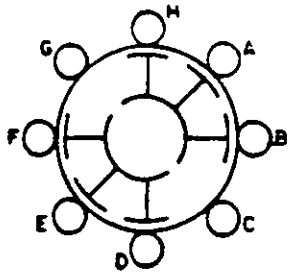


TYPICAL SECTION
SINGLE POLE
SELECTOR

ROTOR DESIGNATION 3



HANDLE POSITIONS



TYPICAL SECTION
DOUBLE POLE
TRIPLE BREAK

ROTOR DESIGNATION 4 1/

1/ WITH ROTOR CONFIGURATION 4, CONTACT H-1 BREAKS BEFORE C-3 MAKES, AND CONTACTS A-1 AND B-2 MAKE BEFORE A-2 AND B-3 BREAK

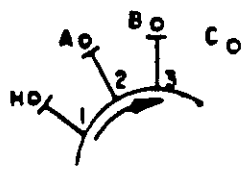


FIGURE 107-26. Switch, rotary, multipole and selector, 10 amperes (MIL-S-21604/5) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

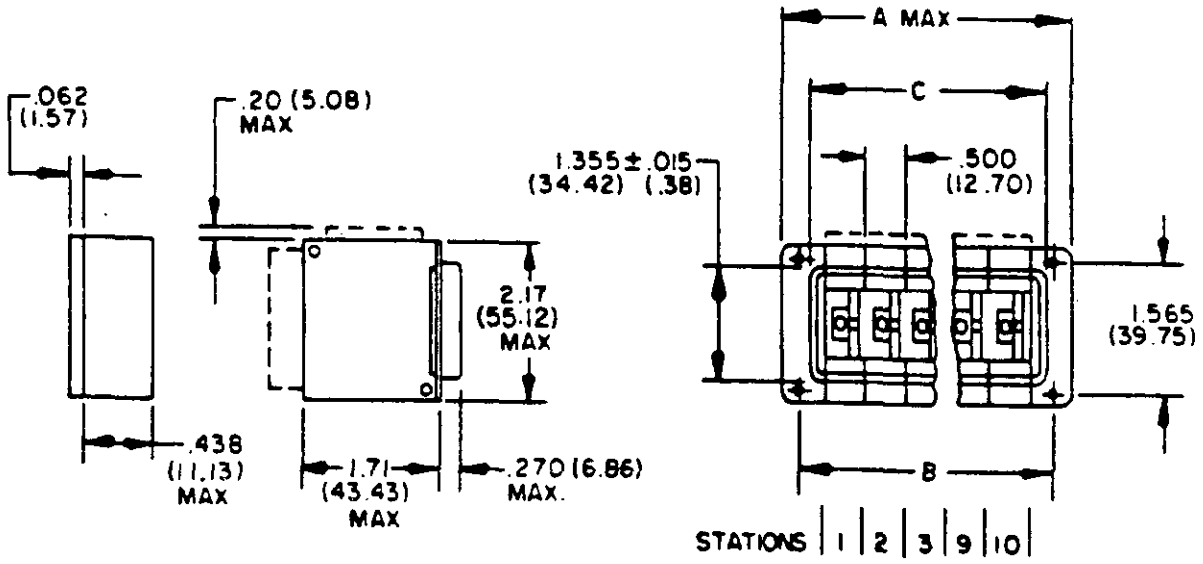
SECTION 109

SWITCHES, ROTARY, PRINTED CIRCUIT (THUMBWHEEL AND PUSHBUTTON)
(Applicable specification: MIL-S-22710)

Figure number	Switch characteristics					Page
	Applicable specification sheet	Operating force		Shock	Vibration	
		Min	Max			
		(oz)	(oz)			
109-1	MIL-S-22710/11	4	22	Medium and high	15G 10 to 2,000 Hz	109.2
109-2	MIL-S-22710/12	4	22	Medium and high	15G 10 to 2,000 Hz	109.6
109-3	MIL-S-22710/14	4	22	Medium	15G 10 to 2,000 Hz	109.10
109-4	MIL-S-22710/15	6	20	Medium and high	15G 10 to 2,000 Hz	109.14
109-5	MIL-S-22710/18	20	40	Medium	15G 10 to 2,000 Hz	109.20
109-6	MIL-S-22710/20	6	22	Medium and high	15G 10 to 2,000 Hz	109.26
109-7	MIL-S-22710/22	20	40	Medium	15G 10 to 2,000 Hz	109.32
109-8	MIL-S-22710/24	15	30	Medium and high	15G 10 to 2,000 Hz	109.37
109-9	MIL-S-22710/17	30	60	Medium and high	15G 10 to 2,000 Hz	109.39

Supersedes page 109.1
of 23 March 1976

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



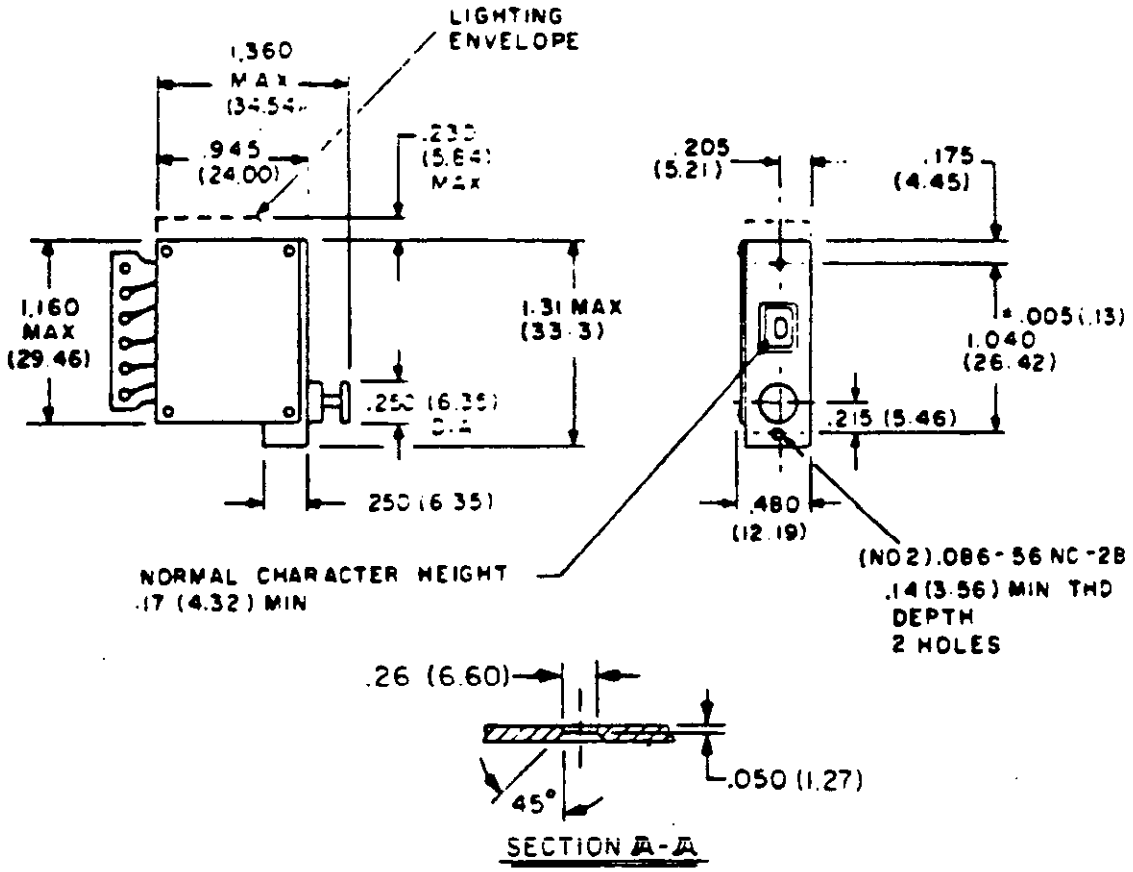
Number of stations	Dimensions			Number of stations	Dimensions		
	A (max)	B	C		A (max)	B	C
1	1.50 (38.10)	1.075 (27.31)	0.870 (22.10)	6	4.00 (101.60)	3.575 (90.81)	3.370 (85.60)
2	2.00 (50.80)	1.575 (40.01)	1.370 (34.80)	7	4.50 (114.30)	4.075 (103.51)	3.870 (98.30)
3	2.50 (63.50)	2.075 (52.71)	1.870 (47.50)	8	5.00 (127.00)	4.575 (116.21)	4.370 (111.00)
4	3.00 (76.20)	2.575 (65.41)	2.370 (60.20)	9	5.50 (139.70)	5.075 (128.91)	4.870 (123.70)
5	3.50 (88.90)	3.075 (78.11)	2.870 (72.90)	10	6.00 (152.40)	5.575 (141.61)	5.370 (136.40)

Number of stations										Part No. M22710/11- Module designation													
1	2	3	4	5	6	7	8	9	10	A	A1	A2	B	B1	B2	C	C1	C2	D	D1	D2	E	E1
x										001	011	021	031	041	051	061	071	081	091	101	111	121	131
x	x									002	012	022	032	042	052	062	072	082	092	102	112	122	132
x	x	x								003	013	023	033	043	053	063	073	083	093	103	113	123	133
x	x	x	x							004	014	024	034	044	054	064	074	084	094	104	114	124	134
x	x	x	x	x						005	015	025	035	045	055	065	075	085	095	105	115	125	135
x	x	x	x	x	x					006	016	026	036	046	056	066	076	086	096	106	116	126	136
x	x	x	x	x	x	x				007	017	027	037	047	057	067	077	087	097	107	117	127	137
x	x	x	x	x	x	x	x			008	018	028	038	048	058	068	078	088	098	108	118	128	138
x	x	x	x	x	x	x	x	x		009	019	029	039	049	059	069	079	089	099	109	119	129	139
x	x	x	x	x	x	x	x	x	x	010	020	030	040	050	060	070	080	090	100	110	120	130	140

Number of stations										Module designation													
1	2	3	4	5	6	7	8	9	10	E2	F	F1	F2	G	G1	G2	H	H1	H2	J	J1	J2	
x										141	151	161	171	181	191	201	211	221	231	241	251	261	
x	x									142	152	162	172	182	192	202	212	222	232	242	252	262	
x	x	x								143	153	163	173	183	193	203	213	223	233	243	253	263	
x	x	x	x							144	154	164	174	184	194	204	214	224	234	244	254	264	
x	x	x	x	x						145	155	165	175	185	195	205	215	225	235	245	255	265	
x	x	x	x	x	x					146	156	166	176	186	196	206	216	226	236	246	256	266	
x	x	x	x	x	x	x				147	157	167	177	187	197	207	217	227	237	247	257	267	
x	x	x	x	x	x	x	x			148	158	168	178	188	198	208	218	228	238	248	258	268	
x	x	x	x	x	x	x	x	x		149	159	169	179	189	199	209	219	229	239	249	259	269	
x	x	x	x	x	x	x	x	x	x	150	160	170	180	190	200	210	220	230	240	250	260	270	

FIGURE 109-1. Switches, rotary (printed circuit) (thumbwheel switch assembly, integral bezel, nonsealed, illuminated or nonilluminated), style SRPC11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



PANEL CUTOUT DETAIL WITH SWITCH INSTALLED

Number of modules	A REF	B	Number of modules	A REF	B
1	.50(12.7)		6	3.00(76.2)	2.500(63.50)
2	1.00(25.4)	.500(12.70)	7	3.50(88.9)	3.000(76.20)
3	1.50(38.1)	1.000(25.40)	8	4.00(101.6)	3.500(88.90)
4	2.00(50.8)	1.500(38.10)	9	4.50(114.3)	4.000(101.60
5	2.50(63.5)	2.000(50.80	10	5.00(127.0)	4.500(114.30

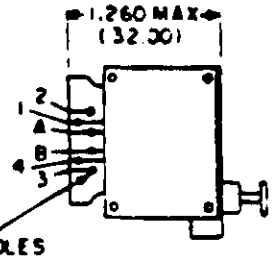
FIGURE 109-9. Switches, rotary (printed circuit) (pushbutton, sealed, illuminated or nonilluminated), style SRPC17 (MIL-S-22710/17).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

DIAL READS ⚠	COMMON 'A' CONNECTED TO		COMMON 'B' CONNECTED TO	
	1	2	3	4
0	●		●	
1		●		●

⚠ REPEATING DIAL

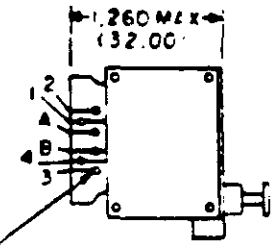
NO. 1



DIAL READS ⚠	COMMON 'A' CONNECTED TO		COMMON 'B' CONNECTED TO	
	1	2	3	4
-	●		●	
+		●		●

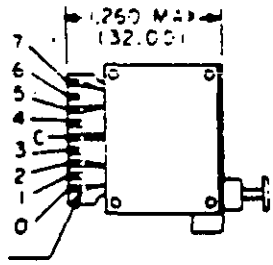
⚠ REPEATING DIAL

NO 2



DIAL READS	DIAL POS	COMMON 'C' CONNECTED TO
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7

NO. 3



DIAL READS	DIAL POS	COMMON 'C' CONNECTED TO
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

NO. 4

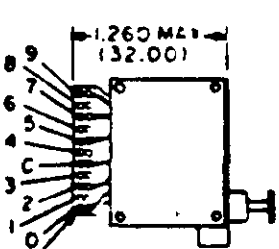


FIGURE 109-9. Switches, rotary (printed circuit) (pushbutton, sealed, illuminated or nonilluminated), style SRPC 17 (MIL-S-22710/17) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

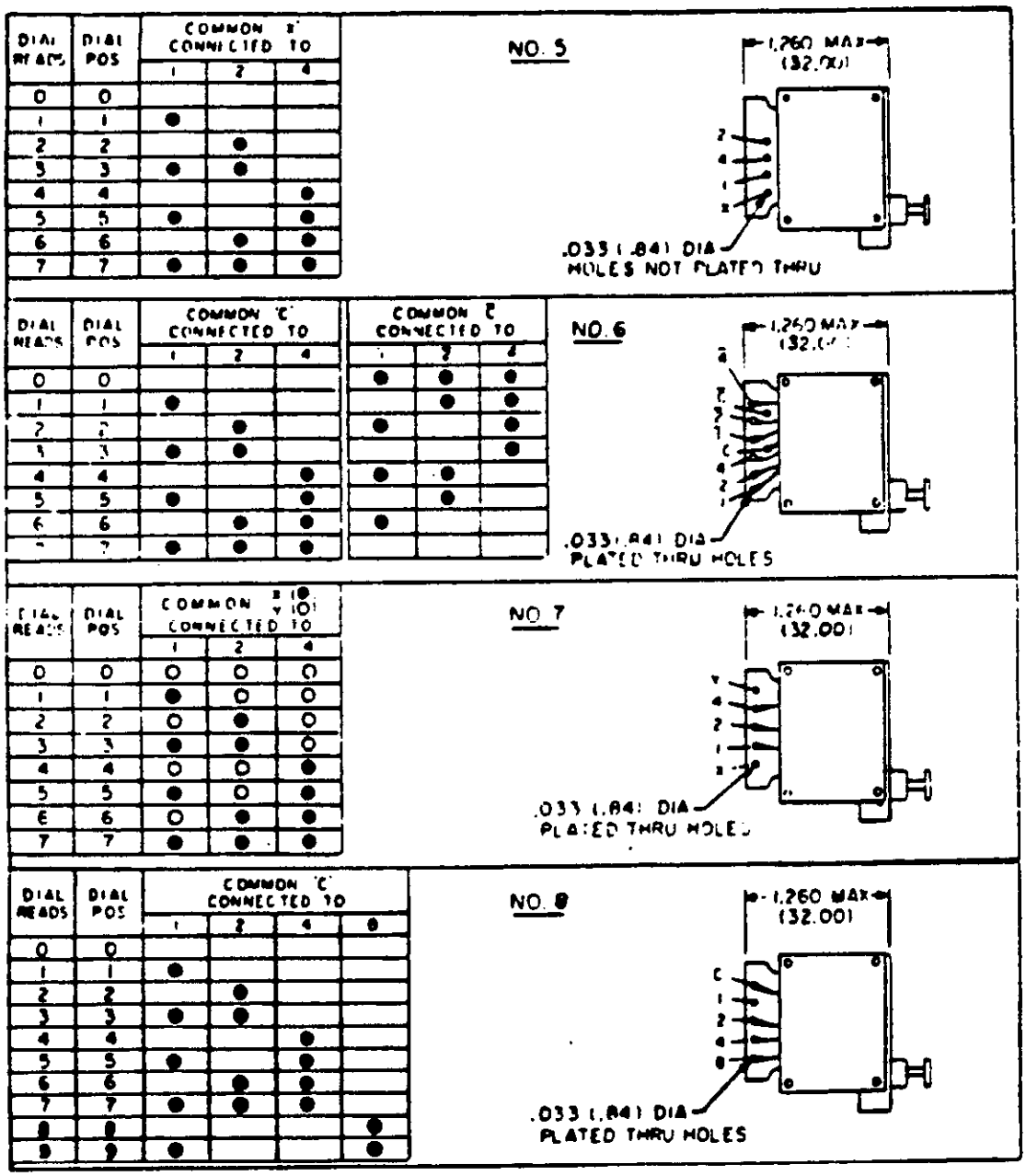
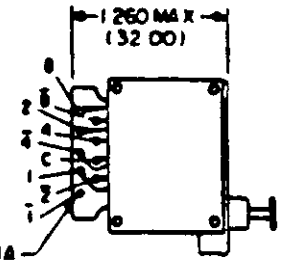


FIGURE 109-9. Switches, rotary (printed circuit) (pushbutton, sealed, illuminated or nonilluminated), style SRPC 17 (MIL-S-22710/17) - Continued.

Source: https://assist.dla.mil -- Downloaded: 20260-26T07:41Z
Check the source to verify that this is the current version before use.

DIAL READS	DIAL POS	COMMON 'C' CONNECTED TO							
		1	2	4	8	$\bar{1}$	$\bar{2}$	$\bar{4}$	$\bar{8}$
0	0					●	●	●	●
1	1	●					●	●	●
2	2		●			●		●	●
3	3	●	●					●	●
4	4			●		●	●		●
5	5	●		●			●		●
6	6		●	●		●			●
7	7	●	●	●					●
8	8				●	●	●	●	
9	9	●			●		●	●	

NO 9



DIAL READS	DIAL POS	COMMON [illegible] CONNECTED TO			
		1	2	4	8
0	0	○	○	○	○
1	1	●	○	○	○
2	2	○	●	○	○
3	3	●	●	○	○
4	4	○	○	●	○
5	5	●	○	●	○
6	6	○	●	●	○
7	7	●	●	●	○
8	8	○	○	○	●
9	9	●	○	○	●

NO 10

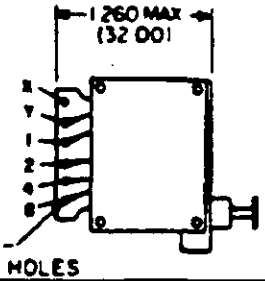


FIGURE 109-9. Switches, rotary (printed circuit) (pushbutton, sealed, illuminated or nonilluminated), style SRPC 17 (MIL-S-22710/17) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Part number	Terminal design and circuit configuration	Seal method 1/	Illuminated	Sand and dust
M22710/17-001	1	II	None	Not applicable
M22710/17-002	"	II	Red	Not applicable
M22710/17-003	"	II	White	Not applicable
M22710/17-004	"	I and II	None	Applicable
M22710/17-005	"	I and II	Red	Applicable
M22710/17-006	"	I and II	White	Applicable
M22710/17-007	2	II	None	Not applicable
M22710/17-008	"	II	Red	Not applicable
M22710/17-009	"	II	White	Not applicable
M22710/17-010	"	I and II	None	Applicable
M22710/17-011	"	I and II	Red	Applicable
M22710/17-012	"	I and II	White	Applicable
M22710/17-013	3	II	None	Not applicable
M22710/17-014	"	II	Red	Not applicable
M22710/17-015	"	II	White	Not applicable
M22710/17-016	"	I and II	None	Applicable
M22710/17-017	"	I and II	Red	Applicable
M22710/17-018	"	I and II	White	Applicable
M22710/17-019	4	II	None	Not applicable
M22710/17-020	"	II	Red	Not applicable
M22710/17-021	"	II	White	Not applicable
M22710/17-022	"	I and II	None	Applicable
M22710/17-023	"	I and II	Red	Applicable
M22710/17-024	"	I and II	White	Applicable
M22710/17-025	5	II	None	Not applicable
M22710/17-026	"	II	Red	Not applicable
M22710/17-027	"	II	White	Not applicable
M22710/17-028	"	I and II	None	Applicable
M22710/17-029	"	I and II	Red	Applicable
M22710/17-030	"	I and II	White	Applicable
M22710/17-031	6	II	None	Not applicable
M22710/17-032	"	II	Red	Not applicable
M22710/17-033	"	II	White	Not applicable
M22710/17-034	"	I and II	None	Applicable
M22710/17-035	"	I and II	Red	Applicable
M22710/17-036	"	I and II	White	Applicable
M22710/17-037	7	II	None	Not applicable
M22710/17-038	"	II	Red	Not applicable
M22710/17-039	"	II	White	Not applicable
M22710/17-040	"	I and II	None	Applicable
M22710/17-041	"	I and II	Red	Applicable
M22710/17-042	"	I and II	White	Applicable
M22710/17-043	8	II	None	Not applicable
M22710/17-044	"	II	Red	Not applicable
M22710/17-045	"	II	White	Not applicable
M22710/17-046	"	I and II	None	Applicable
M22710/17-047	"	I and II	Red	Applicable
M22710/17-048	"	I and II	White	Applicable
M22710/17-049	9	II	None	Not applicable
M22710/17-050	9	II	Red	Not applicable

FIGURE 109-9. Switches, rotary (printed circuit) (pushbutton, sealed, illuminated or nonilluminated, style SRPC 17 (MIL-S-22710/17) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Part number	Terminal design and circuit configuration	Seal method 1/	Illuminated	Sand and dust
M22710/17-051	9	II	White	Not applicable
M22710/17-052	"	I and II	None	Applicable
M22710/17-053	"	I and II	Red	Applicable
M22710/17-054	"	I and II	White	Applicable
M22710/17-055	10	II	None	Not applicable
M22710/17-056	"	II	Red	Not applicable
M22710/17-057	"	II	White	Not applicable
M22710/17-058	"	I and II	None	Applicable
M22710/17-059	"	I and II	Red	Applicable
M22710/17-060	"	I and II	White	Applicable

1/ Seal method II provides a sealed panel.
Seal method I and II provides both a sealed panel and sealed switch.

FIGURE 109-9. Switches, rotary (printed circuit) (pushbutton, sealed, illuminated or nonilluminated), style SRPC 17 (MIL-S-22710/17) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

SECTION 110

SWITCHES, SENSITIVE

(Applicable specification: MIL-S-8805)

TABLE 110.I. Switches, sensitive, characteristics.

Figure number	Applicable specification sheet MIL-S-8805	Operation	Circuit	Electrical ratings (28 Vdc resistive) (amperes)	General remarks
110-1	/1	Momentary	SPDT	2	
110-2	/2	Momentary	SPDT	5	
110-3	/4	Momentary	SPDT	7	Low level ckt.
110-4	/5	Latching - reset button	4 circuit	15	
110-5	/7	Momentary	2 circuit	10	
110-6	/8	Momentary	SPDT	4	Hermetic seal
110-7	/11	Momentary	2 circuit	10	
110-8	/14	Latching - reset button	2 circuit	10	
110-9	/15	Momentary	4 circuit	15	
110-10	/16	Latching - reset button	4 circuit	15	
110-11	/17	Latching - reset button	6 circuit	15	
110-12	/18	Momentary	6 circuit	15	
110-13	/19	Momentary	2 circuit	40	
110-14	/27	Momentary	SPST, SPDT	10	
110-15	/28	Momentary	SPDT	7.5	
110-16	/29	Momentary	SPDT double break	10	
110-17	/31	Momentary	DPDT	10	
110-18	/32	Momentary	SPDT	7	Watertight seal
110-19	/34	Latching - reset button	2 circuit	40	
110-20	/39	Momentary	2PDT 4PDT	7	Plunger - resilient seal
110-21	/40	Momentary	2PDT 2 circuit 4 circuit	4,10 15 15	Plunger - resilient seal
110-22	/41	Momentary	2PDT	5	Plunger - hermetic seal
110-23	/42	Momentary	SPDT	6	Plunger - watertight seal
110-24	/43	Momentary	2PDT 4PDT	7	Pin and roller plunger - resilient seal
110-25	/44	Momentary	SPDT	5	Resilient seal
110-26	/45	Momentary	SPDT	5	Resilient seal
110-27	/47	Momentary	SPDT	25	Hermetic seal
110-28	/48	Roller lever - self return	2PDT 2 circuit 4 circuit	5,10 15 15	Resilient seal
110-29	/49	Rotary linkage lever (nonself-return)	2PDT 2 circuit 4 circuit	4,10 15	Resilient seal
110-30	/65	Momentary	2PDT	5	Plunger - resilient seal
110-31	/68	Momentary	SPDT	7	Hermetic seal low level ckt.

Supersedes page 110.1
of 23 March 1976

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE 110.1. Switches, sensitive, characteristics.

Figure number	Applicable specification sheet MIL-S-8805	Operation	Circuit	Electrical ratings (28 Vdc resistive) (amperes)	General remarks
110-32	/70	Momentary	2PDT 4PDT	7	Pin and roller plunger-hermetic seal
110-33	/72	Momentary	SPDT	7	Plunger-hermetic seal 1/
110-34	/73	Momentary	SPDT	7	Plunger-hermetic seal
110-35	/74	Momentary	SPDT	7	Plunger-hermetic seal
110-36	/76	Momentary	2 circuit	5	
110-37	/77	Momentary	4 circuit	10	
110-38	/78	Momentary	6 circuit	10	
110-39	/80	Momentary	2PDT	5	Plunger-hermetic seal
110-40	/84	Momentary	2 circuit	10	Watertight seal
110-41	/85	Rotary - nonself-return	2PDT 4PDT	4	Resilient seal
110-42	/90	Momentary	SPDT	10	
110-43	/101	Momentary	2 circuit	5	Low level ckt.
110-44	/100	Momentary	2PDT 4PDT	10 7	Pin and roller plunger-resilient seal
110-45	/104	Momentary	4PDT	7	Pin and roller Plunger-resilient seal 2/
110-46	/46	Momentary	SPDT	2,4	Hermetic seal - low level ckt.
110-47	/82	Momentary	SPDT	2,4	Hermetic seal - low level ckt.
110-48	/106	Momentary	SPDT	2	High temperature - low level ckt.

1/ Minimum current applicable.

2/ Low level circuit and icing applicable.

Supersedes page 110.2
of 20 June 1977

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.

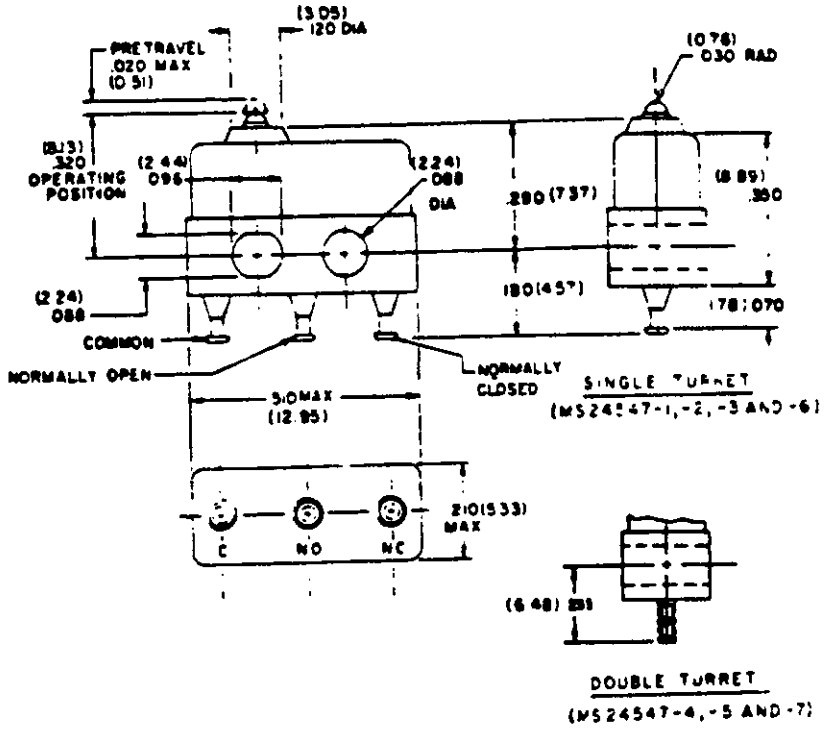


FIGURE 110-3. Switches, sensitive, SPDT, unsealed.

Procurement document: MIL-S-8805/4.

Part numbers and electrical ratings: See tables 110.IV through 110.VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Operating characteristics:
Actuating force:
MS24547-1, -2, -4, and -5: 5 ounces maximum.
MS24547-3: 4 ounces.
Releasing force: 1 ounce minimum.
Movement differential:
MS24547-1, -2, -4, and -5: .005 inch maximum.
MS24547-3: .0015 inch maximum.
Pretravel: .020 inch maximum.
Overtravel: .004 inch minimum.
Differential force: MS24547-3: 2 ounces maximum.
Low level circuit: Applicable to MS24547-6 and -7.

TABLE 110.IV. Silver contacts (MS24547-1 and -4).

Load	Sea level		50,000 feet
	28 Vdc	115 vac, 60 Hz	28 Vdc
	(amperes)	(amperes)	(amperes)
Resistive - - -	7	7	4
Inductive - - -	4	7	2.5
Lamp- - - - - -	2.5	2	2.5

TABLE 110.V. Gold contacts, solid gold, gold filled, or gold diffused (MS24547-2, -5, -6, and -7.

Load	Sea level 28 Vdc	50,000 feet 28 Vdc
	(amperes)	(amperes)
Resistive - - - - - -	1	1
Inductive - - - - - -	.5	.5

TABLE 110.VI. Fine silver (gold plated) (MS24547-3).

Load	Sea level	
	28 Vdc	115 vac, 60 Hz
	(amperes)	(amperes)
Resistive - - - - - -	5	5
Inductive - - - - - -	3	5
Lamp- - - - - - - - -	2.5	2

Supersedes page 110.10
of 23 March 1976

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

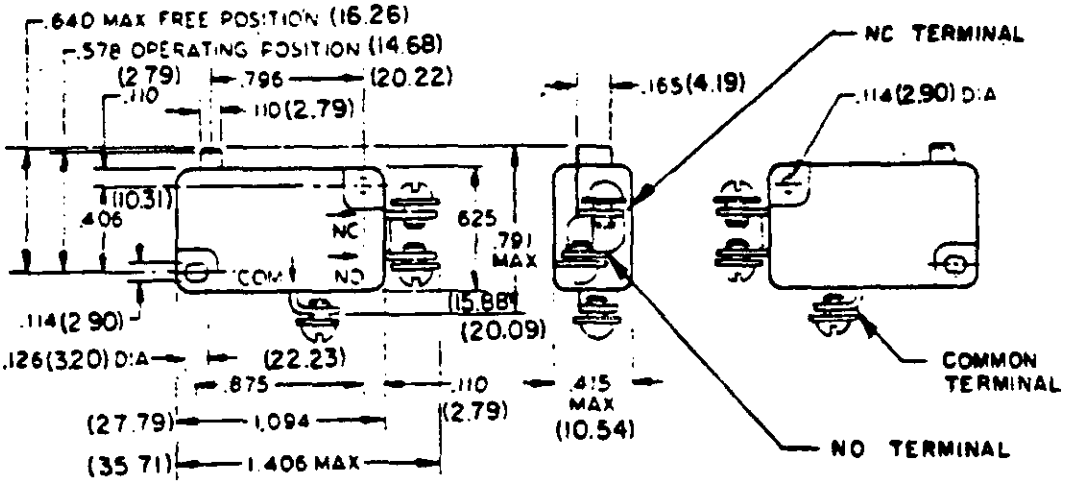


FIGURE 110-42. Switch, sensitive, SPDT, 10 amperes, unsealed.

Procurement document: MIL-S-8805/90.

Part numbers: See table 110.XXV.

Electrical ratings.

Load	Sea level		50,000 feet	
	28 Vdc	115 Vac, 60 Hz	28 Vdc	115 Vac, 60 Hz
	amperes	amperes	amperes	amperes
Resistive	10	10	10	10
Inductive	6	10	6	10
Motor	6	3	---	---

Operating characteristics:

Actuating force:	75 grams maximum.
Pretravel:	0.050 inch maximum.
Movement differential:	0.010 inch maximum.
Overtravel:	0.050 inch maximum.

TABLE 110.XXV. Part numbers and characteristics.

Part number M8805/90	Terminals	Switching characteristics
-001	NC, NO & Common	SPDT
-002	NO & Common	SPNO
-003	NC & Common	SPNC

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

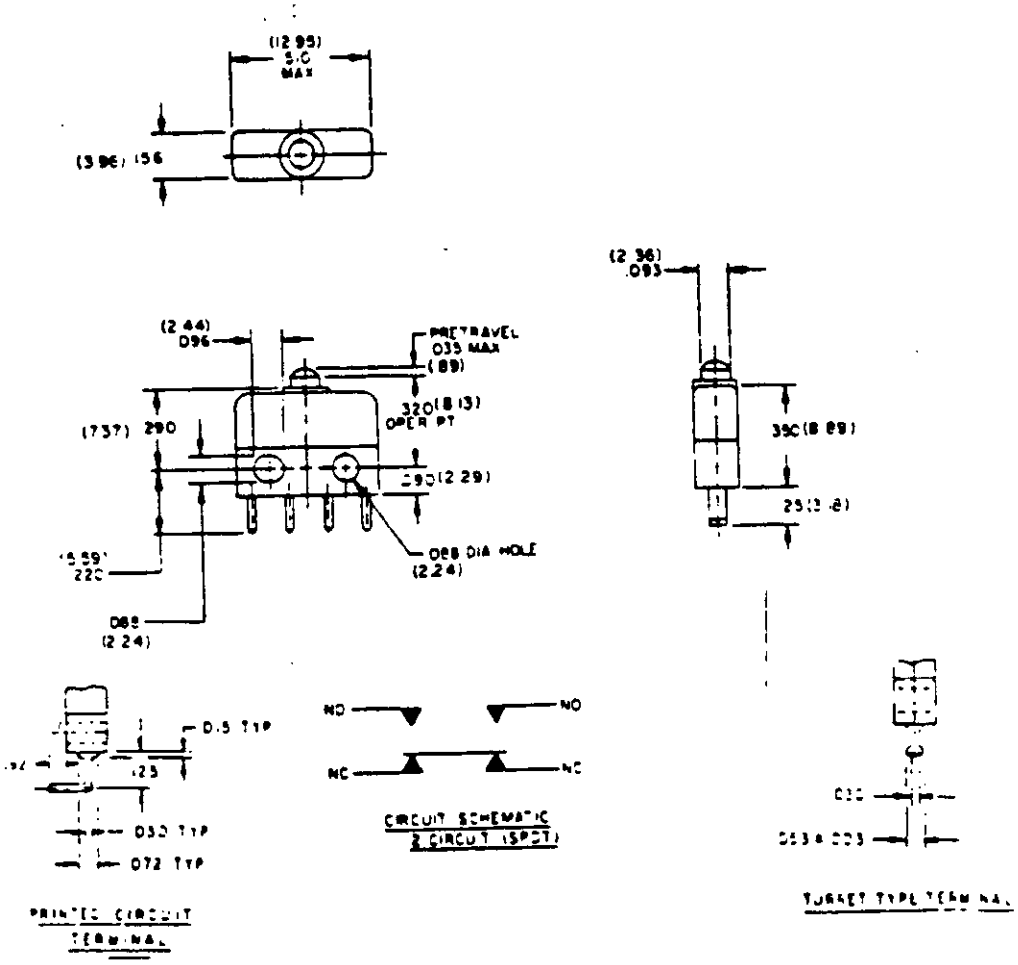


FIGURE 110-43. Switch, sensitive, 2 circuit (5 amperes) or SPDT (8 amperes), unsealed.

Procurement document: MIL-S-8805/101.

Part number: M8805/101 (Dash number from table XXVa).

Supersedes page 110.100
of 23 March 1976

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Electrical ratings - silver contacts.

Load	Sea level, 28 Vdc		70,000 feet, 28 Vdc	
	NO or NC	2 circuit	NO or NC	2 circuit
	amperes	amperes	amperes	amperes
Resistive - - -	8	5	8	5
Inductive - - -	5	3	3	2
Motor - - - - -	5	---	---	---
Lamp- - - - - -	1	---	---	---

Electrical ratings - gold plated contacts.

Load	28 Vdc Sea level	28 Vdc Altitude
	amperes	amperes
Resistive	5	5
Inductive	3	3
Low level circuit requirements apply		

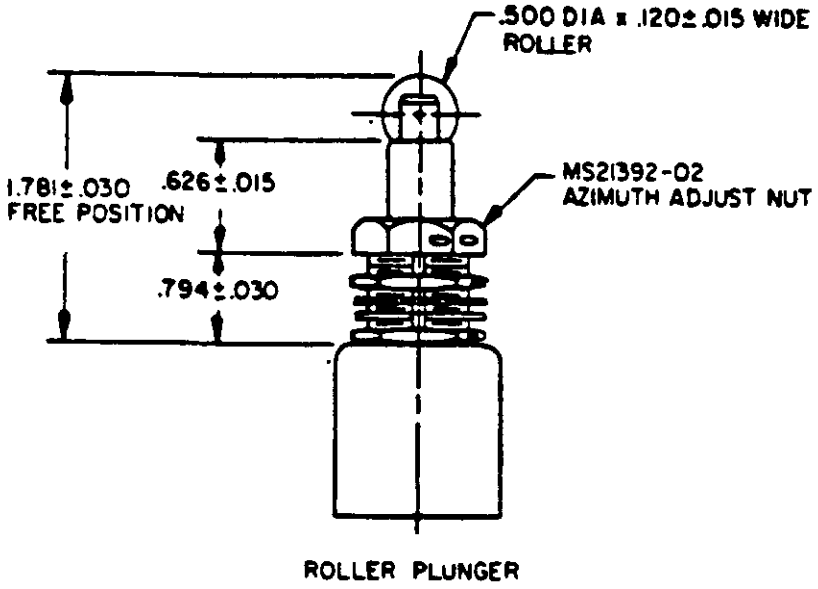
Operating characteristics:
Actuating force: 8 ounces maximum.
Pretravel: .035 inch maximum.
Overtravel: .004 inch minimum.
Movement differential: .010 inch minimum.

TABLE 110.XXVa. Dash numbers.

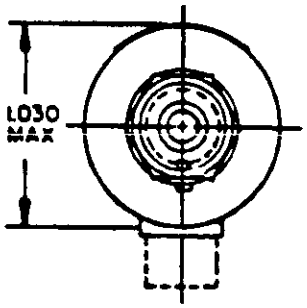
Contacts		Terminal	Dielectric withstanding voltage	
Silver	Gold plated		Sea level	Altitude
007	008	Spade	1050 Vrms	400 Vrms at 70,000 ft
009	010	Turret		
011	012	Printed circuit		

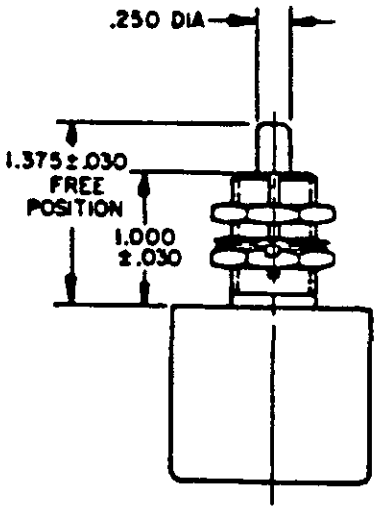
Supersedes page 110.101
of 23 March 1976

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



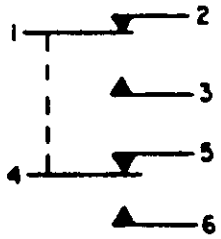
ROLLER PLUNGER



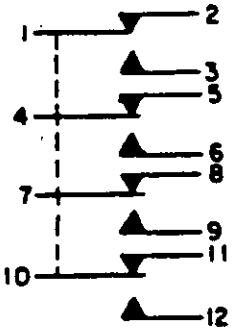


PIN PLUNGER

INCHES	MM
.015	.38
.030	.76
.120	3.05
.250	6.35
.500	12.70
.626	15.90
.794	20.17
1.030	26.16
1.375	34.92
1.781	45.24



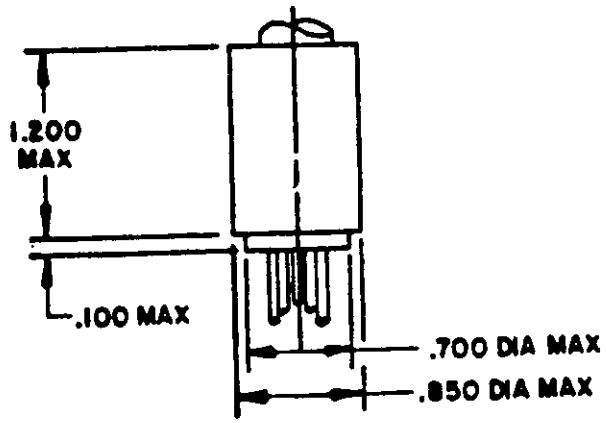
2PDT

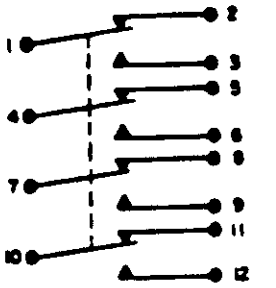


4PDT

FIGURE 110-44. Switches, sensitive, plunger, resilient seal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.





CIRCUIT SCHEMATIC

INCHES	MM
.100	2.54
.700	17.78
.850	21.59
1.200	30.48

M8805/104-003,004
AXIAL LEAD UNITS

WIRE LEADS: M22710/7-20, 72^{+2}_{-0} inches long.

FIGURE 110-45. Switches, sensitive, plunger, resilient seal - Continued.

Procurement document: MIL-S-8805/104.

Operating characteristics:

Actuating force:	9 +3 pounds.
Full overtravel force:	30 pounds maximum.
Releasing force:	5 pounds minimum.
Pretravel:	.050 inch maximum.
Movement differential:	.030 inch maximum.
Overtravel:	.125 inch minimum.

Electrical ratings. 1/

Load	28 Vdc		115 Vac, 400 Hz	
	Sea level	80,000 ft.	Sea level	80,000 ft.
	Amperes	Amperes	Amperes	Amperes
Resistive	7	7	1	1
Inductive	3	3	1	1
Motor	4	4	1	1
Lamp	2	1	1	1

1/ Low level circuit applicable.

TABLE 110.XXVII. Part numbers.

Part number M8805/104	Actuator	Wire exit
-001	Pin plunger	Side
-002	Roller plunger	
-003	Pin plunger	Bottom
-004	Roller plunger	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

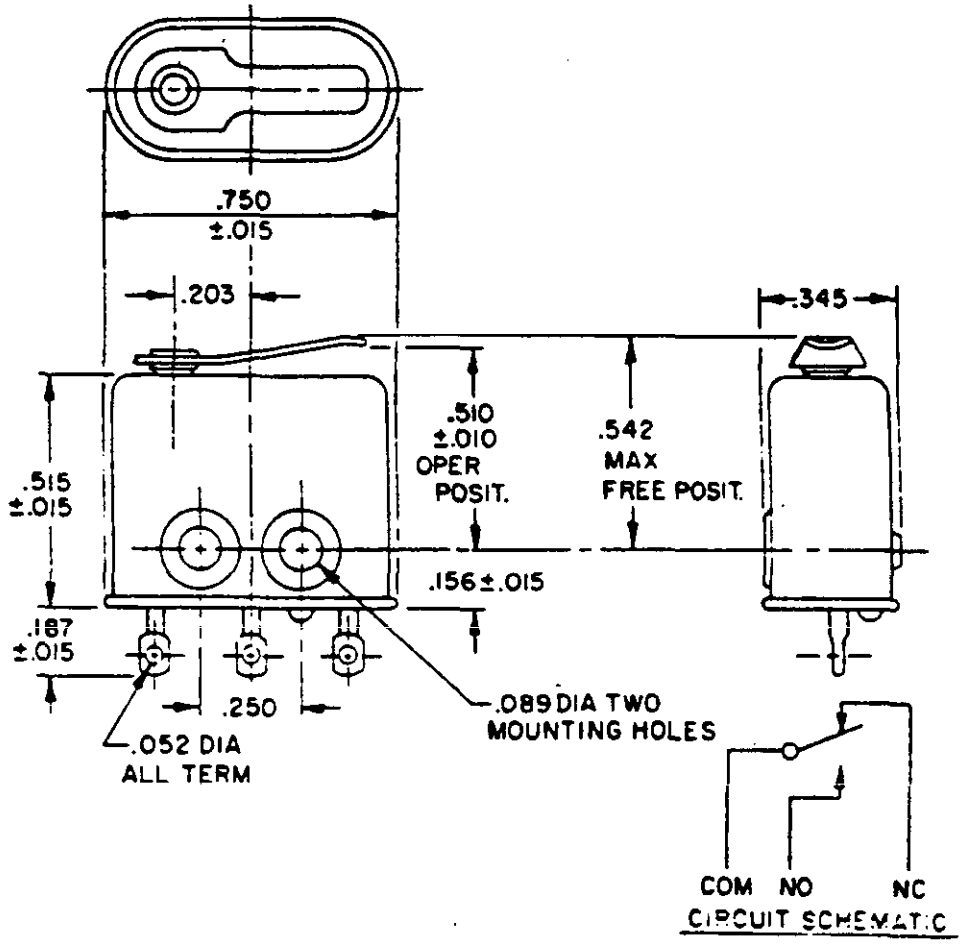


FIGURE 110-46. Switches, sensitive, leaf, SPDT, hermetic (MIL-S-8805/46).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Electrical ratings: See table 110.XXVIII.

Military part number: See table 110.XXIX.

TABLE 110.XXVII. Part numbers and operating characteristics.

Part number	Actuating force	Release force	Differential travel	Over travel	Maximum weight	Termination
M8805/46-001	26 oz max	6 oz min	.020 inch max	.006 inch min	.020 lb	Solder
M8805/46-002	26 oz max	6 oz min	.020 inch max	.006 inch min	.020 lb	Solder
M8805/46-003	32 oz max	6 oz min	.020 inch max	.006 inch min	.020 lb	Solder

TABLE 110.XXIX. Electrical ratings.

Part number	Load	Sea level and 70,000 ft 115 Vac, 400 Hz and 28 Vdc
		Amperes
M8805/46-001 and M8805/46-002	Resistive	5
	Inductive	3
	Lamp	1.5
M8805/46-003	Resistive	2
	Inductive	.5
	Lamp	.5
	Low level circuit test applicable	

FIGURE 110.46. Switches, sensitive, leaf, SPDT, hermetic (MIL-S-8805/46) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

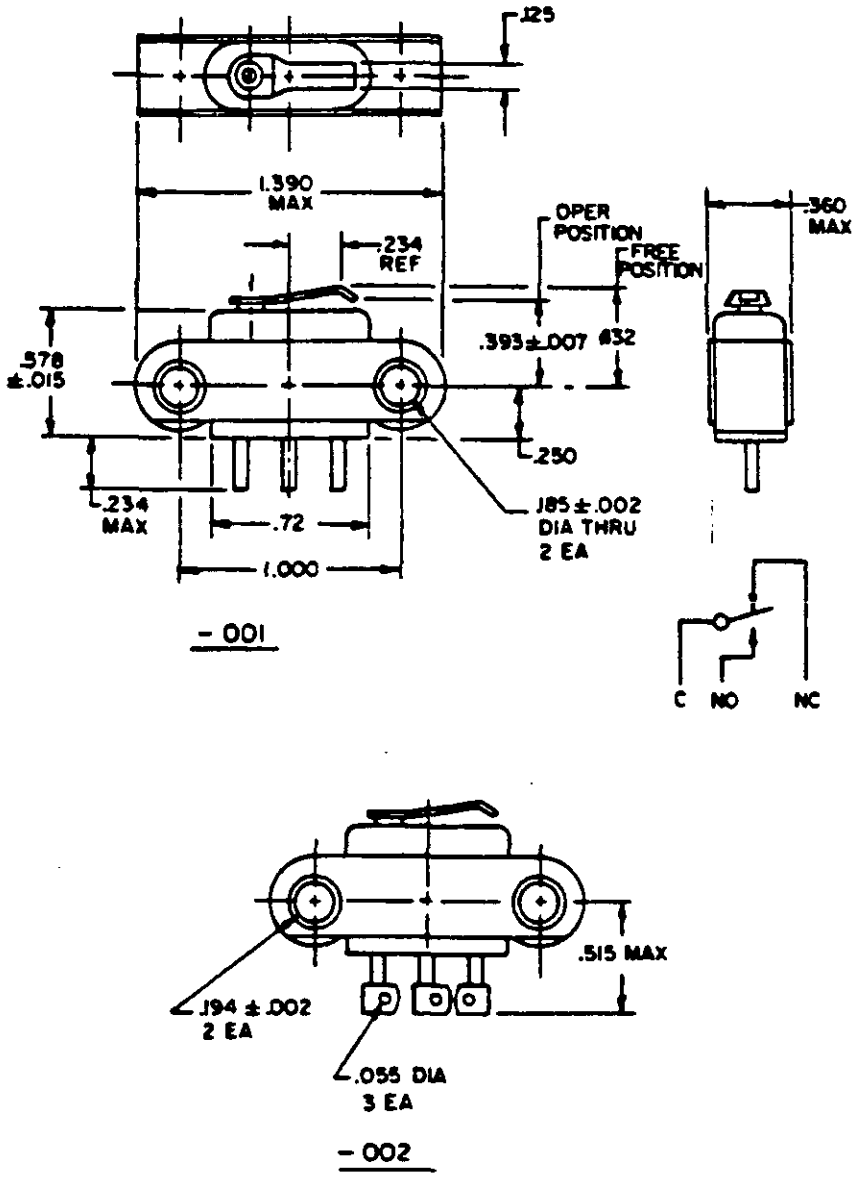
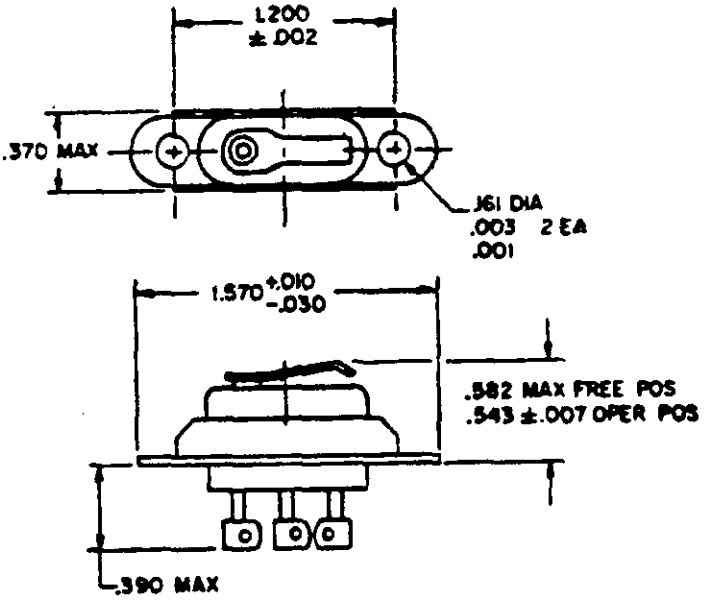


FIGURE 110-47. Switches, sensitive, leaf, SPDT, hermetic (MIL-S-8805/82).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



-003

Electrical ratings: See table 110.XXX.

Military part number: See table 110.XXXI.

TABLE 110.XXX. Part numbers and operating characteristics.

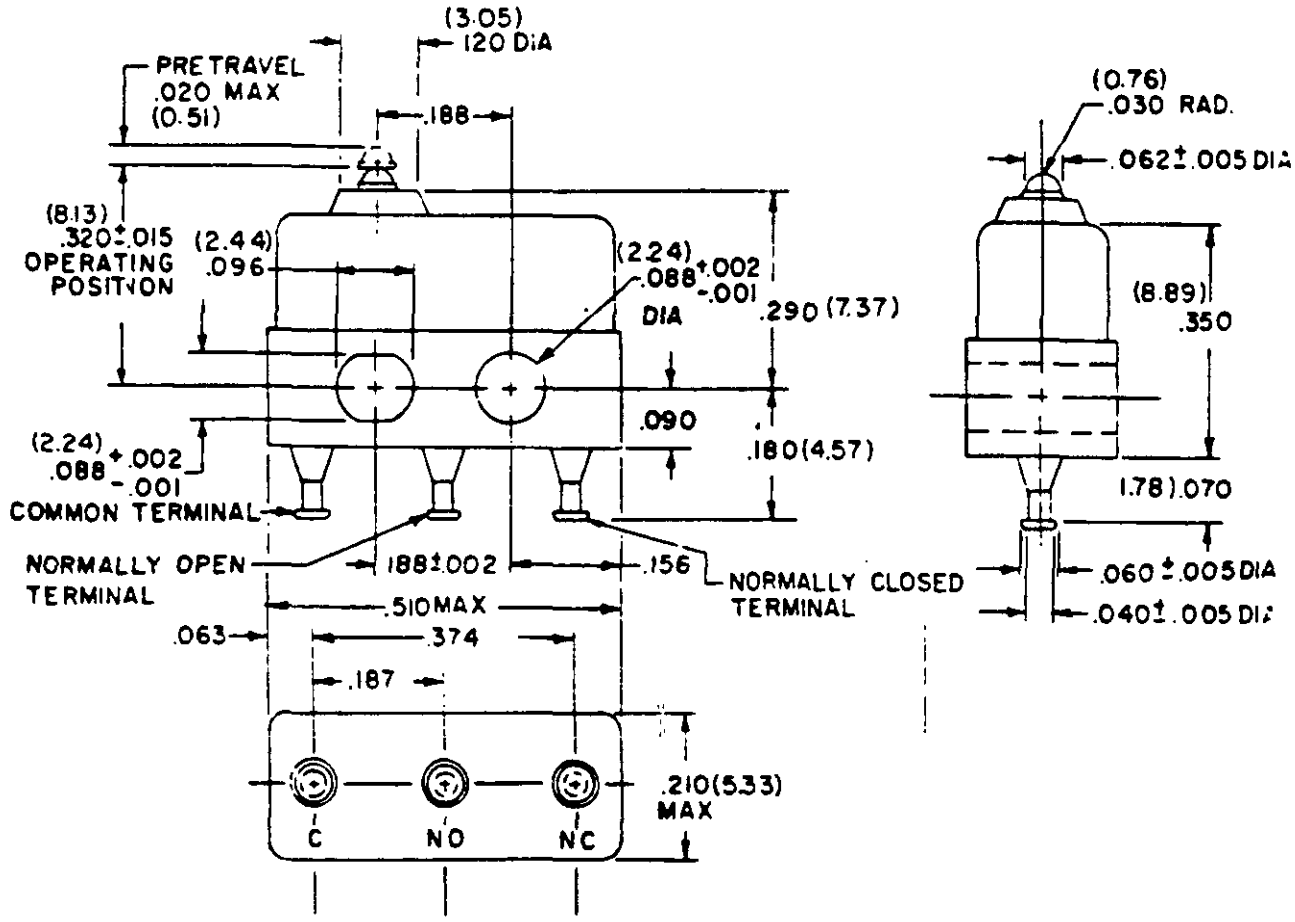
Part number	Actuating force	Release force	Differential travel	Over travel	Maximum weight	Termination
M8805/82-001	26 oz max	6 oz min	.020 inch max	.006 inch min	.025 lb	Braze or weld
M8805/82-002	32 oz max	6 oz min	.020 inch max	.006 inch min	.025 lb	Solder
M8805/82-003	32 oz max	6 oz min	.020 inch max	.006 inch min	.025 lb	Solder

TABLE 110.XXXI. Electrical ratings.

Part number	Load	Sea level and 70,000 ft 115 Vac, 400 Hz and 28 Vdc
		Amperes
M8805/82-001	Resistive	4
	Inductive	3
	Lamp	1.5
M8805/82-002 and M8805/82-003	Resistive	2
	Inductive	.5
	Lamp	.5
	Low level circuit test applicable	

FIGURE 110-47. Switches, sensitive, leaf, SPDT, hermetic (MIL-S-8805/82) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



Electrical ratings: See table 110.XXXII.

Part number: See table 110.XXXII.

TABLE 110.XXXII. Electrical ratings and part numbers.

Load at 28 Vdc - Sea Level	Part number M8805/106-	
	-01	-02
Resistive	2.0	1.0
Inductive	1.0	0.5
Lamp	0.25	---
Low level circuit requirement applies to -02 only.		

FIGURE 110-48. Switch, sensitive, SPDT, low level to 2 amps, 400°F, unsealed (MIL-S-8805/106).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

SECTION 111A

SWITCHES, THERMOSTATIC

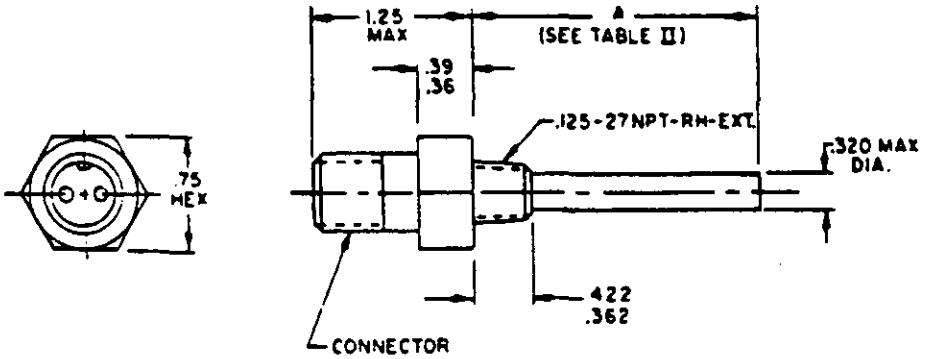
(Applicable specifications: MIL-S-24236 and MIL-S-28827)

Figure no.	Switch characteristics			Page
	Specification sheet number	Circuitry	Electrical rating	
111-1	MIL-S-24236/11	SPST	1 ampere	111.2
111-2	MIL-S-24236/13	SPST	1 ampere	111.4
111-3	MIL-S-24236/19	SPST	1 ampere	111.6
111-4	MIL-S-24236/20	SPST	2 amperes	111.8
111-5	MIL-S-24236/2	SPST	5 amperes	111.10
111-6	MIL-S-24236/1	SPST	5 amperes	111.12
111-7	MIL-S-24236/24	SPDT	2 amperes	111.20
111-8	MIL-S-24236/25	SPST	2 amperes	111.23
111-9	MIL-S-24236/26	SPST-SPDT	2 amperes	111.26
111-10	MIL-S-28827/1	SPDT	1 ampere	111.28
111-11	MIL-S-24236/15	Single terminal	3 amperes	111.30
111-12	MIL-S-24236/16	Single terminal	3 amperes	111.31

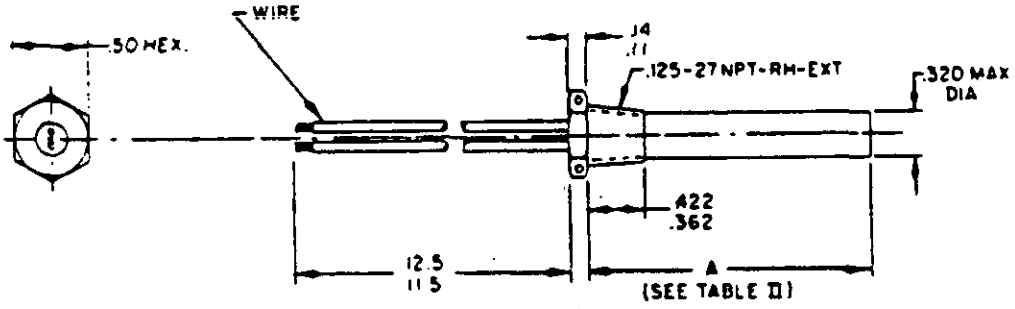
This page supersedes page 111.1
of 20 June 1977

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

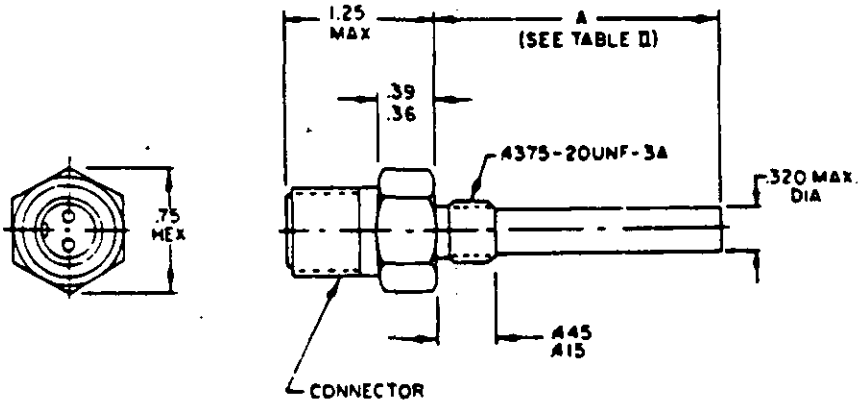
LOAD	ELECTRICAL RATINGS			ENDURANCE
	Sea level		70,000 ft	
	28 V	115 V, 60 Hz	28 V	
	(ampere)	(ampere)	(ampere)	(cycles)
Resistive - - -	1.0	1.0	1.0	10,000
Inductive - - -	.5	.5	.5	10,000
Lamp- - - - - -	.2	.2	.2	10,000



CONFIGURATION 1



CONFIGURATION 2



CONFIGURATION 3

INCHES	MM
.11	2.79
.14	3.56
.320	8.13
.36	9.14
.362	9.19
.39	9.91
.415	10.54
.422	10.72
.445	11.30
.50	12.70
.75	19.05
1.25	31.75
11.5	292.10
12.5	317.50

FIGURE 111-1. Switches, thermostatic (MIL-S-24236/11).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

PART NUMBER: Consists of "M28827/1" followed by a dash and a five-letter code as shown in the following example:

M28827/1- A CD AB

- CONFIGURATION AND SETTING MODE (see table I)
- HIGH TEMPERATURE SETTING (first code, within 20°F; second code, within 2°F (see table II) or DIFFERENTIAL SETTING (first code, low end; second code, high end) (see table III)
- LOW TEMPERATURE SETTING (first code, within 20°F; second code, within 2°F (see table II) or DIFFERENTIAL SETTING (first code, low end; second code, high end) (see table III)

TABLE I. Configuration and setting modes.

	Configuration I	Configuration II	High setting	Low setting
Code - - -	A	J	At temperature with applicable tolerance of table II	At temperature with applicable tolerance of table II
Code - - -	B	K	At temperature with applicable tolerance of table II	By maximum or minimum temperature of table II
Code - - -	C	L	At temperature with applicable tolerance of table II	Differential (see table III)
Code - - -	D	M	By maximum or minimum temperature of table II	At temperature with applicable tolerance of table II
Code - - -	E	N	By maximum or minimum temperature of table II	By maximum or minimum temperature of table II
Code - - -	F	P	By maximum or minimum temperature of table II	Differential (see table III)
Code - - -	G	Q	Differential (see table III)	At temperature with applicable tolerance of table II
Code - - -	H	R	Differential (see table III)	By maximum or minimum temperature of table II

TABLE II. Temperature settings.

TEMPERATURE SETTINGS TO WITHIN 20 °F

Code	Temperature (°F)	Code	Temperature (°F)	Code	Temperature (°F)	Code	Temperature (°F)
A - - -	100	F - - -	200	L - - -	300	R - - -	400
B - - -	120	G - - -	220	M - - -	320	S - - -	420
C - - -	140	H - - -	240	N - - -	340	T - - -	440
D - - -	160	J - - -	260	P - - -	360	U - - -	460
E - - -	180	K - - -	280	Q - - -	380	V - - -	480
						W - - -	500

TEMPERATURE SETTINGS TO WITHIN 2°F

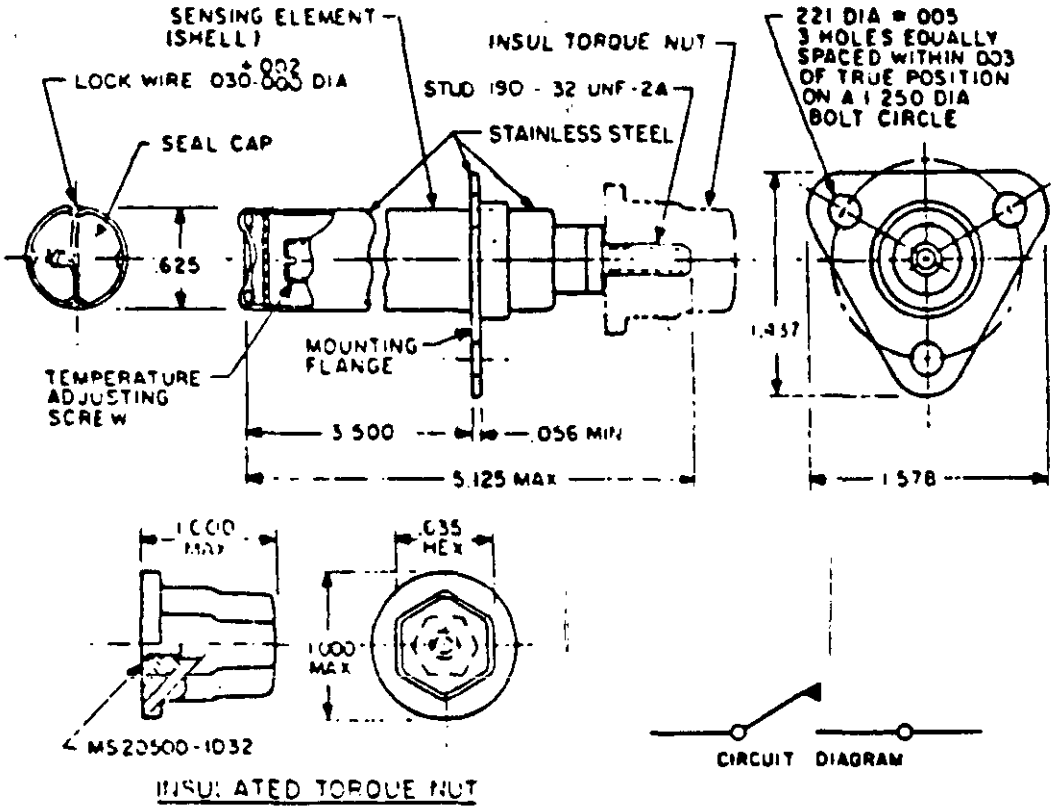
	Maximum, minimum, or ±5°F tolerance	±10°F tolerance	±25°F tolerance	Temperature		Maximum, minimum, or ±5°F tolerance	±10°F tolerance	±25°F tolerance	Temperature
Code - -	A	L	W	0	Code - -	F	R	2	10
Code - -	B	M	X	2	Code - -	G	S	3	12
Code - -	C	N	Y	4	Code - -	H	T	4	14
Code - -	D	P	Z	6	Code - -	J	U	5	16
Code - -	E	Q	1	8	Code - -	K	V	6	18

TABLE III. Differential temperature settings.

Code	Temperature (°F)	Code	Temperature (°F)	Code	Temperature (°F)
A - -	0	E - -	20	J - -	40
B - -	5	F - -	25	K - -	45
C - -	10	G - -	30	L - -	50
D - -	15	H - -	35	M - -	60
				N - -	75

FIGURE 111-10. Switches, thermostatic (MIL-S-28827/1) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



INSULATED TORQUE NUT

Part number: M24236/15-(dash number from table I).

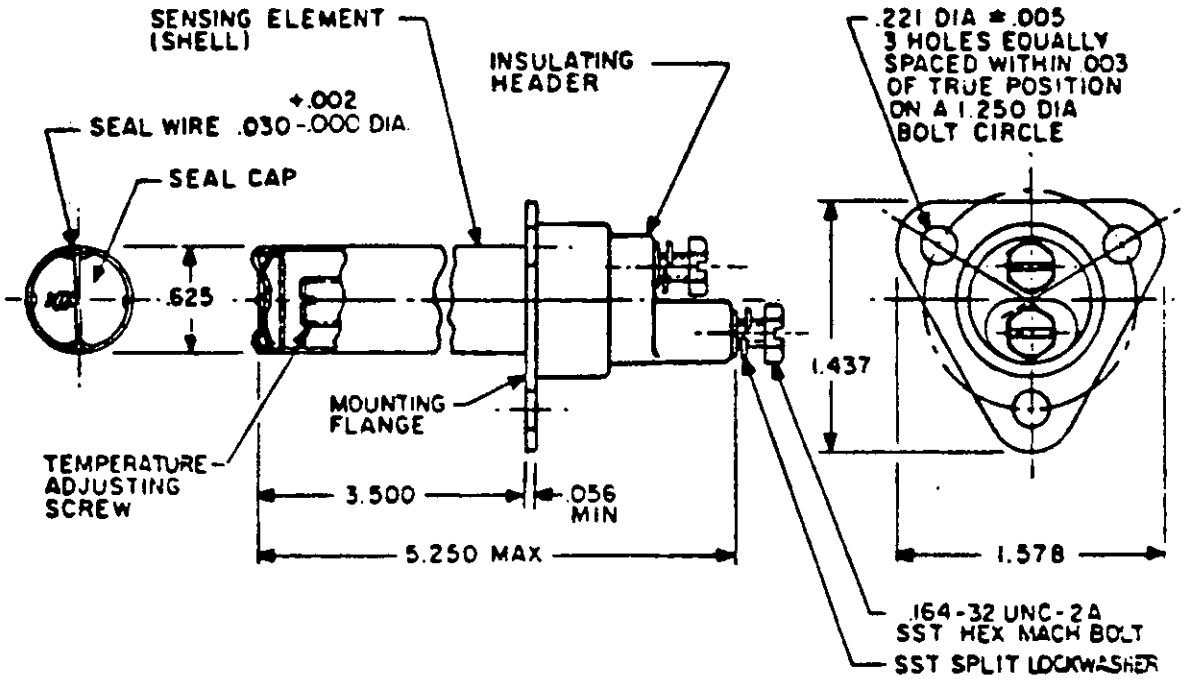
TABLE I. Operating temperature.

Dash number	Temperature setting		Dash number	Temperature setting	
	°F	Tolerance °F		°F	Tolerance °F
135	135	±20	550	550	±20
215	215	"	600	600	"
250	250	"	650	650	"
300	300	"	675	675	"
325	325	"	700	700	"
350	350	"	725	725	"
400	400	"	750	750	"
425	425	"	825	825	±25
450	450	"	900	900	"
500	500	"	925	925	"
525	525	"	950	950	"
535	535	"	1000	1000	"

Electrical ratings: 3 amperes (lamp) at 28 Vdc.

FIGURE 111-11. Switches, thermostatic (MIL-S-24236/15).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



Part number: M24236/16-(dash number from table I).

TABLE I. Operating temperature.

Dash number	Temperature setting		Dash number	Temperature setting	
	°F	Tolerance °F		°F	Tolerance °F
135	135	±20	535	535	±20
180	180	"	550	550	"
215	215	"	600	600	"
240	240	"	650	650	"
250	250	"	675	675	"
300	300	"	700	700	"
325	325	"	725	725	"
350	350	"	750	750	"
400	400	"	825	825	±25
425	425	"	900	900	"
450	450	"	925	925	"
500	500	"	950	950	"
525	525	"	1000	1000	"

Electrical ratings: 3 amperes (lamp) at 28 Vdc.

FIGURE 111-12. Switches, thermostatic (MIL-S-24236/16).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

SECTION 112

SWITCHES, TOGGLE
(Applicable specifications: MIL-S-3950, MIL-S-8834, and MIL-S-8805)

FIGURE AND SPEC NUMBER	MS NUMBER (see NOTE)	SWITCH CHARACTERISTICS			PAGE
		Dash No. for:		Circuit characteristics	
		3 Position	2 Position		
112-1 - MIL-S-3950	MS24523	21,27,31	22,23,26,29,30	One pole	112.3
112-2 - MIL-S-3950	MS24524	21,27,31	22,23,26,29,30	Two pole	112.4
	MS24525	21,27,31	22,23,26,29,30	Four pole	
	MS27406	1,2,3	---	Four pole	
	MS27407	1,2,3	---	Two pole	
112-3 - MIL-S-3950	* MS24658	21,27,31	22,23,26,28,29, 30,32	One pole	112.5
112-4 - MIL-S-3950	* MS24659	21,27,31	22,23,26,28,29, 30,32	Two pole	112.6
	* MS24660	21,27,31	22,23,26,28,29, 30,32	Four pole	
	* MS27408	1,2,3	---	Two pole	
	* MS27409	1,2,3	---	Four pole	
112-5 - MIL-S-3950	MS27722	21,27,31	22,23,26,28,29, 30,32	One pole	112.7
	MS27784	21,27,31	22,23,26,28,29, 30,32	One pole	
112-6 - MIL-S-3950	MS27723	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Two pole	112.8
	MS27724	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Four pole	
112-7 - MIL-S-3950	MS27785	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Two pole	112.9
	MS27786	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Four pole	
112-8 - MIL-S-3950	* MS27782	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Two pole	112.10
	* MS27783	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Four pole	
112-9 - MIL-S-3950	* MS27788	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Two pole	112.11
	* MS27789	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Four pole	
112-10 - MIL-S-3950	* MS27781	21,27,31	22,23,26,28,29, 30,32	One pole	112.12
	* MS27787	21,27,31	22,23,26,28,29, 30,32	One pole	
112-11 - MIL-S-3950	MS27734	21,27,31	22,23,26,28,29, 30,32	One pole	112.13
112-12 - MIL-S-3950	MS27735	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Two pole	112.14
	MS27736	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Four pole	
112-13 - MIL-S-3950	* MS27737	21,27,31	22,23,26,28,29, 30,32	One pole	112.15
112-14 - MIL-S-3950	* MS27738	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Two pole	112.16
	* MS27739	1,2,3,21, 27,31	22,23,26,28,29, 30,32	Four pole	
112-15 - MIL-S-8834	MS14001	212,272,312	222,232,242, 262,282,292,302	One pole	112.17
	MS14002	212,272,312	222,232,242, 262,282,292,302	Two pole	
	MS14003	212,272,312	222,232,242, 262,282,292,302	Four pole	

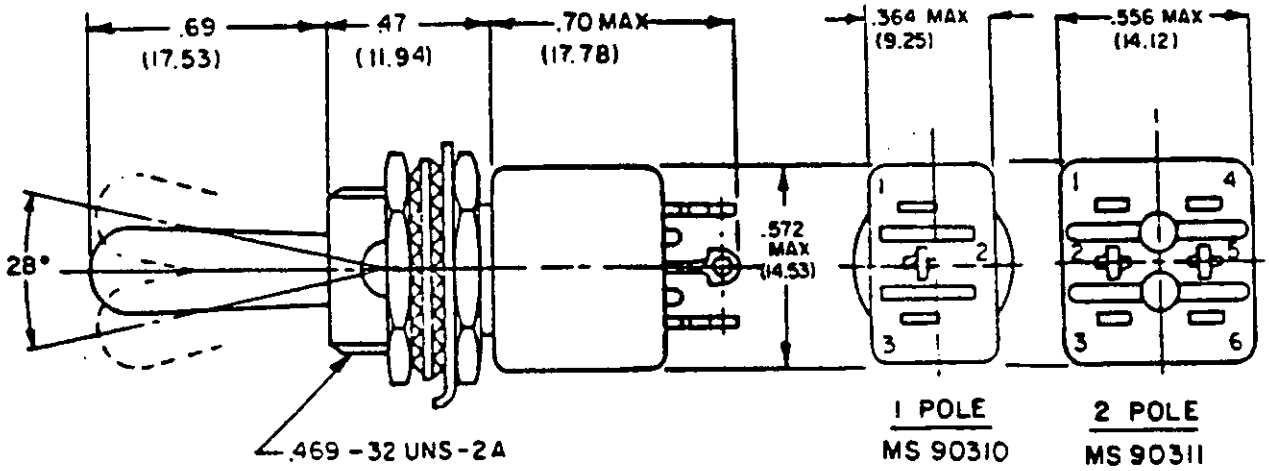
Source: https://assist.dla.mil -- Downloaded: 202640-26T07:41Z
Check the source to verify that this is the current version before use.

Figure and spec number	MS number (see note)	Switch characteristics			Page
		Dash no. for:		Circuit characteristics	
		3 Position	2 Position		
112-16 - MIL-S-8834	* MS21026	211,271,311	221,231,241,281	One pole	112.18
	* MS21027	211,311,371,711,771,811,87	221,231,241,381,721,731,741,781,881	Two pole	
112-17 - MIL-S-8834	MS21346	211,271,311	221,231,241,281	One pole	112.19
	MS21347	211,271,311,711,711,811	221,231,241,281,721,731,741,781	Two pole	
112-18 - MIL-S-8834	MS21350	---	321	One pole	112.20
	MS21351	---	321	One pole	
112-19 - MIL-S-8834	MS21352	331,341,351	321	Two pole	112.21
	MS21353	331,341,351,831,841,851	321,821	Two pole	
112-20 - MIL-S-8834	MS21354	211,271,311	221,231,241,281	One pole	112.22
	MS21355	211,271,311	221,231,241,281	Two pole	
112-21 - MIL-S-8834	MS21356	211,271,311	221,231,241,281	One pole	112.23
	MS21357	211,271,311	221,231,241,281	Two pole	
112-22 - MIL-S-8834	* MS24612	212,272,312	222,232,242,262,282,292,302	One pole	112.24
	* MS24613	212,272,312	222,232,242,262,282,292,302	Two pole	
	* MS24614	212,272,312	222,232,242,262,282,292,302	Four pole	
112-23 - MIL-S-8834	MS24655	211,271,311	221,231,241,281	One pole	112.25
	MS24656	211,271,311	221,231,241,281	Two pole	
112-24 - MIL-S-8834	MS25306	212,272,312	222,232,242,262,282,292,302	One pole	112.26
	MS25307	212,272,312	222,232,242,262,282,292,302	Two pole	
	MS25308	212,272,312	222,232,242,262,282,292,302	Four pole	
112-25 - MIL-S-8834	MS90310	211,271,311	221,231,241,281	One pole	112.27
	MS90311	211,271,311,711,771,811	221,231,241,281,721,731,741,781	Two pole	
112-25A - MIL-S-8834	MS21433	211,271,311	221,231,241,281,321	One pole	112.27A
	MS21434	211,271,311,331,341,351	221,231,241,281,321	Two pole	
112-25B - MIL-S-8834	MS21436	211,271,311	221,231,241,281	One pole	112.27E
	MS21437	211,311,331,341,351,371	221,231,241,321,381	Two pole	
112-26 - MIL-S-8805/26	*	005,006,007,008,009,010	001,003	Two pole	112.28
		011,012,013,014	---	Three pole	
		---	002,004	Four pole	
112-27 - MIL-S-8805/92	*	005,006,007,008,009,010,011,012,013,014,015,016,017,018	001	Two pole	112.30
		019	---	Three pole	
		---	002,003,004	Four pole	
112-28 - MIL-S-8805/93	*	001 thru 007	---	Two pole	112.32
		008 thru 014	---	Four pole	
		015 thru 021	---	Six pole	
		022 thru 028	---	Eight pole	
112-29 MIL-S-8805/98	*	001 thru 093	---	Four pole Six pole Eight pole	112.34

NOTE: The part number consists of the MS number followed by the dash number; however, an asterisk (*) indicates that the "MS" part number will require an additional letter, or an "M" part number is applicable, as shown in the figure.

This page supersedes page
112.2 of 6 June 1979

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T07:41Z
Check the source to verify that this is the current version before use.



MS dash number 1/		2/ Circuit with toggle lever in			Current capacity (amperes)						Life low current level switching 5 mV
					Resistive load			Inductive load			
		Opposite keyway side	Center	Keyway side	28 Vdc	115 Vac		28 Vdc	115 Vac		
						60 Hz	400 Hz		60 Hz	400 Hz	
											(-A)
-711	-211	ON	OFF	ON							
-721	-221	ON	NONE	OFF							
-731	-231	ON	NONE	ON							
-741	-241	ON	OFF	NONE	5	2	3	1	1	2	25
-771	-271	MOM-ON	OFF	MOM-ON							
-781	-281	NONE	OFF	MOM-ON							
-811	-311	ON	OFF	MOM-ON							

1/ 700 and 800 series have silicone bonded seals (applicable to MS90311 only); 200 and 300 series have mechanical seals.
2/ Applicable terminal numbers are marked on the switch.

PART NUMBER: Consists of the applicable MS number followed by the dash number as shown in the tabulation.

FIGURE 112-25. MS90310 and MS90311 toggle switches, miniature, toggle sealed, solder lug, .469 mounting bushing (MIL-S-8834).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

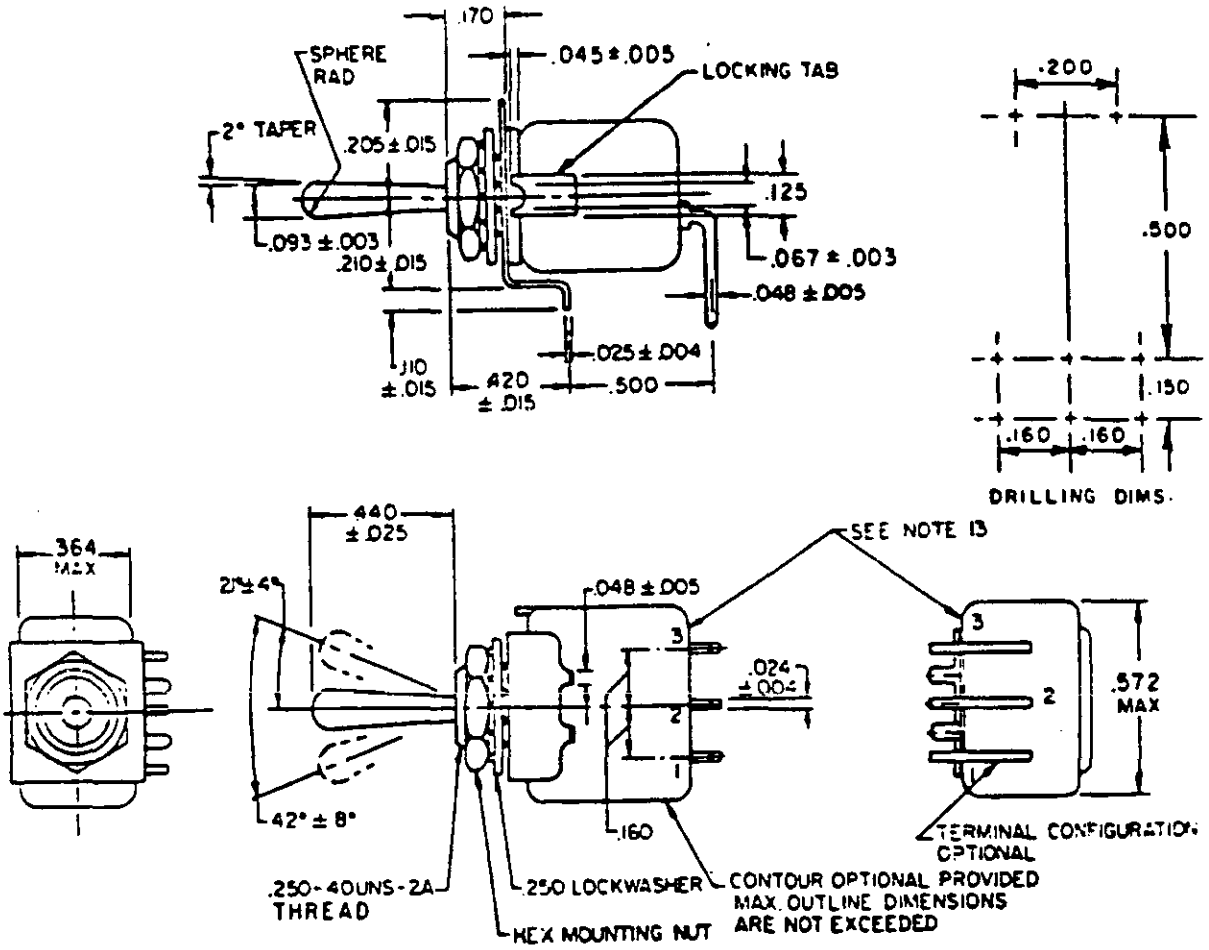


FIGURE 112-25A. Switches, toggle, positive break, miniature, toggle sealed, printed circuit board terminals, right angle mount (MS 21433).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

DETAIL REQUIREMENTS:

MS dash no.	Circuit made between terminals as indicated with toggle lever in these positions (a)			Current capacity in amperes						Life, low current level switching 5 MV
				Resistive load			Inductive load			
	Opposite locking tab side (d)	Center position	Locking tab side (d)	28 volts dc	115 volts 60 hertz ac	115 volts 400 hertz ac	28 volts dc	115 volts 60 hertz ac	115 volts 400 hertz ac	
-211	ON 2-3	OFF	ON 1-2	5	2	3	1 (b)	1	2	25 microamperes (c)
-221		NONE	OFF							
-231			ON 1-2							
-241		OFF	NONE							
-271	MOM-ON 2-3		MOM-ON 1-2							
-281	NONE									
-311	ON 2-3									
-321	NONE	ON 2-3	MOM-ON 1-2							

(a) Direction of movement of internal mechanism is opposite to the direction of the toggle movement.
(b) With the time constant of 020 ±.002 seconds.
(c) Contact resistance not to exceed 50 ohms during life, low current level switching.
(d) Non-functional terminals shall not be supplied.

FIGURE 112-25A Switches, toggle, positive break, minature, toggle sealed, printed circuit board terminals, right angle mount, (MS 21433) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

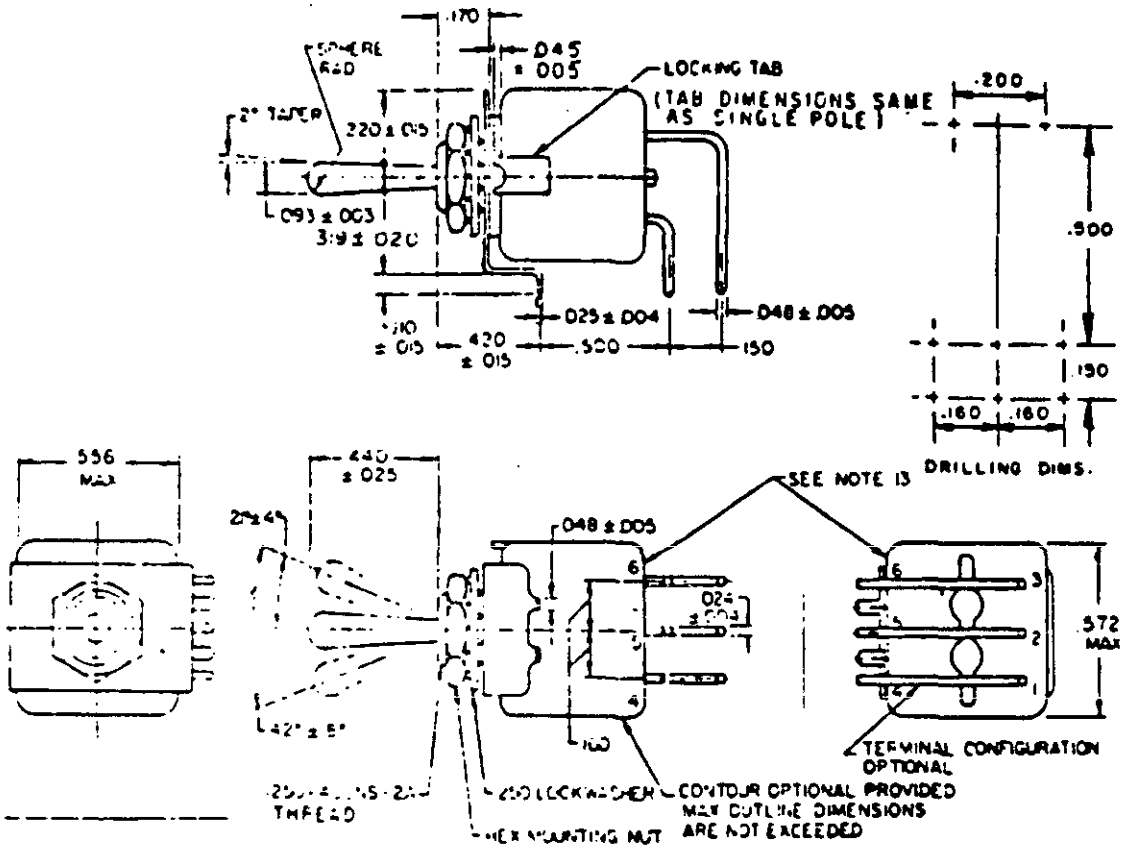


FIGURE 112-25A. Switches, toggle, positive break, miniature, toggle sealed, printed circuit board terminals, right angle mount (MS 21434).

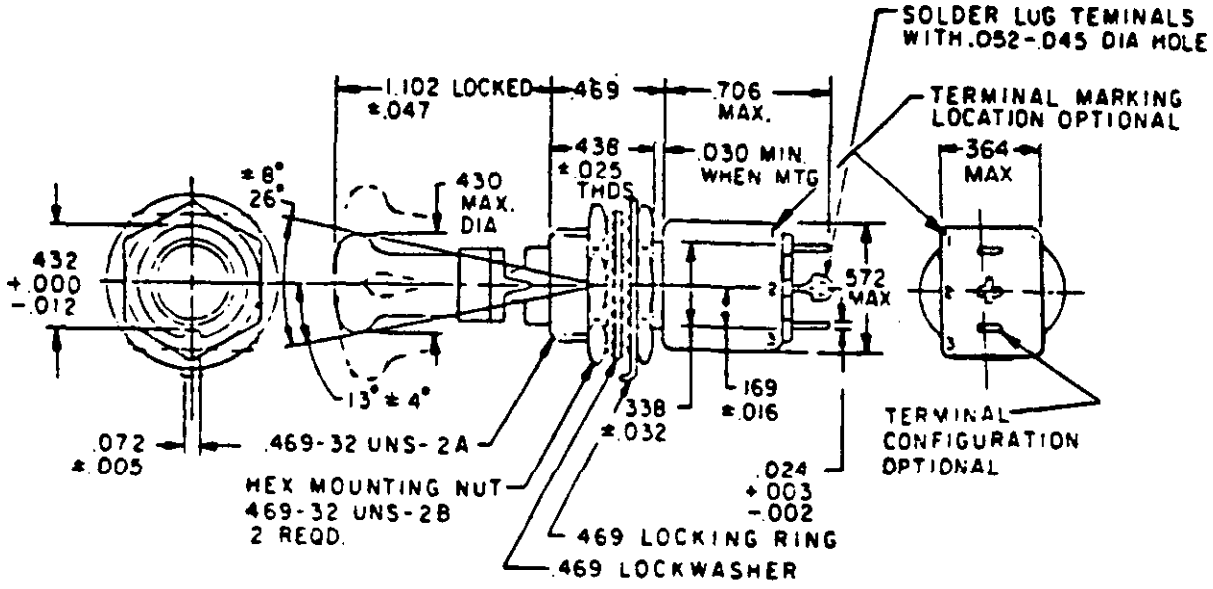
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

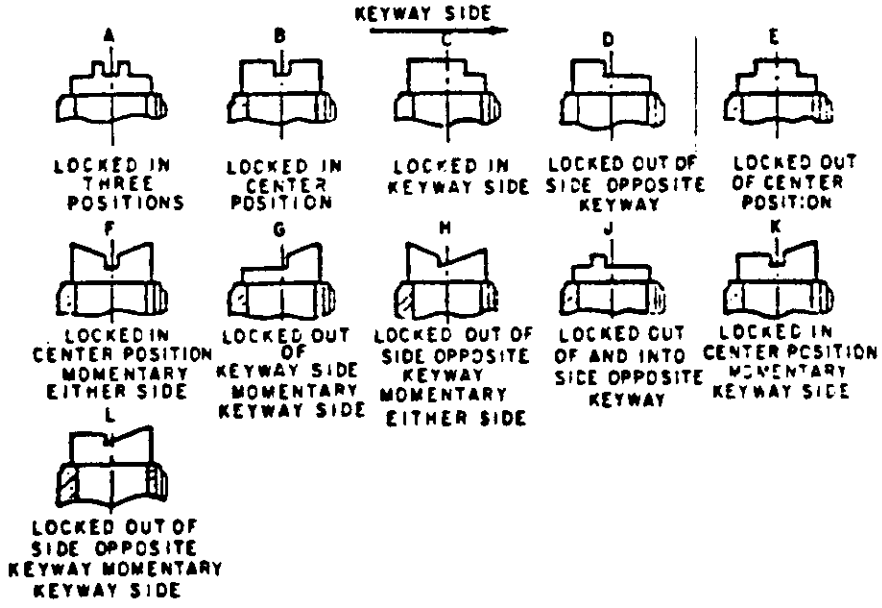
MS dash no.	Circuit made between terminals as indicated with toggle lever in these positions (a)			Current capacity in amperes						Life, low current level switching 5 mV
				Resistive load			Inductive load			
	Opposite locking tab side (d)	Center position	Locking tab side (d)	28 volts dc	115 volts 60 hertz ac	115 volts 400 hertz ac	28 volts dc	115 volts 60 hertz ac	115 volts 400 hertz ac	
-211	ON 2-3,5-6	OFF	ON 1-2,4-5	5	2	3	1 (b)	1	2	25 microamperes (c)
-221		NONE	OFF							
-231			ON 1-2,4-5							
-241		OFF	NONE							
-271	MOM-ON 2-3,5-6		MOM-ON 1-2,4-5							
-281	NONE									
-311	ON 2-3,5-6									
-331	ON 2-3,5-6	ON 2-3,4-5	ON 1-2,4-5							
-341	MOM-ON 2-3,5-6	ON 2-3,4-5	MOM-ON 1-2,4-5							
-321	NONE	ON 2-3,5-6	MOM-ON 1-2,4-5							
-351	ON 2-3,5-6	ON 2-3,4-5	MOM-ON 1-2,4-5							

(a) Direction of movement of internal mechanism is opposite to the direction of the toggle movement.
(b) With the time constant of .020 ±.002 seconds.
(c) Contact resistance not to exceed 50 ohms during life, low current level switching.
(d) Non-functional terminals shall not be supplied.

FIGURE 112-25A. Switches, toggle, positive break, miniature, toggle sealed, printed circuit board terminals, right angle mount (MS 21434) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.





FIGURES A THRU L ILLUSTRATE THE LOCKING AND MOMENTARY CONFIGURATION AND NOT THE DETAIL CONSTRUCTION OF THE LOCKING DESIGN.

FIGURE 112-25B. Switch, toggle, positive break, lever lock, miniature, toggle sealed, solder lug, .469 mounting bushing (MS21436).

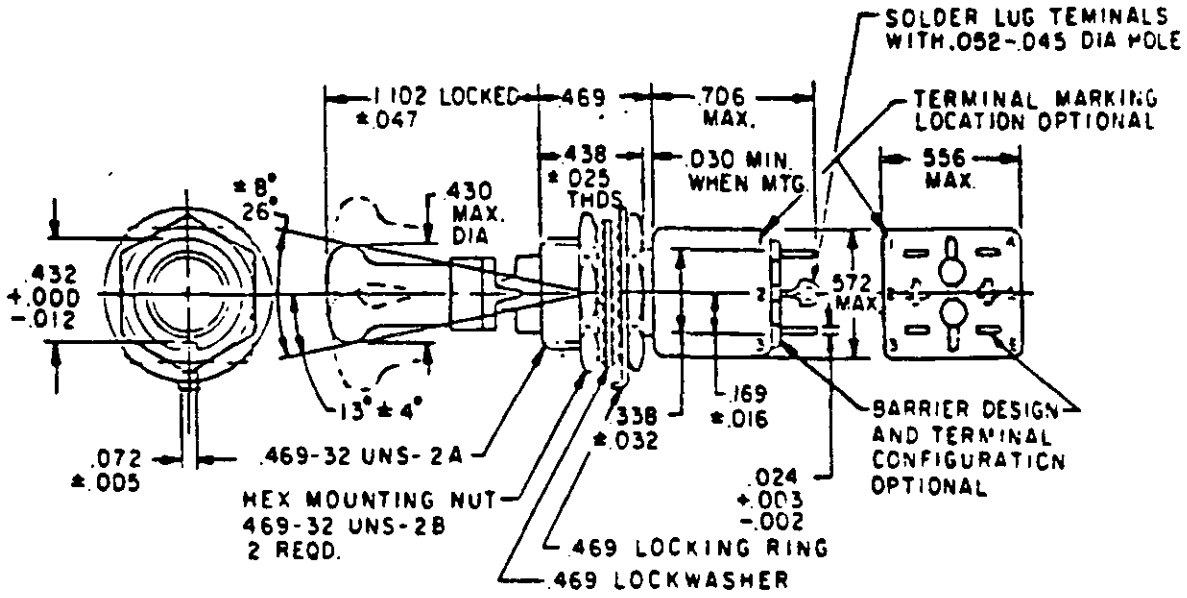
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

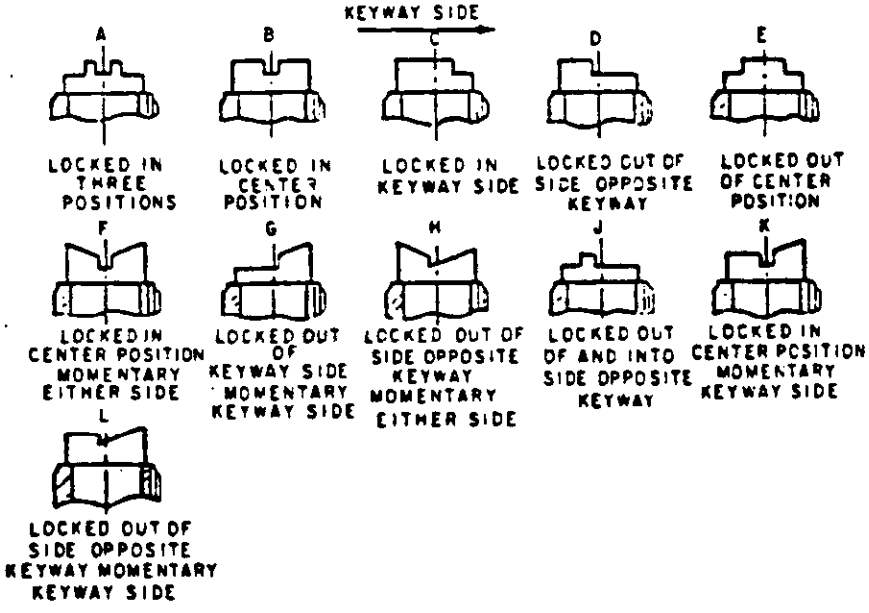
MS dash no.	Locking comb.	Circuit made between terminals as indicated with the toggle lever in these positions: (a)			Current cacacity in amperes						Life low current level switching 5 mV
					Resistive load			Inductive load			
					28 volts dc	115 volts		28 volts dc	115 volts		
		Opposite keyway side (d)	Center position	Keyway side (d)		60 hertz ac	400 hertz ac		60 hertz ac	400 hertz ac	
-211	A,B,D	ON 2-3	OFF	ON 1-2	5	3	2	1	1	2	25 A
-221	C,E		NONE	OFF							
-231				ON 1-2							
-241	J,B		OFF	NONE							
-271	F,H	MOM-ON 2-3	OFF	MOM-ON 1-2							
-281	K	NONE									
-311	C,K,L	ON 2-3									

(a) Direction of movement of internal mechanism is opposite to the direction of the toggle movement.
(b) With time constant of .020 ±.002 (.51 mm ±.05 mm) seconds.
(c) Contact resistance not to exceed 50 ohms during life, low current level switching.
(d) Non-functional terminals shall not be supplied.

FIGURE 112-25B. Switch, toggle, positive break, lever lock, miniature, toggle sealed, solder lug, .469 mounting bushing (MS 21436) - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.





[1] FIGURES A THRU L ILLUSTRATE THE LOCKING AND MOMENTARY CONFIGURATION AND NOT THE DETAIL CONSTRUCTION OF THE LOCKING DESIGN.

FIGURE 112-25B. Switch, toggle, positive break, lever lock, miniature, toggle sealed, solder lug, .469 mounting bushing (MS 21437) - Continued.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T07:41Z
Check the source to verify that this is the current version before use.

MS dash no.	Locking comb.	Circuit made between terminals as indicated with the toggle lever in these positions: (a)			Current capacity in amperes						Life low current level switching 5 mV
					Resistive load			Inductive load			
					28 volts dc	115 volts		28 volts dc	115 VOLTS		
		Opposite keyway side (d)	Center position	Keyway side (d)		60 hertz ac	400 hertz ac		60 hertz ac	400 hertz ac	
-211	A,B,D	ON 2-3,5-6	OFF	ON 1-2,4-5	5	2	3	1	1	2	25 A
-221	C,E		NONE	OFF				(b)			(c)
-231				ON 1-2,4-5							
-241	J,B		OFF	NONE							
				MOM-ON 1-2,4-5							
-311	G,K,L										
-321	K	NONE	ON 2-3,5-6	MOM-ON 1-2,4-5				(b)			(c)
-331	A,B,C,D	ON 2-3,5-6	ON 2-3,4-5	ON 2-3,4-5							
-341	F,H	MOM-ON 2-3,5-6	ON 2-3,4-5	MOM-ON 2-3,4-5	5	2	3	1	1	2	25 A
-351	G,K,L	ON 2-3,5-6	ON 2-3,4-5	MOM-ON 2-3,4-5							
-371	F,H	MOM-ON 2-3,5-6	OFF	MOM-ON 2-3,4-5							
-381	K	NONE	OFF	MOM-ON 1-2,4-5							

(a) Direction of movement of internal mechanism is opposite to the direction of the toggle measurement.
(b) With time constant of .020 ±.002 seconds.
(c) Contact resistance not to exceed 50 ohms during life, low current level switching.
(d) Non-functional terminals shall not be supplied.

FIGURE 112-25B. Switch, toggle, positive break, lever lock, miniature, toggle sealed, solder lug, .469 mounting bushing (MS 21437) - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.

LOAD	ELECTRICAL RATINGS		
	Sea level		50,000 feet
	28 Vdc	115 Vac, 60 Hz	28 Vdc
	amperes	amperes	amperes
Resistive - - -	5	5	5
Inductive - - -	3	5	2.5
Lamp- - - - - -	2.4	1.5	2.4

INCHES	MM
.060	1.52
.115	2.92
.36	9.14
.470	11.94
.500	12.70
.720	18.29
.860	21.84
.990	25.15
1.125	28.58
1.160	29.46

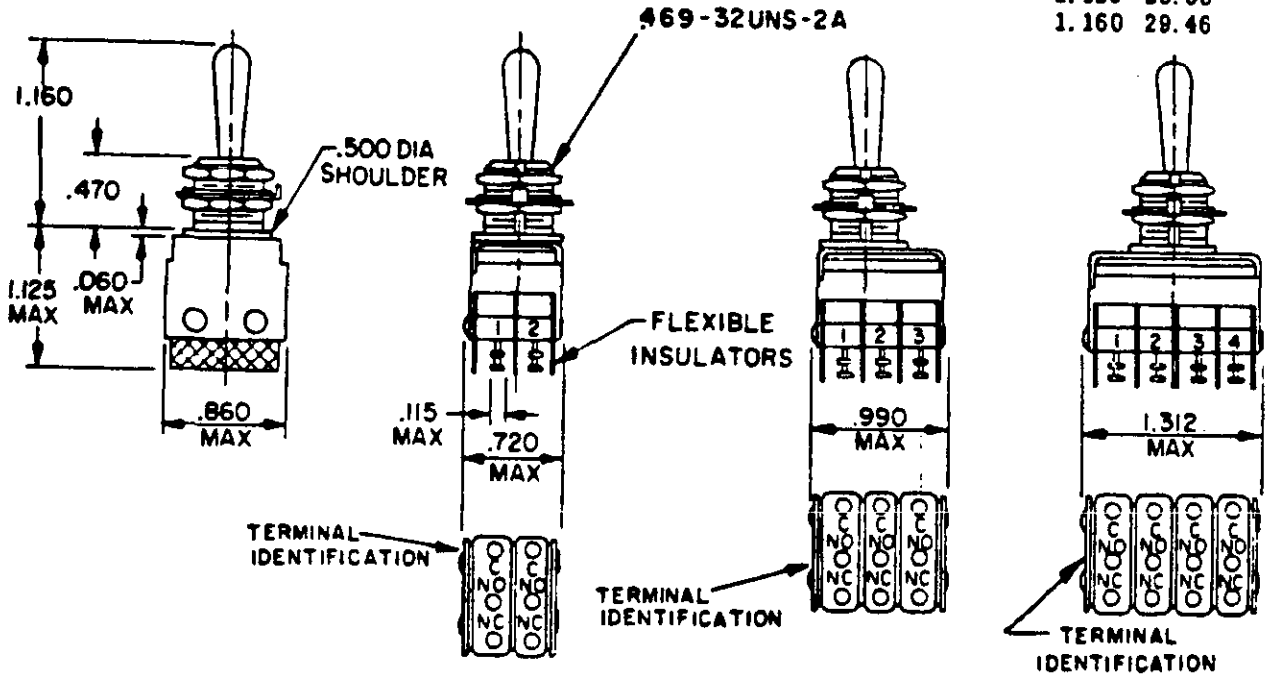
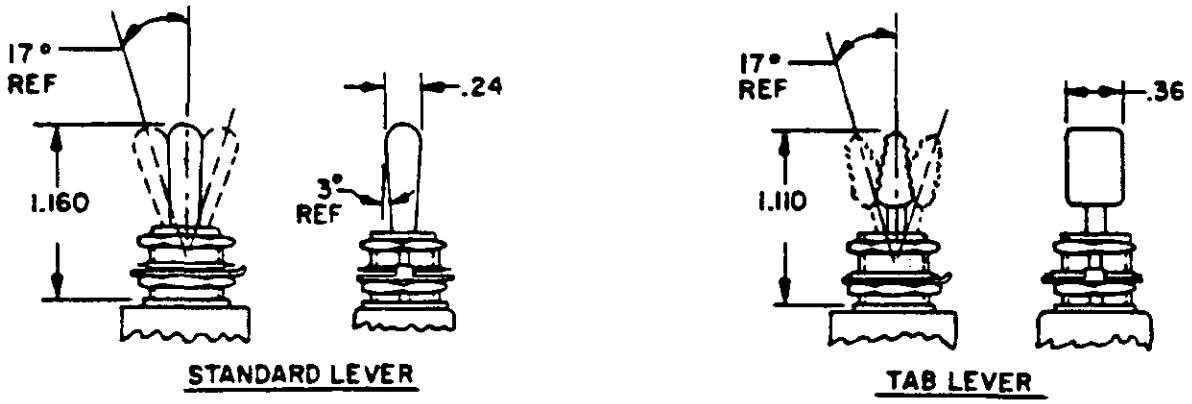


FIGURE A FIGURE B FIGURE C



STANDARD LEVER TAB LEVER

FIGURE 112-26. Switch assemblies, sensitive, toggle lever (with integral leaf), 5 amperes, unsealed (MIL-S-8805/26).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

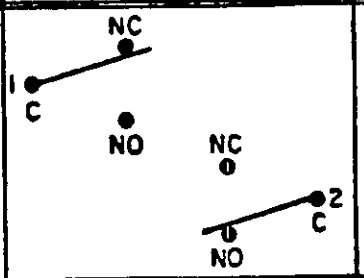
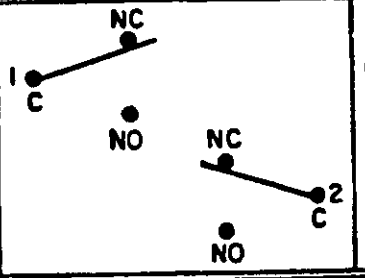
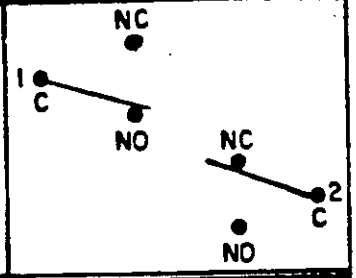
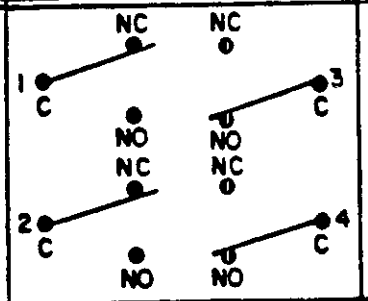
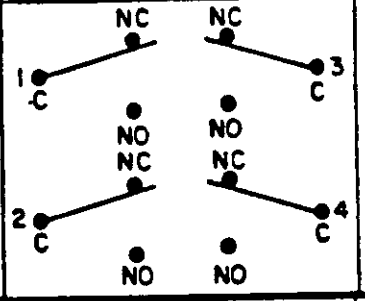
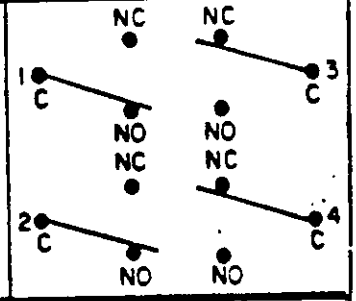
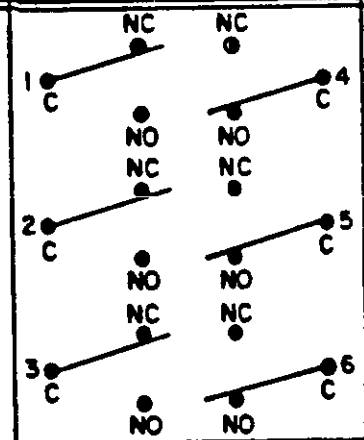
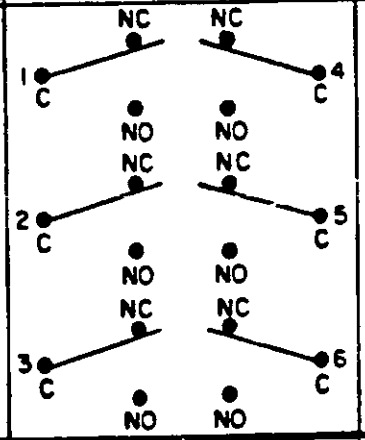
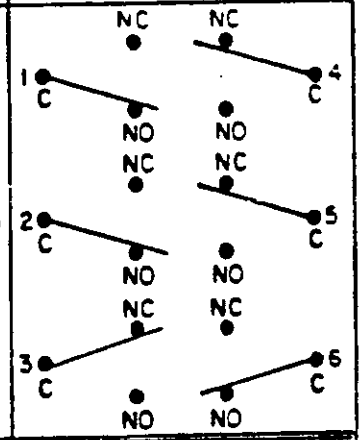
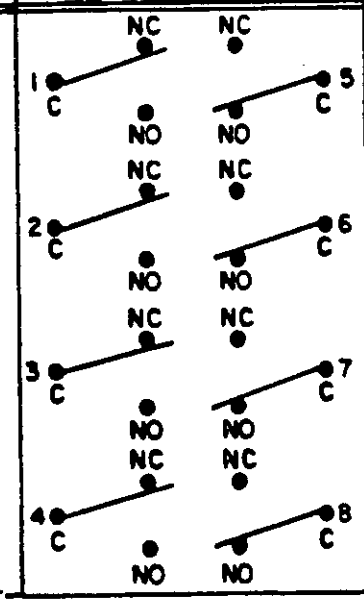
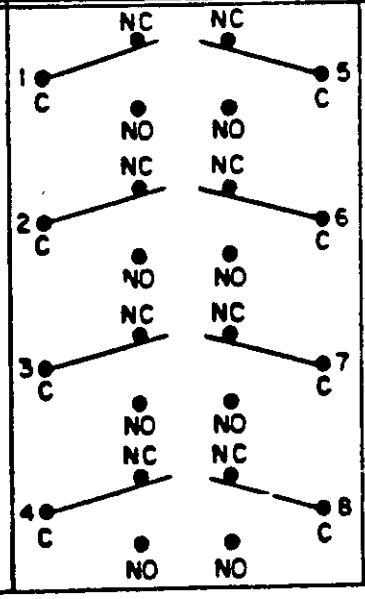
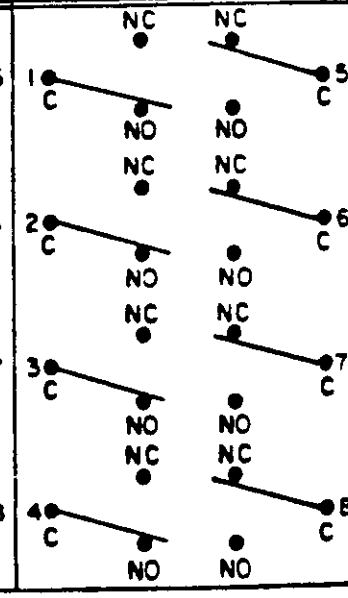
TYPE	SCHEMATIC DIAGRAM		
	POSITION NO.1 ("D" FLAT)	POSITION NO.2 (CENTER)	POSITION NO.3 (OPPOSITE)
I			
II			
III			
IV			

FIGURE 112-28. Switch assemblies, sensitive, lever, unsealed, 7 amperes (MIL-S-8805/93) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Load	Sea level		50,000 ft
	28 Vdc	115 Vac, 60 Hz	28 Vdc
	(Amperes)	(Amperes)	(Amperes)
Resistive	7	7	4
Inductive	4	7	2.5

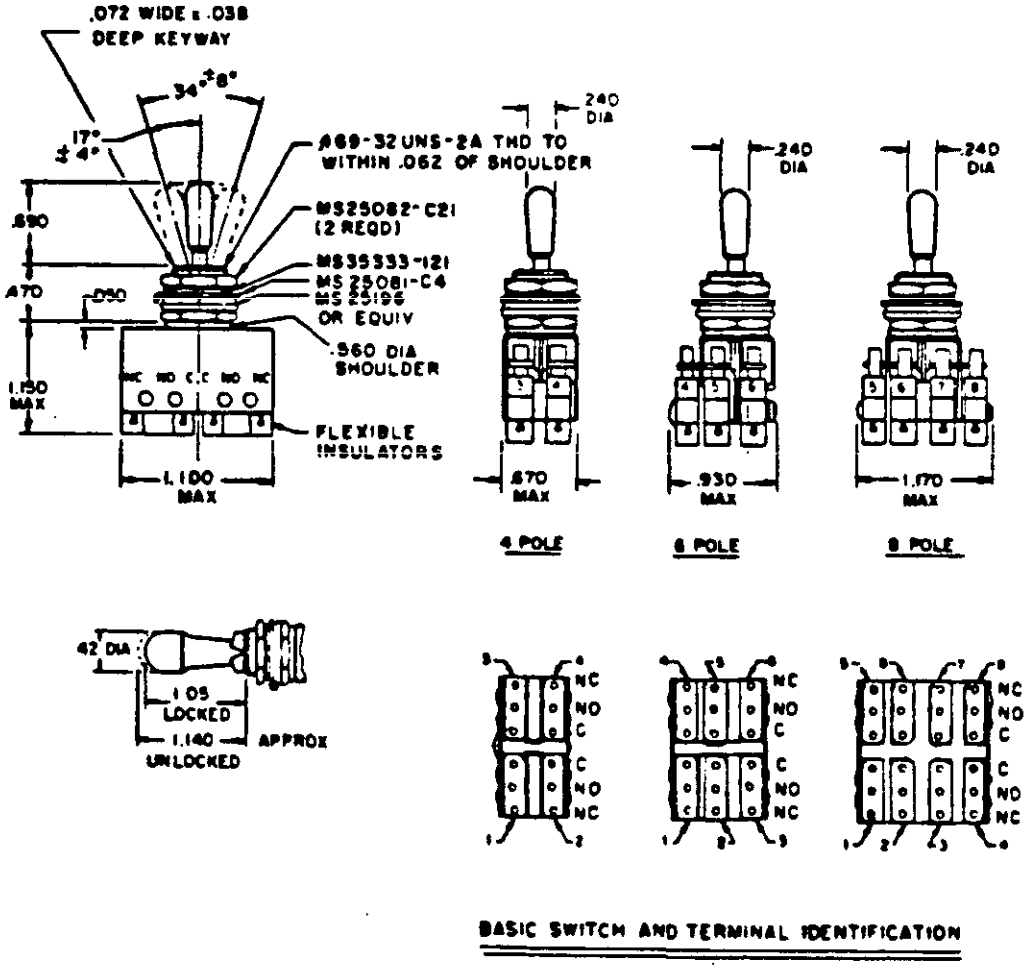
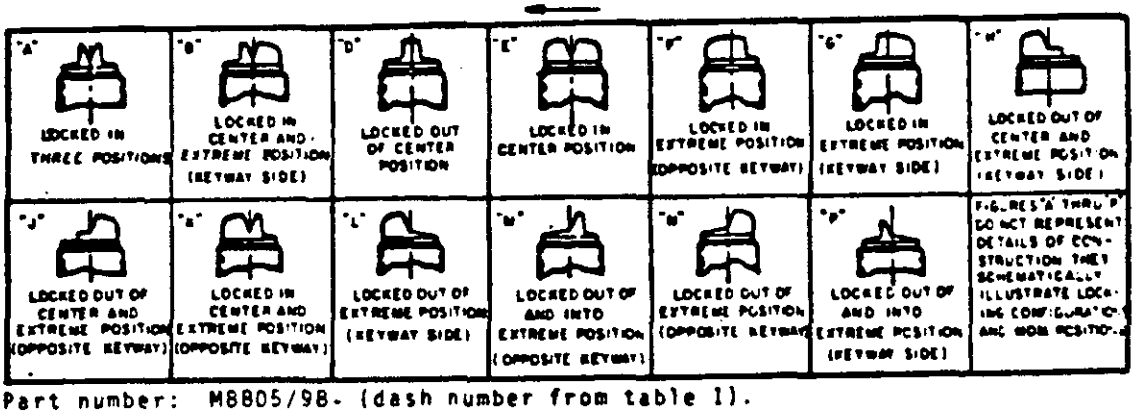


FIGURE 112-29. Switch assembly, sensitive, toggle, 7 amperes, panel seal (MIL-S-8805/98).

Supersedes page 112.34
of 6 June 1979

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

(KEYWAY SIDE)



Part number: M8805/98- (dash number from table I).

TABLE I. Dash number and characteristics.

Dash number	No. of poles	Level action and circuitry (table II)	Locking combination
-001	4 pole	1	None
-002			A
-003			B
-004			D
-005			E
-006			F
-007			G
-008			H
-009			J
-010			K
-011			L
-012			M
-013			N
-014			P
-015		3	None
-016			D
-017			F
-018			G
-019		5	None
-020			E
-021			F
-022			K
-023			L
-024			M
-025			N
-026		7	None
-027			E
-028			L
-029			N
-030		8	None
-031			F
-032	6 pole	1	None
-033			A
-034			B
-035			D
-036			E
-037			F
-038			G
-039			H
-040			J
-041			K
-042			L
-043			M
-044			N
-045			P

Supersedes page 112.35 of 6 June 1979

FIGURE 112-29. Switch assembly, sensitive, toggle, 7 amperes, panel seal (MIL-S-8805/98) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

TABLE I. Dash number and characteristics - Continued.

Dash number	No. of poles	Level action and circuitry (table II)	Locking combination
-046	6 pole	3	None
-047			D
-048			F
-049			G
-050		5	None
-051			E
-052			F
-053			K
-054			L
-055			M
-056			N
-057		7	None
-058			E
-059			L
-060			N
-061		8	None
-062			F
-063	8 pole	1	None
-064			A
-065			B
-066			D
-067			E
-068			F
-069			G
-070			H
-071			J
-072			K
-073			L
-074			M
-075			N
-076			P
-077		3	None
-078			D
-079			F
-080			G
-081		5	None
-082			E
-083			F
-084			K
-085			L
-086			M
-087			N
-088		7	None
-089			E
-090			L
-091			N
-092		8	None
-093			F

FIGURE 112-29. Switch assembly, sensitive, toggle, 7 amperes, panel seal (MIL-S-8805/98) - Continued.

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:41Z
Check the source to verify that this is the currentversion before use.

TABLE II. Lever action and circuitry.

Type number	Basic switch numbers			With toggle level in position					
				Keying side		Center		Opposite keying side	
	For 4 pole	For 6 pole	For 8 pole	Toggle action	Circuit closed	Toggle action	Circuit closed	Toggle action	Circuit closed
1	1-2	1-2-3	1-2-3-4	Maintain	C-NC	Maintain	C-NC	Maintain	C-NO
	3-4	4-5-6	5-6-7-8		C-NO		C-NC		C-NC
3	1-2	1-2-3	1-2-3-4	Maintain	C-NC	(None)		Maintain	C-NO
	3-4	4-5-6	5-6-7-8		C-NO				C-NC
5	1-2	1-2-3	1-2-3-4	Momentary	C-NC	Maintain	C-NC	Maintain	C-NO
	3-4	4-5-6	5-6-7-8		C-NO		C-NC		C-NC
7	1-2	1-2-3	1-2-3-4	Momentary	C-NC	Maintain	C-NC	Momentary	C-NO
	3-4	4-5-6	5-6-7-8		C-NO		C-NC		C-NC
8	1-2	1-2-3	1-2-3-4	Momentary	C-NC	(None)		Maintain	C-NO
	3-4	4-5-6	5-6-7-8		C-NO				C-NC

FIGURE 112-29. Switch assembly, sensitive, toggle, 7 amperes, panel seal (MIL-S-8805/98) - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.

SECTION 113

SWITCH ASSEMBLIES, MODULAR

(Applicable specification: MIL-S-22885)

TABLE 205.I. Switch assemblies, parameters and characteristic.

Figure number	Applicable specification sheet	Part number	Operation	Circuit characteristic	Operating force
					(lbs max)
113-1	MIL-S-22885/11	M22885/11			
		-01	Momentary	2PDT	4
		-02	Momentary	4PDT	4
		-03	Alternate	2PDT	6
		-04	Alternate	4PDT	7
113-2	MIL-S-22885/26	M22885/26			
		-01	Momentary	2 two-circuit	4
		-02	Alternate	2 two-circuit	6
		-03	Momentary	4 two-circuit	4
		-04	Alternate	4 two-circuit	7
		-05	Momentary	2 four-circuit	4
		-06	Alternate	2 four-circuit	7
113-3	MIL-S-22885/88	M22885/88			
		-01	Momentary	2PDT	4
		-02	Momentary	4PDT	4
		-03	Alternate	2PDT	6
		-04	Alternate	4PDT	7

This page supersedes page
113.1 of 23 March 1976

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

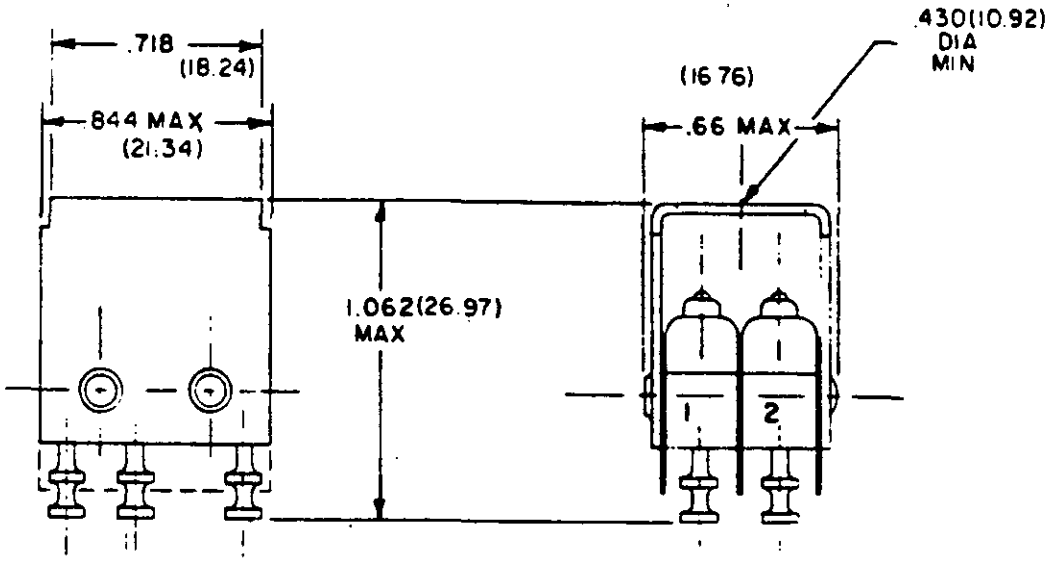


FIGURE 113-1A. Two pole, double throw, momentary action switch, part no. M22885/11-01.

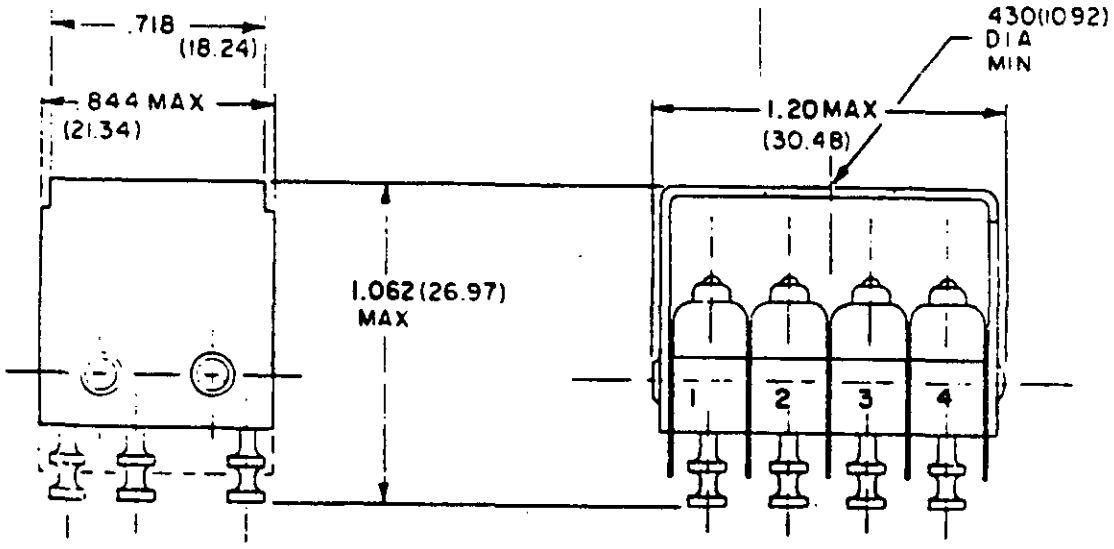


FIGURE 113-1B. Four pole, double throw, momentary action switch, part no. M22885/11-02.

FIGURE 113-1. Switch assemblies (MIL-S-22885/11).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

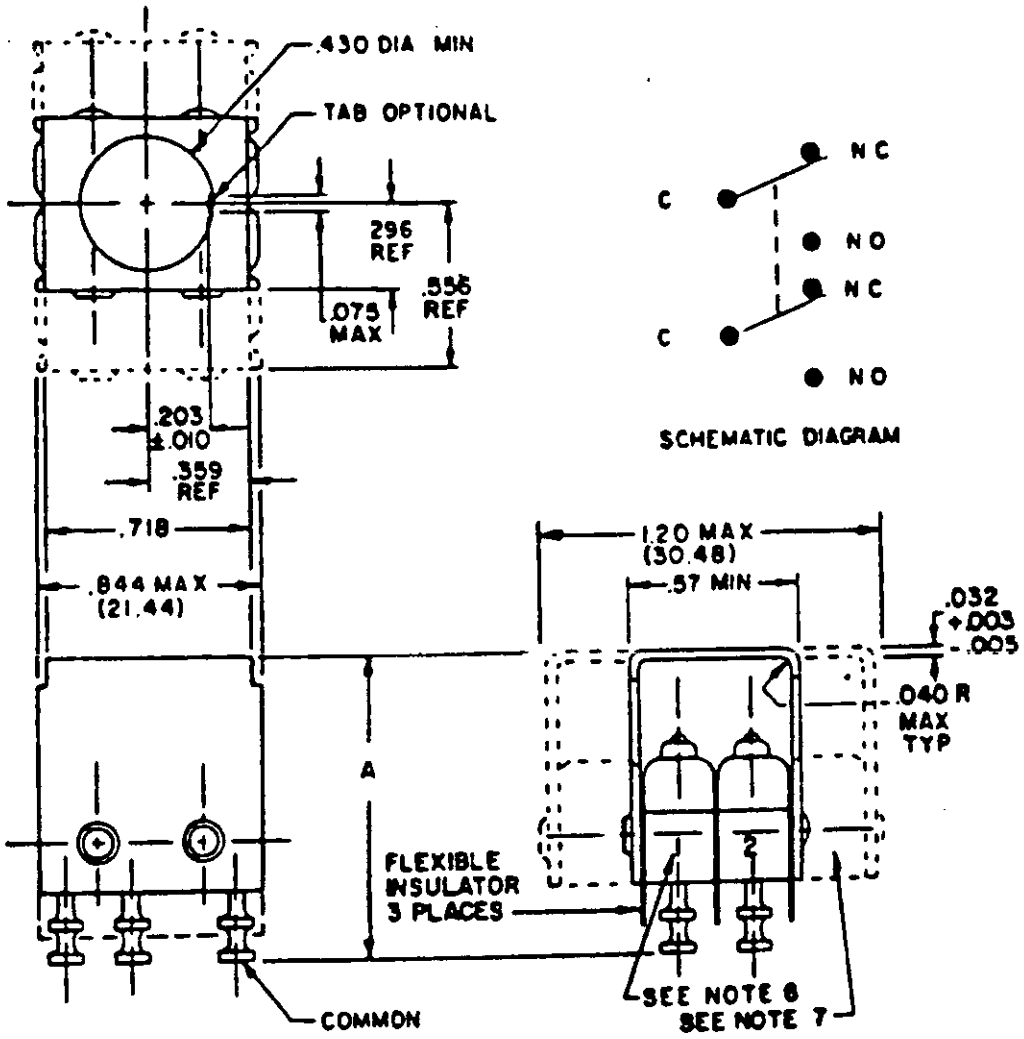


FIGURE 113-3A. Two pole, double throw switch assembly.

FIGURE 113-3. Switch, push button, illuminated, high impact shock, modular switch assemblies for

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

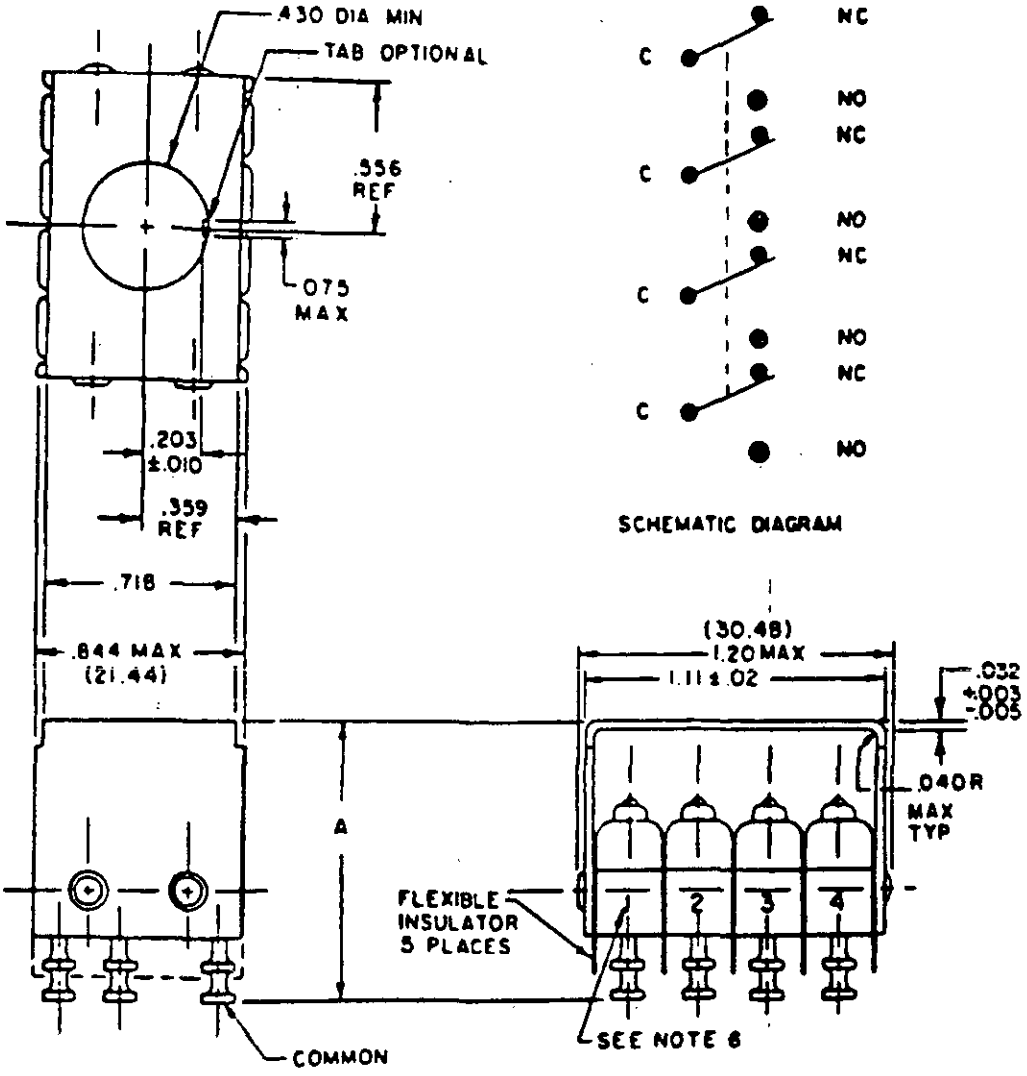


FIGURE 113-1B. Four pole, double throw switch assembly.

FIGURE 113-3. Switch, push button, illuminated, high impact shock, modular switch assemblies for

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Electrical ratings for basic switch.

Load	Sea level		50,000 feet
	28 Vdc	115 Vac	28 Vdc
	(amperes)	(amperes)	(amperes)
Resistive	5.0	5.0	5.0
Inductive	3.0	5.0	2.5
Lamp	2.4	1.5	2.4

Procurement Document: MIL-S-22885/88

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

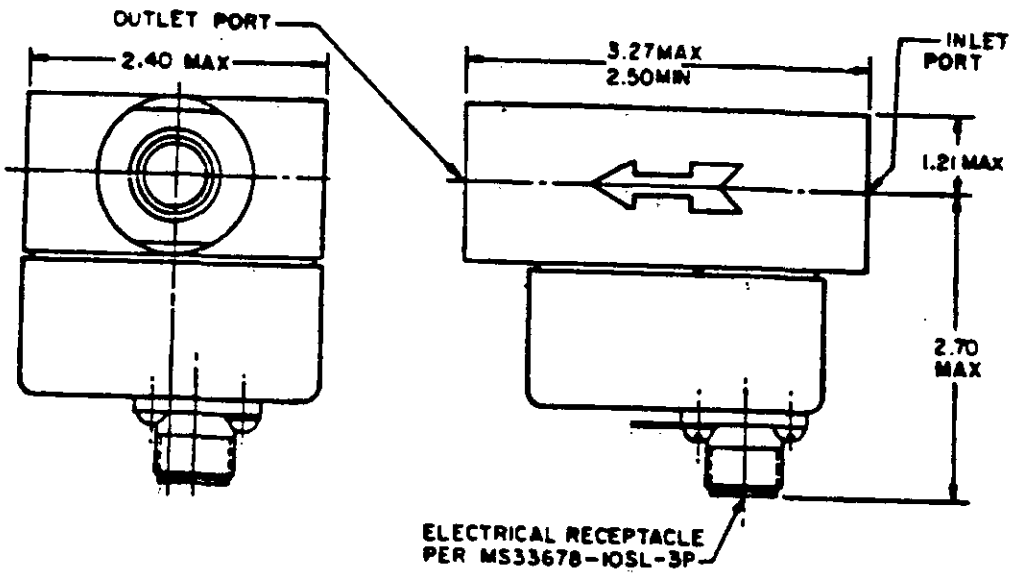
SECTION 114A

SWITCHES, AIR AND LIQUID FLOW
(Applicable specification: MIL-S-28788)

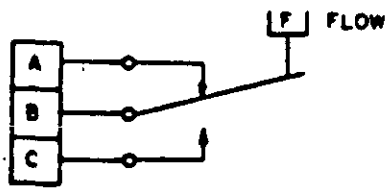
Figure and spec sheet no.	Switch characteristics				Page
	Electrical rating at 28 Vdc (amperes)	Operating temperature	Proof pressure	Burst pressure	
114-1 - MIL-S-28788/1	3 inductive	-32° to 185°F	150 psig	300 psig	114.2

This section supersedes
section 114 dated 6 June 1979

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.



INCHES	MM
1.21	30.7
2.70	68.6
2.40	61.0
2.50	63.5
3.27	83.1



WIRING DIAGRAM
(SHOWN WITH ZERO FLOW)

Military part number	Port size per MS16142		Switch flow rates		Pressure drop during flow	
	Inlet	Outlet	Actuation increasing GPM (max) at 104° ±9°F	Deactuation decreasing GPM (min) at 104° ±9°F	PSIG (max)	Rated flow GPM
M28788/1-1	-6	-6	.85	.60	5	1.00
-2	-6	-6	.42	.27	5	.50
-3	-6	-6	.60	.40	4.5	.66
-4	-6	-6	.65	.45	5	.75
-5	-8	-8	3.70	2.90	5	4.0
-6	-6	-6	1.30	1.00	5	1.50
-7	-6	-6	1.80	1.50	5	2.00
-8	-6	-6	2.30	2.00	5	2.50
-9	-8	-8	2.70	2.40	5	3.00
-10	-8	-8	3.70	3.20	7	4.00
-11	-8	-8	4.50	4.00	7	5.00
-12	-8	-8	5.40	4.80	7	6.00

FIGURE 114-1. Switches, liquid flow (MIL-S-28788/1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

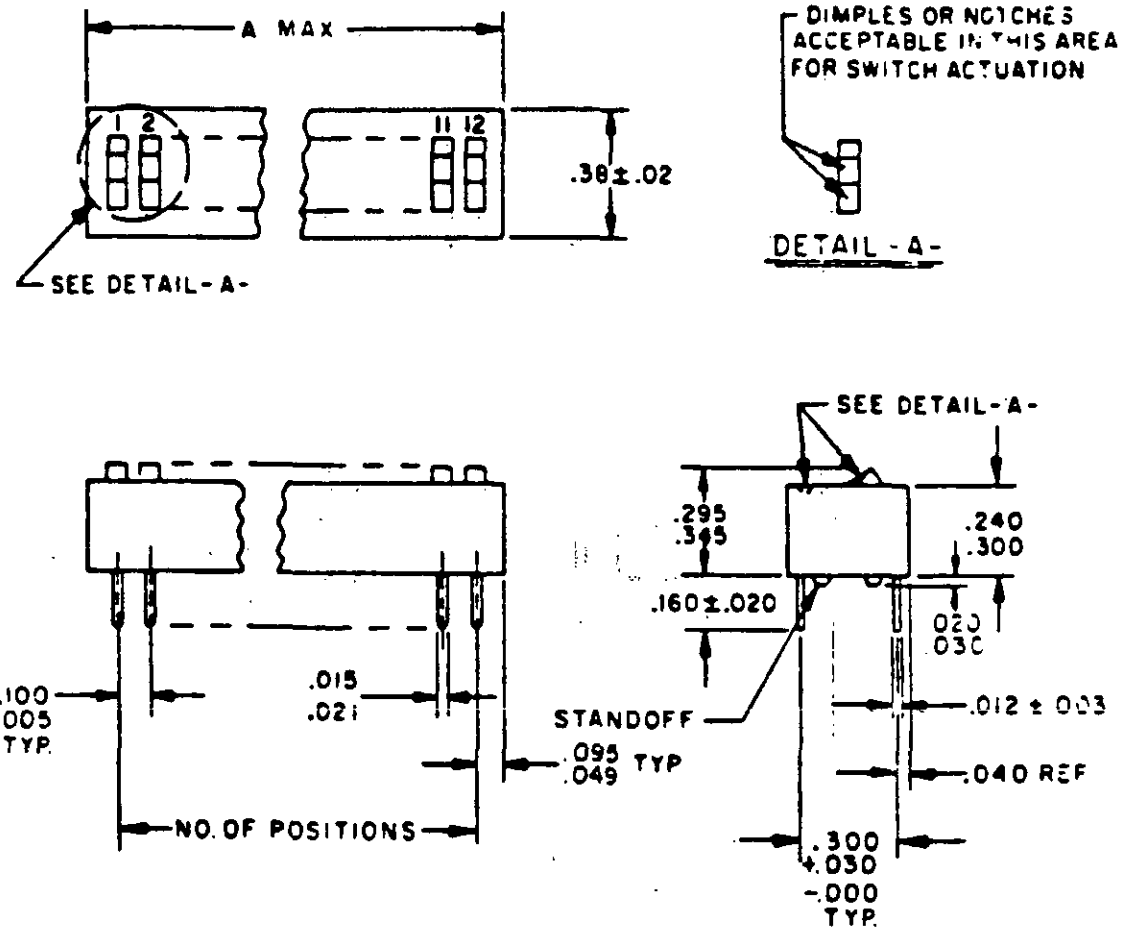
SECTION 115

SWITCHES, DUAL IN-LINE PACKAGE (DIP)
(Applicable specification: MIL-S-83504)

Figure number	Switch characteristics				
	Applicable specification sheet	Operating force		Base	Actuator
		Min	Max		
		(oz.)	(oz.)		
115-1	MIL-S-83504/1	6	32	Sealed	Rocker
115-2	MIL-S-83504/2	6	32	Sealed	Rocker

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:41Z
Check the source to verify that this is the current version before use.

Electrical ratings: 10 mA maximum at 50 mV, switching (resistive load).
100 mA maximum at 6 Vdc, switching (resistive load).
25 mA maximum at 24 Vdc, switching (resistive load).



Part number: M83504/01- (and dash number from table I).

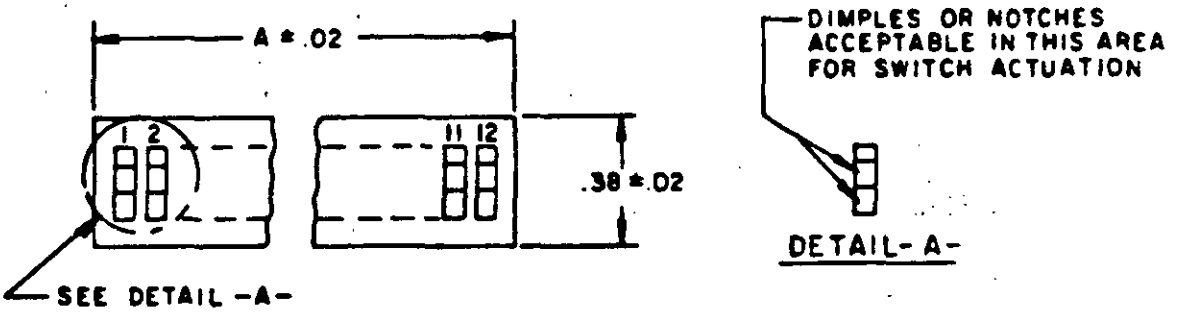
TABLE I. Dash numbers and characteristics.

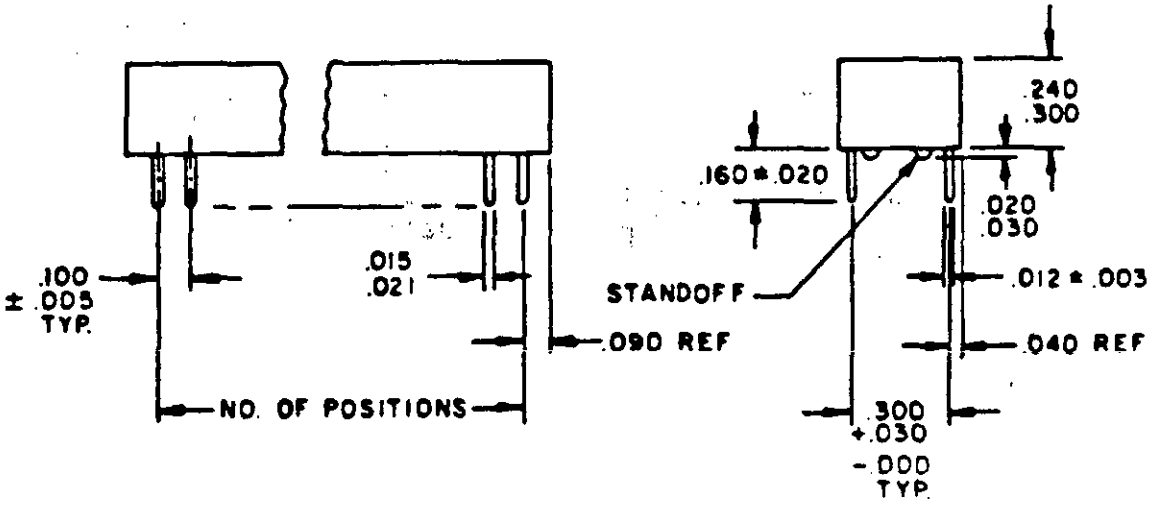
Dash number		Number of stations	Dimension "A" inches (mm)
Gold	Solderable		
002	022	2	.300 (7.62)
003	023	3	.400 (10.16)
004	024	4	.500 (12.70)
005	025	5	.600 (15.24)
006	026	6	.700 (17.78)
007	027	7	.800 (20.32)
008	028	8	.900 (22.86)
009	029	9	1.000 (25.40)
010	030	10	1.100 (27.94)
		11	1.200 (30.48)
	032	12	1.300 (33.02)

FIGURE 115-1. Switches, dual-in-line package (DIP), rocker actuator, standard profile, single pole, single throw stations, sealed base.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:41Z
Check the source to verify that this is the current version before use.

Electrical ratings: 10 mA maximum at 50 mV, switching (resistive load).
100 mA maximum at 6 Vdc, switching (resistive load).
25 mA maximum at 24 Vdc, switching (resistive load).





Part number: M83504/02- (and dash number from table I).

TABLE I. Dash numbers and characteristics.

Dash number		Number of stations	Dimensions "A" inches (mm)
Gold	Solderable		
002	022	2	.280 (7.11)
003	023	3	.380 (9.65)
004	024	4	.480 (12.19)
005	025	5	.580 (14.73)
006	026	6	.680 (17.27)
007	027	7	.780 (19.81)
008	028	8	.880 (22.35)
009	029	9	.980 (24.89)
010	030	10	1.080 (27.43)
		11	1.180 (29.97)
	032	12	1.280 (32.51)

FIGURE 115-2. Switches, dual-in-line package (DIP), rocker actuator, low profile, single pole, single throw stations, sealed base.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:41Z
Check the source to verify that this is the current version before use.